

Supercharge Your IBM Storage FlashSystem: Performance Tips and Tricks

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**Supercharge Your IBM Storage FlashSystem:
Performance Tips and Tricks**

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Note: Before using this information and the product it supports, read the information in “Notices” on page ix.

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Preface

IBM® Storage FlashSystem is an enterprise-class all-flash storage system that delivers high performance, scalability, and data protection for demanding workloads.

To fully realize the potential of IBM Storage FlashSystem storage, a thorough understanding of their architecture, features, and optimal configuration is essential. This book serves as your comprehensive guide to maximizing IBM Storage FlashSystem performance, providing practical techniques, best practices, and real-world examples to help you fine-tune your environment and achieve the speed and efficiency your organization demands.

This book is intended for experienced storage, SAN, IBM Storage FlashSystem, and IBM SAN Volume Controller (SVC) administrators and technicians. Understanding this book requires advanced knowledge of these environments.

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IBM Storage FlashSystem performance fundamentals

This chapter provides an overview of the fundamental aspects of IBM Storage FlashSystem (FlashSystem) performance, encompassing monitoring methodologies, available tools, established best practices, and relevant techniques as of the publication date of this IBM Redbooks.

The chapter includes the following topics:

- ▶ 1.1, “Key performance metrics” on page 2
- ▶ 1.2, “FlashSystem architecture overview” on page 3
- ▶ 1.3, “Workload characterization and its importance” on page 11
- ▶ 1.4, “Performance monitoring tools” on page 11

The IBM Storage FlashSystem product family currently comprises seven distinct models, each possessing unique hardware specifications and software capabilities:

- ▶ [IBM Storage FlashSystem 5015 and 5045](#)
- ▶ [IBM Storage FlashSystem 5300](#)
- ▶ [IBM Storage FlashSystem C200](#)
- ▶ [IBM Storage FlashSystem 7300](#)
- ▶ [IBM Storage FlashSystem 9500](#)
- ▶ [IBM SAN Volume Controller SV3](#)

Notably, all members of this family are unified by their usage of the [IBM Storage Virtualize](#) software platform.

This shared software foundation not only helps ensure a consistent user experience across the entire IBM Storage FlashSystem family but also provides unified management of performance and performance monitoring. Therefore, all models can be administered by using the same methodologies for performance analysis and monitoring.

The objective of this chapter is to provide a foundational understanding of the performance monitoring capabilities within FlashSystem environments. Upon completion, you should possess a basic comprehension of how to monitor performance and identify the critical performance metrics essential for evaluating your environment's workload characteristics.

1.1 Key performance metrics

Performance metrics are quantifiable measures that are used to evaluate the performance of IBM FlashSystem storage solutions. Key performance metrics are described in Table 1-1.

Table 1-1 Key performance metrics

Performance metric	Description
Latency	Latency denotes the time delay between a data request and the system's response, also known as response time. Shorter latencies signify superior performance and are typically measured in milliseconds (ms) or even microseconds (µs).
IOPS (Input/Output operations per second)	This metric quantifies the number of read and write operations a storage system can process per second. Although higher values might suggest improved performance, it is crucial to consider the associated latency, which significantly impacts overall efficiency.
Throughput	This metric, representing the data transfer rate, measures the volume of data that is moved within a specific time frame between the storage system and its host. Typically expressed in megabytes per second (MBPS), it prioritizes throughput over latency. High MBPS is crucial for streaming I/O, where data buffering mitigates the impact of latency.
I/O transfer size	This metric, known as I/O block size, represents the size of a single input/output operation. Typically measured in kilobytes (KB), it is determined by factors such as the hypervisor, the application, or the file system. The values of input/output operations per second (IOPS), MBPS (Megabytes Per Second), and block size are interrelated. If you know two of the measurements, then you can calculate the third. The formula is $\text{MBPS} = \text{IOPS} * \text{Block Size} / 1024$.
Read/Write ratio	This metric quantifies the read/write ratio, expressed as a percentage, to characterize the workload's nature (for example, read-intensive or write-intensive). This ratio establishes general performance expectations for the storage system.
Read cache hits	This metric quantifies the effectiveness of the I/O cache (DRAM) by measuring the percentage of read requests that are served from it. A higher percentage indicates a greater use of cache, and it results in lower average read latency. Although the relationship between cache size and read cache hit percentage is complex and nonlinear, read-intensive applications benefit significantly from a larger read cache. Workload characterization often includes a ratio such as "70/30/50" representing 70% writes, and 50% read cache hits.
CPU usage	This indicates the percentage of time the storage controller's CPU is actively processing tasks.

It is important to not only have a view on these metrics in real-time. The ability to collect historical performance metrics is also essential to properly monitor and manage storage subsystems in any I/T infrastructure. During troubleshooting and performance tuning, the historical data can be used as a baseline to identify unwanted behavior or indicate a point in time where something has suddenly changed.

1.2 FlashSystem architecture overview

IBM Storage FlashSystem and SAN Volume Controller are an award-winning range of storage platforms that all run IBM Storage Virtualize software.

IBM Storage Virtualize software was first introduced in 2003 for SAN Volume Controller. For more than 20 years it has been continually improved to give the latest advanced features, and to gain the optimum performance from all of the hardware platforms that it runs on.

This section gives an overview of the key hardware and software concepts when planning for optimum performance.

1.2.1 IBM Storage FlashSystem hardware

All products in the IBM Storage FlashSystem family are designed to give comprehensive hardware redundancy to prevent the failure of a single hardware component from causing loss of access to data.

Control enclosure

A control enclosure contains all of the hardware that is used by a single I/O group of the system. The control enclosure includes two node canisters, redundant power supplies, and drive bays where the NVMe drives for FlashSystem 9x00, 7x00 and 5x00 models or SAS drives for FlashSystem 50xx models are located.

Node canister

Each FlashSystem control enclosure contains two node canisters. A node canister is a self-contained server that runs the IBM Storage Virtualize software and contains CPUs, memory, and interface adapters. In a fully operational system where both nodes are online, the I/O workload of the system is distributed between the two node canisters. If one of the node canisters suffers a hardware failure that causes it to be taken offline, paths to volumes fail over to the remaining canister, which continues to process all host I/O requests.

Redundancy between node canisters allows nondisruptive software upgrades, by upgrading each node canister while the other node handles I/O processing.

Although node canister redundancy prevents loss of access to data during a hardware failure, there might be some increase in I/O response times because of the following factors:

- ▶ With an offline node, the amount of CPU resource available is halved. In a correctly sized system this has a minimal performance impact as there is available processing capacity to handle the extra workload on the remaining node. This is one reason why it is important to accurately estimate the expected peak workload for the system when deciding which product is suitable for a given solution.
- ▶ To protect data integrity, write caching is disabled unless both node canisters are online. This prevents the node being a single point of failure for cached writes that have not been written to backend storage.

Figure 1-1 on page 4 shows the FlashSystem 9500 internal architecture.

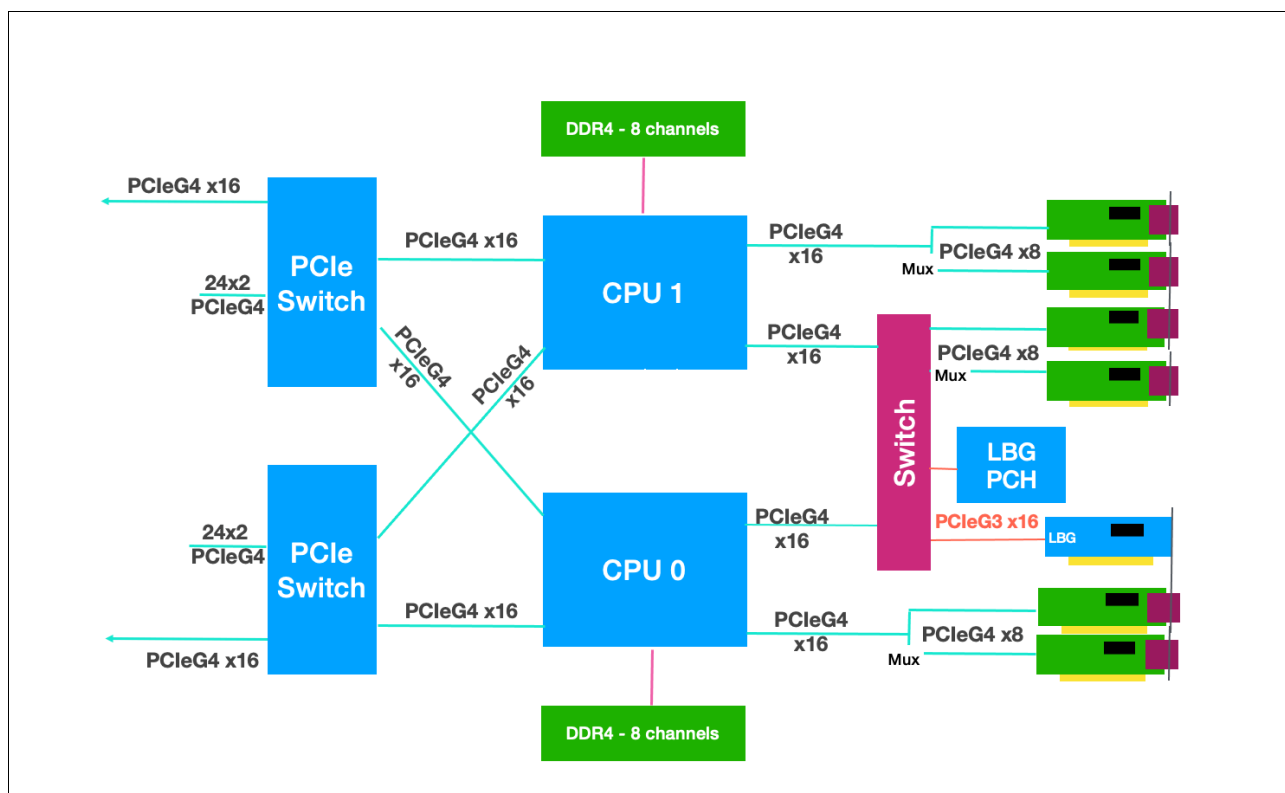


Figure 1-1 FlashSystem 9500 internal architecture

High performance PCIe bus

The main hardware components within the node canister (CPU, DIMMs, interface adapters, NVMe) are connected by a high-performance PCIe bus, which allows each component to perform to its maximum capability.

All inter-node communication within the control enclosure also uses a direct PCIe bus connection between the two nodes, which helps ensure a reliable and high-performance communication channel to ensure maximum I/O performance.

FlashCore Modules

All NVMe-based FlashSystem products (FlashSystem 9x00, 7x00, and 5x00) support FlashCore Modules. These drives are exclusive to IBM storage products, and offer the following features:

- ▶ Physical capacity that ranges from 4.8 TiB to 38.4 TiB of high-performance Flash storage.
- ▶ Effective capacity of 3x the physical capacity, enabled by drive compression.
- ▶ Hardware compression within the drive is always-on, and compresses data that is written to the drive without affecting I/O performance.

1.2.2 IBM SAN Volume Controller hardware

SAN Volume Controller (SVC) SV3 nodes are based on the same hardware as FlashSystem 9500, meaning that they benefit from the same redundancy and performance features that were previously outlined.

Each SVC node is contained in its *own control enclosure*, which means that communication with the partner node is done through an external Fibre Channel (FC) connection instead of through an inter-node PCIe link. This means that I/O performance is dependent on reliable and high-performance FC connectivity between the nodes. These connections can be either direct-attached or attached via a SAN Switch.

1.2.3 IBM Storage Virtualize software concepts

This section describes IBM Storage Virtualize software concepts.

Storage pools

A storage pool is a collection of Managed Disks (MDisks) that provides the storage from which volumes are provisioned.

Each MDisk's capacity in the storage pool is divided into extents. The size of the extent (between 16 MiB and 8192 MiB) is selected by the administrator when the storage pool is created and cannot be changed later.

Each MDisk has a "tier" attribute, which indicates the rate of I/O requests that drives in that tier can handle. The IBM Easy Tier® feature (explained in "Easy Tier" on page 8) promotes frequently-access data from low tiers to higher tiers, and moves less frequently accessed data from high tiers to lower tiers. MDisks that are in the same tier of the same pool typically have similar performance characteristics. Otherwise, the performance of volumes that have extents in the same pool might be inconsistent.

Volumes

A volume is a logical disk that the system presents to attached hosts.

Volumes can be combined into volume groups for use with policy-based replication and volume-group snapshots. The replicated volumes or snapshots within the group are kept consistent with each other, which helps ensure that all of the volumes that are used by an application provide a valid point-in-time copy.

Managed disks (MDisks)

A managed disk (MDisk) is a logical unit of physical storage. MDisks are not visible to host systems. MDisks are either arrays (RAID) from internal storage or volumes from external storage system.

RAID arrays use Distributed RAID (DRAID) to optimize performance of rebuild operations by spreading the rebuild write workload across rebuild areas on all drives in the array instead of on just one spare drive.

Drives

FlashSystem RAID arrays are composed of FlashCore Modules, industry-standard NVMe drives, or SAS drives. For more information about the differences between supported drive types, see Chapter 3, "Performance optimization for FlashSystem storage tiers and arrays" on page 43.

Hosts

A host is a logical object that represents a list of worldwide port names (WWPNs), NVMe qualified names (NQN), or iSCSI or iSER names that identify the interfaces that the host system uses to communicate with the IBM SAN Volume Controller and IBM Storage FlashSystem. Fibre Channel connections use WWPNs to identify host interfaces to the

system. iSCSI or iSER names can be iSCSI qualified names (IQNs) or extended unique identifiers (EUIs). NQNs are used to identify hosts that use FC-NVMe connections.

Volumes can be mapped to a host to enable access to a set of volumes.

1.2.4 IBM Storage Virtualize software features

The Storage Virtualize software provides many advanced features to enable easy management of your data. This section provides a summary of the main features that are relevant to performance.

For a more detailed explanation of the software features and how to use them, see *Implementation Guide for IBM Storage FlashSystem and IBM SAN Volume Controller dSG24-8542*.

Cache

Each FlashSystem or SAN Volume Controller node contains RAM, much of which is used to cache read and write data, which benefits performance. Some RAM is also used as working memory for the Storage Virtualize software and to store deduplication hashes if data reduction pools are used. Each product offers a choice of RAM capacities. Specifying a larger amount of RAM can improve performance, depending on the I/O workload submitted by hosts, and the features being used.

Upper and lower cache

IBM Storage Virtualize software uses a two-tiered cache system, as depicted in the software stack in Figure 1-2 on page 7.

The two levels of cache provide distinct benefits:

- ▶ *Upper cache* provides a small amount of write cache, and is above most of the other software components in the I/O stack. Those other components provide advanced features such as snapshots, data reduction, and volume mirroring. Therefore, it can complete host write requests quickly, which improves the volume write response times.
- ▶ *Lower cache* provides the majority of write cache and a read cache with pre-fetch capability. It is below most of the other software components in the I/O stack, so it can improve the performance of those software features.

A cache partitioning algorithm helps ensure that no single storage pool can occupy all of the write cache to minimize the effect of slow storage in one pool from impacting write performance in another pool.

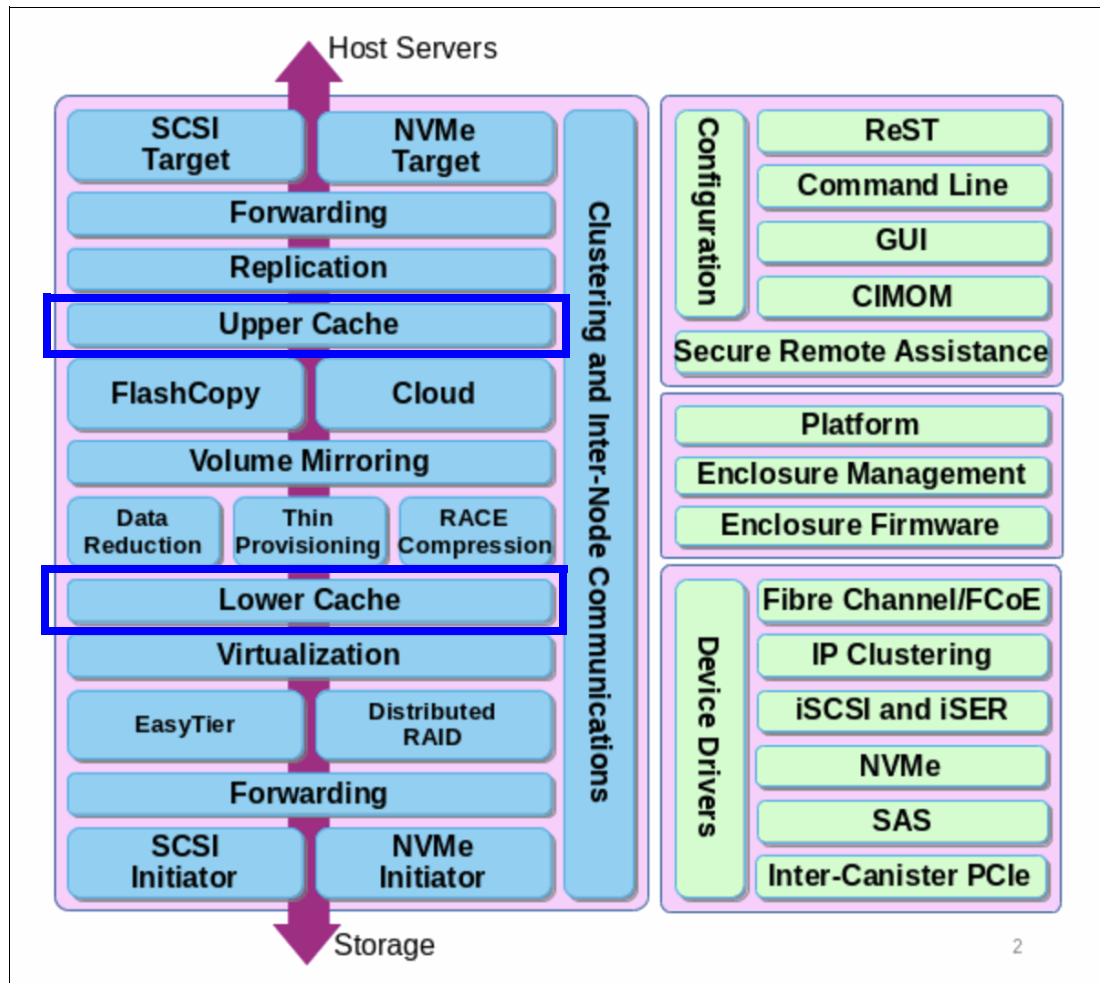


Figure 1-2 IBM Storage Virtualize software stack

I/O statistics for upper and lower cache

In IBM Storage Insights and Spectrum Control, upper cache is referred to as *Volume Cache (VC)*, and lower cache is referred to as *Volume Copy Cache (VCC)*. Each statistic is reported separately, which helps narrow down the potential cause of a performance issue.

Tip: VC destage latency denotes the time taken to destage from VC cache to VCC cache. For instance, if the overall volume write latency is high, VC destage latency is minimal, but VCC destage latency is high, then this suggests an issue between the RAID array and the drives themselves, and a potential overloading of the disks. Conversely, if VCC destage latency is low but VC destage latency is high, the back-end storage can be ruled out as a cause of the high write response time.

For further details of how to interpret cache I/O statistics, see Chapter 6, “Performance monitoring and analysis” on page 135.

Snapshots

The user can use snapshots to take point-in-time copies of data from a volume group. Data is only copied to a snapshot when the original data is overwritten (known as Copy-on-Write). Snapshots are also used to enable cycling mode for policy-based replication.

Snapshots cannot be mapped to a host. This protects the point-in-time copy of data from being overwritten. To access a snapshot, you can create either a clone (which contains a full copy of all data in the volume group), or a thin-clone (which contains only the data that changed since the snapshot was originally taken). Either clone can be mapped to a host and used to restore an application to the point-in-time corresponding to the snapshot.

The 8.7.0 software release introduces a performance optimization that greatly improves I/O performance at the point that a snapshot is triggered. In previous versions, I/O requests were paused to allow components in the software stack to prepare for the snapshot, but the pause is no longer needed in 8.7.0. Therefore, in environments where snapshots are frequently triggered (either on a schedule, or as a result of policy-based replication in cycling mode), upgrading to 8.7.0 or later can improve overall production volume performance.

For more information, refer to *IBM Redbooks Data Resiliency Designs: A Deep Dive into IBM Storage Safeguarded Snapshots*, [REDP-5737](#).

Easy Tier

With multiple types of storage with different performance characteristics (for example, an NVMe Flash array and a SAS drive array), you can use IBM Easy Tier to automatically distribute data within a storage pool. This provides maximum performance for data that is frequently accessed and moves less critical data to more cost-effective storage.

Easy Tier monitors the host I/O activity and latency on the extents of all volumes with the Easy Tier function that is turned on in a multitier storage pool over a 24-hour period. Then, it creates an extent migration plan that is based on this activity and dynamically moves high-activity, or hot, extents to a higher disk tier within the storage pool. It also moves extents whose activity dropped off, or cooled down, from the high-tier MDisk to a lower-tiered MDisk.

The automatic load-balancing function is enabled by default on each volume and cannot be turned off by using the GUI. This load-balancing feature is not considered an Easy Tier function, although it uses the same principles.

The management GUI supports monitoring Easy Tier data movement in graphical reports. The data in these reports helps you understand how Easy Tier manages data between the different tiers of storage, how tiers within pools are used, and the workloads among the different tiers. Charts for data movement, tier composition, and workload skew comparison can be downloaded as comma-separated value (CSV) files

Encryption

FlashSystem and SAN Volume Controller provide option encryption of data at rest, which protects against the potential exposure of sensitive user data and user metadata that is stored on discarded, lost, or stolen storage devices.

If encryption is to be used, it must be enabled when creating a storage pool or RAID array. It is not possible to perform in-place encryption of data at a later point. The option to make a storage pool or RAID array encrypted is only available if the relevant license has been activated on the system.

Methods of encryption

There are three ways that data can be encrypted, and the performance of volumes in an encrypted pool depends on which method is being used:

- Self-encrypting drives

All NVMe drives supported by FlashSystem products are self-encrypting, including FlashCore Modules. The IBM Storage Virtualize software provides a key to the drive, and the drive then encrypts the data in hardware.

- ▶ **SAS adapter encryption**

For RAID arrays that are composed of SAS drives, the SAS adapter within the control enclosure encrypts the data in hardware.

- ▶ **Software encryption of external virtualized MDisk**

It is also possible to encrypt data that is hosted on external virtualized MDisk. In this case, the data must be encrypted in software by the Storage Virtualize software running on the CPUs within the node canister.

Performance implications of data encryption at rest

Hardware encryption either by self-encrypting drives or by the SAS adapter has no impact on performance. The I/O performance seen by the host is the same whether or not the data is encrypted.

When using software encryption of a virtualized MDisk, there can be a minor performance impact because of the extra CPU resources needed to perform encryption. If you use a SAN Volume Controller to virtualize a FlashSystem controller, it is strongly recommended to encrypt at the FlashSystem level. Encrypting at the FlashSystem level provides optimal performance and ensures that the hardware compression feature within FlashCore Modules can provide data savings. Encrypted data is inherently uncompressible, so data encrypted by the SAN Volume Controller or any other host results in a 1:1 compression ratio.

Data reduction

IBM Storage Virtualize offers a comprehensive range of features to help reduce the amount of physical storage needed for host data. The decision of which features to use depends on the preferred balance between maximum data savings and maximum performance. Often, hardware compression in FlashCore Modules provides sufficient data savings and provides the maximum performance under the most demanding I/O workloads.

When creating a storage pool, it can be either a standard pool or a data reduction pool:

- ▶ Standard pools can contain either fully-allocated or thin-provisioned volumes. SCSI Unmap can be used to free FCM physical capacity. Unmapped capacity in a standard pool is not automatically freed back to the storage pool - there is no garbage collection process. However, capacity can be freed back to the pool by using volume mirroring, or in 8.7.0 and later by using the reclaim_se tool.
- ▶ Data reduction pools can contain fully-allocated, thin-provisioned, compressed, and deduplicated volumes. Unmapped capacity in a data reduction pool is eventually made available for newly written data by a background garbage collection process.

The extra capacity savings provided by data reduction pools are enabled by maintaining a log-structured array. A log-structured array is a data structure that allows the system to track where host data is located within the pool's storage. Writes to the pool are always stored in empty space, and any data that is overwritten (or unmapped by the host) is marked as no longer in use. Garbage collection runs in the background and allows this unused MDisk capacity to be made available for new data.

SCSI UNMAP support

SCSI UNMAP is a command that can be sent by hosts to a storage system to indicate that data in a region of a volume is no longer needed, and that data can be discarded. This allows greater data reduction savings.

Enable host unmap support for any system where data reduction pools are in use or where a RAID array contains FlashCore Modules. Both of these technologies are designed to work most effectively when there is at least 15% free capacity. An internal garbage collection process runs in the background and makes capacity that is no longer occupied by useful data available for use. If there is very little free capacity, the rate of garbage collection must increase to accommodate incoming write I/O requests, which can increase I/O response times.

By enabling host unmap, the amount of free capacity in the pool or FlashCore Modules increases, which reduces the amount of garbage collection work that is needed. It also helps to prevent an out-of-space condition, which causes a loss of access to data.

Host unmap is enabled by default on FlashSystem 9100, FlashSystem 9200, and FlashSystem 9500. On other systems where data reduction pools or FlashCore Modules are used, enable host unmap to help ensure maximum performance.

Replication

Policy-based replication, first released in Storage Virtualize 8.5.2, provides asynchronous replication between two storage systems. It simplifies configuring and managing replication by using replication policies and volume groups with simple reporting of the replication status and recovery point. It delivers an asynchronous replication solution that can dynamically adapt to the available link bandwidth and write throughput to maintain the disaster recovery copy without impacting application performance.

Policy-based high availability is the preferred high availability solution for Storage Virtualize. It was first released in 8.6.1 and provides a high-performance, high-availability (HA) storage solution between two independent storage systems. For version 8.7.1 and later, it can be combined with policy-based replication to deliver an HA between two systems with asynchronous replication to a third independent system for disaster recovery.

The earlier remote copy features (Metro Mirror, Global Mirror, Global Mirror with Change Volumes) are supported up to 8.7.0 but not in any later software versions. If you use these features, investigate changing to policy-based replication or high availability to benefit from higher performance and easier management. Upgrade from 8.7.0 is not possible until legacy remote copy features are no longer used.

For more information on policy-based high availability and policy-based replication, refer to IBM Redbooks *Ensuring Business Continuity: A Practical Guide to Policy-Based Replication and Policy-Based High Availability for IBM Storage Virtualize Systems*, [SG24-8569](#).

For more details of possible migration approaches, see the [IBM support website](#).

1.3 Workload characterization and its importance

To effectively analyze FlashSystem performance, begin by understanding the workload of the storage controller. In most environments, a blend of application workloads, not a single application, impacts the controller. Databases typically generate small block I/Os. File servers and ESXi clusters produce larger ones. Therefore, observed performance metrics reflect a weighted average of these mixed I/O sizes.

Large block I/Os generally exhibit higher response times, potentially including significant host-attributed latency. Therefore, aligning performance expectations with typical workload block sizes is crucial.

Workload characterization requires monitoring the Storage Virtualize I/O stack, specifically upper cache and lower cache performance metrics. For read-intensive workloads, prioritize the read cache hit ratio and prefetch cache hits. For write-intensive workloads, although the upper cache write cache hit ratio is usually 100%, it is necessary to monitor “write efficiency” (also known as “dirty writes” or “block overwrites”). This metric indicates the frequency of lower cache block overwrites before destaging and directly impacts write I/O response times.

Similar to read cache hit ratios, write efficiency is workload-dependent and cannot be tuned at the storage controller. However, both metrics are essential for performance analysis.

A comprehensive discussion of workload-specific tuning within IBM Storage Virtualize can be found in Chapter 2, “Workload specific tuning” on page 13.

1.4 Performance monitoring tools

Performance monitoring tools for FlashSystem are crucial for maintaining optimal system performance, ensuring data integrity, and troubleshooting issues. To achieve this, the tools must include the following features:

- ▶ **Real-time monitoring.** These tools provide real-time insights into the system's health and performance. Administrators can identify any anomalies or potential issues before the issues worsen.
- ▶ **Performance optimization.** By tracking key performance metrics, administrators can identify bottlenecks and optimize the system's configuration for better performance.
- ▶ **Capacity planning.** These tools provide data on storage usage, helping administrators plan for future capacity needs and prevent potential storage shortages.
- ▶ **Troubleshooting.** For performance issues or system failures, these tools provide valuable data for diagnosing and resolving problems.
- ▶ **Proactive maintenance.** By continuously monitoring the system, administrators can perform proactive maintenance, which reduces the likelihood of unexpected downtime.
- ▶ **Cost efficiency.** Efficient system performance means better resource usage, which leads to cost savings.

FlashSystem storage systems are equipped with a suite of built-in real-time performance monitoring functions.

A more comprehensive view, which includes historical data and predictive analytics, external on-premises or cloud-based tools, is often employed. These advanced solutions not only offer a broader perspective on system performance but also incorporate predictive features, which

helps administrators anticipate potential issues and optimize system configuration for enhanced efficiency and reliability.

These tools provide an immediate overview of the system's health and performance, enabling administrators to swiftly identify and address any anomalies.

Chapter 6, “Performance monitoring and analysis” on page 135 contains real-time performance monitoring capabilities for FlashSystem. It explores the built-in features of the system, providing insights into how they facilitate performance tracking. That chapter also covers IBM Spectrum Control and IBM Storage Insights, detailing how these tools monitor and manage storage performance in real-time, as well as based on historical data.

Chapter 7, “Isolating issues by using IBM Storage Insights” on page 155 provides a case study on isolating performance issues with IBM Storage Insights.

Though not covered in this book, several free or open-source performance monitoring tools are also available:

- ▶ [Prometheus](#)
- ▶ [Quick Performance Overview \(qperf\)](#)
- ▶ [STOR2RRD](#)



Workload specific tuning

Efficient performance relies on fine-tuning storage and operating system settings to match the unique demands of each workload. Whether you are managing databases, virtual machines, or other enterprise applications, each system exhibits distinct I/O characteristics and performance needs. By aligning storage configurations with these requirements, organizations can avoid bottlenecks and achieve optimal efficiency. In today's competitive landscape, even minor adjustments can lead to significant performance gains.

In this chapter, the following topics are discussed and described:

- ▶ 2.1, “Database workloads on Linux” on page 14
- ▶ 2.2, “VMware workloads” on page 19
- ▶ 2.3, “IBM i workloads” on page 28

2.1 Database workloads on Linux

Database workloads running on Linux have complex interactions with I/O patterns, resource management, and file system features. An examination of how databases work on Linux systems can provide practical insights into tuning and configuration for improved performance and reliability.

2.1.1 OLTP and OLAP workloads

This section describes key workload types, such as OLTP versus OLAP, and shows how Linux-specific optimizations can effectively support each.

Characteristics and storage requirements

Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) workloads serve fundamentally different purposes, each requiring different storage characteristics. OLTP systems are designed for high-volume transactional operations where fast, concurrent updates are critical. They require low-latency storage with robust random I/O capabilities, as each transaction often involves small, frequent writes mixed with reads. In contrast, OLAP workloads focus on complex queries over large datasets, often involving aggregations and scans. These systems benefit from storage solutions optimized for high throughput and sequential access patterns, where read performance is prioritized over rapid transactional updates.

For OLTP workloads, which require ultra-low latency and high IOPS, all-flash arrays are optimized through these modern methods.

Read versus write patterns and IOPS considerations

The I/O behavior of OLTP systems is dominated by a high frequency of small, random read/write operations, requiring storage configurations that deliver high input/output operations per second (IOPS) with minimal latency. Conversely, OLAP workloads typically have a higher proportion of sequential reads. Although write performance remains important, the overall design often leans toward storage systems that can efficiently handle large data transfers and sustained throughput. Tailoring IOPS provisioning and storage layouts to the specific needs of each workload type is critical to achieving optimal performance and reliability in Linux-based environments.

Note: For OLTP and OLAP workloads, the XFS file system is generally preferred over ext4 because it is optimized for large-scale, high-performance environments.

Storage layout best practices

A well-planned storage layout is critical for OLTP and OLAP workloads to ensure that different types of I/O operations do not interfere with each other. Use separate data and log files to isolate database files from transaction logs on different filesystems or disks to reduce contention and optimize for different I/O patterns.

OLTP workloads

The following are best practices for OLTP workloads:

- ▶ Optimized block size. Use smaller block sizes that match the typical transaction size to minimize wasted space and improve random read/write efficiency.
- ▶ Journaling and caching. Leverage filesystems with robust journaling and configure caching options to reduce latency.

- Low latency and high IOPS. For scenarios that require low latency and high IOPS, FlashSystem is an excellent choice. This enterprise-class storage solution uses flash technology and NVMe architecture to deliver exceptional performance with minimal latency. Its advanced features, such as built-in data reduction, resiliency, and efficient caching help ensure that the storage system can handle the high-frequency, small, random I/O operations typical of transactional databases while providing scalability and reliability for growing workloads.

OLAP workloads

The following are best practices for OLAP workloads:

- Large sequential reads. Design storage layouts that favor large, sequential read performance. Filesystems like XFS can be tuned for high throughput with larger block sizes.
- High throughput and scalability. Use DRAID6 on FlashSystem to efficiently scale capacity and aggregate throughput.

Comparison of OLTP and OLAP performance

OLTP and OLAP workloads have different performance profiles. OLTP is highly latency-sensitive, and even small delays can disrupt transaction processing and impact user experience. In contrast, OLAP is throughput sensitive. Its performance is measured by its ability to process large volumes of data. Consequently, it is natural to observe higher latencies in OLAP workloads due to the nature of large, sequential I/O operations. Consider the differences when you monitor and tune performance.

2.1.2 SAP HANA workloads

This section includes a description of SAP HANA workloads.

In-memory database characteristics and impact on storage performance

SAP HANA relies on an in-memory computing architecture that enables real-time analytics and transaction processing. But even in an in-memory environment, persistent storage plays a critical role for recovery, logging, and data durability. FlashSystem is designed to complement these workloads with ultra-low latency and high IOPS. Its flash-based architecture significantly reduces the latency gap between volatile and persistent storage. This means that even when data is written to storage (for example, for logging or snapshots), the performance impact is minimized, which maintains the overall responsiveness of the SAP HANA system.

Storage requirements (persistent versus non-persistent data)

SAP HANA distinguishes between persistent data, such as transaction logs, system snapshots, and backups and non-persistent, in-memory data that is used during active processing. FlashSystem addresses these dual storage requirements with the following functions:

- Providing fast, reliable write operations that are critical for capturing persistent data changes.
- Supporting advanced caching and data reduction techniques to manage the different I/O patterns between persistent and transient data.
- Ensuring that both fast transactional log writes and bulk data snapshots are handled efficiently without impacting system performance.

Recommended file system

For SAP HANA, the XFS file system is generally preferred over ext4 because it is optimized for large-scale, high-performance environments.

Designed to meet the requirements of SAP HANA, the combination of FlashSystem and XFS delivers reliable performance even under the heavy load typical of in-memory analytics.

Storage layout best practices

A well-planned storage layout is crucial for SAP HANA to ensure that different types of I/O operations do not interfere with each other.

- ▶ Workload isolation. This minimizes I/O contention by using separate volumes for data, logs, shared files, and backups. This isolation ensures that numerous log writes and data read performance do not each other.
- ▶ Using Logical Volume Management (LVM) for enhanced performance. For the /data and /log filesystems, deploying multiple volumes under LVM can further boost performance. This approach enables parallel I/O operations and improved scalability, which helps ensure that the high transactional demands of SAP HANA are met efficiently.
- ▶ Enhanced monitoring with IBM Storage Insights. IBM provides IBM Storage Insights, a cloud-based monitoring tool that helps to simplify performance monitoring of SAP HANA volumes within the FlashSystem environment. By using IBM Storage Insights enables administrators can define thresholds and receive alerts, which ensures proactive monitoring and rapid response to any performance anomalies.

2.1.3 Storage volumes and LVM tuning

Consider the following aspects of storage volumes and LVM tuning.

Using multiple storage volumes for database performance

Distributing database workloads across multiple storage volumes can significantly improve performance by reducing I/O contention. By isolating different components, such as separating data files from transaction logs, each volume can be individually optimized for its specific access patterns. For example, logs, which typically involve sequential writes, benefit from a dedicated volume tuned for write efficiency. Conversely, data volumes focus on fast random access. In FlashSystem environments, when you configure multiple volumes the system can intelligently balance workloads, which ensures that high throughput and low latency are maintained even under heavy transactional loads.

LVM settings and their impact

Logical Volume Manager (LVM) on Linux provides flexible storage management, so administrators can tailor logical volumes to meet the needs of specific workloads. An important first step is to understand the two main types of logical volumes:

Linear logical volumes

In a linear logical volume, physical volumes are concatenated end-to-end to form a single contiguous storage space. This is the default logical volume type when you create a new logical volume in LVM, which makes it a straightforward option for many scenarios. However, linear logical volumes do not provide any performance benefits from parallel I/O operations and are used when simplicity is key and there is no need for increased throughput.

Striped logical volumes

Striped volumes spread data evenly across multiple physical volumes by splitting I/O operations across multiple disks. This approach can significantly improve performance by

enabling parallel reads and writes. To maximize the benefits of striping, it is important to carefully tune the stripe size and ensure proper alignment with the underlying hardware characteristics. Figure 2-1 shows linear versus striped logical volumes.

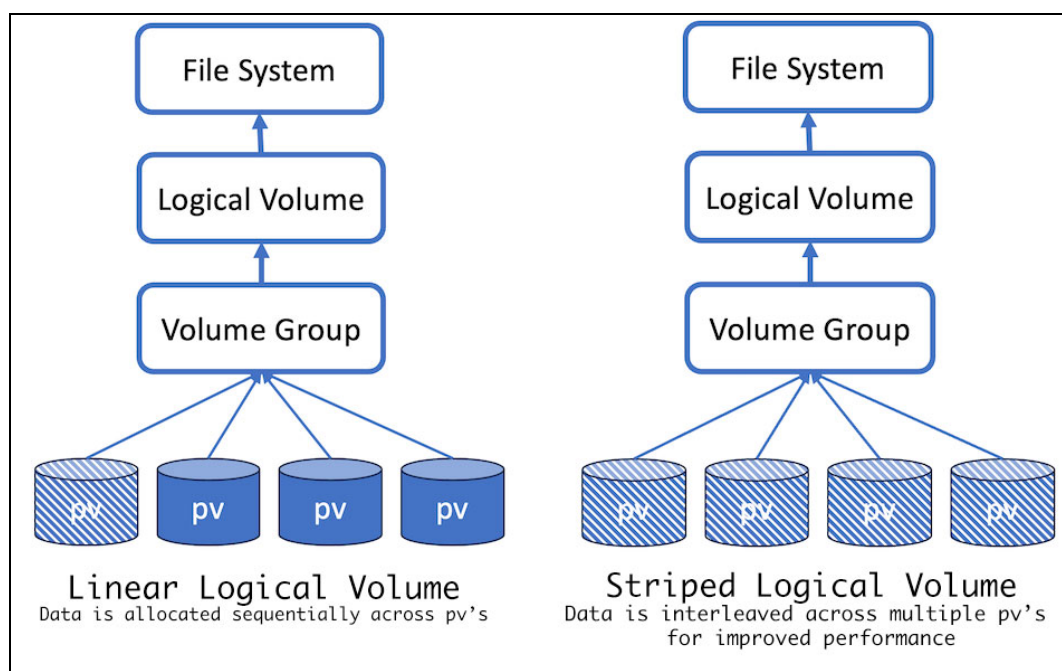


Figure 2-1 Linear versus striped logical volumes

Building on the concept of striped logical volumes, tuning specific LVM settings is critical for database performance.

Stripe size and alignment

Adjusting the stripe size in striped logical volumes is essential to align the logical configuration with the physical characteristics of flash storage. Proper alignment ensures that I/O operations are evenly distributed across FlashSystem volumes, which helps to minimize latency and avoid bottlenecks.

Stripe size and alignment for Oracle databases

Logical volume managers such as Oracle ASM stripe by default, but native OS LVM implementations do not always do so. Some implementations join multiple LUNs together as a concatenated device, resulting in data files that reside on a single LUN. This setup can create hotspots. Other LVM variants use distributed extents by default, which is a method similar to striping, but on a coarser scale. In this approach, the LUNs in the volume group are divided into large segments called extents, typically measured in megabytes, and the logical volumes are spread across these extents. Although this can help distribute random I/O across multiple LUNs effectively, it might not optimize sequential I/O operations as efficiently as a finely tuned, striped configuration.

Performance-intensive application I/O is almost always either in units of base block size or in 1 MB segments. The primary goal of a striped configuration is to ensure that single-file I/O can be performed as a contiguous unit and that multi-block I/O, ideally 1 MB in size, is parallelized evenly across all LUNs in the striped volume. Consequently, the stripe size should not be smaller than the database block size and the product of the stripe size and the number of LUNs should ideally be 1 MB.

Figure 2-2 shows three possible options for tuning stripe size and width. The number of LUNs is chosen to meet the performance requirements described previously, but in all cases the total data within a single stripe is 1 MB.

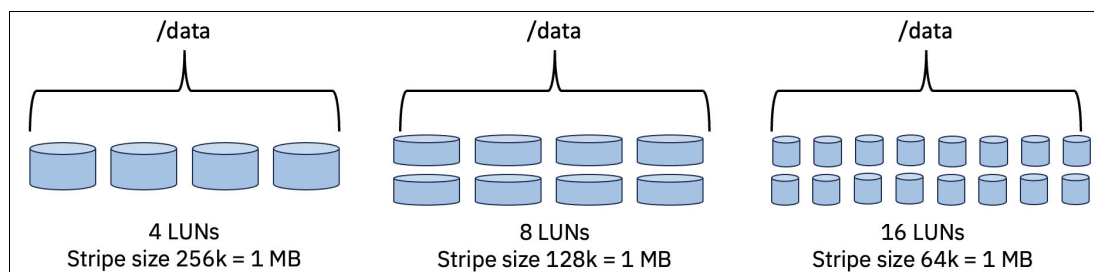


Figure 2-2 Number of LUNs and stripe sizes

Stripe size and alignment for SAP HANA

For SAP workloads, incorporating specific stripe size recommendations can help fine-tune performance:

- ▶ **Data volumes.** A stripe size of 256 KB is typically recommended for data volumes. This size helps maximize throughput by aligning well with the storage subsystem's optimal I/O block sizes, which helps to ensure efficient handling of large, sequential operations.
- ▶ **Log volumes.** For log volumes that primarily handle sequential writes, a smaller stripe size, such as 64 KB, might be more appropriate. This setting helps to optimize write performance by reducing latency and ensuring that log records are written efficiently.

2.1.4 File system choices for database data and logs

File systems serve as a critical interface between the operating system and storage hardware and have a direct impact on the performance and reliability of database workloads. When selecting a file system for FlashSystem environments, it is important to consider workload characteristics and the specific requirements of data versus log processing.

Ext4

As a mature and widely used file system, ext4 provides robust performance and reliability. Its features, such as delayed allocation and extents, work well for general-purpose database environments, although it may require additional tuning to take full advantage of all-flash storage.

XFS

Known for its scalability and high performance, XFS excels in environments with heavy parallel I/O and large files. Its allocation group architecture can significantly improve concurrent access performance, making it a strong candidate for high-throughput database workloads. XFS is widely recommended for Oracle and SAP HANA workloads. Its scalability and efficient handling of concurrent operations make it well-suited to handle the heavy I/O and large file sizes typical of Oracle databases.

Separating file systems for data and log

Separating database data and log files onto dedicated filesystems hosted on distinct physical volumes and logical volumes is a best practice. This approach isolates the divergent I/O patterns, which are random access for data or sequential writes for logs, which reduces contention and enables targeted tuning. By optimizing each file system individually, overall

system performance can be significantly improved. The following list describes some of the key benefits of separate file systems:

- ▶ I/O isolation. Logs typically involve sequential write operations, but data files often demand random I/O. Isolating these workloads prevents their operational characteristics from interfering with each other, which results in a more consistent performance profile.
- ▶ Tailored tuning. You can tune dedicated filesystems based on the specific needs of data or log processing. For example, the file system that is hosting logs can be optimized for sustained sequential writes, and the file system hosting data can be tuned for fast random access.
- ▶ Simplified management. Distinct file systems facilitate targeted backup strategies, recovery processes, and maintenance routines, to ensure that a problem in one area (for example, log corruption) does not affect data integrity in the other.

2.2 VMware workloads

VMware environments present unique challenges and opportunities for optimizing storage performance. By understanding the characteristics of different workload types and the associated storage provisioning options, administrators can fine-tune their systems to meet performance and reliability goals.

2.2.1 Storage performance considerations for VMware

VMware workloads vary widely from high-transaction databases to virtual desktop infrastructures (VDI) and general-purpose virtual machines, each with different storage performance requirements.

Workload types and their storage demands

VMware environments support a variety of workloads, each with unique storage subsystem requirements.

I/O-intensive workloads (for example, databases)

Databases like Microsoft SQL Server or Oracle run in VMs with heavy transactional activity. These typically have high IOPS requirements, with predominantly random I/O patterns. Workloads in this category often involve small, frequent transactions where low latency is critical. The focus is on fast, random read/write operations rather than large sequential transfers.

Virtual Desktop Infrastructures (VDI)

VDIs generate heavy random I/O due to multiple concurrent sessions that are accessing virtual desktops. The operations are usually small and random, resulting from user interactions such as application launches, desktop rendering, and frequent context switching. This workload is highly sensitive to latency and benefits from high-concurrency performance.

General purpose virtual machines

General purpose workloads have mixed I/O characteristics. They can have a mix of sequential and random operations depending on the applications running on them. Although they typically do not require the extreme performance of database VMs or VDIs, they do require balanced performance to efficiently handle both read and write operations.

Application and file servers

Application servers can vary widely. Transactional applications tend to have moderate to high IOPS with a mix of random and sequential I/O, and file servers often deal with larger sequential file transfers. File servers typically prioritize throughput over latency due to the larger block sizes involved. Understanding these I/O profiles is critical when configuring storage performance settings such as provisioning strategies, path selection policies, and queue depth adjustments to ensure that each workload receives optimal performance and reliability in a VMware environment.

VM Storage provisioning: Thin versus thick provisioning

When you provision storage for virtual machines in a VMware ESXi environment, choosing between thin and thick provisioning requires a careful evaluation of performance, capacity usage, and management requirements. Some best practices are tailored for VM storage provisioning.

Assess workload requirements

Assess workload requirements first:

- ▶ High performance and predictability. For performance-critical VMs (such as databases or high-transaction workloads), thick provisioning, specifically thick eager zeroed, is recommended. This method pre-allocates storage, to help ensure consistent performance and eliminating delays caused by on-the-fly allocation.
- ▶ Flexibility and efficient usage. Thin provisioning is ideal for less performance-intensive applications or environments where over-provisioning is acceptable. It allows for more efficient capacity utilization by allocating storage as needed, but requires careful monitoring to avoid over-commitment of storage.

Choose the right thick provisioning type

You have the following options for thick provisioning type:

- ▶ Thick Eager Zeroed. Pre-allocates and zeroes out disk space during VM creation for predictable performance and increased security, especially important for I/O-intensive workloads.
- ▶ Thick Lazy Zeroed. Allocates storage space upfront but defers zeroing until it's needed. This can be faster to provision, but can introduce latency during initial writes.

Monitor and manage capacity

With thin provisioning, it is critical to closely monitor datastore usage to prevent over-commitment, which can lead to performance degradation and potential operational issues. Implement alerting and capacity planning processes to ensure that free space is available.

Performance impacts

Consider the following performance impacts:

- ▶ Allocation overhead. Understand that thin-provisioned disks can affect performance during allocation bursts, especially when memory is near full.
- ▶ I/O impact. Performance impact is generally minimal with modern all-flash arrays, but thorough testing in your specific environment is recommended to verify that thin provisioning meets your performance criteria.

By aligning your VM storage provisioning strategy with the specific performance needs and capacity goals of your workloads, you can ensure optimal storage efficiency and reliability within your VMware ESXi deployments.

2.2.2 Path Selection policies

In VMware vSphere, storage performance and reliability depend heavily on how ESXi manages multiple paths to storage arrays. With the FlashSystem, optimizing I/O paths has become even more critical to make use of its low-latency and high-throughput capabilities. VMware's multipathing framework provides two key options: the traditional Native Multipathing Plugin (NMP) with its Path Selection policies (PSPs) and the newer High Performance Plug-in (HPP), which is designed for modern storage systems. This section explores both approaches, their mechanics, performance impacts, and practical guidance for VMware administrators.

Native Multipathing Plug-in

Native Multipathing Plug-in (NMP) is VMware's default multipathing solution, part of the Pluggable Storage Architecture (PSA). It includes two features:

1. Storage Array Type Plugin (SATP) that manages array-specific logic (for example, ALUA support).
2. Path Selection Policy (PSP) that controls path usage for I/O. NMP has been the standard for years but does not always take advantage of the performance of modern all-flash arrays.

High Performance Plug-in

Introduced in vSphere 7.0, High Performance Plug-in (HPP) is a next-generation multipathing framework that is optimized for high-speed storage, such as NVMe and all-flash arrays like FlashSystem. Unlike NMP, HPP bypasses the layered approach of the PSA and provides a streamlined I/O path with reduced resource usage and better scalability.

Types of Path Selection Policies (PSPs) with NMP

With the Native Multipath Plug-in, VMware provides the following PSPs, each tailored to specific storage scenarios:

- ▶ Fixed:
 - Behavior. Uses a single preferred path, switching to an alternate only in the event of a failure and reverting back after recovery.
 - Use case. Suitable for local disks and active/passive arrays without ALUA.
- ▶ Most Recently Used (MRU):
 - Behavior. Stays on the first working path, and switches on failure without reverting.
 - Performance. Single-path throughput, no load balancing.
 - Use case. Legacy active/passive arrays without ALUA.
- ▶ Round Robin (RR)
 - Behavior. Rotates I/O across all active paths. Default is 1000 I/Os per path.
 - Performance. Scales IOPS and throughput.
 - Use case. Active/active or ALUA arrays, which are ideal for high I/O workloads.

The default Round Robin IOPS limit is 1000. When the ESXi host requests 4000 IOPS or more, the Round Robin algorithm uses all paths, as previously described. However, if the host's I/O is below 1000 IOPS, only a single path is active because of this default setting. IBM recommends changing this limit to 1. This value uses all available paths for each I/O request. For example, at an ESXi host load of 800 IOPS, all paths are actively used, with I/O requests evenly distributed across them, as illustrated in Figure 2-3 on page 22.

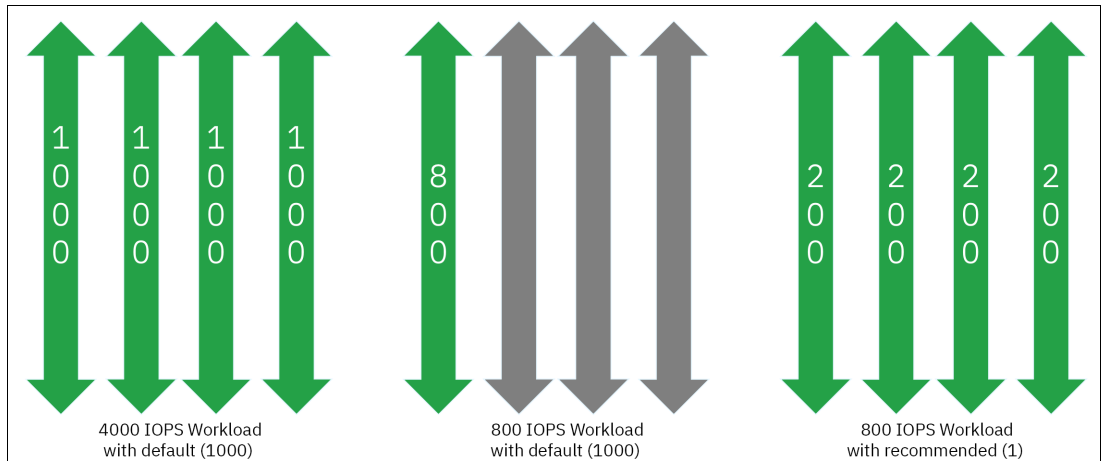


Figure 2-3 Round Robin IOPS limit setting and workloads

The following **esxccli** command changes the IOPS limit of all FlashSystem volumes;

```
for i in `esxcfg-scsidevs -c |awk '{print $1}' | grep naa.6005076`; do esxccli
storage nmp psp roundrobin deviceconfig set --type=iops --iops=1 --device=$i;
done
```

High Performance Plug-in: A modern alternative

High Performance Plug-in (HPP), as shown in Figure 2-4, replaces NMP's PSP model with a simplified, highly efficient multipathing engine. It dynamically selects paths based on latency and throughput, with built-in load balancing tailored for all-flash arrays and NVMe.

- **Performance.** Reduces I/O latency (often less than 50 μ s with NVMe) and scales IOPS beyond RR limits by minimizing software overhead. For FlashSystem, HPP can unlock microsecond-level performance, which is unattainable with NMP.
- **Use case.** Optimized for all-flash arrays (such as FlashSystem) and NVMe-over-Fabrics, especially in latency-sensitive or high-throughput environments, such as databases or VDI.

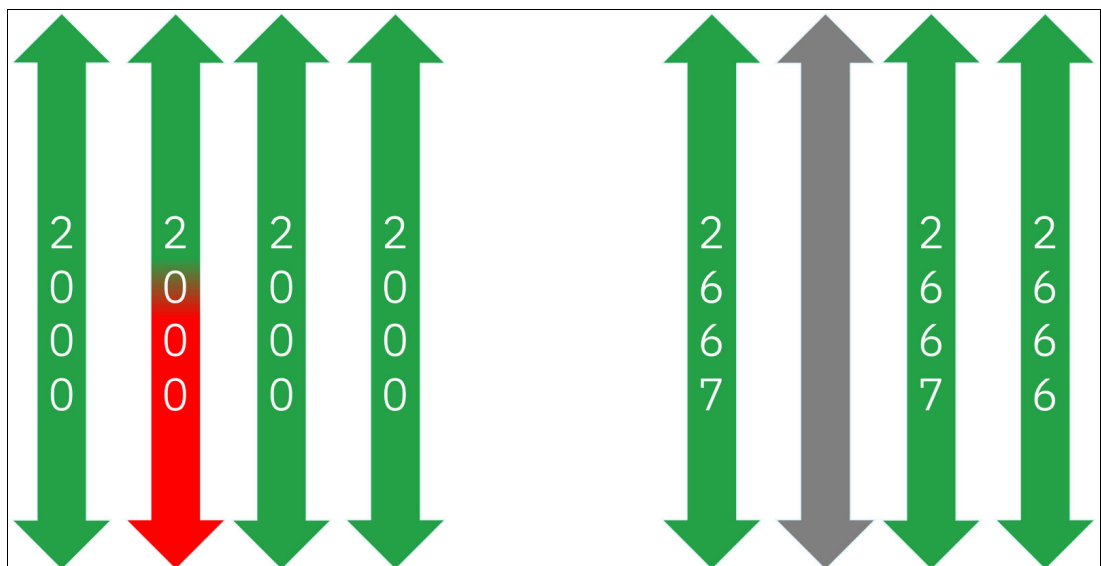


Figure 2-4 High Performance Plug-in load balancing latency path selection policy

If a path's latency exceeds the defined threshold (for example, 1 ms), HPP dynamically redirects new I/O requests to a lower-latency path without stopping existing operations on the high-latency path. This is a seamless, kernel-level adjustment that ensures minimal disruption. I/O already in progress completes on its original path, and subsequent I/Os are rerouted based on real-time latency data.

How PSP and HPP affect performance

The following considerations describe how PSP and HPP affect performance:

- ▶ Load balancing and throughput:
 - NMP with round robin. Distributes I/O across paths but is constrained by PSA overhead and static switching (for example, 1000 I/Os). A VM can achieve 15,000 IOPS on a 4-path round robin setup.
 - HPP. Dynamically balances I/O based on real-time metrics, potentially doubling IOPS (for example, exceeding 30,000) on the same configuration by fully leveraging FlashSystem capabilities.
- ▶ Failover and redundancy:
 - NMP. Fixed and MRU prioritize simplicity, and round robin adapts but introduces failover delays.
 - HPP. Provides faster failover (< 1 s) and seamless path recovery, which is critical for HA workloads.
- ▶ Latency considerations:
 - NMP. Congested paths in fixed or MRU configurations increase latency, and round robin reduces latency but introduces overhead.
 - HPP. Minimizes latency through direct I/O handling, which is optimized for FlashSystem microsecond response times.

2.2.3 Advanced storage optimization techniques

Optimizing storage performance in VMware requires more than basic configuration. Advanced tools and techniques are essential for efficient, scalable, and fast-recovering workloads. This section describes key VMware storage optimization features such as Storage I/O Control (SIOC), Storage DRS (SDRS), and queue depth adjustments. When combined with FlashSystem, an all-flash platform that has low latency, high IOPS, and advanced multipathing (like HPP), these techniques can significantly enhance performance and efficiency. The following sections detail each technique and its integration with FlashSystem.

VMware storage I/O control

VMware storage I/O control (SIOC) dynamically manages I/O prioritization across VMs that share a datastore, which prevents any single VM from monopolizing storage resources. It enforces I/O shares and latency thresholds during contention.

Enable SIOC in vSphere client under **Datastore** → **Configure** → **General**. Set a latency threshold, for example, 10 ms, and assign shares based on VM priority, for example, 2000 for critical VMs and 1000 for standard.

Reduces latency spikes and ensures consistent performance for critical workloads like databases or VDI.

FlashSystem's microsecond response times (for example, <50 μ s with HPP) complement SIOC by minimizing baseline latency, allowing tighter thresholds (for example, 5 ms) without overloading the array.

FlashCore modules can deliver over 1 million IOPS, so SIOC can allocate resources to ensure that the underlying storage accommodates peak demands.

Storage DRS

Storage DRS (SDRS) automates datastore load balancing by migrating VMs based on I/O latency and space utilization and maintains optimal performance across a cluster.

From the vSphere Client GUI, enable SDRS by selecting **Datastore Cluster** → **Configure** → **Storage DRS**. Set automation level (for example, Fully Automated) and thresholds (for example, 80% space, 15 ms latency).

SDRS prevents datastore congestion, reduces latency, and ensures even I/O distribution.

Adaptive Queue Depth

Queue depth determines the number of I/O requests that can be queued at the HBA, an ESX system, or a storage array. Default settings (for example, 32 for HBAs) can limit performance under heavy I/O loads. Adaptive Queue Depth Tuning (AQDT) dynamically adjusts this parameter based on real-time workload and device capabilities and includes the following key benefits:

- ▶ Optimize throughput and latency. By tailoring queue depth to the I/O characteristics of the workload, administrators can reduce latency for critical operations while maximizing overall throughput.
- ▶ Dynamic adjustments. Use adaptive mechanisms so the system can respond to real-time changes in workload intensity, which increases queue depth when the storage subsystem is underused and decreases it during periods of high contention to avoid overloading.
- ▶ Balancing multi-VM environments. In shared environments, optimal queue depth settings help ensure that no single VM monopolizes storage resources, which provides a more balanced distribution of I/O requests across all active VMs.

Use SIOC and adaptive queue depth tuning together to create a robust and responsive storage environment in VMware ESXi. By dynamically managing I/O resources and optimizing request queues, these technologies sustain high performance for critical workloads under heavy load, which ensures fairness and stability. When integrated with FlashSystem, which offers exceptionally low latency and high throughput, the advantages of SIOC and adaptive queue depth tuning are amplified, guaranteeing predictable, balanced storage performance even during demanding workloads.

Managing the number of volumes and VMs per VMDK

In VMware ESXi environments, consolidating multiple virtual machines (VMs) onto a single Virtual Machine Disk (VMDK) can significantly impact performance. Although seemingly convenient, this practice often leads to several performance challenges:

- ▶ Performance throttling. Consolidating multiple VMs onto a single VMDK forces all I/O operations through a shared logical volume, which limits the available storage paths. This can lead to performance throttling when the storage subsystem reaches its queue depth limit during peak I/O demands.
- ▶ Limited path usage. Storage arrays often support multiple paths to enhance redundancy and performance. However, consolidating VMs onto a single VMDK restricts the usage of these paths, which can potentially hinder throughput and failover resilience.

- IOPS constraints. The I/O operations per second (IOPS) available to a single VMDK are constrained by its queue depth. Overloading a single VMDK with numerous VMs can quickly saturate its IOPS capacity, which increases latency and degrades overall performance.

To optimize VMware ESXi environments, distribute the VM workloads across multiple, well-configured VMDKs. This approach improves load balancing, enhances performance, and promotes better overall scalability compared to using a single large volume. The following configuration options are recommended best practices:

- Distributed provisioning. Distributing VMs across multiple, smaller VMDKs, rather than a single large one, optimizes performance. For many deployments, using 8 to 16 VMDKs enables better path usage and balanced I/O loads, which minimizes the risk of a single volume overload and ensures consistent performance.
- Multiple datastores for database VMs. For database VMs, consider using multiple VMDKs spread across multiple datastores. This strategy enhances parallelism and fault isolation. By distributing the workload across several datastores, you can make use of additional available paths and further minimize I/O contention, which leads to improved performance and resilience for high-demand database environments.
- Scalability considerations. For database VMs, deploying multiple VMDKs across several datastores enhances parallelism and fault isolation. This distribution makes use of additional paths and minimizes I/O contention, which improves performance and resilience in high-demand database environments.
- Performance tuning. Distributing VMs across multiple VMDKs allows for targeted volume tuning. Administrators can optimize individual volumes for specific workload patterns, enhancing IOPS performance and reducing latency.

2.2.4 Monitoring and troubleshooting storage performance

In VMware environments, storage performance is crucial for workload efficiency. Proactive monitoring detects issues, and effective troubleshooting resolves them. The use of the right tools, including VMware's built-in options, IBM Storage Insights, and the unique capabilities of IBM Storage FlashSystem, ensures optimal performance.

- vSphere performance charts:
 - Access. **vSphere Client** → **Host/Datastore/VM** → **Monitor** → **Performance** → **Advanced**.
 - Metrics. Latency (Device latency (DAVG), kernel latency (KAVG), queue latency (QAVG)), IOPS, throughput (MB/s).
 - Use. Real-time and historical views to spot trends (for example, latency spikes during backups).
- esxtop:
 - Access. SSH to ESXi, run *esxtop*, press *d* for disk view.
 - Metrics. Device latency, kernel latency, queue latency, IOPS (CMDS/s).
 - Use. Granular, real-time analysis for pinpointing bottlenecks (for example, high QAVG indicating queue saturation).
- vRealize Operations Manager:
 - Access. Deployed as a vSphere plugin or standalone.
 - Metrics. Predictive analytics, capacity planning, and detailed I/O breakdowns.

- Use. Long-term monitoring and alerting (for example, Datastore latency > 10 ms for 5 minutes).
- IBM Storage Virtualize Plugin for vSphere:
 - Access. Installed by using the vSphere Client, which is downloaded from [IBM Fix Central](#); integrated into Hosts and Clusters or Storage views.
 - Metrics. Real-time FlashSystem performance data (for example, IOPS, latency, throughput) at the datastore and array level, and ransomware detection alerts.
 - Use. Directly integrates FlashSystem monitoring into the vSphere GUI, so that administrators can view storage health (for example, latency below 50 μ s) within vCenter. Simplifies VM performance correlation with array metrics and provides actionable alerts (for example, Volume latency exceeding 1 ms) for rapid response. Complements Storage Insights by emphasizing vSphere-integrated management.

For details about IBM Storage Virtualize Plugin for vSphere see [IBM Docs](#).

- IBM Storage Insights

IBM Storage Insights is a powerful cloud-based tool for end-to-end storage performance monitoring in VMware environments, especially when diagnosing SAN-wide issues that span from the VM level to the storage array.

 - Metrics. End-to-end performance from VM to host HBA, SAN switch connectivity, storage ports, and volumes. Tracks latency, IOPS, throughput, and path health.
 - Use. Provides a single-pane view of the SAN stack, ideal for diagnosing complex, cross-layer issues (for example, VM slowdowns because of SAN switch congestion). IBM Storage Insights Pro retains historical performance data and supports VMware host/VM monitoring to correlate storage performance with application impact.
 - Strength. Identifies elusive SAN-wide issues, such as HBA saturation or switch port errors, that are difficult to isolate with siloed tools.

Note: For more information on IBM Storage Insights refer to the following publications:

- *Maximize Your Storage Investment: Leverage IBM Storage Insights for Observability and Artificial Intelligence for IT Operations*, [SG24-8584](#)
- *Intelligent Storage Management with Artificial Intelligence for IT Operations (AIOps)*, [REDP-5755](#)

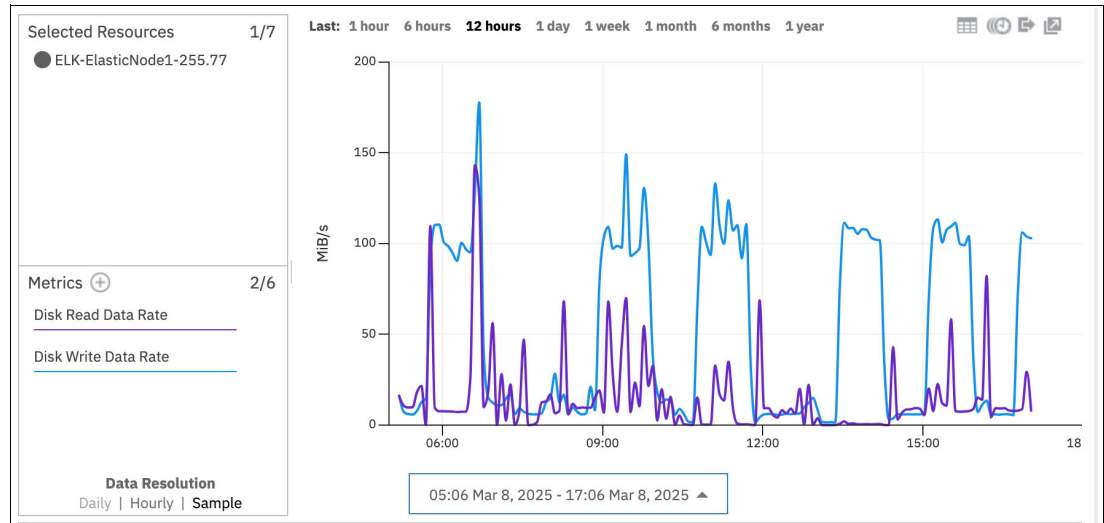


Figure 2-5 IBM Storage Insights single VM performance

2.3 IBM i workloads

IBM i has unique storage management concepts compared to typical open systems hosts. This section introduces these concepts and makes recommendations in 2.3.9, “Recommendations based on field experience” on page 39.

This section includes the following topics:

- ▶ “IBM i storage management” on page 28
- ▶ “Single-level storage” on page 29
- ▶ “Planning for IBM i storage capacity” on page 35
- ▶ “Planning for IBM i storage capacity” on page 35
- ▶ “Storage connection to IBM i” on page 36
- ▶ “IBM i unmap support” on page 36
- ▶ “Defining LUNs for IBM i” on page 37
- ▶ “Data layout” on page 38
- ▶ “Recommendations based on field experience” on page 39

2.3.1 IBM i storage management

Because of the unique IBM i storage architecture, special considerations for planning and implementing a SAN storage solution are required with FlashSystem. This section describes how IBM i storage management manages its available disk storage.

Many host systems require the user to take responsibility for how information is stored and retrieved from the disk units. An administrator must also manage the environment to balance disk usage, enable disk protection, and maintain balanced data to be spread for optimum performance.

The IBM i architecture is different in the way that the system takes over many of the storage management functions, which are the responsibility of a system administrator on other platforms.

IBM i, with its Technology Independent Machine Interface (TIMI), largely abstracts the underlying hardware layer from the IBM i operating system and its users and manages its system and user data in IBM i disk pools, which are also called *auxiliary storage pools* (ASPs).

Figure 2-6 on page 29 demonstrates that when you create a file, you do not assign it to a storage location. Instead, the IBM i system places the file in the location that ensures the best performance from an IBM i perspective

Note: When a program presents instructions to run to the machine interface, the interface appears to the program as the system hardware, but it is not. The instructions that are presented to TIMI pass through a layer of microcode before they are understood by the hardware.

As a component of the IBM i SLIC, IBM i storage management typically spreads the data in the file across multiple disk units (logical unit numbers (LUNs) when external storage is used). When you add records to the file, the system automatically assigns more space on one or more disk units or LUNs.

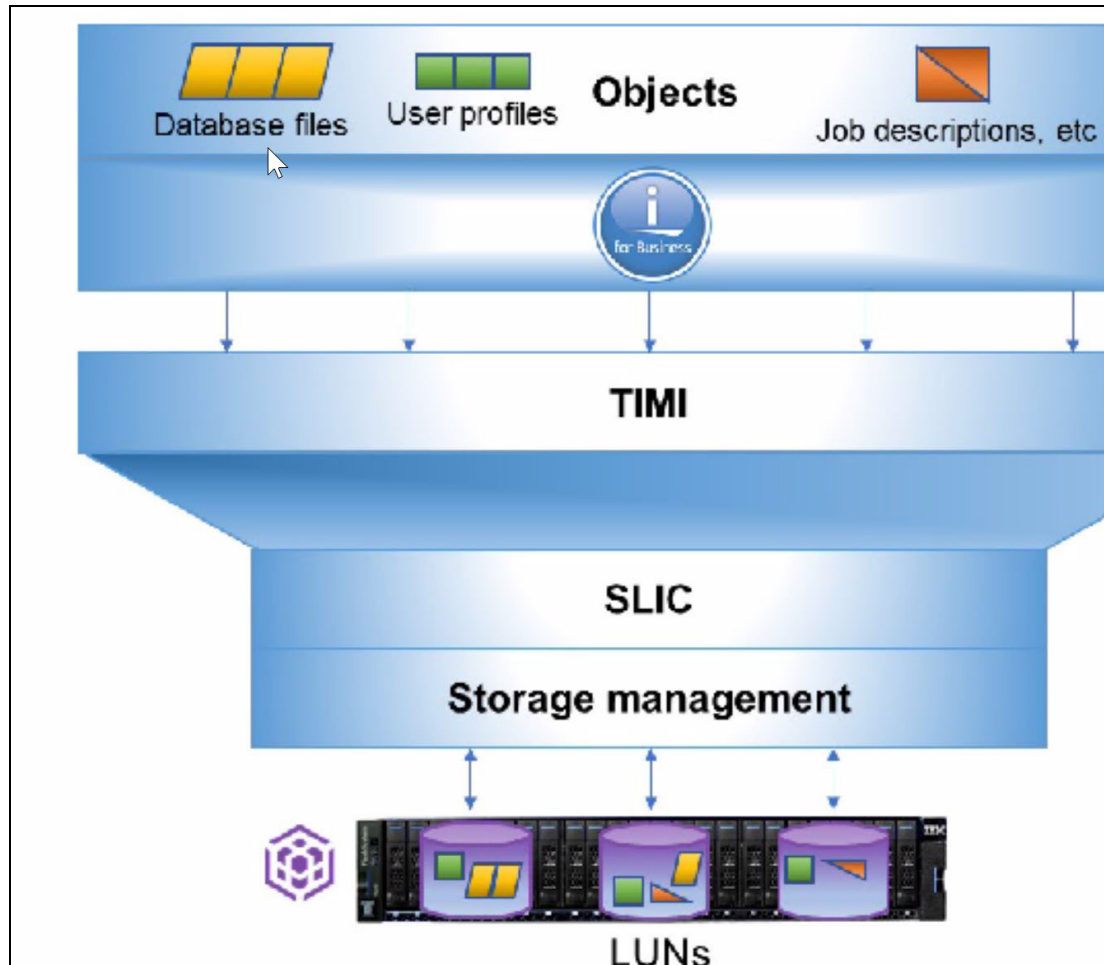


Figure 2-6 IBM i storage management spreads objects across logical unit numbers

2.3.2 Single-level storage

IBM i uses a single-level storage, object-orientated architecture. It sees all disk space and the main memory or main storage as one address space. It also uses the same set of virtual addresses to cover main memory and disk space. Paging the objects in this virtual address space is performed in 4 KB pages, as shown in Figure 2-7 on page 30.

After a page is written to disk, it is stored with metadata, including its unique virtual address. For this purpose, IBM i originally used a proprietary 520 bytes per sector disk format.

Note: The system storage that is conformed with main storage or main memory and auxiliary storage is addressed in the same way. This single, device-independent addressing mechanism means that objects are referred to by name or name and library, and *never* by disk location. The virtual addressing of IBM i is independent of the physical location of the object, type, capacity, and the number of disks units or LUNs on the system.

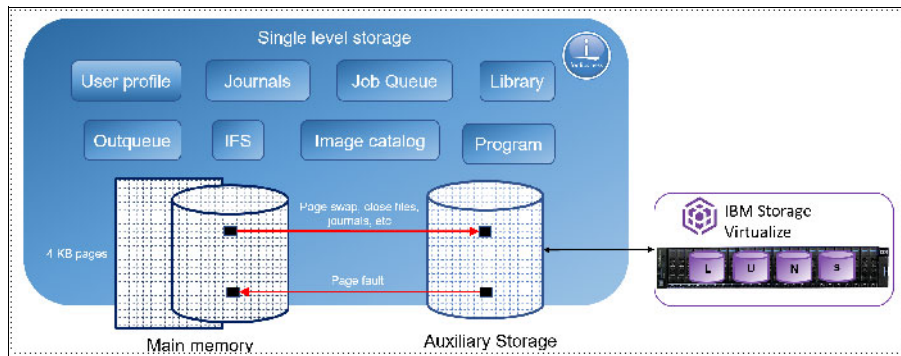


Figure 2-7 Virtual address space

The IBM i disk storage space is managed by using ASPs. Each IBM i system has a system ASP (ASP 1), which includes the load source (also known as *boot volume* on other systems) as disk unit 1, and optional user ASPs (ASPs 2–33). The system ASP and the user ASPs are designated as SYSBAS, and they constitute the system database.

The single-level storage with its unique virtual addresses also implies that the disk storage that is configured in SYSBAS of an IBM i system must be available in its entirety for the system to remain operational. It cannot be shared for simultaneous access by other IBM i systems.

To allow for sharing of IBM i disk storage space between multiple IBM i systems in a cluster, switchable independent auxiliary storage pools (IASPs) can be configured. The IBM i ASPs' architecture is shown in Figure 2-8.

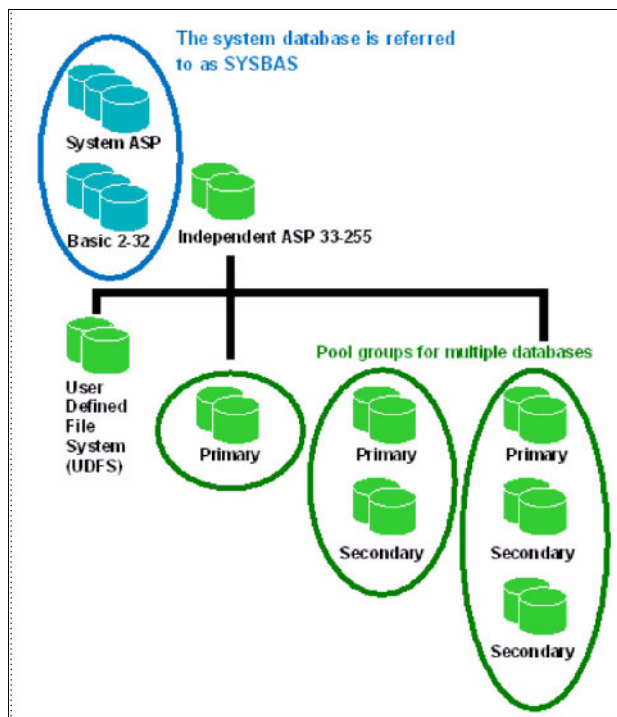


Figure 2-8 IBM i auxiliary storage pools architecture

Single-level storage makes main memory work as a large cache. Reads are done from pages in main memory, and requests to disk are done only when the needed page is not there yet.

Writes are done to main memory or main storage, and write operations to disk are performed as a result of swap, file close, or forced write. Application response time depends on disk response time and many other factors.

Other storage-related factors include the IBM i storage pool configuration for the application, how frequently the application closes files, and whether it uses journaling.

An example is shown in Figure 2-9.

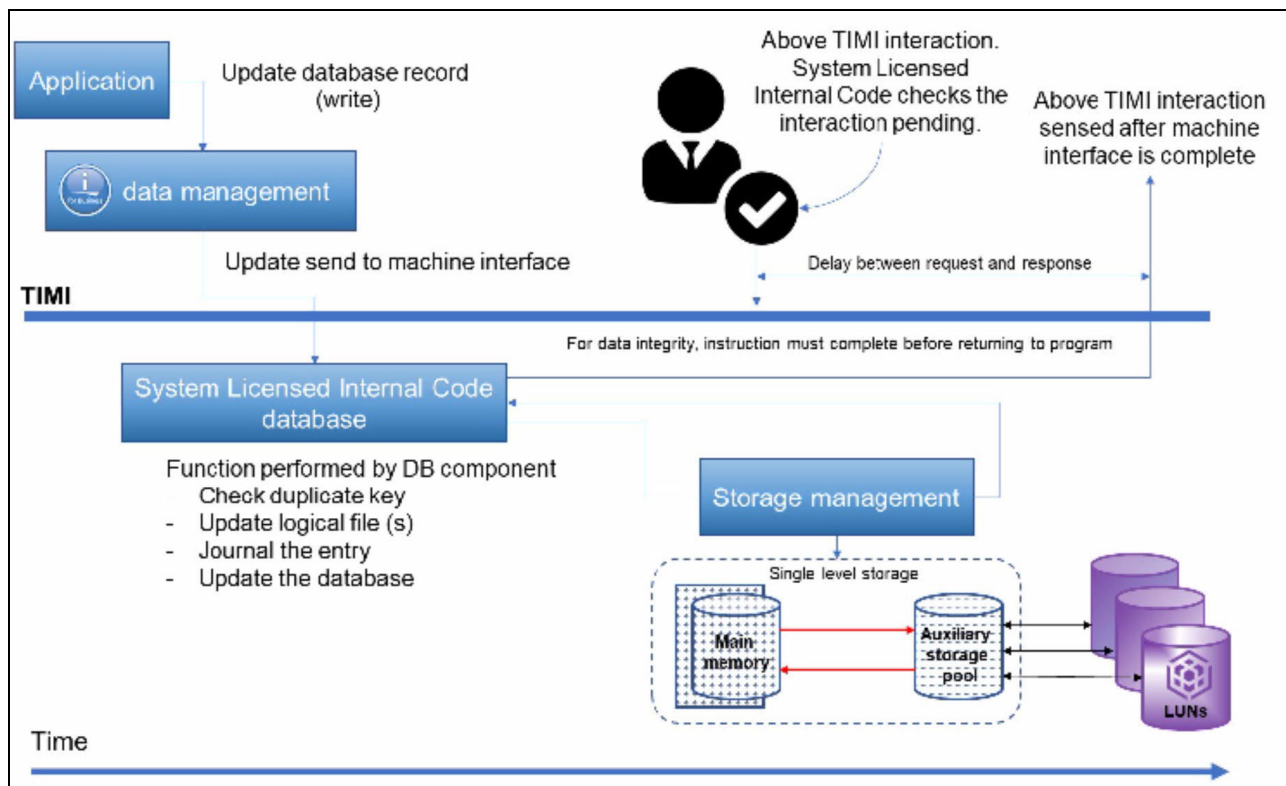


Figure 2-9 TIMI atomicity

Note: Figure 2-9 illustrates the ASP, composed of assigned LUNs from IBM Storage Virtualize to the IBM i system. It depicts an application request and the subsequent update to a database record. While the TIMI task is in progress, interactions above TIMI can occur, but they will not proceed until the TIMI task completes.

2.3.3 IBM i response time

IBM i customers are often concerned about the following types of performance:

- ▶ Application response time. The response time of an application transaction. This time is often critical for the customer.
- ▶ Duration of batch jobs. Batch jobs are often run during the night or other off-peak periods. The duration of a batch job is critical for the customer because it must be finished before regular daily transactions start.

- ▶ **Disk response time.** Disk response time is the time that is needed for a disk I/O operation to complete. It includes the service time for I/O processing and the wait time for potential I/O queuing on the IBM i host.

Disk response time can influence application response time and the duration of a batch job. Because the performance of the disk subsystem affects overall system performance, this issue is described in “Disk performance considerations” on page 32.

Disk performance considerations

Disk subsystem performance affects overall IBM i system performance, especially in a commercial data processing environment where often a large volume of data is processed. Disk drives or the LUNs’ response times contribute to a major portion of the overall response time (online transaction processing (OLTP)) or run time (batch).

Also, disk subsystem performance is affected by the type of protection (redundant array of independent disks (RAID), distributed RAID (DRAID), or mirroring).

The amount of free space (GB) on the drives and the extent of fragmentation also has an effect. The reason is the need to find suitable contiguous space on the disks to create objects or extend objects. Disk space often is allocated in extents of 32 KB. If a 32 KB contiguous extent is not available, two extents of 16 KB are used.

The following disk performance considerations are described in the following sections:

- ▶ Disk I/O requests
- ▶ Disk subsystems
- ▶ Disk operation
- ▶ Asynchronous I/O wait
- ▶ Disk protection
- ▶ Logical database I/O versus physical disk I/O

Disk I/O requests

Many disk requests often occur if a request for information cannot be satisfied by what is in memory. Requests to bring information into memory also result in disk I/O. Memory pages also can be purged periodically, which results in disk I/O activity.

Note: The Set Object Access (**SETOBJACC**) command on IBM i temporarily changes the speed of access to an object by bringing the object into a main storage pool or purging it from all main storage pools. An object can be kept in main storage by selecting a pool for the object that has available space and does not have jobs that are associated with it.

Disk subsystems

Typically, an external disk subsystem (storage system) connects a server through a SAN, as shown in Figure 2-10 on page 33.

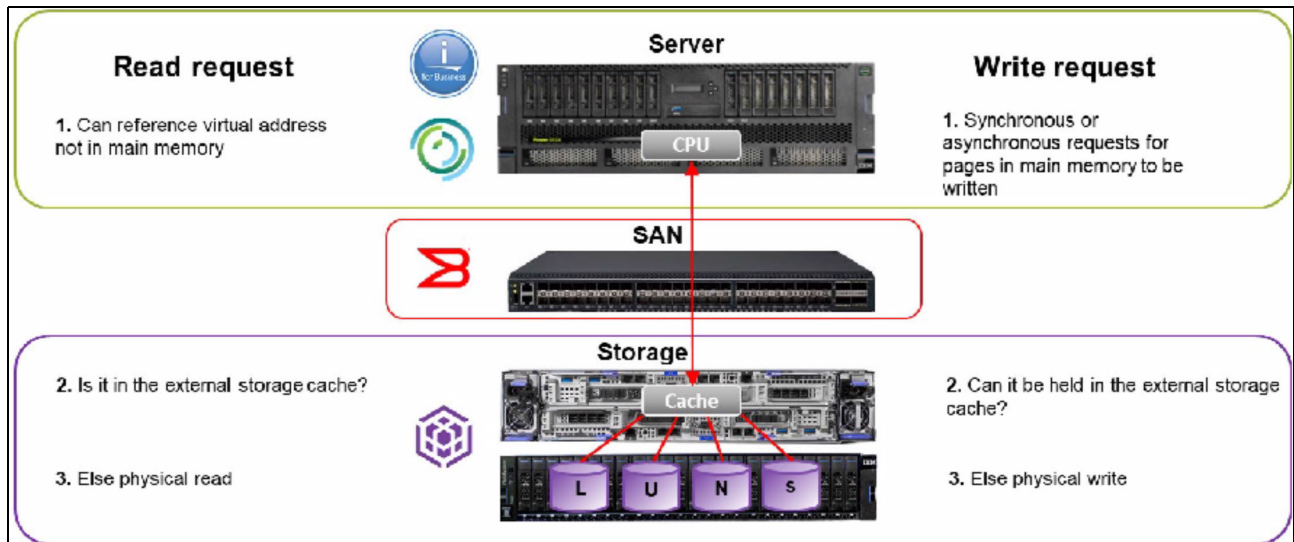


Figure 2-10 Disk subsystem

An information request (data or instructions) from the CPU that is based on user interactions is submitted to the disk subsystem if it cannot be satisfied from the contents of main memory. If the request can be satisfied from the disk subsystem cache, it responds or forwards the request to the disk drives or LUNs.

Similarly, a write request is retained in memory unless the operating system determines that it must be written to the disk subsystem. Then, the operating system attempts to satisfy the request by writing to the controller cache.

Note: The QAPMDISKRB from the collections services data files in IBM i includes disk file response bucket entries. It also contains one record for each device resource name. It is intended to be used with the QAPMDISK file.

For more information, see [Collection Services data files: QAPMDISKRB](#).

Disk operation

On IBM i, physical disk I/O requests are categorized as database (physical or logical files) or non-database I/Os, as shown in Figure 2-11.

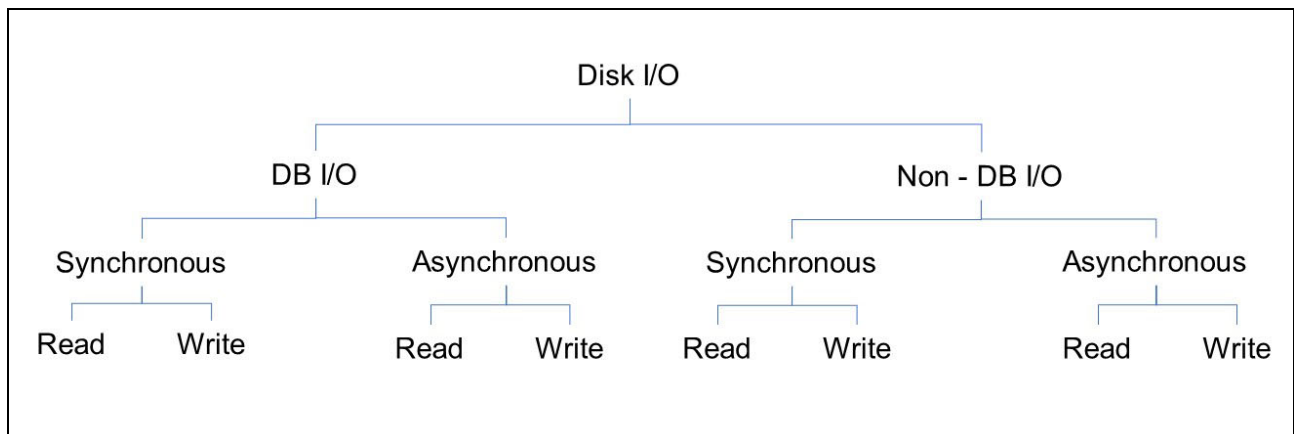


Figure 2-11 Disk I/O on IBM i

The time that it takes to respond to synchronous disk I/Os contributes to the OLTP response time or batch run time. With asynchronous I/O, the progress of a request does not wait for the completion of I/O.

Often, write requests are asynchronous, including journal deposits with commitment control. However, the writes become synchronous if journaling is active without commitment control.

Asynchronous I/O wait

On IBM i, jobs might have to wait for asynchronous I/O requests to complete. The job issues a request but requires the data sooner than it can be made available by the disk subsystem. When a job waits for asynchronous I/O requests to complete, the I/O portion of the operation becomes synchronous. The time is recorded as asynchronous disk I/O wait in the QAPMJOBL file.

JBWIO is the number of times that the process waited for outstanding asynchronous I/O operations to complete. For more information, see [Collection Services data files: QAPMJOBS and QAPMJOBL](#).

This issue might be caused by faster processors that are running with relatively poor disk subsystems performance. Disk subsystem performance can be affected by busy or slow disks or small I/O cache.

Disk protection

For more information about external storage considerations for setting up your RAID protection, see Chapter 3, “Drives, RAID and Arrays”.

Note: If you need high I/O performance on your IBM i workload, you can create a DRAID 1 array on your supported storage system. In this configuration, the rebuild area is distributed over all member drives. The minimum extent size for DRAID1 is 1024 MB.

Logical database I/O versus physical disk I/O

Information in partition buffer memory is available for use by any job or thread. Commonly, information is available in the partition buffer as a block of data rather than individual records. Data in a job buffer is available for use by the job only.

When an application program requests data, storage management determines whether the data is available in memory. If it is available in memory, the data is moved to the open data path in the job buffer. If the data is not in memory, the request is submitted to the disk subsystem as a read command.

In that context, logical database I/O information is moved between the open data path of the user program and the partition buffer. This information is a count of the number of buffer movements, and not a reflection of the records that are processed.

For more information, see [Sharing an Open Data Path](#).

Physical disk I/O occurs when information is read or written as a block of data to or from the disk. It involves the movement of data between the disk and the partition buffer in memory. For more information, see [IBM i 7.5: Performance](#)

2.3.4 Planning for IBM i storage capacity

To correctly plan the storage capacity that is provided by IBM Storage Virtualize family systems for IBM i, you must be aware of IBM i block translation for external storage that is formatted in 512-byte blocks. IBM i internal disks use a block size of 520 or 4160 bytes.

IBM Storage Virtualize storage for hosts is formatted with a block size of 512 bytes. Therefore, a translation or mapping is required to attach the storage to IBM i. IBM i changes the data layout to support 512-byte blocks (sectors) in external storage by using an extra ninth sector to store the headers for every page.

The eight 8-byte headers from each 520-byte sectors of a page are stored in the ninth sector, which is different than 520-byte sector storage where the 8 bytes are stored continuous with the 512 bytes of data to form the 520-byte sector.

The data that was stored in eight sectors is now stored by using nine sectors, so the required disk capacity on IBM Storage Virtualize based systems is 8/9ths of the IBM i usable capacity. Similarly, the usable capacity in IBM i is 8/9ths of the allocated capacity in these storage systems.

When attaching IBM Storage Virtualize storage to IBM i, plan for extra capacity on the storage system so that the 8/9ths of the effective storage capacity that is available to IBM i is enough for the capacity requirements for the IBM i workload.

The performance impact of block translation in IBM i is small or negligible.

Tip: When the attachment type is Virtual Fibre Channel (VFC), additional optimization might be required. See section 2.3.9, “Recommendations based on field experience” on page 39 for more details.

Figure 2-12 shows the byte sectors for IBM i.

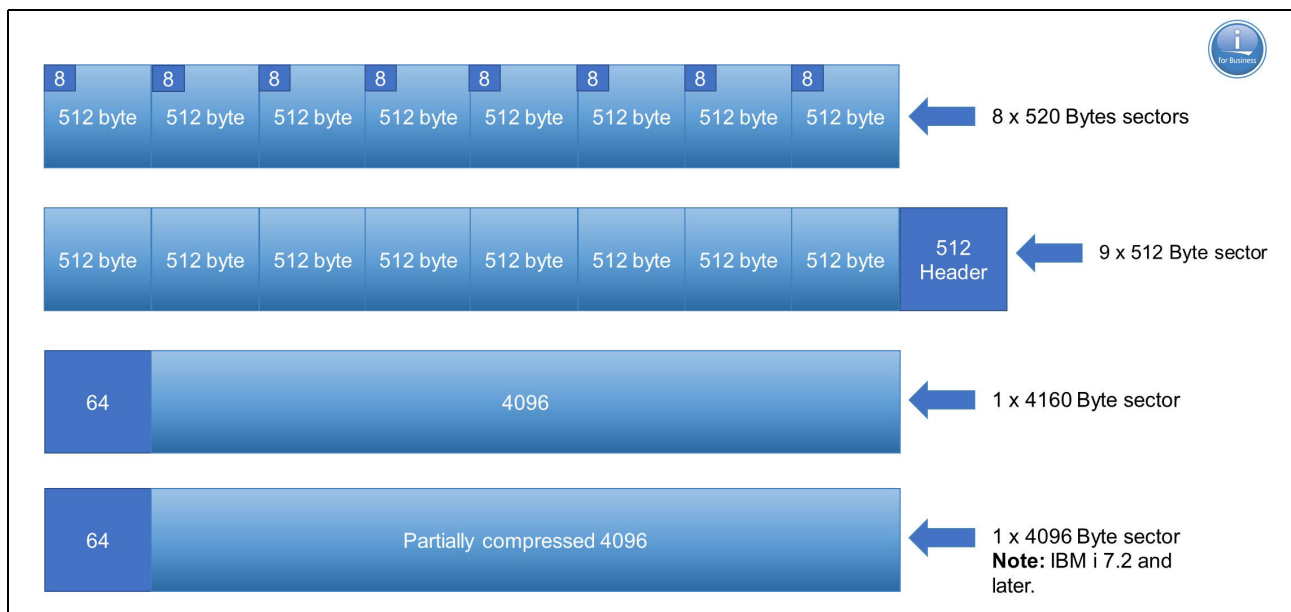


Figure 2-12 IBM i with different sector sizes

2.3.5 Storage connection to IBM i

IBM Storage Virtualize storage can be attached to IBM i in the following ways:

- ▶ Native connection without using the IBM PowerVM® Virtual I/O Server (VIOS)
- ▶ Connection with VIOS in N_Port ID Virtualization (NPIV) mode
- ▶ Connection with VIOS in virtual Small Computer System Interface (SCSI) mode

The decision for IBM i native storage attachment or VIOS attachment is based on the customer's requirements. Native attachment has its strength in terms of simplicity, and it can be a preferred option for static and smaller IBM i environments with only a few partitions. It does not require extra administration and configuration of a VIOS environment. However, it also provides the least flexibility and cannot be used with PowerVM advanced functions, such as Live Partition Mobility (LPM) or remote restart.

Table 2-1 lists the key criteria to help you with the decision of selecting an IBM i storage attachment method.

Table 2-1 Comparing IBM i native and Virtual I/O Server attachment

Criteria	Native attachment	VIOS attachment
Simplicity (configuration, maintenance, and failure analysis)	✓	More complex
Performance	✓	✓ (with NPIV)
Consolidation (storage and network adapters)	More limited	✓
PowerVM advanced functions (partition mobility, suspend and resume, remote restart, and private cloud deployment)	Not available	✓
Hardware support (storage and network adapters, and entry level servers)	More limited	✓

2.3.6 IBM i unmap support

To better use IBM Storage Virtualize storage flash technology with efficient storage space allocation and deallocation, IBM i supports storage system unmap capabilities by using corresponding host unmap functions.

Initially, IBM i unmap support that was implemented by using the SCSI Write Same command was introduced with IBM i 7.2 TR8 and IBM i 7.3 TR4 for LUN initialization only, that is, for the Add Disk Units to ASP function.

With IBM i 7.3 TR9 and IBM i 7.3 TR5, runtime support was added, which also supports synchronous unmap for scenarios, such as object deletion and journal clearance. The runtime unmap algorithm was further enhanced supported by IBM i 7.3 TR7 and IBM i 7.4 TR1, which implements an asynchronous periodic free-space cleaning.

IBM Storage Virtualize version 8.1.1 and later can use the unmap function to efficiently deallocate space, such as for volume deletion, on their back-end storage by sending SCSI

unmap commands to specific supported internal solid-state drives (SSDs) and FCMs, and selected virtualized external flash storage.

Space reclamation that is triggered by host **unmap** commands is supported by IBM Storage Virtualize 8.1.2 and later for DRP thin-provisioned volumes, which can increase the free capacity in the storage pool so that it becomes available for use by other volumes in the pool.

For more information about IBM Storage Virtualize storage SCSI unmap support, see [SCSI Unmap support in Spectrum Virtualize systems](#).

2.3.7 Defining LUNs for IBM i

LUNs for an IBM i host are defined from IBM Storage Virtualize block-based storage. Similar to open-system hosts, the LUNs are created from available extents within a storage pool.

Although IBM i supports a usable, large-size LUN of up to 2 TB, the usage of only a few large-size LUNs for IBM i is not recommended for performance reasons.

In general, the more LUNs that are available to IBM i, then the better the performance for the following reasons:

- ▶ If more LUNs are attached to IBM i, then storage management uses more threads, which improves performance.
- ▶ More LUNs provide a higher I/O concurrency. Higher I/O concurrency reduces the likelihood of I/O queuing and reduces the likelihood of disk wait time in the disk response time. All of which results in lower latency of disk I/O operations.

For planning purposes, many LUNs might also require more physical or vFC adapters on IBM i based on the maximum number of LUNs per FC adapter port that is supported by IBM i.

The sizing process helps to determine a reasonable number of LUNs that is required to access the needed capacity and meet performance objectives. As a suggested best practice, you can use the following guidelines:

- ▶ For any IBM i disk pool (ASP), define all the LUNs as the same size.
- ▶ 40 GB is the preferred minimum LUN size.
- ▶ You should not define LUNs larger than about 200 GB.

Note: This rule is not fixed because it is important that enough LUNs are configured, with which this guideline helps. Selecting a larger LUN size might lead to configurations with fewer LUNs being configured, which might have detrimental effects on performance.

- ▶ For optimal performance, allocate a minimum of eight LUNs per ASP for small IBM i partitions, approximately two dozen for medium partitions, and up to several hundred for large partitions.

When defining LUNs for IBM i, consider the following required minimum capacities for the load source (boot disk) LUN:

- ▶ With IBM i 7.1, the minimum capacity is 20 GB.
- ▶ With IBM i 7.2 before TR1, the minimum capacity is 80 GB in IBM i.
- ▶ With IBM i 7.2 TR1 and later, the minimum capacity is 40 GB in IBM i.

IBM Storage Virtualize dynamic volume expansion is supported for IBM i with IBM i 7.3 TR4 and later. An IBM i initial program load (IPL) is required to use the extra volume capacity.

Tip: For more information about cross-referencing IBM i disks units with IBM Storage Virtualize LUNs by using NPIV, see [Cross-Referencing \(Device Mapping\) IBM i Disks with SAN Disks Using N-Port ID Virtualization \(NPIV\)](#).

Disk arms and maximum LUN size

Selected limits that are related to disk arms and LUN sizes were increased in IBM i 7.4, as listed in Table 2-2.

Table 2-2 Limits increased for maximum disk arms and LUN sizes.

System limits	IBM i 7.2	IBM i 7.3	IBM i 7.4	IBM i 7.5
Disk arms in all basic ASPs (ASPs 1–32) per logical partition (LPAR)	2047	2047	3999	3999
Disk arms in all IASPs (IASPs 33–255) in all nodes in a cluster	2047	2047	5999	5999
Maximum combined number of disk arms and redundant connections to disk units	35.600	35.600	35.600	35.600
512-byte block size LUNs ^a	2 TB	2 TB	2 TB	2 TB
4096-byte block size LUNs ^b	2 TB	2 TB	16 TB	16 TB

a. The limit is one block short of the maximum that is listed in Table 2-2. For all 512 block LUNs, the maximum is still 2 TB, including FlashSystem LUNs.

b. This size includes FlashSystem LUNs, and 4 K block serial-attached SCSI (SAS) disks (VSCSI-attached).

For more information about these limits and other limits, see [Miscellaneous limits](#).

2.3.8 Data layout

Spreading workloads across all IBM Storage Virtualize storage components maximizes the usage of the hardware resources in the storage subsystem. I/O activity must be balanced between the two nodes or controllers of the IBM Storage Virtualize storage system I/O group, which often is addressed by alternating the preferred node volume assignments at LUN creation.

However, performance problems might arise when sharing resources because of resource contention, especially with incorrect sizing or unanticipated workload increases.

Some isolation of workloads, at least regarding a shared back-end storage, can be accomplished by using a configuration in which each IBM i ASP or LPAR has its own managed storage pool. Such a configuration with dedicated storage pools results in a

trade-off between accomplishing savings from storage consolidation and isolating workloads for performance protection. This result occurs because a dedicated storage pool configuration likely requires more back-end storage hardware resources because it cannot use the averaging effect of multiple workloads, typically showing their peaks at different time intervals.

Consider the following data layouts:

- ▶ For all-flash storage configurations, if the back-end storage is sized correctly, usually no reason exists for not sharing the disk pool among multiple IBM i workloads.
- ▶ For hybrid configurations with Easy Tier on mixed HDD and flash disks, the storage pool can be shared among IBM i workloads. Only large performance-critical workloads are configured in isolated disk pools.
- ▶ For HDD only pools, isolate performance-critical IBM i workloads in separate storage pools.
- ▶ Avoid mixing IBM i LUNs and non-IBM i LUNs in the same disk pool.

Apart from the usage of Easy Tier on IBM Storage Virtualize for managing a multi-tier storage pool, an option is available to create a separate storage pool for different storage tiers on IBM Storage Virtualize storage and create different IBM i ASPs for each tier. IBM i applications that have their data in an ASP of a higher storage tier experience a performance boost compared to the ones that use an ASP with a lower storage tier.

IBM i internal data relocation methods, such as the ASP balancer hierarchical storage management function and IBM Db2 media preference, are not available to use with IBM Storage Virtualize flash storage.

2.3.9 Recommendations based on field experience

- ▶ The maximum number of disk paths for VIOS-attached 512-byte sector storage is approximately 6000 for VFC/NPIV and approximately 3000 for VSCSI.

Important: Do not exceed these limits.

- ▶ The storage management layer distributes data placement and workload across all volumes in the ASP. If the ASP contains volumes of different sizes, this can cause unbalanced workloads.

Recommendation: Ensure that all volumes are the same size.

- ▶ IBM i can meet write performance requirements with internal storage because of architecture and locality. However, with external storage there can be delays in the transport layer (SAN) and the virtualization layer (VFC/VSCSI). These small delays can sometimes negatively affect applications that are affected by only a small latency. Additionally, sometimes even moderate workloads can overload HDD arrays. For example, small random writes are unable to be coalesced as full stride writes on the FlashSystem and can incur read/modify/write (RMW) penalties at the array level. An example of a workload that may exhibit this behavior is file-level journaling.

Recommendation: Consider enabling IBM i Journal cache so that the system can bundle or cache journal entries in main storage before writing them to disk. Thereby reducing the applications dependency on disk I/O performance for some types of workloads.

- When you trigger a snapshot, peak response time is proportional to the number of volumes in the volume group. For consistency from the IBM i perspective, volumes in the same ASP must be in the same volume group. This dictates the number of volumes in the volume group and therefore the magnitude of peak response times as well. Policy-based replication also uses the snapshot function while in cycling mode. The pauseless snapshot feature introduced in code level 8.7.0.0 reduces peak response times to negligible levels. Refer to Chapter 8, “Pauseless snapshots case study” on page 167 for more details.

Recommendation: If volume group snapshots or policy-based replication is used, the FlashSystem storage system must be running a recommended code level later than 8.7.0.0.

- Section 2.3.4, “Planning for IBM i storage capacity” on page 35 discusses the capacity requirements to translate to or from the 520-byte addressing scheme. FlashSystem systems that have IBM i volumes in a data reduction pool (DRP) where the IBM i capacity requirements were not factored into the capacity sizing might cause the pool to exceed 85% usage. This can cause aggressive DRP garbage collection and can create a potential performance impact.

Recommendation: Ensure that capacity sizing for the IBM i capacity requirements is considered and that there is sufficient margin to the suggested maximum DRP usage of 85%

- Internal storage and IBM DS8000® support a 520-byte block size natively, so no translation is required. Native Fibre Channel and Virtual Fibre Channel (VFC/NPIV) IBM i attachment does require address translation because FlashSystem supports a 512-byte block size. For native Fibre Channel, this is seamless because the native Fibre Channel drivers are able to manage the user data blocks and metadata blocks together, interfacing directly with the Fibre Channel adapter to gather the appropriate data. Virtual Fibre Channel drivers are not able to directly interface with the physical Fibre Channel adapters because the physical adapters reside on VIOS and because VIOS uses a different structure to organize storage. As a result, the IBM i Virtual Fibre Channel (VFC) driver must build and populate a buffer that contains user data and metadata. This task consumes CPU time and adds to the overall completion time for each disk I/O. Each of these tasks is linked with an internal resource called an interrupt handler. Contention for interrupt handlers can negatively affect performance.

Recommendation: Instead of creating a single host object on the FlashSystem, add more VFCs and break up the IBM i host object into multiple pseudo host objects and map a subset of volumes to each pseudo host object. This helps achieve a path count of 8 (in combination with zoning) and allocates more interrupt handlers to reduce the CPU usage caused by translation.

- Maximum IBM i limits are documented at 127 volumes per VFC adapter. However, field experience shows that less than 32 volumes per Storage Virtualize host is best for performance. Depending on peak I/O workload, you might need to further reduce those values to maintain adequate performance. This does not apply to native attached storage.

Recommendation: For optimal performance with currently supported IBM i OS versions, aim for a total volume count per host within the lower half of the supported range of the FlashSystem.

- ▶ Certain IBM i levels can generate an aggressive unmap workload. This might increase usage of FlashSystem resources and affect performance. This is addressed by [IBM i APAR MA48078](#).

This issue can also be mitigated on the FlashSystem by applying an offload throttle. For more information, see [mkthrottle](#).

Refer to the following IBM i Documentation For more information see the following IBM i documentation:

- ▶ [IBM i Virtual Fibre Channel Performance Best Practices](#)
- ▶ [IBM i on Power - Performance FAQ](#)



Performance optimization for FlashSystem storage tiers and arrays

This chapter describes the different IBM Storage FlashSystem storage tiers, the drive technologies behind them, and performance considerations.

This chapter has the following sections:

- ▶ 3.1, “Internal storage types” on page 44
- ▶ 3.2, “Performance and scaling considerations” on page 49
- ▶ 3.3, “Arrays” on page 54

3.1 Internal storage types

IBM Storage FlashSystem (FlashSystem) is a part of the IBM Storage Virtualize family. IBM Storage Virtualize is the common name for IBM Storage FlashSystem and IBM SAN Volume Controller. FlashSystem supports the following three types of devices that are attached by using the Non-Volatile Memory Express (NVMe) protocol:

- ▶ Storage-class memory (SCM) drives
- ▶ IBM FlashCore® Module (FCM)
- ▶ Industry-standard NVMe flash drives

With a Serial Attached SCSI (SAS) attachment, flash (solid-state drives (SSDs)), and hard disk drives (HDDs) are supported. The set of supported drives depends on the platform. SAS is a primary interface for connecting drives in expansion enclosures.

3.1.1 NVMe support

FlashSystem offers end-to-end NVMe and provides feature-rich, enterprise-grade storage solutions that help support crucial workloads and applications.

The Non-Volatile Memory Express NVMe protocol

Non-Volatile Memory Express (NVMe) is a high-speed interface that is optimized for fast PCIe-based storage like SSDs.

NVMe has the following key performance highlights:

- ▶ Designed for speed. Built for low latency and high-performance storage.
- ▶ Direct PCIe connection. Reduces latency by connecting directly to the host.
- ▶ High throughput. Offers significant scalability in data transfer rates.
- ▶ Multiple queues. Can run more concurrent operations for better performance.
- ▶ RDMA support. Enhances performance through efficient data access.
- ▶ High bandwidth. Uses dedicated PCIe lanes for much faster data transfer in GiBps compared to older interfaces like SAS that have transfer speeds in MiBps.

The adoption of NVMe reduces overall latency through greater parallelism and modern drivers. NVMe often employing efficient polling instead of interrupt-driven interfaces. Although storage-side latency within IBM Storage Virtualize might not drastically change with a protocol switch from SCSI to NVMe, bandwidth increases are noticeable with PCIe-attached drives. The main performance advantage of end-to-end NVMe with IBM Storage Virtualize is a reduction in host server CPU cycles for interrupt handling. Note that improvements in parallelism and reduced CPU cycles have been part of IBM Storage Virtualize since 2003.

3.1.2 Overview of the different Flash devices used in the FlashSystem

Figure 3-1 on page 45 illustrates the Flash devices that are used across the different FlashSystem models.

				
	SAS SSD	NVMe SSD	FlashSystem Module	SCM SSD
Type of Flash	3D NAND	3D NAND	3D NAND	Optane, zSSD
Protocol	Serial Attached SCSI	NVMe	NVMe	NVMe
Physical Size	2.5"	2.5"	2.5"	2.5"
Physical Capacities	1.92TB – 30TB	800GB – 30.72TB	4.8TB – 38.4TB	1.6TB
Differentiator	None	None	Hardware Comp and Encryption	Ultra low Latency
	FlashSystem 5015 FlashSystem 5045 FlashSystem 5300/7300/9500*	FlashSystem 5300 FlashSystem 7300 FlashSystem 9500	FlashSystem 5300 FlashSystem 7300 FlashSystem 9500	FlashSystem 5300 FlashSystem 7300 FlashSystem 9500
* Expansion enclosure required				

Figure 3-1 Different Flash devices used in FlashSystem

3.1.3 Storage Class Memory

Storage Class Memory (SCM) provides significantly faster nonvolatile storage than traditional NAND SSDs with latency times of approximately 10 μ s instead of 100 μ s. However, it is slower than DRAM, which has a latency of approximately 100 ns. This technology is available in select FlashSystem models (FS5300, FS7300, FS9500). Although it is typically more expensive than NAND SSDs but less than DRAM, SCM offers a compelling balance of speed and cost. For example, the FlashSystem 5300, 7300, and 9500 support low-latency, high-speed SCM NVMe drives, with a control enclosure capable of housing up to twelve 1.6 TB drives.

Note: SCM and other NVM Express (NVMe) drive types can be installed in any drive slot. However, the highest capacity drives must be installed in the lowest available drive slots.

SCM and RAID configurations

Use the following rules for the SCM drive RAID configuration:

- ▶ Best practice. Use Distributed RAID 1 (DRAID 1) with 2 or more drives (including distributed spare).
- ▶ Supported. Distributed RAID 6 (DRAID 6) is supported with 6 or more drives (including distributed spare).
- ▶ Drive types. SCM drives have their own specific technology and drive class. See section 3.3.3, “Array geometry and considerations” on page 56.
- ▶ No Intermixing. SCM and Standard NVMe/SAS drives cannot be mixed in the same array.
- ▶ Easy Tier:
 - Tier_scm arrays can use lower-tech drives as spares if no tier_scm spares are available.
 - Lower-tier arrays can accept tier_scm drives as superior drives.

3.1.4 NVMe FlashCore module

FlashCore Module (FCM) is a family of high-performance flash drives that provides performance-neutral, hardware-based data compression and self-encryption.

IBM FlashCore technology (Figure 3-2) is at the heart of the FlashSystem if the IBM FlashCore Module NVMe type drives are ordered, which consists of the following key elements:

- ▶ Hardware-accelerated architecture that is engineered for flash, with a hardware-only data path.
- ▶ The FlashSystem data compression and decompression algorithm is a modified dynamic gzip algorithm. Because it is implemented completely in hardware, no processor intervention is required.
- ▶ IBM FlashCore Modules, which are designed for low latency, high density, and reliability.
- ▶ IBM Advanced Flash Management, which improves flash endurance over standard implementations without sacrificing latency.
- ▶ Depending the machine type and model of the control enclosure, the largest enclosure can contain up to 48 FCM drives.

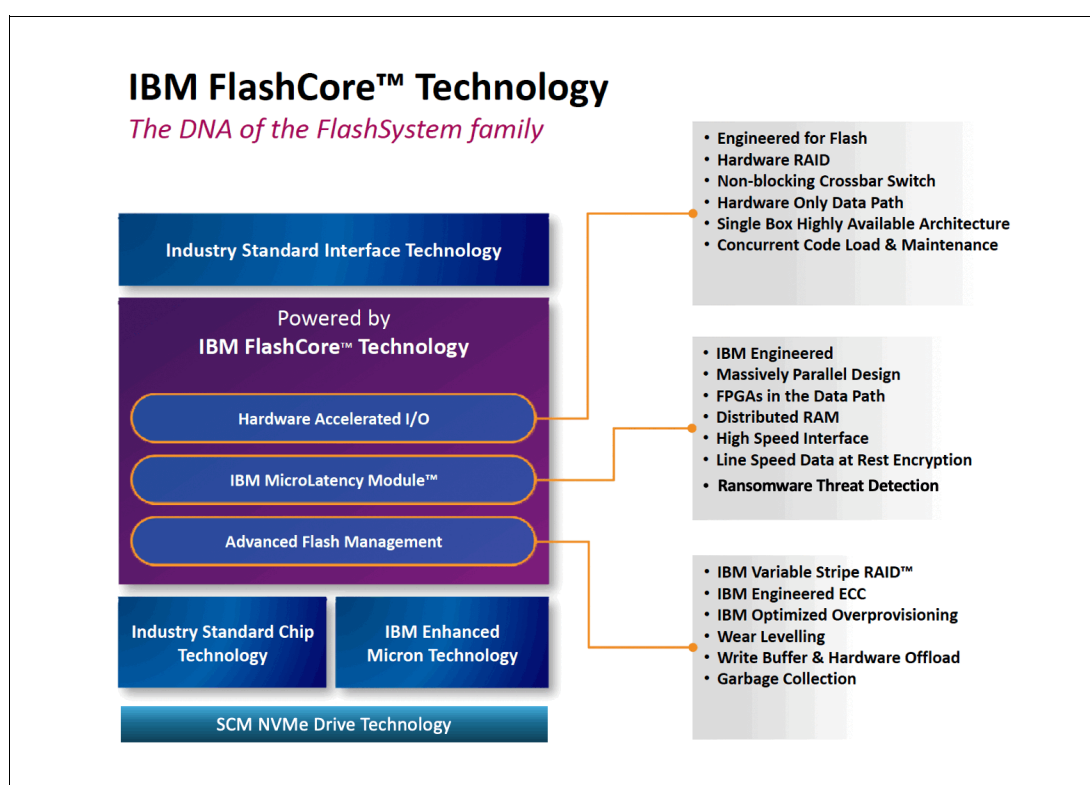


Figure 3-2 IBM FlashCore technology

For more information about IBM FlashCore technology, see this [IBM Video](#).

Note: At the time of writing, IBM is the only vendor offering Variable Stripe RAID (VSR) for comprehensive RAID protection while maintaining peak performance. FCM drives implement VSR at the FCM level. This multi-dimensional protection incorporates system-level RAID. The benefit is that many issues that typically require system-level RAID intervention are handled at the module level in IBM solutions.

Bit errors that are caused by electrical interference are continually scanned for, and if any errors are found, they are corrected by an enhanced Error Correcting Code (ECC) algorithm. If an error cannot be corrected, then the DRAID layer on the FlashSystem is used to rebuild the data.

Note: FCMs use inline hardware compression to reduce the amount of physical space that is required. Compression cannot be disabled. If the written data cannot be compressed further or compressing the data causes it to grow in size, the uncompressed data is written. In either case, because the FCM compression is done in the hardware, there is no performance impact.

FCMs are available with a physical or usable capacity of 4.8 TB, 9.6 TB, 19.2 TB, and 38.4 TB. The *usable capacity* is a factor of how many bytes the flash chips can hold.

Because FCMs are compression capable, FCMs have a maximum effective capacity (or virtual capacity) beyond which they cannot be filled.

Effective capacity is the total amount of user data that can be stored on a module, assuming that the compression ratio (CR) of the data is at least equal to (or higher than) the ratio of effective capacity to usable capacity. Each FCM contains a fixed amount of space for metadata, and the maximum effective capacity is the amount of data it takes to fill the metadata space.

IBM FlashCore Gen4 modules

IBM offers the fourth generation of FCMs. FCM4, which provides better performance and lower latency than earlier FCMs.

The FCM4 includes the following features:

- ▶ Logical to Physical Table (LPT) paging enables increased effective capacity. Increased effective capacity stems from the FCM storing the LPT more efficiently in DRAM. In prior iterations of FCMs, IBM allocated the entire LPT in DRAM, increasing the cost and limiting the amount of effective capacity. LPT paging enables the new FCM4 to increase the effective capacity.
- ▶ IBM Storage Virtualize and FCM4 hints improve performance and reduce latency:
 - FCM4 has some significant performance and latency improvements.
 - Large and extra-large performance is significantly increased:
 - PCIe Gen 4 technology.
 - In comparison with FCM2, read throughput increases from 2.25 GBps to 3.52 GBps, and write throughput doubles.
- ▶ New optimizations between IBM Storage Virtualize and FCM reduce system-level latency and enhance performance. The drive can be sent hints regarding the access frequency of an I/O that it is receiving so that hot data can be placed in faster pages, and colder data can be placed in slower pages. This process is also referred to as *smart data placement*.
- ▶ Data reduction pool (DRP) metadata hints to prioritize DRP metadata.
- ▶ RAID parity hints to prioritize RAID parity.
- ▶ Parity scrub hints to de-prioritize background scrubber reads.
- ▶ RAID rebuild enhancements.

Table 3-1 shows the values of FCM4.

Table 3-1 Values of FCM4

Capacity Point	Small	Medium	Large	X-Large	C200 X-Large
NAND Dies	2DP	4DP	8DP	16DP	16DP
Flash Technology	SLC / QLC	SLC / QLC	SLC / QLC	SLC / QLC	SLC / QLC
Raw Capacity	5.75 TB	11.50 TB	23 TB	46 TB	46 TB
Usable Capacity	4.8 TB	9.6 TB	19.2 TB	38.4 TB	38.4TB
Effective Capacity	22 TB	28.8 TB	57.6 TB	115.2 TB	115.2 TB
Endurance	1 DWPD	1 DWPD	1 DWPD	1 DWPD	1 DWPD
PCIe	Gen 4	Gen 4	Gen 4	Gen 4	Gen 4
Read Bandwidth	2,200 MB/s	2,200 MB/s	3,500 MB/s	3,500 MB/s	1,000 MB/s
Write Bandwidth	750 MB/s	750 MB/s	1,500 MB/s	1,500 MB/s	700 MB/s
Read IOPS (16K)	150k	150k	200k	200k	150k
Power Envelope	Up to 25W	Up to 25W	Up to 30W	Up to 30W	Up to 24W
Ransomware detection	Yes	Yes	Yes	Yes	Yes
Supported on	FS5300, FS7300, FS9500 FS9500R	FS5300, FS7300, FS9500, FS9500R	FS5300, FS7300, FS9500 FS9500R	FS5300, FS7300, FS9500 FS9500R	FlashSystem C200

Notes: For information about the usage of FCMs for your FlashSystem model see [IBM Documentation](#), and search for your model. For example, searching for FlashSystem 9500 in the search bar provides a number of links for you to view.

3.1.5 Industry-standard NVMe drives

FlashSystem control enclosures provide an option to use self-encrypting industry-standard NVMe flash drives, which are available with 1.92–30.72 TB capacity.

3.1.6 SAS drives

FlashSystem control enclosures have only NVMe drive slots, but systems can be scaled up by attaching SAS expansion enclosures with SAS drives.

FlashSystem 5015, 5035 and 5045 control enclosures have twelve 3.5-inch large form factor (LFF) or twenty-four 2.5-inch small form factor (SFF) SAS drive slots. They can be scaled up by connecting SAS expansion enclosures. Figure 3-3 on page 49 shows you all the available SAS expansion enclosures.

Scale up expansion

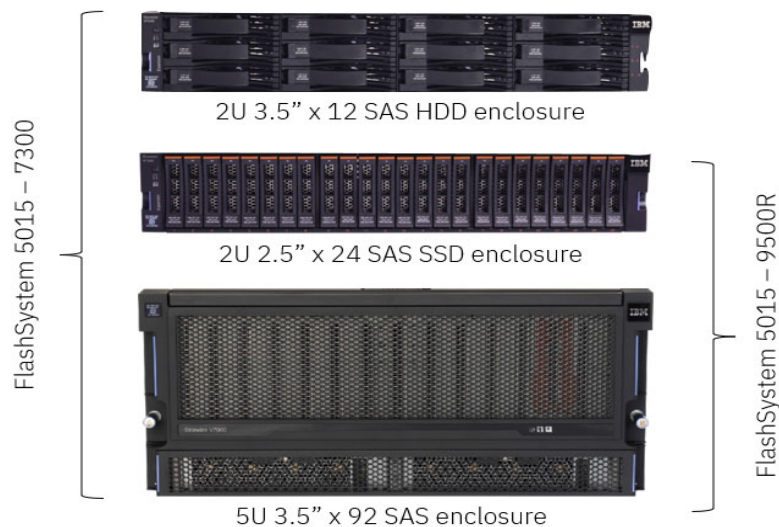


Figure 3-3 Scale up expansion

To find out exact and recent details regarding FlashSystem and expansion possibilities, refer to the following IBM Documentation links:

- ▶ [IBM Storage FlashSystem 50X0](#)
- ▶ [IBM Storage FlashSystem 5200](#)
- ▶ [IBM Storage FlashSystem 5300](#)
- ▶ [IBM Storage FlashSystem 7200](#)
- ▶ [IBM Storage FlashSystem 7300](#)
- ▶ [IBM Storage FlashSystem 9100](#)
- ▶ [IBM Storage FlashSystem 9200](#)
- ▶ [IBM Storage FlashSystem 9500](#)
- ▶ [IBM Storage FlashSystem C200](#)

Note: There can be a mixture of different drive types, so determine if drives are on the latest software level. This check is done by the Software Upgrade Test Utility Software. The latest Software Upgrade Test Utility can be downloaded from the [Fix Central Webpage](#).

For information about manually verifying the drive firmware levels, see [Drive Types and Firmware levels](#).

3.2 Performance and scaling considerations

This section includes a description of best practices for planning and managing FlashSystem internal storage.

Select the capacity and performance level that meets the client's application needs. The performance data presented in Figure 3-4 on page 50 uses a 16 KB block size, a 70% read and 30% write workload, and a 50% cache hit rate to demonstrate the comparative performance of each system under a client-relevant benchmark. For application-specific platform optimization, it is recommended to model workload requirements by using the [IBM](#)

[Storage Modeller Tool](#). This link is accessible to IBM representatives and IBM Business Partners. IBM Business Partners can access this link with an IBM Internet ID.

Figure 3-4 provides an overview of performance in relation to capacity.

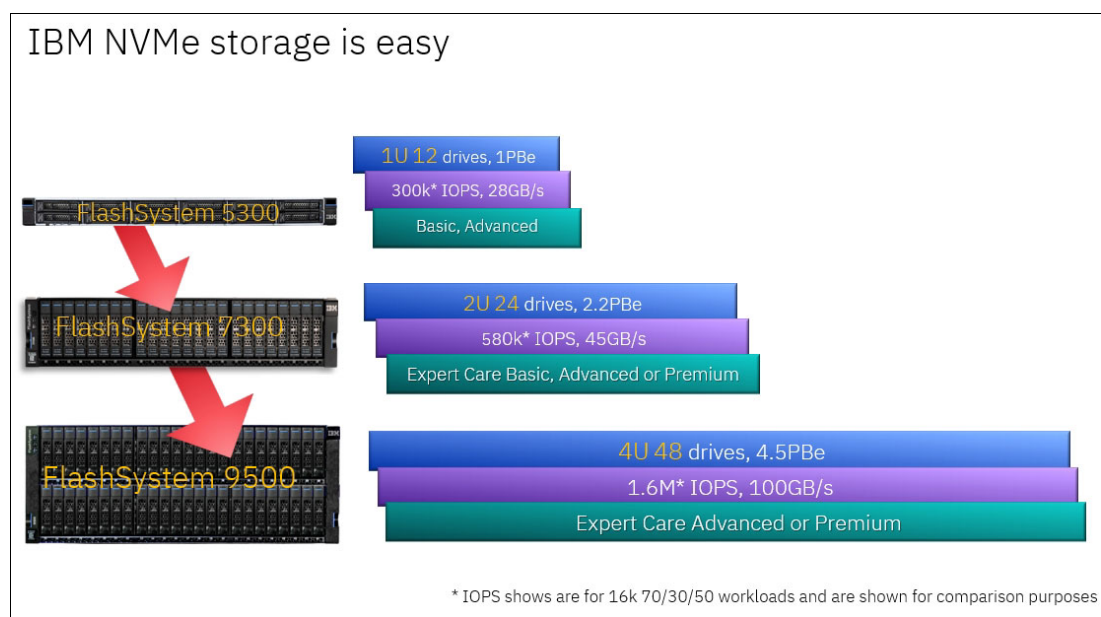


Figure 3-4 Choose right level of capacity and performance

Planning for performance and capacity

FlashSystem models 50x0, 5300, 7300, and 9500 use SAS enclosures for scaling capacity within a single controller's performance limits. Starting with code level 8.7.0, the FlashSystem grid enables further performance scaling with additional NVMe storage. For optimal performance, it is recommended to maintain physical capacity usage at approximately 85% or less.

Flash drives require free space for efficient write operations and garbage collection. Insufficient free space can lead to drive overwork, potentially causing proactive software failure or a hard failure. A hard failure means that the flash drive changes to a status of write-protected. Sufficient free space also facilitates efficient rebuild scenarios and mitigates write amplification.

If using data reduction, it is best to keep the system below ~85% capacity. This allows for responsiveness to sudden changes in data reduction (like host encryption activation). Additionally, running near full capacity strains garbage collection and new write processing, potentially slowing the system and increasing host latency.

For more information about physical flash provisioning, see [Do not provision 100% of the physical flash](#).

Tip: Implementing DRPs on storage systems composed of only Near-Line SAS (NL-SAS) drives is *not* recommended because of performance limitations and potential inefficiencies. Although NL-SAS drives offer high capacity at a lower cost, they inherently possess fewer IOPS compared to faster drive types like SSDs. The additional demand introduced by data reduction techniques, such as compression and deduplication, can further exacerbate these performance constraints.

Intermix rules

Within an expansion enclosure, drives of the same form factor and connector type can be intermixed. For systems supporting NVMe drives, both NVMe and SAS drives can be used in the same system, but with specific placement. Only NVMe drives can be used in the control enclosure, and SAS drives are limited to SAS expansion enclosures.

Within an NVMe control enclosure, different types and capacities of NVMe drives are intermixable. Furthermore, industry-standard NVMe drives and SCMs can be intermixed with FCMs.

For additional information about mixing different drives in a single DRAID array, see “Considerations for different drive types intermix in the arrays” on page 60.

3.2.1 Specific drives related activities and considerations

This section describes specific drives-related activities and considerations.

Formatting

Drives and FCMs require formatting before use. The chosen format is crucial because array members must have zero used capacity upon creation. Drives are automatically formatted when they become candidates.

FCM formatting typically completes within 70 seconds. Formatting SCM drives takes significantly longer than FCM or standard NVMe drives. For example, Intel Optane drives can take up to 15 minutes.

During formatting, a drive appears as an offline candidate, and array creation commands are queued until all formatting is complete. If a drive fails to format, it goes offline and requires a manual format to restore it.

Securely erasing

All SCM, FCM, and industry-standard NVMe drives that are used in the system are self-encrypting. For SAS drives, encryption is performed by a SAS chip in the control enclosure.

Note: Because of the methods and commands that are used to securely delete data from drives, the system is in compliance with [European Regulation EU2019/424](#).

For industry-standard NVMe drives, SCMs, and FCMs, formatting the drive completes a cryptographic erase of the drive. After the erasure, the original data on that device becomes inaccessible and cannot be reconstructed.

To securely erase SAS or NVMe drives, use the `chdrive -task erase <drive_id>` command.

Monitoring FCM capacity

You can use the FlashSystem GUI and CLI to monitor effective and physical capacity for each FCM. Physical used capacity is expected to be nearly the same on all modules in one array. When FCMs are used, data compression rate (CR) should be thoroughly planned and monitored.

Even with highly compressible data, an FCM can fill when its maximum effective capacity is reached. Any remaining spare space is used to enhance module performance and extend wear-leveling.

Example: A total of 20 TiB of data that is compressible 10:1 is written to a 4.8 TB module.

The maximum effective capacity of the module is 21.99 TB, which equals 20 TiB.

The usable capacity of the module is 4.8 TB, which equals 4.36 TiB.

After 20 TiB of data is written, the module is 100% full for the array because it has no free effective (logical) capacity. At the same time, the data uses only 2 TiB of the physical capacity. The remaining 2.36 TiB cannot be used for host writes, only for drive internal tasks and to improve the module's performance.

If non-compressible or low-compressible data is written, the module fills until the maximum physical capacity is reached.

Example: A total of 20 TiB of data that is compressible 1.2:1 is written to a 19.2 TB module.

The maximum effective capacity of the module is 43.99 TB, which equals 40 TiB.

The usable capacity of the module is 19.2 TB, which equals 17.46 TiB.

After 20 TiB is written, only 50% of effective capacity is used. With 1.2:1 compression, it occupies 16.7 TiB of physical capacity, which makes the module physically 95% full, and potentially affects the module's performance.

Pool-level and array-level warnings can be set to alert and prevent compressed drive overflow.

Drive writes per day

Drive writes per day (DWPD) is a term that is used to express the number of times that the total capacity of a drive can be written per day within its warranty period. This metric shows drive write endurance.

If the drive write workload is continuously higher than the specified DWPD, the system alerts that the drive is wearing faster than expected.

Although DWPD measurement is largely irrelevant for FCMs and large SSDs, it is important to consider DWPD during system sizing. A DWPD alert usually indicates that the workload differs from what was expected on the array, necessitating a revision of the configuration or workload distribution.

DWPD numbers are especially important with SSD drives of smaller sizes. With drive capacities below 1 TB, it is possible to write the total capacity of a drive several times a day. When a single SSD provides tens of terabytes, it is unlikely that you can exceed the DWPD parameter.

Consider the following points:

- SAS-attached Tier1 flash drives support up to 1 DWPD, which means that full drive capacity can be written on it every day and it lasts the 5-year lifecycle.

Example: A total of 3.84 TB read-intensive (RI) SAS drive is rated for 1 DWPDP, which means 3,840,000 MB of data can be written on it each day.

Each day has $24 \times 60 \times 60$ seconds = 86400 seconds.

The average daily write workload required to reach 1 DWPDP is $3840000 \text{ MB} / 86400 \text{ seconds} = 44.4 \text{ MBps}$ of average daily write workload is required to reach 1 DWPDP.

Total cumulative writes over a 5-year period are $3.84 \times 1 \text{ DWPDP} \times 365 \times 5 = 6.8 \text{ PB}$.

- FCM2 and FCM3 drives are rated with two DWPDP over five years, which is measured in usable capacity. Therefore, if the data is compressible (for example, 2:1), the DWPDP doubles.

Example: A total of 19.2 TB FCM is rated for 2 DWPDP. Its effective capacity is nearly 44 TB = 40 TiB, so if you use 2.3:1 compression, to reach the DWPDP limit, the average daily workload over 5 years is approximately 1 GBps. Total cumulative writes over a 5-year period are more than 140 PB.

- FCM4 are rated with one DWPDP.
- SCM drives are rated with 30 DWPDP over five years.

The system monitors the number of writes for each drive that supports the DWPDP parameter, and it logs a warning event if this amount is above the DWPDP for the specific drive type.

The write endurance usage rate has high warnings, which indicates that the write data rate exceeds the expected threshold for the drive type during the initial phase of system implementation or stress testing. Afterward, when the system's workload is stabilized, the system recalculates usage rate and removes the warnings. Calculation is based on a long-run average, so it can take a month for them to be automatically cleared.

Cumulative writes that are based on possible DWPDP numbers are listed in Table 3-2. It provides an overview of the total cumulative writes over a 5-year period with various DWPDP numbers.

Table 3-2 Cumulative writes based on possible DWPDP

Capacity of flash drive	One DWPDP	Two DWPDP	Five DWPDP
3.84 TB SAS or NVMe SSD	7.0 PB	14.0 PB	35.0 PB
7.68 TB SAS or NVMe SSD	14.0 PB	28.0 PB	70.0 PB
15.36 TB SAS or NVMe SSD	28.0 PB	56.0 PB	140.0 PB
4.8 TB IBM FCM3/4	26.3 PB	52.6 PB	131.5 PB
9.6 TB IBM FCM3/4	52.5 PB	105.0 PB	262.5 PB
19.2 TB IBM FCM3/4	105.1 PB	210.2 PB	525.6 PB
38.4 TB IBM FCM3/4	210.2 PB	420.4 PB	1.05 EB

To understand the value behind FCMs, consider the following comparison of three drives that share similar-sized physical capacities:

- ▶ A 3.84 TB NVMe SSD and a 4.8 TB FCM. Because of the no-penalty compression that is built into the FCM, it delivers up to 3.75 times the cumulative capacity of the 4.8 TB FCM.
- ▶ A 7.68 TB NVMe SSD and a 9.6 TB FCM. The built-in compression means that the FCM delivers 3.75 times the cumulative capacity.
- ▶ A 15.36 TB NVMe SSD and a 19.2 TB FCM. With FCM, you can achieve approximately 3.75 times the cumulative capacity of NVMe SSDs.

3.3 Arrays

The FlashSystem family provides storage capacity for host applications:

- ▶ It virtualizes capacity of one or more external storage systems that are serving as back-end storage and that are presented as managed disks (MDisks) to FlashSystem or SAN Volume Controller to form capacity pools,
- ▶ The FlashSystem has its internal drives in the control and expansion enclosures. These devices must be configured in arrays to form MDisks (also known as managed disks) for storage pools. This configuration is essential to achieve the wanted performance and data storage reliability. Internal storage capacity is provided by conventional drives, SSDs, or FCMs.

Depending on array arrangement and devices that are members of the arrays, you can achieve desired performance of the storage configuration with appropriate storage reliability.

3.3.1 Supported array (RAID) types

The correct choice of the RAID array type and level for an application is an important foundational step for further storage capacity configuration and management, storage reliability, and system performance.

There are different array types and levels, but all have similar purposes:

- ▶ Organization of several drives or FCMs to provide continuous capacity that is larger than a single drive
- ▶ Increased storage reliability because of increased tolerance of array member drive failures without data loss
- ▶ Increased I/O performance in comparison with single drive capabilities because of I/O load distribution

The FlashSystem family supports two RAID types.

Attention: Beginning with code levels 8.7.1 and later (including 8.7.3), support for Traditional RAID (TRAID) has been fully removed from FlashSystem storage. The final terminal code release that still supports TRAID is 8.7.0. For a better understanding of the technology and the rationale behind this removal, information about TRAID is included.

Traditional RAID (TRAID)

Traditional RAID uses member drives for data storage, often employing striping, and has dedicated, inactive spare drives. If a member drive fails, data is accessed from a mirror (RAID 10) or recalculated by using parity and is written to a spare drive (RAID 5/6). Although TRAID enhances data reliability and I/O performance by distributing the load across members, it presents drawbacks.

Specifically, in TRAIID5 and TRAIID6, a spare drive can become a performance bottleneck during recovery. Large arrays and drives can also cause prolonged recovery times, potentially increasing the risk of further failures. To address these limitations, IBM introduced Distributed RAID (DRAID).

Notes:

- ▶ TRAIID 5 and TRAIID 6 are not supported by FlashSystem.
- ▶ TRAIID is not supported on IBM Spectrum Virtualize 8.7.1 and later.

Distributed RAID

The main feature that distinguishes distributed RAID (DRAID) from TRAIID is the change in approach to spare drives usage for maintaining redundancy of the array.

In distributed RAID there are no spare drives. Instead, a rebuild area is introduced for redundancy.

A rebuild area, or areas, is distributed across all members of the array, and this eliminates the need for idle spare drives. Therefore, all drives in the system contribute to storage space and the array's workload performance.

If a member drive fails, the rebuild activity load is distributed across multiple drives and resolves any rebuild-related bottlenecks that are common in TRAIIDs.

Rebuild time is reduced and array performance is less affected during rebuild times. Starting with version 8.6.2, the enhancement to DRAID rebuild with smart extents allocation was introduced which improved rebuild time and performance even more.

Additionally, because of the implementation of DRAID in FlashSystem storage, you can effectively spread the workload across multiple CPU cores of clustered nodes, providing significant performance improvement over single threaded TRAIID array implementation.

The FlashSystem family supports various traditional and distributed RAID levels, depending on drive technology and code level.

3.3.2 Supported RAID levels for FlashSystem models

Refer to Table 3-3 on page 56 for supported array types and RAID levels in the FlashSystem family.

Important: Because of improvements in RAID technology, the DRAID 6 array type and level is highly recommended to achieve the stability, performance, and reliability objectives with FlashSystem.

Consider the following key factors of RAID type, level, drive interface, and drive technology:

- ▶ Flash Core Modules (FCMs) can be used *only* within DRAIDs levels 1, 5 and 6.
- ▶ Storage Class Memory drives are supported in DRAIDs levels 1, 5, 6 and TRAIID 0.
- ▶ Industry Standard NVMe SSDs are supported in DRAID 6, DRAID 5, TRAIID 1, and TRAIID 10.
- ▶ SAS HDDs and SAS SSDs are supported for DRAID 6, DRAID 5, TRAIID 1, and TRAIID 10 but only in expansion enclosures for specific models.

Table 3-3 RAID level support by FlashSystem model

IBM Storage FlashSystem Family	RAID type				
	Traditional RAID ^a		Distributed RAID		
	TRAID 0	TRAID 1	DRAID 1	DRAID 5	DRAID 6
FlashSystem 5010,5030,5100	Yes	Yes	-	Yes	Yes
FlashSystem 7200	Yes	Yes	Yes	Yes	Yes
FlashSystem 9100	Yes	Yes	-	Yes	Yes
FlashSystem 5015,5035,5045	-	-	Yes	Yes	Yes
FlashSystem 5200,5300	-	-	Yes	Yes	Yes
FlashSystem FS7300	-	-	-	Yes	Yes
FlashSystem FS9200 and 9200R	Yes	Yes	Yes	Yes	Yes
FlashSystem 9500 and 9500R	-	-	Yes	-	Yes

a. Traditional RAID is not supported on code level 8.7.1 and later.

3.3.3 Array geometry and considerations

The FlashSystem management GUI automatically applies default best practice RAID settings when storage capacity is added to a pool. These recommendations are based on available system resources and drive technology. To create a RAID array, select the pool and choose **Add Storage** from the right-click menu (Figure 3-5).

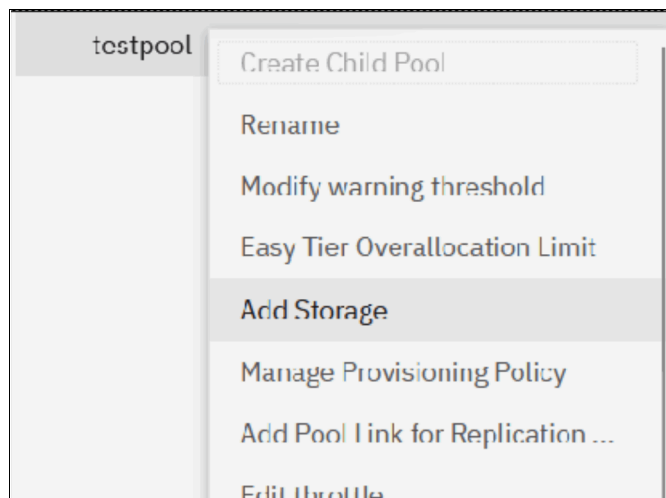


Figure 3-5 Right-click menu to Add Storage to the pool

As previously noted, using distributed RAID is highly recommended, and the management GUI defaults to this option with the currently possible RAID configurations (Figure 3-6).

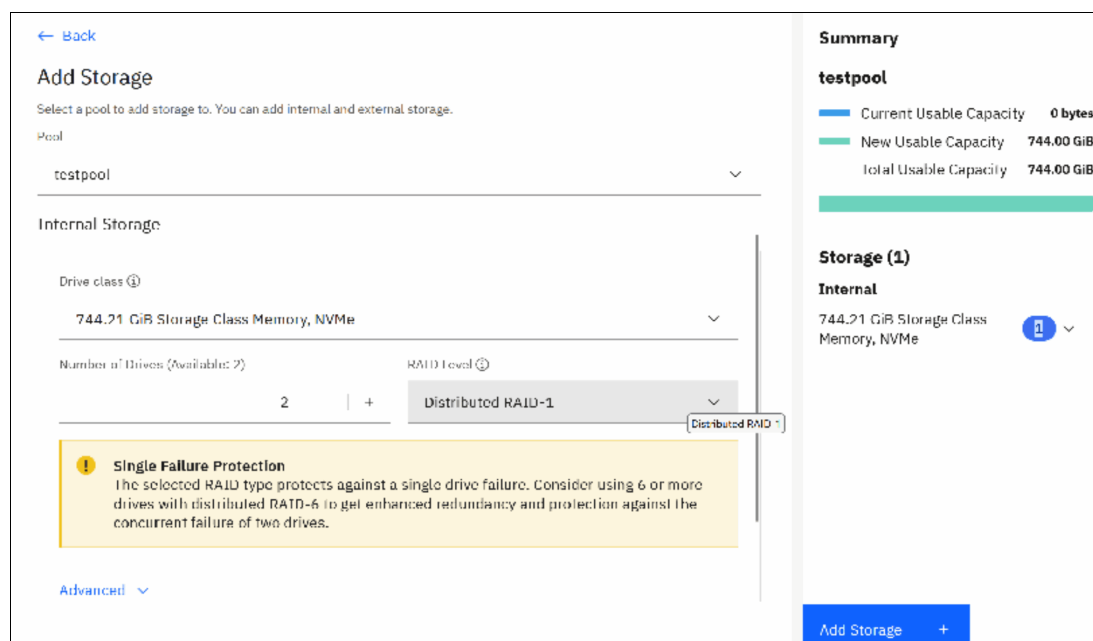


Figure 3-6 Management GUI recommendations based on available resources

In the Add Storage interface you can select the question mark icon to access in-context help for parameters and recommended settings.

Parameters and their extended information

Consider each of the fields in the Add Storage panel and the related information for each field.

Drive class

Depending on the available resources, drives with similar characteristics are grouped together under the same class to ensure optimal performance of the array.

The system assigns drives to a specific class by using following information:

- ▶ Block size of that drive class can handle (512 or 4096 bytes)
- ▶ Capacity
- ▶ I/O group that is associated with the drive class
- ▶ RPM speed if applicable, depending on specific drive technology

Important: Starting with software code level 8.7.1, all FlashSystem models support only a single I/O group. Configuration scaling in this case can be achieved by using a FlashSystem grid. Do not build multi-I/O group clusters.

Technology

Indicates the drive technology with the following supported technology types:

- ▶ Storage Class Memory (SCM). Flash storage devices with NVMe interface, reduced latency and improved endurance.
- ▶ Tier 0 flash. High-performance flash drives that process read and write operations and provide faster access to data than Enterprise or nearline drives. For most Tier 0 flash drives, the system monitors their wear level and issues warnings when the drive should be

replaced. Some NVMe-attached drives, such as IBM FC M drives, are considered Tier 0 flash drives.

- ▶ Tier 1 flash: Lower-cost flash drives, typically with larger capacities, but slightly lower performance and write endurance characteristics. The system monitors their wear level and issues warnings when the drive should be replaced.
- ▶ Enterprise disks. High-performance drives optimized for demanding workloads, typically operating at 15K RPM or 10K RPM.
- ▶ Nearline disks. Drives that are optimized for capacity. Usually 7.2K, 10K RPM with high capacity.

Number of Drives

Displays the number of drives available within the selected drive class. This parameter can be adjusted based on your configuration, and the system automatically suggests the recommended RAID type and level. See “Considerations related to parameters” on page 59 for additional information.

RAID Level

The GUI suggests a recommended RAID type and level. Although alternative options are available, it is strongly recommended to follow the suggestion provided in the GUI. For advanced customization, you can use an interface section for configuration of detailed parameters, which are automatically calculated based on the selected drive count and RAID level.

Rebuild Areas

Rebuild areas are reserved drive capacities within a distributed RAID array that is used exclusively for data reconstruction after a drive failure. The rebuild areas provide no usable storage. This rebuild area for the array is distributed across all drives in the array. During recovery, the rebuild area enhances performance by using all member drives for I/O operations.

Stripe Width

This defines the number of data strips written concurrently during a rebuild after a drive failure. Also referred to as the redundancy unit width, a single stripe (redundancy unit) represents the smallest addressable data unit. The default strip size for a distributed RAID array is 256 KiB.

Stripe width can also be represented as $N+P+Q$, where N is the data drive count, and P and Q represent the distributed parity capacity (for example, a DRAID 6 array with 10 data drives would have a stripe width of $10+P+Q = 12$).

You can increase the array width beyond the minimum (more drives than $N+P+Q$) to create a larger stripe width. A larger stripe width increases usable capacity but also extends rebuild time because of more data recalculation per stripe. Therefore, you can balance rebuild time and usable capacity when designing a distributed array.

Important: The FlashSystem Management GUI optimizes parameters based on the number of drives in the array. For example, the default stripe width settings (12 for DRAID 6) provide an optimal balance between usable capacity and rebuild time.

Strip Size

This parameter, always set to 256 KiB, determines the Full Stride Write (FSW) size in conjunction with the stripe width. It is also relevant to the host file system block size.

Important: The cache in the FlashSystem storage is designed to prioritize Full Stride Writes (FSW) to improve performance.

Considerations related to parameters

After reviewing the information about the parameters, the following considerations apply to each:

- Optimal array drive counts vary based on drive class. For Flash Core Modules (FCMs) and NVMe industry-standard drives, 16–24 drives are recommended, and the FlashSystem 9500 supports up to 48. These configurations balance performance, rebuild time, and usable capacity. Specifically, some FlashSystem 5XXX and 7XXX models support up to 24 NVMe drives, and the FlashSystem 9500 supports 48.

Note: The current number of drive members per DRAID for a certain interface and that latest code level, can be found at [V8.7.3.x Configuration Limits for IBM FlashSystem® and SAN Volume Controller](#).

For version 8.6.3 and later, the FS7300 system can be ordered with a minimum of 12 drives, and the FS9500 can be ordered with minimum of 18 drives.

FlashSystem 9500 (FS9500) supports 48 NVMe drives in distributed RAID 6 that includes the following configurations:

- A DRAID 6 of industry standard NVMe drives supports a maximum of 48 member drives with a maximum of 4 distributed rebuild areas. You can expand an existing array to 48 drives.
 - A DRAID 6 of Flash Core Module (FCM) NVMe drives up to 48 member drives with one distributed rebuild areas. You can expand an existing array to 48 drives.
- Because of the single controller and limited PCIe lanes inherent in each FCM and SSD, insufficient drive population can restrict bandwidth allocation to the FlashSystem CPUs, which can cause performance bottlenecks. This limitation affects both SSDs and FCMs. Consequently, a minimum drive count is now mandated for each FlashSystem model to ensure optimal resource usage.
 - For SAS HDDs, configure a single DRAID array with at least 40 drives to optimize rebuild times, rather than creating multiple smaller arrays. Ideally, a DRAID 6 array with 48–64 HDDs provides the best performance benefits by making use of the optimized settings to enhance parallelism and efficiency in I/O operations.
 - For SAS SSD drives, the optimal array size is 24–36 drives per DRAID 6 array.
 - For SCM, the maximum number of drives in an array is 12.
 - RAID type and level considerations. For optimal efficiency, it is a best practice to configure DRAID 6 arrays with a larger number of drives. This approach minimizes the effect of parity and rebuild areas relative to the total capacity. The approach also increases the bandwidth during rebuild operations. Maximum system performance is often achieved with a single DRAID 6 array.
 - RAID level recommendations based on drive count:
 - More than six drives. Always create DRAID 6. It is a best practice because of its superior protection against a second drive failure, and DRAID 5 generally does not offer significant performance gains over DRAID 6.
 - Two or three drives (where DRAID 1 is supported). DRAID 1 is recommended. Its mirrored geometry provides exact copies across the array members.

- Four or five drives (where DRAID 1 is supported). It is a best practice to use DRAID 1 over DRAID 5. Although DRAID 5 offers more capacity, DRAID 1 provides better performance, particularly during rebuild operations with the same number of drives.
 - Six drives. Both DRAID 1 and DRAID 6 are viable options.
 - Four to five drives (platforms without DRAID 1 support for NVMe SCM). The recommended level is DRAID 5.
- Recommended number of Rebuild Areas (GUI guidelines):
- DRAID 1:
 - With 2 member drives: 0 rebuild areas.
 - With 3 to 16 member drives: 1 rebuild area.
 - DRAID 5 and 6:
 - 1 rebuild area for every 36 drives.
 - Optimal configuration: 1 rebuild area for every 24 drives.
 - FCM Drive Arrays: Only 1 rebuild area per array is supported.

Considerations for different drive types intermix in the arrays

It is important to note the following restrictions when combining different drive types within an array:

- Incompatible drive types within a single array:
- Compressing and non-compressing drives. Drives with onboard compression (such as FCMs) and non-compressing drives (with NVMe or SAS interfaces) cannot be combined within the same array.
 - NVMe drives from different control enclosures. Because of the specific implementation of their interfaces and for optimal performance, NVMe drives from different control enclosures are not allowed in the same array.

Note: Starting with software code level 8.7.1, all FlashSystem models support only a single I/O group.

- SCM drives are not compatible for mixing within the same array with other NVMe and SAS drive types.
- Rules related to physical and logical capabilities (including capacity):
- For all NVMe interface drives, members of the same array must have same physical and logical capacity. It is not possible to replace an NVMe drive of one capacity with an NVMe drive of greater capacity.
 - For SAS interface drives, array members do not require identical physical capacity. When you create a SAS array, you can include drives with enhanced characteristics. However, the array's capacity and performance is limited to the base drive type.
- For example, you can create a DRAID 6 array with four 1.6 TB SSDs and two 3.2 TB SSDs. However, the 3.2 TB drives use only 1.6 TB of their capacity. The remaining capacity is unused. With SAS drives, you can combine drives from expansion enclosures into a single array, and DRAID 6 arrays can span multiple enclosures.

Array capacity and expansion planning

Use the following guidelines for array capacity and expansion planning:

- ▶ Initial and pre-order system configuration can be planned by using the IBM Storage Modeler tool. It is available for IBMers and IBM Business Partners.
- ▶ For deployed systems, FlashSystem offers command-line tools for array creation and capacity planning in addition to the Management GUI's in-context recommendations.
- ▶ The **lsarrayrecommendation** command provides array recommendations. The output of the command provides the recommended and best practice configuration for a given drive class and number of drives. For more information, see [lsarrayrecommendation](#).
- ▶ The **lspotentialarraysize** command determines the potential capacity of a planned array based on the specified drive count, drive class, and RAID level within a given pool. For more information, see [lspotentialarraysize](#).
- ▶ To calculate the approximate capacity of the DRAID 6, you can use the following formula:

$$\text{Array Capacity} = D / ((W \times 256) + 16) \times ((N - S) \times (W - 2) \times 256)$$

where

D is Drive capacity

N is Drive count

S is Rebuild areas (spare count)

W is Stripe width

Example 3-1 Approximate capacity calculation

For the capacity of a DRAID 6 array of sixteen 9.6 TB FCMs, use the following values:

D = 9.6 TB = 8.7 TiB

N = 16

S = 1

W = 12

Array capacity =

$$8.7 \text{ TiB} / ((12 \times 256) + 16) \times ((16 - 1) \times (12 - 2) \times 256) =$$

$$8.7 \text{ TiB} / 308800 \times 38400 = 108.2 \text{ TiB}$$

You can expand distributed array capacity by adding additional drive members to an existing array, which is one way to increase usable storage space in the pools.

Distributed RAID expansion is done through the management GUI from menu **Pools** → **MDisks by Pools** section. Choose the **MDisk** to expand the menu and follow the instructions and recommendations in the GUI.

Consider the following aspects of DRAID expansion:

- ▶ Array expansion can affect host performance and latency. It is recommended to expand the array when the I/O load to this array is below 50%. If I/O load from the hosts is higher the amount of time that is needed to complete the expansion operation increases significantly.
- ▶ Array expansion can take a significant amount of time.
- ▶ Expansion is dynamic and non-disruptive, data on the expanding DRAID is fully available during the process. All other functionality, such as Easy Tier, data migrations are also fully available.

- It is possible to expand an array by adding one or more drives. During single expansion it is possible to add up to 12 drives. New drives are integrated to the array and data is re-striped to maintain the placement of strips across the existing and new components.

Considerations for expansion operation

When you expand an array, consider the following aspects of the expansion operation:

- When you expand a thin-provisioned NVMe array with FCM members, the new physical capacity is not immediately usable, and its availability does not match the timing of the logical expansion.
- Array expansion is not possible when the array is in write-protected mode. This can happen if it has no more physical capacity. Any capacity issues must be resolved first.
- During array expansion, the stripe width remains constant. Consequently, the parity space (N+P+Q) is allocated proportionally to the initial array configuration. From a capacity perspective, this means an expanded array has less usable capacity than a similarly sized array created with a larger initial stripe width, because the ratio of data strips to parity strips is lower.
- It is not possible to change DRAID geometry (level) or stripes width.
- After expansion is started it cannot be canceled. Only one expansion process can run on an array at a time.
- It is not possible to shrink the array. Array member count cannot be decreased.
- Expanding a distributed array can take several weeks, particularly when adding high-capacity nearline SAS drives and if a large number of drives are included. The maximum possible expansion is 12 drives. In such cases, increasing the pool's capacity by creating a new array (MDisk) and adding it might be more practical. Although this could be sub-optimal depending on the new array's performance, Easy Tier balancing and performance-aware cache can mitigate this.

DRAID capacity and performance at drive failure and replacement

It is crucial to carefully monitor and plan the capacity of dynamic arrays. Arrays with FCM members can be over-provisioned, use compression, and are susceptible to running out of space.

Therefore, when planning array capacity, you must determine the data compression rate and factor it in with future scaling needs based on application requirements. It is recommended to monitor array capacity and maintain usage below 85% of physical capacity.

For monitoring array capacity, use IBM Spectrum Control or IBM Storage Insights. Both offer configurable alerts. Capacity information is available in the management GUI and by using the command `lsarray <array_id>`.

On the storage system, it is strongly recommended to enable the UNMAP capability, which can be done by using the `hostunmap` parameter. This feature, which reclaims unused capacity from the host, is enabled by default for all-flash models (FlashSystem 9x00 and FlashSystem C200) for versions 8.7.0 and later. There is no performance impact when UNMAP is used with FCM and SCM based MDisk. For other FlashSystem models with standard HDDs, UNMAP is disabled by default.

When an array's physical capacity reaches 99%, the FlashSystem storage system issues a critical warning with event 1242, which indicates 1% physical space left for compressed array.

Configured notifications are sent, demanding immediate action.

Take the following corrective actions either in combination or separately:

- ▶ Add more storage to the pool and allow Easy Tier to automatically rebalance data across the managed disks and arrays.
- ▶ Migrate volumes with extents from the managed disk with low physical space to another storage pool. Alternatively, migrate extents from the array with low physical space to other managed disks with sufficient space.
- ▶ Delete or migrate data from volumes using a host that supports UNMAP commands. Ensure the UNMAP capability is enabled on the storage system. You can check the setting with the command `lsystem -hostunmap on`.

Note: UNMAP is enabled by default on all-flash FlashSystem models that are running version 8.7.0 or later.

It is crucial to recognize that arrays are most vulnerable to running out of space during a rebuild or when degraded. Although DRAID's distributed spare capacity is available during normal operation, reducing overall drive fullness, this changes during a rebuild.

During a DRAID array rebuild, the dedicated rebuild area is employed. Upon completion, this reserved space is reclaimed, potentially resulting in full drive usage. This scenario can induce substantial write amplification and performance degradation. Critically, if an array is already operating at over 99% capacity prior to a rebuild, a physical out-of-space condition might occur during the rebuild process.

Write amplification: Write amplification means flash memory storage, such as SSDs, writes more physical data than the host intends. It is the ratio of physical writes to logical writes.

Important: Monitor compression-capable and over-provisioned arrays, ensuring their capacity remains below 85%. Use monitoring tools and configure alerts on your storage system.

When a member drive fails, arrays recover redundancy by rebuilding data to designated rebuild areas. After the failed drive is replaced, a copyback operation frees the rebuild area and restores the array's full redundancy when the copyback finishes.

DRAID distinguishes between two types of rebuilds:

- ▶ Non-critical rebuilds. If a single drive fails in a DRAID 6 array, the array retains redundancy. A non-critical rebuild then occurs with limited throughput to minimize performance impact.
- ▶ Critical rebuilds. If an array loses all redundancy (due to a single drive failure in DRAID 5 or a double drive failure in DRAID 6), a critical rebuild is initiated to restore redundancy as quickly as possible. Critical rebuilds are typically performed at nearly twice the speed of non-critical rebuilds.

Note: IBM Storage Virtualize code level 8.6.2 and later includes smart rebuild functionality specifically for DRAID arrays that use FCM drives.

Rebuild, Rebuild-in-place, Copyback and Scrub operations act only on parts of the array that have extents allocated to volumes, volume copies or snapshots and includes the following features:

- Progress reporting is based on the amount of data that needs rebuilt only.
- If a rebuild is active during the upgrade, the existing “rebuild everything” process finishes.
- Smart Rebuild, is supported with only Flash Core Modules (FCMs) and is available for DRAID1, DRAID5, and DRAID6 arrays. Smart Rebuild improves rebuild performance. It reduces the critical rebuild period and enhances overall storage system performance during rebuild activity.

3.3.4 External storage resources utilization and considerations

Certain FlashSystem models support external storage virtualization. External storage from back-end controllers or back-end storage can be connected to FlashSystem or SAN Volume Controller systems via Fibre Channel (FC) or iSCSI. NVMe-FC back-end attachment is *not* supported because of the lack of performance gains.

External storage systems capacity resources or logical units (LU) are detected by FlashSystem or SAN Volume Controller as managed disks (also known as MDisks). These external MDisks are used in a same manner as FlashSystem's internal MDisks formed by arrays and can be added to the pools for further capacity allocation and this capacity can be allocated and presented to an application hosts in a form of volumes called VDisks.

FlashSystem or SAN Volume Controller acts in between an application host and an external storage system by virtualizing the storage. It plays the role of the host for external storage systems and a role of storage for application hosts.

External storage connectivity and path selection

Logical units from back-end storage are mapped to multiple storage ports of a FlashSystem or a SAN Volume Controller. These LUs are recognized by FlashSystem as MDisks. External storage is recognized as a controller. FlashSystem and SAN Volume Controller are clustered systems, requiring a minimum of two nodes. Mapped resources from external storage must be presented to all nodes in the cluster, according to the cluster topology.

IBM Storage Virtualize nodes and node canisters can be configured to access external storage via specific portsets. For path selection to MDisks, IBM Storage Virtualize employs two algorithms. The back-end storage controller determines the algorithm, which cannot be modified. Connectivity strategies are advised based on the selected path algorithm.

Round-robin path selection algorithm

When the back-end storage controller supports round-robin path selection, each MDisk uses one path per target port per IBM Storage Virtualize node. For storage systems without a preferred controller, such as the DS8000, FlashSystem and SAN Volume Controller use all available or configured Fibre Channel (FC) ports to communicate with MDisks.

Round-robin path selection significantly enhances performance by distributing MDisk access across multiple back-end Fibre Channel ports. This parallel I/O workload benefits sequential I/O within a single extent.

For enhanced resilience, round-robin path selection offers a significant advantage in the event of performance issues on a back-end storage system's Fibre Channel ports. By cycling I/O across all configured FC ports, this method allows for timely detection of specific port failures. In some cases, IBM Storage Virtualize systems can automatically exclude problematic paths while maintaining good performance to facilitate investigation.

Note: For optimal performance with IBM Storage Virtualize and back-end storage systems supporting round-robin path selection, zone as many external storage FC ports as possible.

Each IBM Storage Virtualize storage controller supports up to 16 FC ports. Consult your IBM Storage Virtualize model's documentation for specific zoning details.

To verify MDisk and controller round-robin path selection support, use the `lsmdisk <mdisk_id/name>` command. If supported, value of the `active_WWPN` parameter in the output is many.

MDisk group balanced and controller balanced paths selection

Some external storage systems require manual configuration for optimal performance. In these cases, with active-passive communication, IBM Storage Virtualize accesses the mapped MDisks through a preferred back-end controller's port. The system prioritizes specific ports on the back-end controller for accessing the MDisks.

To optimize back-end storage usage, ensure the number of logical units (MDisks) mapped to the IBM Storage Virtualize system is a multiple of the configured FC ports used for communication. Additionally, assign all MDisks from the external storage to a single storage pool (MDisk group).

If the MDisk is serviced from the controller that supports the MDisk group balanced path selection, then the `lsmdisk` command specifically shows the dedicated World Wide Port Number (WWPN) in the `active_WWPN` parameter of the command output.

Guidelines for creating an optimal back-end configuration

Most of the back-end controllers (external storage systems) aggregate HDDs or SSDs into a RAID, and then join arrays into pools. Logical volumes or units are created from these pools and assigned to hosts.

When virtualizing back-end storage resources, IBM Storage Virtualize acts as a host. Therefore, it is crucial to configure back-end storage for optimal performance and resiliency.

If the back-end controller has homogeneous storage, create the required number of RAIDs (RAID 6 or RAID 10 are usually advised) with an equal number of drives. The type and geometry of an array depends on the recommendations of vendor of the back-end controller. If your back-end controller supports load striping across multiple arrays within a resource pool, create a single pool encompassing all arrays.

On back-end systems with mixed drives, create a separate resource pool for each type of drive (HDD or SSD). The drive type is important because you must assign the correct tier for an external MDisk when it is virtualized by IBM Storage Virtualize.

Create a set of fully allocated logical volumes from the back-end storage pools. Each volume that is created is detected as an MDisk by IBM Storage Virtualize.

The number of logical volumes to create, depends on the type of drives that are used by your back-end controller.

Back-end controller with HDDs

If your back-end storage uses HDDs, the volume number calculation must be based on a queue depth. Queue depth is the number of outstanding I/O requests of a device.

For optimal performance, HDDs need 8–10 concurrent I/O at the device, which does not change with drive rotation speed. For a highly loaded system make sure that any given IBM Storage Virtualize MDisk can queue up approximately eight I/O per back-end controller drives. In the IBM Storage Virtualize system, queue depth per MDisk is approximately 60. The exact maximum for a real system might vary. However, for this calculation, 60 is a good approximation.

The queue depth per MDisk number leads to the HDD rule of 8.

According to this rule, to achieve eight I/Os per drive with a queue depth of 60 per MDisk from IBM Storage Virtualize, a back-end array with $60/8 = 7.5$ that is approximately equal to eight physical drives is optimal, or one logical volume per every eight drives in an array.

All-flash back-end controllers

When considering all-flash controllers, the primary focus shifts from per-drive queue depth to optimizing I/O distribution across IBM Storage Virtualize ports and processing threads. Ensure that IBM Storage Virtualize has the best opportunity to balance the load and use its internal resources effectively, given the high I/O capabilities of most all-flash arrays used in virtualization. Due to the lower latency per I/O, queue depths are less of a critical factor in this scenario.

Important: Automatic configuration is the recommended best practice for FlashSystem systems that are virtualized as Fibre Channel back-end controllers for SAN Volume Controller or other FlashSystem systems. This approach applies best practice settings and streamlines the configuration process.

Automatic configuration intended for new system setups can be enabled during initial System Setup. FlashSystem defaults to a storage layer configuration, automatically adjusting settings for use as a back-end for virtualizing systems.

For more details, refer to “*Section 3.3.8 Automatic configuration for IBM SAN Volume Controller back-end storage*” of the IBM Redbooks *Implementation Guide for IBM Storage FlashSystem and IBM SAN Volume Controller*, [SG24-8542](#).

Considerations for compressing and deduplicating back-end controllers

Many FlashSystem models support over-provisioning. If the back-end storage performs data compression (as with FlashSystem using FCMs) or deduplication on its volumes (LUs), these LUs, when virtualized as external MDisk by systems like SAN Volume Controller, benefit from these data reduction techniques. As stated in “All-flash back-end controllers” on page 66, automatic configuration on FlashSystem is recommended for best practices.

Also, note that the SAN Volume Controller can detect the following properties:

- ▶ If the external MDisk is thin-provisioned.
- ▶ The total physical capacity of the MDisk.
- ▶ The used and remaining physical capacity of the MDisk.
- ▶ Whether the unmap commands are supported by the back-end. This is crucial for thin-provisioned MDisk capacity management. These commands inform the back-end system when storage capacity is no longer needed so that it can be released and reallocated.

Automatic configuration simplifies the setup of thin-provisioned or compression-enabled back-end storage resources. It generates a .csv file with volume configuration data, facilitating accurate configuration and mapping of corresponding MDisk on the SAN Volume Controller.

Accurate initial sizing of the required space for your host applications, along with an estimate of the potential data reduction ratio, is crucial. Your IBM technical sales representative can assist you in obtaining this information.

It is highly recommended to carefully monitor thin-provisioned back-end storage. IBM Storage Insights or Spectrum Control are strongly recommended for this purpose. In addition to enhanced performance monitoring capabilities, these tools can collect and store historical I/O load data for analysis and offer other valuable features.

As a best practice, employ data reduction features in a single virtualization layer.

Additional considerations for FlashSystem as a back-end

FlashSystem storage systems can be configured with a mix of Flash Core Modules (FCMs), SSDs, and HDDs. These can serve as a virtualized back-end storage layer, for example, when managed by an SAN Volume Controller.

It is important to note that SAN Volume Controller does not monitor the Easy Tier activity of the back-end storage. Therefore, if Easy Tier is enabled on both the front-end (SAN Volume Controller) and the back-end (for example, FlashSystem), each system independently rebalances hot data based on their own heat maps. This independent operation does not typically result in improved overall performance. Therefore, it is generally recommended to enable Easy Tier primarily on the front-end (SAN Volume Controller). The same principle applies if both the front-end and back-end are FlashSystem devices.

It is important to note that the SAN Volume Controller does *not* monitor the Easy Tier activity of the back-end storage. For example, if Easy Tier is enabled on both the front-end SAN Volume Controller and the back-end FlashSystem, the systems independently rebalance hot data based on their own heat maps. This independent operation does not typically result in improved overall performance. Therefore, it is generally recommended to enable Easy Tier primarily on the front-end SAN Volume Controller.

3.3.5 Main concepts of Easy Tier

The purpose of Easy Tier is to optimize the response time and reduce the latency of the storage system on a pool level. Easy Tier monitors how frequently a specific data extent is accessed and is used when pools contain MDisk of different tiers to balance I/O load among MDisk of same tier in the pool.

Storage Tiering

A Storage Virtualize storage system uses storage tiering to automatically balance load between MDisk in a pool and deliver optimal performance.

There are five storage tiers defined for MDisk, starting with best performing ones:

1. SCM tier
2. Tier 0 flash
3. Tier 1 flash
4. Enterprise tier
5. Nearline (NL)

An MDisk consists of specific drives, and Easy Tier recognizes the specific storage tier and performance capabilities of the drives.

For MDisk created from internal storage, the system assigns tiers automatically as it knows what drives comprise the MDisk and their technology types (see 3.3.3, “Array geometry and considerations” on page 56).

For external MDisk, the storage administrator must assign the tier to the MDisk when it is introduced to the front-end system for virtualization.

Although MDisk can be classified into five tiers, Easy Tier operates with its own three-tier system within each storage pool. Extents are migrated based on access frequency to these three tiers:

- ▶ Hot tier
- ▶ Middle tier
- ▶ Cold tier

Easy Tier determines the assignment of MDisk to Hot, Middle, or Cold tiers based on their storage technology tier, which is contingent upon the pool's constituent MDisk tier configuration.

Within each pool, the highest-performing, lowest-latency MDisk, such as SCM, are assigned to the Hot tier. If the pool contains other tiered MDisk, they are assigned to the Middle or Cold tiers.

By employing an adaptive AI-like algorithm, Easy Tier dynamically migrates frequently accessed extents to the Hot tier and less frequently accessed extents to lower tiers. This dynamic tiering mechanism yields exponential performance enhancements in heterogeneous pools, which might contain high-capacity, low-performance MDisk and high-performance SSDs or flash drives.

Besides balancing performance in multi-tier pools, Easy Tier features storage pool balancing, which automatically moves extents within the single storage tier pool from overloaded to less loaded MDisk.

Storage pool balancing ensures that your data is optimally placed among all MDisk and tiers within storage pools. If a pool consists of the same tier MDisk, the extents are balanced among these MDisk to avoid a single Mdisk overload in the pool.

Attention: Easy Tier considers only performance and does not consider capacity usage. Therefore, if there are two arrays in the same pool and one of them is nearly out of space, and other is empty, Easy Tier does not attempt to move extents between arrays based on the capacity usage.

Therefore, it is recommended to expand distributed arrays in order to increase capacity rather than add additional arrays (MDisk) to the pool.

Easy Tier activities

Easy Tier has the following features and restrictions:

- ▶ When enabled, Easy Tier continuously monitors volumes I/O statistics.
- ▶ Average workload statistics are continuously monitored for each extent across a rolling 24-hour window, and this data is used to determine extent promotion or demotion between tiers.
- ▶ Based on the preceding decisions, a migration plan is generated. For multi-tier pools, an Automatic Data Placement plan is created to migrate extents across tiers. For single-tier pools, a Rebalance plan is generated to migrate extents within the tier, based on load distribution among MDisk.
- ▶ To maintain performance during migration, extent migration is capped at 12 GiB per 5 minutes, equating to approximately 3.4 TiB per day between MDisk tiers or MDisk.

- ▶ Extent migration is restricted to adjacent tiers. Hot to Middle, Middle to Hot, Middle to Cold, or Cold to Middle.
- ▶ Image mode and sequential volumes are ineligible for Easy Tier automatic data placement because of the immutability of their extents, which necessitates collocation on the same MDisk.

Easy Tier modes, control and status

Depending on control settings, Easy Tier operates in three modes:

- ▶ Evaluation mode is enabled by default for single-tier storage and gathers usage statistics for all volumes within the pool. For multi-tier storage pools, it can be activated when the Easy Tier attribute is set to measure.
- ▶ Automatic data placement is enabled depending on Easy Tier attributes for pool and volume, and automatically migrates extents depending on analysis of the workload across tiers or MDisk for balance in case of single tier storage.
- ▶ Acceleration mode is off by default, and it is a global setting on the storage system level. It is used to change rates of extent migration, see details in “Easy Tier Acceleration” on page 71.

Easy Tier activity mode is controlled per storage pool and per volume copy through their respective settings. Its activity status also depends on the number of tiers within the storage pool.

Easy Tier at the storage pool level

At the storage pool level, Easy Tier has two primary states:

1. On state. Easy Tier activity depends on how many tiers are in the pool and volume level Easy Tier setting.
2. Off state has two settings:
 - Measure. Easy Tier collects usage statistics of the extents on the volumes but automatic placement (extents migration) is not active
 - Auto. Easy Tier activity depends on how many tiers are in the pool and volume level Easy Tier setting.

Easy Tier at the volume level

At the volume level, Easy Tier can be controlled more granularly for specific volumes by modifying its Easy Tier parameter to either an On or Off state.

Easy Tier status is displayed in volume (and per volume copy) related information and can be:

- ▶ Inactive - indicates that Easy Tier functionality is disabled for that specific volume copy.
- ▶ Measured - Easy Tier gathers volume usage statistics but does not perform any actions, resulting in the deactivation of automatic extent migration.
- ▶ Balanced - enables performance-based pool balancing for the volume copy and also means that it is working in single tier pool, thus balancing is performed among same MDisk storage tier to spread the load, automatic data placement is enabled.
- ▶ Active - automatic data placement is enabled, and extents are migrated according with the workload profile for the specific volume.

Easy Tier pool settings override volume settings

The storage pool's Easy Tier setting applies globally to all volumes within the pool. Therefore, if Easy Tier is disabled at the pool level, all volumes exhibit an inactive Easy Tier status, irrespective of their individual volume parameter configurations.

When the storage pool's Easy Tier setting is set to measure, the Easy Tier volume status becomes measured, regardless of the number of tiers in the pool or volume-level Easy Tier settings.

A detailed table of corresponding settings, storage pool tier counts, and statuses is available on the IBM documentation site. The following settings are for 8.7.x continuous delivery:

- ▶ For FlashSystem 9X00 family and latest code level 8.7.x_cd, see [9x00 Easy Tier settings](#).
- ▶ For FlashSystem 5X00 family and latest code level 8.7.x_cd, see [5x00 Easy Tier settings](#).
- ▶ For SAN Volume Controller it can be found [SAN Volume Controller Easy Tier settings](#).

Easy Tier monitoring activity and performance

Easy Tier activity and performance monitoring for IBM Storage Virtualize can be done by viewing graphical reports in the storage system's management GUI. These reports can be accessed from **Monitoring** → **Easy Tier Reports or Pools** → **Easy Tier Reports**. Also, these reports can be exported in .csv format.

There are three reports:

1. Data movement statistics
2. Tier composition statistics
3. Workload skew statistics

For the most current details regarding report content and their parameters, see the official IBM Documentation for the relevant product and code level:

- ▶ For FlashSystem 9X00 family and latest code level 8.7.X_cd, see [9x00 Easy Tier reports](#).
- ▶ For FlashSystem 5X00 family and latest code level 8.7.X_cd, see [5x00 Easy Tier reports](#).
- ▶ For SAN Volume Controller, see [SAN Volume Controller Easy Tier reports](#).

Easy Tier best practices and considerations

Various configurations and usages can affect how you implement Easy Tier:

- ▶ When creating a single-tier pool with external storage MDisk, ensure that all MDisk possess identical RAID geometry (disk count). This requirement helps maintain consistent performance and reliability within the pool.
- ▶ For optimal SAN Volume Controller performance, pools should not contain more than three different storage tiers of MDisk.
- ▶ In all other scenarios that involve external storage devices or expansion enclosures with SAS drives, it is a best practice to limit pools to a maximum of two storage tiers of MDisk.
- ▶ By default, the Easy Tier setting is configured to Auto for storage pools and On for volumes.
- ▶ Automatic data placement is enabled by default for all striped volumes in a storage pool with two or more tiers.
- ▶ For external MDisk that are used in external storage virtualization, the storage tier assignment must be performed by an administrator. By default, all external MDisk are assigned to the enterprise tier.
- ▶ When you create a single-tier pool by using external storage MDisk, ensure that all MDisk possess identical RAID geometry, meaning the same number of drives per array.
- ▶ When creating a single-tier pool with internal MDisk, ensure that all MDisk possess identical hardware characteristics.

- ▶ Easy Tier and storage pool balancing are inoperative when 100% of the storage pool capacity is allocated to volumes. This is because Easy Tier makes use of the extent migration capabilities of IBM Storage Virtualize, which requires some free capacity for extent movement. Therefore, avoid provisioning 100% of the pool's capacity when Easy Tier is enabled.
- ▶ For external storage virtualization, best practices recommend enabling Easy Tier on only one layer: either the front-end or the back-end. Front-end (virtualizer) Easy Tier offers greater granularity and volume-level control for enablement and disablement so that administrators can select specific volumes. Furthermore, it provides statistics and analysis that are aligned with application host workloads and supports Easy Tier across heterogeneous back-end storage controllers. Adjusting extent sizes can further optimize Easy Tier efficiency. These benefits are not available when Easy Tier is implemented on the back-end controller.
- ▶ Do not intermix data reduction-capable and non-data reduction capable storage in the same tier of a pool with Easy Tier enabled.

Easy Tier tuning

When tuning Easy Tier, consider the following points.

Easy Tier Acceleration

Easy Tier Acceleration is a system-wide setting, and can be set with the command **chsystem -easytieracceleration on**.

This parameter increases the extent migration throughput from 12 GiB to 48 GiB within a 5-minute interval. The feature is set to off by default because of it creates a high demand for resources and should be enabled during periods of low system activity. It accelerates the distribution of existing volume extents to newly added storage pool capacity. Acceleration is also advantageous when migrating multiple volumes between pools, particularly if the target pool has more storage tiers than the source.

Note: Accelerated mode should be used temporarily, ideally for a few days, so that Easy Tier can rapidly achieve a steady state.

MDisk Tier Load

It can be set for specific MDisk and can be changed only for external MDisk. It is used to provide additional granularity for external MDisk performance tuning by advanced users.

The system uses a default setting based on external storage performance profile and the MDisk tier setting.

The following values can be assigned to each MDisk for Easy Tier load levels:

- ▶ Default
- ▶ Low
- ▶ Medium
- ▶ High
- ▶ Very high

This setting can be changed by following command:

```
chmdisk -easytierload <Easy_Tier_load_level> <mdisk_name>
```

Modify the default setting only when you have confirmed that an MDisk is either underutilized and capable of handling increased load, or overutilized and requires a load reduction. Set this value to very high for only SSDs and flash MDisk.

Additional detailed information about Easy Tier functionality can be found in “Section 4.7” of the IBM Redbooks *Performance and Best Practices Guide for IBM Storage FlashSystem and IBM SAN Volume Controller Updated for IBM Storage Virtualize Version 8.6*, [SG24-8543](#).



Performance considerations for volumes

In IBM Storage Virtualize, a volume, which is also called a virtual disk (vdisk), is storage space that is provisioned out of a storage pool and presented to a host as a logical unit (LU). This chapter discusses strategies and optimization guidance for configuring volumes on IBM Storage Virtualize systems. For more information on volumes and the various volume types, see [Volumes](#).

This chapter includes the following topics:

- ▶ 4.1, “Volume configuration guidelines” on page 74
- ▶ 4.2, “Volume groups” on page 76
- ▶ 4.3, “Thin provisioning” on page 78
- ▶ 4.4, “Compression and deduplication” on page 82
- ▶ 4.5, “Sharing workloads within the same pool” on page 84
- ▶ 4.6, “Volume placement guidelines” on page 85

4.1 Volume configuration guidelines

This section describes best practices for volume configuration.

4.1.1 Volume size versus quantity

A frequent point of discussion during volume provisioning is *optimal size*. However, performance is determined primarily by the number of volumes provisioned. Host resources, including memory and processing cycles, are allocated per mapped storage LUN. Each additional path incurs increased demand of memory and processing resources. Because each LUN maintains an independent I/O queue, provisioning multiple volumes maximizes parallelism and resource usage, thereby optimizing performance.

Recommendation: For optimal performance, the minimum volume count within a system should be at least equal to, and preferably a multiple of, the system's CPU core count.

If you use a high number of volumes and if you build volume groups for snapshots or replication, it is recommended not to have too many volumes in a single group. For example, if configuration changes are implemented simultaneously across all volumes in a volume group, such as during a snapshot creation or replication start process, then having a large number of volumes in the volume group can affect resource availability.

A storage administrator can control this effect by using fewer, larger LUNs rather than many small ones. However, you might need to tune queue depths and I/O buffers to support controlling memory and processing time efficiently. Wherever possible, based on the overall capacity requirement of the application, use volume groups with no more than 128 volumes.

Larger volume sizes can also help to reduce the number of replication relationships, which can lead to a performance benefit in larger environments. For more information on the configuration limits, see [Configuration Limits for IBM Storage FlashSystem and IBM SAN Volume Controller](#).

4.1.2 Volume queue depth

Queue depth is used to control the number of concurrent I/O operations that occur in different storage resources. It applies at various levels of the environment, which includes disk or flash drive, storage controller, or per volume and Host Bus Adapter (HBA) level on the host.

No set rule is available for setting a queue depth per volume. The requirements for your environment are driven by the intensity of each workload. To make sure all storage resources are evenly used, distribute the volumes evenly among the I/O groups in a clustered system or among the storage systems part of a FlashSystem grid.

Improperly sized queue depth can cause a high number of outstanding I/O requests and can impact system performance. When response time is already high, a higher queue depth can cause even higher response times and possibly even lead to queue full condition, which then creates additional complications because the host must run error recovery procedures.

If a system node reaches the maximum number of queued commands, it starts to reject new I/O requests that start to queue on the host end. Many operating systems cannot recover if the situation persists for more than 15 seconds. This can cause path or application failures in one or more servers that present errors to applications.

Ensure that one application or host does not use the entire system node queue. However, if you have a specific host application that requires the lowest latency and highest throughput, consider giving it a proportionally larger share than others.

Note: Most host volume queue depth default values are around 32, which is typically what IBM recommends.

When tuning the queue depth size at volume or host level, consider the following points:

- ▶ A single IBM Storage Virtualize Fibre Channel port supports a maximum concurrent queue depth of 2048.
- ▶ Each IBM Storage Virtualize node has a maximum concurrent queue depth capacity of 10,000. Upon exceeding this threshold, the queue reports a full status condition.
- ▶ The host queue depth must be controlled through the per volume value:
 - A typical random workload volume should use a value of approximately 32.
 - To limit the workload of a volume, use a value of 4 or less.
 - To maximize throughput and give a higher share to a volume, use a value of 64.

As a general rule of thumb, when tuning queue depth, it can be increased incrementally as long as the response time on the storage system remains within acceptable values for the given workload. However, as performance improves by using more parallel resources, and volumes in particular, it might prove that a lower queue depth (for example, 16) achieves better performance results, depending on the number of volumes used by the host or application.

For more information about queue depth, see the following IBM Documentation articles:

- ▶ [FC hosts](#)
- ▶ [iSCSI hosts](#)

Non-Volatile Memory Express (NVMe) transport protocols, such as NVMe over Fabrics (NVMeoF), NVMe over Transmission Control Protocol (NVMe/TCP), and NVMe over Remote Direct Memory Access (RDMA) over Converged Ethernet (NVMe/RoCE), manage the queue depth in-band. When you provision volumes attached to hosts using those protocols, IBM recommends using default values of the queue depths.

4.1.3 I/O size

I/O size is the volume of data that a single I/O request transfers between a host and a storage system. This is typically dictated by the application, and you might not be able to influence this easily.

- ▶ Online transaction processing (OLTP) workloads, such as databases and business intelligence (BI) applications, typically use smaller I/O sizes, such as 16 KiB for both read and write requests.
- ▶ Online analytical processing (OLAP) workloads, such as data warehouses or data lakes, usually generate larger read requests of a maximum of 256 KiB but can still use smaller 16 KiB write I/O sizes.
- ▶ Backup servers and appliances and media streaming workloads use larger I/O sizes, which can range between 64 KiB and 4 MiB.

I/O transfer size affects the usage of internal system resources and has an impact on overall performance. Larger I/O sizes affect performance more than smaller block sizes.

Understanding and considering the I/O size in the design, optimization, and operation of the storage system-sizing leads to more predictable behavior of the entire environment.

IBM Storage Virtualize allocates cache resources in 32 KiB blocks. Therefore, aligned I/O sizes generally yield better performance. For IOPS-intensive workloads, 16 KiB or 32 KiB I/O sizes are recommended, if supported by the application and host. For throughput-intensive workloads, 128 KiB or 256 KiB I/O sizes are preferred, so that transfers can be split evenly into 4 or 8 parallel 32 KiB transfers, which aligns with the recommended maximum paths per volume.

Recommendation: Where possible, limit the maximum I/O transfer size that is sent to FlashSystem storage to no more than 256 KiB.

Figure 4-1 illustrates the positive effect on the response time by reducing the I/O size from 1 MiB to 256 KiB on a backup workload.

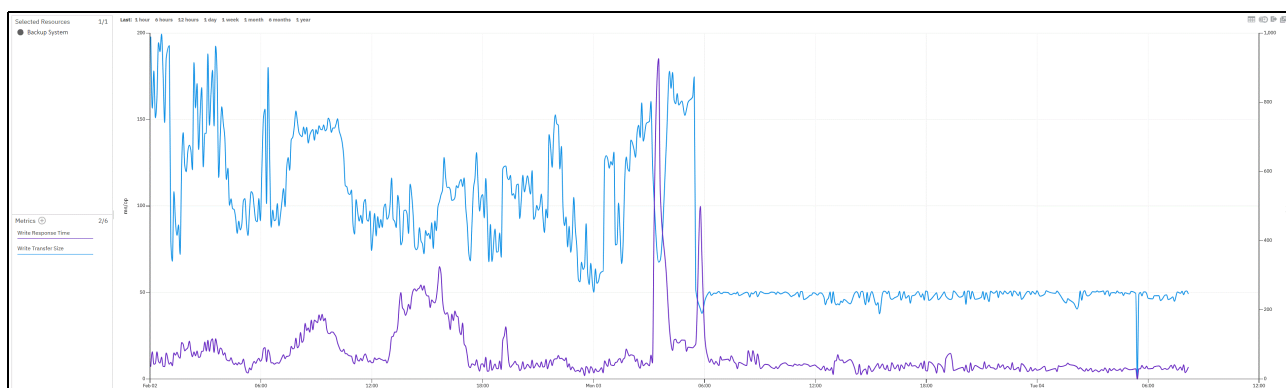


Figure 4-1 I/O size impact on performance

4.2 Volume groups

A *volume group* is a collection or grouping of volumes that share common characteristics or are organized together for specific reasons. Volume groups can be formed based on various factors:

- ▶ Volumes with similar requirements, such as performance targets, availability, or data protection policies can be grouped within a volume group. This helps ensure that the volumes within the group are managed and treated according to the same service level guidelines.
- ▶ Some applications or data sets might require mutual consistency among multiple volumes. In such cases, these volumes can be grouped together to ensure that they are synchronized and maintain consistency in terms of data updates or access.
- ▶ Volumes residing on the same server or cluster of servers can be grouped within a volume group. This grouping facilitates efficient management and administration of the volumes within the environment and provides centralized control.

For example, for a database instance it is recommended that you create separate volumes for data, transaction logs, archive logs, backups, and software installation binary files. Also, for the database data, create 16–32 volumes, depending on the database size. However, as all of those volumes are part of the same database, they require mutual consistency, so it is beneficial to configure them as a volume group.

Volume groups are recommended when using the following functions:

- Volume group snapshots (VGS)

Snapshots are read-only point-in-time copies of a volume group that are not directly accessible from the hosts. To access the snapshot contents, you can create a clone or thin clone of a VGS. You can use the management GUI or the command-line interface (CLI) to configure volume groups to use snapshot policies for multiple volumes for consistent management.

- Safeguarded Copy)

Another implementation of volume groups is to group volumes to be configured as Safeguarded. Safeguarded Copy is a cyber resiliency feature that creates immutable copies of data that cannot be changed or manipulated.

A Safeguarded volume group describes a set of volumes that can span different pools and are protected collectively with the Safeguarded Copy function. Safeguarded snapshots are supported on the system through an internal scheduler that is defined in the snapshot policy or can be configured with an external snapshot scheduling application, such as IBM Copy Services Manager. Each volume group can be assigned internal or external schedule or both, when you need to apply different retention policies.

- Policy-based high availability

Another implementation of volume groups is to add them to a storage partition that is associated with a high availability replication policy. Volumes in that volume group, hosts, and volume mappings contained in the partition are automatically configured across both systems associated with the replication policy.

- Asynchronous policy-based replication

Similarly, a volume group can be assigned an asynchronous replication policy. The system automatically replicates the data and configuration for the volumes in the volume group based on the values and settings in the replication policy. As part of asynchronous replication, a recovery volume group is created automatically on the recovery system. Recovery volume groups cannot be created, changed, or deleted manually. A single replication policy can be assigned to multiple volume groups to simplify replication management. When additional volumes are added to the group, replication is automatically configured for these new volumes.

When provisioning volumes and using volume groups for the aforementioned use cases, plan them around your concept of a *minimum viable business* (MVB). It is the smallest possible version of an organization that can still function and serve customers should an incident bring down part of the operation and systems. Ensure that replication, VGS, and safeguarded copy are part of your business continuity and disaster recovery plans. Use volume groups and storage partitions to help ensure that all dependencies are included in your plans so that you can recover your MVB in the shortest amount of time.

Provisioning policies

In addition to volume groups, storage administrators can make use of provisioning policies to help simplify storage management and enhance volume configuration efficiency. A provisioning policy establishes a set of rules for capacity allocation from a storage pool. New volumes provisioned within the pool adopt the capacity-saving techniques specified in the policy. Modifications to the pool's provisioning policy do not affect the capacity savings of existing volumes. A provisioning policy solely dictates the capacity savings applied during the creation of new volumes in the pool.

The system supports several methods to optimize capacity. These methods can be used separately, but when used together they can provide greater capacity savings and efficiencies for the system. Provisioning policies support the following capacity-saving options:

- ▶ **None (-capacitysaving none)**

Use the none option when you do not want volumes to use additional capacity savings. When this option is set, volumes use only the built-in capacity-saving functions of the storage pool. For example, pools containing certain drive types, such as FlashCore Module (FCM) drives, are self-compressing.

- ▶ **Thin provisioning (-capacitysaving thin)**

Use the thin provisioning option to indicate that all volumes in a pool use thin provisioning to save capacity. Thin provisioning allocates capacity only when data is written to storage.

- ▶ **Compression (-capacitysaving compressed)**

Use the compression option to indicate that all volumes in a pool use compression to save capacity. With compression enabled, data is compressed as it is written to disk. This applies to all types of drives in the pool, including FCM drives.

- ▶ **Deduplication (-deduplicated)**

Use it when you create a new provisioning policy to add deduplication to your capacity savings for a data reduction pool (DRP).

Both compression and thin provisioning can be used with deduplication to increase capacity savings for data reduction pools. Standard pools do not support deduplication.

A provisioning policy can be renamed after the policy is created. However, the capacity savings option cannot be changed. To change the capacity savings to be used for new volumes, create and assign a new provisioning policy to the storage pool, replacing any existing policy. The old policy can be deleted if it is no longer required. A provisioning policy can be associated with multiple pools. However, a pool can have only one provisioning policy.

The capacity savings for a volume cannot be changed after a volume is created. A provisioning policy for a storage pool is applied only when new volumes or volume copies are created. If you want to change the capacity savings for an existing volume, use a new provisioning policy with the desired options and create a volume copy.

After you add a provisioning policy to a pool, or assign a different provisioning policy, you can use the **addvolume** command to convert existing volumes so that they use the capacity savings of the new provisioning policy. The **-autodelete** parameter can be used to automatically delete the original copy of the volume after the new copy is synchronized.

For more information, see [Creating provisioning policy and assigning to pools](#).

4.3 Thin provisioning

A *thin-provisioned volume* presents a different capacity to mapped hosts than the capacity the volume consumes in the storage pool. With thin provisioning, storage administrators can configure volumes that can grow into the usable capacity they need based on the storage requirements of the host. This flexibility saves on the costs to manage the storage capacity for host applications. The system supports thin provisioned volumes in both standard pools and DRPs.

Note: IBM does not recommend the use of thin provisioned volumes in a DRP with FCMs.

In standard pools, thin-provisioned volumes are created based on capacity savings criteria. These properties can be manually managed at a volume level or automatically on a pool level when a provisioning policy is used. In DRPs, all the benefits of thin provisioning are available to all the volumes that are assigned to the pool.

Thin-provisioned volumes in a standard pool are available in two operating modes: *autoexpand* and *noautoexpand*. You can switch the mode at any time. Thin-provisioned volumes in a DRP always have *autoexpand* enabled.

If you select the *autoexpand* feature, the system automatically adds a fixed amount of real capacity to the thin volume as required. Therefore, the *autoexpand* feature attempts to maintain a fixed amount of unused real capacity for the volume. It is a best practice to use *autoexpand* by default to avoid volumes going offline because of an out-of-capacity condition.

This amount of extra real capacity, also known as *contingency capacity*, is initially set to the real capacity that is assigned when the volume is created. If the real capacity is modified, the contingency capacity is reset to be the difference between the used capacity and real capacity.

Note: A thin-provisioned volume created without auto-expansion (and zero contingency capacity) goes offline upon reaching its real capacity limit. Manual expansion is required for recovery.

For thin provisioned volumes in a DRP, you can also configure compression and data deduplication, which increases the capacity savings for the entire pool. When you use all of the capacity-saving options, you must be careful about out-of-space issues in the volume and in the pool where the volume is created. Set the warning threshold in the pool that contains thin-provisioned volumes and in the volume, also.

Warning threshold: When working with thin-provisioned volumes, enable the warning threshold notification (by e-mail or a Simple Network Management Protocol (SNMP) trap) in the storage pool. If the *autoexpand* feature is not used, enable the warning threshold on the volume level, too. If the pool or volume runs out of space, the volume goes offline, resulting in a loss of access.

When you create a thin-provisioned volume, specify the space that is initially allocated to it by using the management GUI or by using the *-rs* option in the CLI. By default, the real capacity is set to 2% of the volume logical capacity. These values, with the *autoexpand* enabled and warning disabled options, work in most scenarios. However, make sure you understand the change rate pattern of your application as different values might suit your environment better. For example, a volume that is expanding frequently, can benefit from a higher percentage real capacity. If multiple volumes have to frequently expand simultaneously, you might see a transient latency for that period.

Most modern file systems and operating systems can benefit from thin provisioning. However, if the host performs a low-level format of the entire volume or the file system writes to a large percentage of the volume, then it is possible that no significant advantage is gained by using a thin-provisioned volume over a fully allocated volume.

Monitoring capacity with thin-provisioned volumes

Note: It is critical to monitor both the physical and logical capacity when thin-provisioned or compressed volumes are used. Be sure to add capacity *before* running out of space.

If you run out of space on a volume or storage pool, the host that uses the affected volumes cannot perform new write operations to these volumes. Therefore, an application or database that is running on this host becomes unavailable.

In a storage pool with only fully allocated volumes, the storage administrator can easily manage the used and available capacity because the used capacity increases only when volumes are created or expanded. However, in a pool with thin-provisioned volumes, the used capacity increases any time that the host writes data. For this reason, the storage administrator must consider capacity planning carefully.

It is critical to put volume and pool capacity monitoring in place. Tools such as IBM Spectrum Control and IBM Storage Insights can display the capacity of a storage pool in real time and can graph how it is growing over time. These tools can assist in predicting when a pool runs out of space.

IBM Storage Virtualize also alerts you by including an event in the event log when the storage pool reaches the configured threshold, known as a *warning level*. The GUI sets this threshold to 80% of the storage pool capacity by default. By using enhanced Call Home and IBM Storage Insights, IBM can monitor and identify systems that have low capacity. This ability can generate a predictive support ticket, and the client can be contacted and advised to take proactive measures.

Reclamation process in standard pools

IBM Storage Virtualize version 8.7.0 and later includes a feature that monitors the amount of reclaimable capacity on thin-provisioned volumes in standard storage pools and schedules a background reclamation process if a configurable threshold is reached. This process identifies data that was deleted or unmapped. When reclamation is complete, the thin-provisioned volume consumes only the capacity that is in use by the host. This feature does not operate on volumes in a DRP, which handles reclamation automatically.

The reclaimable capacity is determined by using the thin provisioning (and compression, if used) estimation analysis that is provided by the integrated Compress estimator tool. The reclamation process is started by adding a volume copy to the existing volume by using the autodelete parameter in the CLI. The autodelete parameter automatically deletes the original copy after the synchronization process completes. The new volume copy is created in the same storage pool as the original copy and uses the same configuration parameters (such as grain size and warning).

Figure 4-2 on page 81 illustrates the reclamation process for thin-provisioned volumes in standard pools. While the volume copy synchronization is running in the background, the process temporarily requires additional capacity, which is approximately equal to the sum of 110% of the thin size of the volume, the real size to be used, and a 10 GB safety buffer. A reclamation starts only if the pool has sufficient capacity for this operation. Additional volume mirroring bitmap space is required in the associated I/O group for the duration of the process.

The volume copy synchronization runs with a configurable sync (background copy) rate and can be adjusted to either speed up reclamation or to slow it down to prevent an impact to I/O performance. The default sync rate is 100, which corresponds to 64 MB/s. The allowed range is 1–150, which corresponds to a minimum of 128 KB/s and a maximum of 2 GB/s.

Recommendation: To ensure optimal performance of all IBM Storage Virtualize features, it is a best practice not to exceed a sync rate of 130.

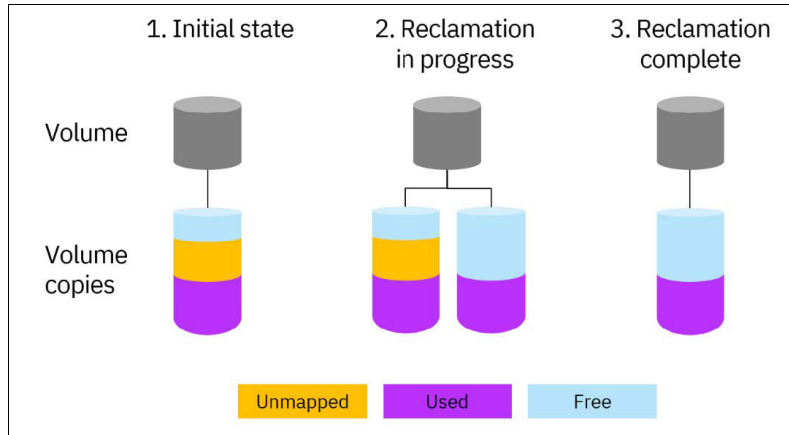


Figure 4-2 Standard pool reclamation process

Volumes in which the reclaimable capacity exceeds a configurable threshold are eligible for reclamation. This threshold is expressed in terms of capacity and is 500 GB by default. Optionally, a percentage-based threshold can be configured, which must be met in addition to the capacity threshold (there is no percentage-based threshold by default).

Note that only volumes with the following characteristics can be reclaimed:

- ▶ The volume has a single thin-provisioned copy in a standard storage pool. Volumes with two copies or a compressed copy are not eligible.
- ▶ The copy is configured to auto-expand (unless the `autoexpand-off-mode` is used).
- ▶ The volume is online.
- ▶ The volume is not a change volume for replication or a snapshot volume. There are no restrictions for volumes that are part of policy-based high availability or policy-based replication.
- ▶ The capacity savings were estimated within the last 24 hours.

To enable, configure, disable, and monitor the reclamation feature, use the `reclaim_se` command. The `reclaim_se` command includes the following main command options:

- ▶ `reclaim_se --start [optional configuration]` enables the feature. This starts a daemon running on the configuration node of the system, which checks for eligible volumes and begins the reclamation process after such volumes are detected.
- ▶ `reclaim_se --status [--dry-run] [optional configuration]` monitors the feature
- ▶ `reclaim_se --reload [optional configuration]` changes the configuration
- ▶ `reclaim_se --cancel` disables the feature

Use the following `reclaim_se --help` command to obtain a full list of all parameters and their description.

Table 4-1 lists the available configuration parameters.

Table 4-1 Reclaim process configuration parameters

Parameter	Default	Description
interval	60	Interval in minutes (5–10080)
reclaimable (capacity)	500 GB	Reclaimable capacity threshold

Parameter	Default	Description
reclaimable (percentage)	N/A	Reclaimable percentage threshold (0–100)
syncrate	100	Syncrate for volume copy synchronization (1–150)
reclaims-per-pool	1	Number of concurrent reclamations per parent pool
rsize	N/A	Override of the real size
rsize_fallback	2%	Real size to fall back to if it cannot be guessed
autoexpand-off-mode	disabled	Mode for volumes with autoexpand disabled

You can use the GUI Volume view or the output of the `lsvdiskanalysis` command to see the amount of reclaimable capacity per volume.

4.4 Compression and deduplication

This section includes a discussion of compression and deduplication considerations.

Note: This section relates to the DRP features. For FCM volumes in standard pools, do *not* use the `compress` attribute.

4.4.1 Compressed volumes

When you create volumes, you can specify compression as a method to save capacity. With compressed volumes, data is compressed as it is written to disk. When data is read to hosts, the data is decompressed.

Note: The volume compression attribute is independent of any compression that is performed by FCMs or other compressing backend storage. Although the FCMs cannot further compress data that is already compressed by DRP, they can still provide further capacity savings by compressing the volume metadata.

If you are planning to virtualize volumes that are connected to your hosts directly from any storage subsystem and you want to estimate the potential compression capacity savings, run the IBM Data Reduction Estimator Tool (DRET). For more information, see [IBM Data Reduction Estimator Tool for SVC, Storwize, and FlashSystem products](#).

If you are already using an IBM Storage Virtualize system, it includes an integrated Compressimator tool, which is available through the management GUI and CLI. If you are considering applying compression on noncompressed volumes in a FlashSystem, you can use this tool to evaluate whether compression generates enough capacity savings.

Recommendation: To optimize performance, use compressed volumes only if they provide at least 25% capacity savings.

Volume level compression is not recommended for database redo or transaction logs. Similarly, it is not recommended if compression is already turned on at the database or table level.

IBM Storage Virtualize systems support compressed volumes in a DRP only. The compressed volumes do not display individual compression ratio. As shown in Figure 4-3, customize the GUI Volume view to see the compression savings for a compressed volume and estimated compression savings for a noncompressed volume that you are planning to migrate.

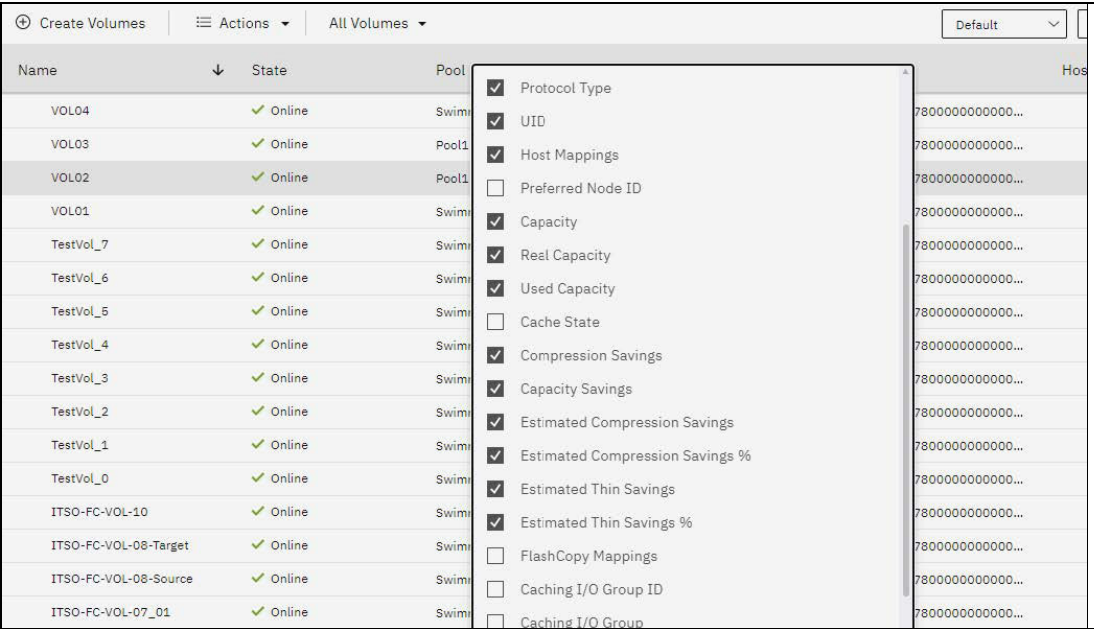


Figure 4-3 Customized view

4.4.2 Deduplicated volumes

Deduplication is a data reduction technique for eliminating duplicate copies of data. It can be configured with thin-provisioned and compressed volumes in a DRP.

The deduplication process identifies unique chunks of data and stores a signature of the chunk for reference when writing new data chunks. If the new chunk's signature matches an existing signature, the new chunk is replaced with a small reference that points to the stored chunk. The same pattern can occur many times, which results in the amount of data that must be stored being greatly reduced.

The scope of deduplication is all deduplicated volumes within the same pool, regardless of the volume's preferred node.

Depending on the data type that is stored on the volume, the capacity savings can be significant. Examples of use cases that typically benefit from deduplication are backup servers and virtual environments with multiple virtual machines (VMs) running the same operating systems (OS), e.g. either virtual desktop infrastructure (VDI) environments or VM datastores containing the OS images.

In both cases, it is expected that multiples copies of identical files exist, such as components of the standard operating system or applications that are used in the organization. Data that is encrypted or compressed at the file-system level does not benefit from deduplication.

4.4.3 Compression and deduplication considerations

You can use IBM Storage Virtualize to create volumes that have both the compression and deduplication capacity savings options enabled. If you are using a compressed backend storage (such as FCMs) and have data reduction (thin provisioning or deduplication) enabled in the pool without compression enabled as well, the system cannot determine the amount of usable capacity as it is not possible to determine the compression ratio of the unmapped or reclaimable data per drive.

All members of the FlashSystem portfolio with the exception of FlashSystem 5035 and FlashSystem 5045 can offload hardware compression, so volume compression can be turned on without affecting performance. FlashSystem 5035 and FlashSystem 5045 use software compression which can negatively affect performance, especially when there is a higher number of compressed volumes in use. FCMs use hardware-assisted compression, which does not create affect performance.

The effect on performance for volumes in a DRP is a result of the additional metadata handling and the resulting I/O amplification. The additional metadata is used for any form of data reduction. Therefore, the overhead of using volumes with deduplication is approximately the same as the overhead of using volumes with both compression and deduplication enabled.

Although using deduplicated and compressed volumes in a DRP brings the greatest capacity savings, it must be balanced against the additional performance cost of metadata management. Workloads that obtain most of the data reduction savings from compression alone or latency-sensitive applications, such as databases, are not ideal scenarios for such a configuration. Therefore, make sure the workload is properly sized before using deduplicated and compressed volumes.

Recommendation: Unless your workload is likely to achieve high deduplication savings, consider using thin-provisioned volumes and FCMs always-on native compression in a standard pool.

The recommendation is to use no more than approximately 85% of the available capacity in a DRP for data and metadata of volume copies, leaving enough room for the system to reclaim space. After that point, the garbage collection process must work harder to deallocate space, which can increase the latency for all volumes in the pool. When you cross the 80% warning threshold, consider adding more capacity to the pool or consider removing some of the volumes to reduce the amount of data stored in the pool.

4.5 Sharing workloads within the same pool

Larger FlashSystem models, especially FlashSystem 9500, are designed for workload consolidation, so using them for multiple purposes is expected. For simplicity of planning and for improved performance, it is recommended that you create a single array and a single pool out of all the drives in the control enclosure of the system. Volumes inherit the performance characteristics of the pool they are created in and the characteristics of its underlying disks and mdisks. For more information, see Chapter 3, “Performance optimization for FlashSystem storage tiers and arrays” on page 43.

If a system provides storage to a wide variety of applications, then production volumes can be competing with volumes that have lower priority operations. A given volume, subset of volumes, or a volume group, might place a load on the system that could have an adverse

effect on the other volumes in the pool, which can create a *noisy neighbor* scenario. For example, volumes that are used for backup or archive operations can have I/O intensive workloads, potentially taking bandwidth from production volumes. Alternatively, you might want to allocate more resources for mission critical applications or databases, knowingly limiting the resources for other volumes.

For either situation, you can use *throttles* as a mechanism to control the amount of resources that are used when the system is processing I/O requests on supported objects. If a throttle is defined, the system either processes the I/O request for that object, or delays the processing of the I/O requests so that resources are available for more critical I/O operations.

Furthermore, for administration simplicity, you can separate volumes and volume groups in child pools within the same parent pool. IBM Storage Virtualize version 8.7.2 and later includes the capability to create a child pool in a standard parent pool without specifying a quota. When a child pool is created without a quota, the child pool can access the entire usable capacity of the parent pool without limit, which reduces the amount of monitoring and management required.

Whether you decide to configure throttles on a volume or child pool level, use the following guidelines:

- ▶ The throttle limit is a per node limit. For example, if a throttle limit is set for a volume at 1000 IOPS, each node on the system that has access to the volume allows 1000 IOPS for that volume. Any I/O operation that exceeds the throttle limit is queued at the receiving node.
- ▶ The multipathing policy on the host determines how many nodes receive I/O operations and the effective throttle limit. For redundancy purposes, configure paths to both nodes in a system.
- ▶ If more than one throttle applies to an I/O operation, the most restrictive throttle is used. For example, if different throttles are applied to a parent pool and a child pool, the throttles work in a hierarchical manner, with the more stringent throttle taking precedence.

For more information, see [Managing throttles for volumes](#) and [Managing throttles for pools](#).

If a given workload has very specific workload pattern, such as IBM i, it is recommended to use a dedicated storage system for it instead of mixing it with other workloads having very different characteristics, such as VMware, for example. For truly isolated workloads, you can use several smaller systems, such as FlashSystem 5300 or FlashSystem 7300, instead of a single larger one.

4.6 Volume placement guidelines

This section includes a discussion of volume placement guidelines.

4.6.1 Volume placement within a single storage system

Volumes are assigned to nodes and paths automatically, but the assignments can be modified as necessary to balance the workload.

Preferred paths to a volume

During volume creation, a *preferred node* is designated to manage the volume's I/O operations. The preferred node serves primarily for load balancing and to determine the node responsible for write destaging.

Preferred node assignment is normally automatic. The system selects the node in the I/O group that includes fewer volumes at the time of the volume creation. However, the preferred node can be specified during the volume creation or changed afterward, if needed.

All modern multipathing drivers support Asymmetric Logical Unit Access (ALUA). By using ALUA, the storage system can mark paths to the preferred node as the preferred path. ALUA-capable multipathing drivers acknowledge preferred paths and send I/O requests to the non-preferred node only if the preferred node is not accessible.

Figure 4-4 shows write operations from a host to two volumes with different preferred nodes.

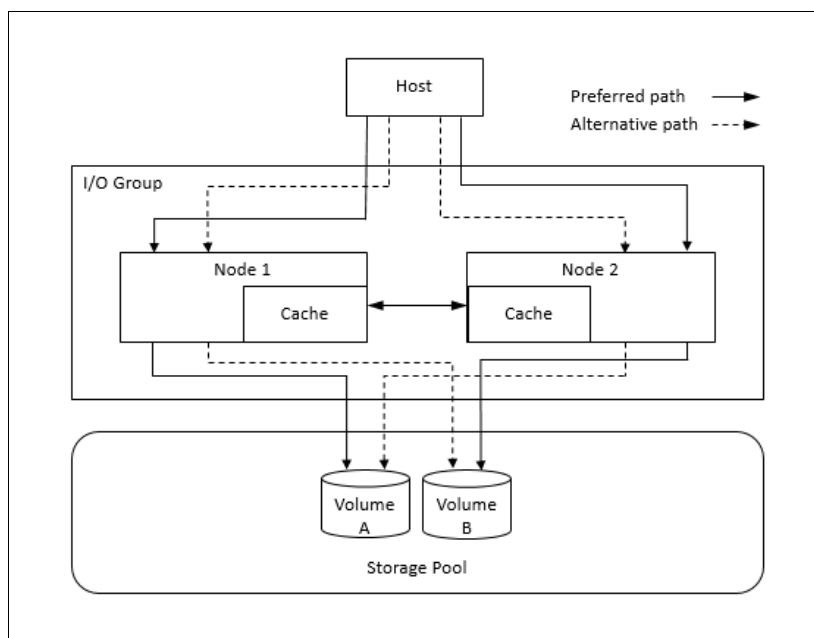


Figure 4-4 Write operations from a host through different preferred nodes for each volume

When analyzing performance problems, it can be useful to review the “Non-Preferred Node Usage Percentage” metric in IBM Spectrum Control or IBM Storage Insights. I/O requests to the non-preferred node might cause performance overhead for the I/O group because of increased inter-node communications, which can be identified by these tools.

For more information about this and other performance metric, see [Performance metrics for resources that run IBM Storage Virtualize](#) (for IBM Spectrum Control) or [Performance metrics for resources that run IBM Storage Virtualize](#) (for IBM Storage Insights).

Balancing volumes between nodes

IBM Storage Virtualize automatically distributes workloads across I/O group nodes by maintaining a balanced volume ownership. However, performance imbalances can occur if volumes on one node experience significantly higher activity than those on its partner. Workload fluctuations, in terms of volume or characteristics, are the primary cause of performance shifts. To rebalance workload, adjust the preferred node of a volume within the I/O group or migrate it to another I/O group.

You can change the preferred node within an I/O group by using concurrent I/O. However, changing the node can lead to a transient delay in performance while preferred path designations are changed and internal cache resources ownership is aligned.

Perform volume migration between I/O groups on only an SVC. During migration, the system provides a recommendation to automatically select the volume's preferred node within the new I/O group. Alternatively, administrators can manually specify the preferred node by using the GUI or the CLI.

I/O operations can continue concurrently during I/O group modification if the host supports nondisruptive volume move (NDVM). A host rescan is required to update the multipathing driver with the preferred node and access port changes. If host mappings exist, the host must also be a member of the target I/O group for successful migration.

Confirm the creation of paths to both I/O groups on the system. Following successful addition of the new I/O group to the volume's access set and subsequent volume migration, detect the updated paths on the host. For optimal performance and to minimize inter-node messaging, be sure the volumes are accessed solely through the caching I/O group, except during migration.

At the time of writing, FlashSystem code levels support a single I/O group. SAN Volume Controller (SVC) supports multiple I/O groups on only long-term support (LTS) code levels. For enhanced scalability, a FlashSystem grid can be deployed.

4.6.2 Balancing volumes across your FlashSystem storage fleet

Starting with IBM Storage Virtualize code 8.7.0, FlashSystem grids replace I/O groups as the scale-out method for FlashSystem. A FlashSystem grid is a scalable storage platform, composed of a maximum of eight mixed-hardware systems that can increase operational simplicity through federated management, AI-powered data placement, and flexible deployment. This architecture enables linear scaling of performance, capacity, and volume and volume group count.

The performance characteristics of each FlashSystem model within the grid significantly impact overall performance. Mixed-hardware grids facilitate workload tiering based on performance requirements so that you can run non-disruptive migration of less demanding workloads to cost-optimized tiers to align with user-defined SLAs.

Integrated with IBM Storage Insights, FlashSystem grids help simplify workload requirement evaluation. IBM Storage Insights analyzes historical capacity and performance trends over the past year and forecasts for the next year to provide ranked recommendations for compatible FlashSystem models.

To determine the optimal FlashSystem for your workload and volume placement by using the IBM Storage Insights Workload Placement Advisor, follow these steps:

1. Log in to IBM Storage Insights.
2. If you are in the classic view, select **Insights** → **Workload placement**. If you are in the carbon UI, select **Planning** → **Workload placement**.
3. By default, IBM Storage Insights ranks storage systems for new workload requirements based on one year of historical capacity growth and forecasts. To modify this time frame, click **Change**, specify the date range, and click **Apply**.
4. Enter all the relevant details in the fields under the **Set Workload Details** section:
 - a. To ease the experience, the system provides the option to choose from pre-defined workload templates (OLAP, OLTP, Backup) and t-shirt sizes (small, medium, large). After a template is selected, you can fine tune the required properties. Alternatively, you can select **Custom** and tune all the properties.

- b. Specify the required capacity for your new workload and the anticipated capacity growth for the forecast period. Optionally, include the desired capacity percentage for snapshots or safeguarded copies in **Snapshot/Safeguarded Copies**.
5. To filter the storage systems for evaluating your workload, enter the relevant details in the **Set Storage System Search Scope** section:
 - a. Select the specific FlashSystem grid from the **Flash Grid** drop-down menu.
 - b. Choose the storage systems from the **Storage Systems** drop-down menu.

Choose the specific FlashSystem storage systems that you want to evaluate for the workload placement. Although all available systems in your environment are considered by default, you can narrow the selection to relevant systems (designated for business unit or application use) to improve the recommendation speed.
6. When you are satisfied with your selections, click **Evaluate**. A list of best compatible systems is displayed by rank based on your workload requirements.

You can expand each storage system from the list to view detailed charts for several key parameters, including capacity, IOPS, data rate, latency, and CPU utilization.

Figure 4-5 shows the IBM Storage Insights Workload Placement Advisor. For a demo of the feature, see [IBM Storage Insights: Workload Placement Advice](#).

Figure 4-5 IBM Storage insights Workload Placement Advisor

If a FlashSystem approaches its limits of capacity or performance, you can use the IBM Storage Insights Workload Placement Advisor to evaluate the best location to rebalance your workloads. After you select the new target system for your volumes, you can use the Partition Migration wizard within the IBM Storage Insights FlashSystem grid view to perform the migration. For more information, see [Migrating storage partitions between systems](#).

With the help of IBM Storage Insights and its integrated AIOps capabilities, you can extract the most value of your systems by storing data in the right system, either performance-optimized for high-demand applications and workloads or capacity-optimized for colder data.

For more information, see *Maximize Your Storage Investment: Leverage IBM Storage Insights for Observability and Artificial Intelligence for IT Operations*, [SG24-8584](#).



Networking and multipathing optimization

This chapter discusses aspects of networking and multipathing configuration that relates to performance optimization and the debugging of performance issues.

This chapter has the following sections:

- ▶ 5.1, “Connection protocols” on page 90
- ▶ 5.2, “SCSI FC general configuration guidelines” on page 92
- ▶ 5.3, “iSCSI host attachment considerations” on page 95
- ▶ 5.4, “iSCSI extensions for RDMA host attachment considerations” on page 98
- ▶ 5.5, “NVMe over Fibre Channel host attachment considerations” on page 98
- ▶ 5.6, “NVME over Ethernet host attachment considerations” on page 99
- ▶ 5.7, “Portsets” on page 100
- ▶ 5.8, “Host multipathing” on page 104
- ▶ 5.9, “I/O queues and queue depths” on page 105
- ▶ 5.10, “Host clusters” on page 106
- ▶ 5.11, “Multipath configuration and optimization” on page 108
- ▶ 5.12, “Network considerations” on page 116
- ▶ 5.13, “Impact of path selection on performance and availability” on page 118

5.1 Connection protocols

There are two main protocols types for accessing volumes on IBM Storage Virtualize:

- ▶ SCSI is a standard protocol and is built for conventional hard disk drives.
- ▶ NVME is a newer protocol with fewer commands and better performance that was purposefully built for all flash arrays.

A volume in IBM Storage Virtualize can be accessed by using only a single protocol, either SCSI or NVMe. This is because the storage controller in the system can present a volume to a host in only one way.

However, a volume can be accessed using different transport protocols (iSCSI, FC, and so on) from different hosts. For example, a volume could be accessed using iSCSI from one host and FC from another host. This is because the transport protocol is the way that the host communicates with the storage system, and the storage system can present a volume to different hosts using different transport protocols.

IBM Storage Virtualize does not support the same host accessing a volume using different transport protocols via the same host bus adapter. This is because the host bus adapter can communicate with the storage system by using only one transport protocol at a time.

Consider the following example:

- ▶ Host A can access a volume by using iSCSI.
- ▶ Host B can access the same volume by using FC.
- ▶ Host C cannot access the same volume by using both iSCSI and FC.

All hosts and cluster connections must use the same protocol type for a volume. Different hosts can have different protocols working with the same storage port. So iSCSI, iWARP, iSER, NVMe/RDMA and NVMe/TCP can coexist on the same port with different host mapping types. This includes the ports that are used for replication.

VMware ESXi supports NPIV and provides FC SCSI and NVMe-oF on the same HBA with separate logical devices. Users can create FC SCSI host and NVMe host on a FlashSystem for the same ESXi host to use NVMe for some specific volumes (such as database volumes).

5.1.1 SCSI over Fibre Channel characteristics

SCSI over Fibre Channel connections include the following characteristics:

- ▶ Currently the most mature technology with arguably the widest deployment in the field.
- ▶ Wide interoperability support for operating systems, host bus adapters (HBA) hardware/driver/firmware, SAN switch vendors, and storage vendors.
- ▶ Comparatively low protocol overheads compared to ethernet based connectivity.
- ▶ Availability of deep technical expertise at vendors, clients, and system integrators.
- ▶ Availability of mature documentation with industry and vendor best practices.
- ▶ Constant improvement of Fibre Channel hardware with regular bandwidth and feature improvements.
- ▶ High bandwidth and low latency.
- ▶ Dedicated for storage.

- ▶ Each FC port does not use multiple CPU cores per port for I/O handling. This is unfavorable when workloads are unbalanced.
- ▶ Queue depth must be configured explicitly on hosts and requires attention to detail in non-homogeneous environments.

5.1.2 iSCSI characteristics

iSCSI connections include the following characteristics:

- ▶ Wide interoperability support for operating systems, network interface cards (NICs) hardware/driver/firmware, and storage vendors.
- ▶ Support for software initiators only reduces interoperability requirements.
- ▶ Can use existing network infrastructure for storage attachment.
- ▶ Availability of deep technical expertise at vendors, clients, and system integrators.
- ▶ Availability of mature documentation with industry and vendor best practices.
- ▶ Continuous changes to ethernet hardware with regular bandwidth and feature improvements.
- ▶ Supports VLAN tagging and priority flow control (PFC) to prioritize iSCSI traffic.
- ▶ Supports a wide range of bring your own (BYO) cables and optics.
- ▶ Code levels earlier than 8.6.0.0 used a small subset of the CPU cores that are available to the I/O process. An improvement in code level 8.6.0.0 improved the multithreadedness of iSCSI.
- ▶ Queue depth must be configured explicitly on hosts and requires attention to detail in non-homogenous environments.
- ▶ It may be necessary to use third party tools to collect packet traces to debug complex cases.
- ▶ Storage traffic potentially competes with traditional network traffic.
- ▶ Limited to 25 Gb/s on 100 Gb/s ports.

5.1.3 iSCSI extensions for Remote Direct Memory Access characteristics

iSCSI extensions for Remote Direct Memory Access (ISER) include the following characteristics:

- ▶ Continuous improvement of ethernet hardware with regular bandwidth and feature improvements.
- ▶ High bandwidth and low latency in well configured environments.
- ▶ Supports VLAN tagging and priority flow control(PFC) to prioritize ISER traffic.
- ▶ To provide greater flexibility, two implementations are available depending on the client use case, network configuration, and host and storage NICs:
 - Remote Direct Memory Access (RDMA) over converged ethernet (RoCE)
 - Internet Wide-area RDMA Protocol (iWARP)
- ▶ iWARP runs on TCP routable networks and has wide hardware and network compatibility.
- ▶ RDMA techniques bypass CPU, which reduces host and storage CPU usage for I/O processing.
- ▶ Supports a wide range of bring your own(BYO) cables and optics.

- ▶ Queue depth must be configured explicitly on hosts and requires attention to detail in non-homogenous environments.
- ▶ RoCE implementation requires supported switches and explicit switch configuration.
- ▶ Adoption in the field is still increasing; maturity is less than that of Fibre Channel.
- ▶ Not supported from code level 8.7.1 and later.

5.1.4 NVME over Fabric characteristics

NVME over Fabric includes the following characteristics:

- ▶ Fiber channel, RDMA and TCP fabrics are supported.
- ▶ High bandwidth and low latency.
- ▶ Greater concurrency and balanced resource usage, with a queue pair per host CPU or namespace with each being able to drive I/O concurrently.
- ▶ Queue depth is dynamically negotiated in-band for queue pairs and explicit tuning and configuration are not required.
- ▶ Halves the number of interactions and round trips for an I/O compared to SCSI.
- ▶ Purposefully designed to achieve the full potential of high performance flash-based arrays.
- ▶ Relatively new technology with low field adoption and maturity.
- ▶ Interoperability is less comprehensive compared to SCSI host attachment.

5.2 SCSI FC general configuration guidelines

This section describes some general configuration guidelines.

5.2.1 Number of paths

Adding paths does not significantly increase redundancy, and can slow down the host response because the host must address path management. Also, too many paths might increase the failover time.

Note: As a best practice, do not use more than four FC paths per volume.

5.2.2 Host ports

Each host uses two ports from two different host bus adapters (HBAs). These ports should go to separate SAN fabrics and be zoned to one target port of each node or node canister. When the volumes are created, they are assigned to an I/O group. Ensure that the resulting path count between the volume and the host is four.

Hosts with a large amount of host ports might require alternative zoning and pseudo host creation. Refer to 2.3.9, “Recommendations based on field experience” on page 39 for an example with IBM i hosts that require this.

Best practice: Keep FC tape (including virtual tape libraries) and FC disks on separate HBAs. These devices have two different data patterns when operating in their optimum mode. Switching between them can cause unwanted processor usage and performance slowdown for the applications.

5.2.3 N_Port ID Virtualization

IBM Storage Virtualize uses N_Port ID Virtualization (NPIV) by default. NPIV reduces failover time, and you can use features such as Hot Spare Nodes (HSNs).

For more information about the IBM Storage Virtualize 8.6 NPIV configuration and details, see “7.5, “*N_Port ID Virtualization support*”, in *Implementation Guide for IBM Storage FlashSystem and IBM SAN Volume Controller Updated for IBM Storage Virtualize Version 8.6*, [SG24-8542](#).

NPIV was optional in code levels earlier than 8.7.0.0. It is mandatory at code level 8.7.0.0 and later. This requires special handling for direct attached IBM i hosts. Refer to “Multipathing support and IBM i configuration recommendations” on page 114.

5.2.4 Host to I/O group mapping

An *I/O group* consists of two nodes or node canisters that share the management of volumes within the cluster. Use a single I/O group (iogrp) for all volumes that are allocated to a specific host. This guideline results in the following benefits:

- ▶ Minimizes port fan-outs within the SAN fabric.
- ▶ Maximizes the potential host attachments to IBM Storage Virtualize because the maximums are based on I/O groups.
- ▶ Reduces the number of target ports that must be managed within the host.

5.2.5 Volume size versus quantity

In general, host resources, such as memory and processing time, are depleted by each storage logical unit number (LUN) that is mapped to the host. For each extra path, more memory is used, and a portion of more processing time is also required.

Each LUN has its own I/O queue, so especially in SCSI environments where you have only one queue per LUN, you can get more throughput with LUNs that work in parallel.

A best practice is to set a minimum number of LUNs in the I/O group to be equal or higher than the number of cores in the I/O group.

When using snapshots, the number of volumes in a volume group was an important consideration on code levels earlier than 8.7.0.0. The pauseless snapshot feature that was introduced in code level 8.7.0.0 makes this consideration less significant. Refer to Chapter 8, “Pauseless snapshots case study” on page 167.

5.2.6 Host volume mapping

Host mapping is the process of controlling which hosts can access specific volumes within the system. IBM Storage Virtualize always presents a specific volume with the same SCSI ID on all ports of the I/O group. When a volume is mapped, IBM Storage Virtualize software

automatically assigns the next available SCSI ID if none is specified. In addition, each volume has a unique identifier (UID).

You can allocate the operating system volume of the SAN boot as the lowest SCSI ID (zero for most hosts), and then allocate the various data disks. If you share a volume among multiple hosts, consider controlling the SCSI ID so that the IDs are identical across the hosts. This consistency ensures ease of management at the host level and prevents potential issues during IBM Storage Virtualize updates and even node restarts, especially for VMware ESXi operating systems.

If you are using image mode to migrate a host to IBM Storage Virtualize, allocate the volumes in the same order that they were originally assigned on the host from the back-end storage.

The **lshostvdiskmap** command displays a list of virtual disks (VDisks) or volumes mapped to a specified host. These volumes are recognized by that host.

Example 5-1 shows the syntax of the **lshostvdiskmap** command that is used to determine the SCSI ID and the UID of volumes.

Example 5-1 The lshostvdiskmap command

```
svcinfo lshostvdiskmap -delim : 0 (host id)
id:name:SCSI_id:host_id:host_name:vdisk_UID:IO_group_id:IO_group_name:mapping_type
:host_cluster_id:host_cluster_name:protocol
0:ESXVOL_01:0:0:HG-ESX6:600507681280000C70000000000004B2:0:io_grp0:private::scsi
```

Example 5-2 shows the results of using the **lsdiskhostmap** command with mismatched SCSI IDs. Example 5-2 shows the same volume that is mapped to five different hosts, but host 110 features a different SCSI ID than the other four hosts.

Example 5-2 Vdisk ID 950 is mapped with mismatched SCSI IDs

```
svcinfo lsdiskhostmap -delim : EEXCLS_HBin01
id:name:SCSI_id:host_id:host_name:vdisk_UID:IO_group_id:IO_group_name:mapping_type:host_cluster_
id:host_cluster_name:protocol
950:EEXCLS_HBin01:14:109:HDMCENTEX1N1:600507680191011D4800000000000466:0:io_grp0:private::scsi
950:EEXCLS_HBin01:13:110:HDMCENTEX1N2:600507680191011D4800000000000466:0:io_grp0:private::scsi
950:EEXCLS_HBin01:14:111:HDMCENTEX1N3:600507680191011D4800000000000466:0:io_grp0:private::scsi
950:EEXCLS_HBin01:14:112:HDMCENTEX1N4:600507680191011D4800000000000466:0:io_grp0:private::scsi
950:EEXCLS_HBin01:14:113:HDMCENTEX1N5:600507680191011D4800000000000466:0:io_grp0:private::scsi
```

Note: Example 5-2 is a *not* a recommended practice because it can cause loss of access in some situations because of the SCSI ID mismatch.

5.2.7 Server adapter layout

If your host system includes multiple internal I/O buses, place the two adapters that are used for IBM Storage Virtualize cluster access on two different I/O buses to maximize the availability and performance. When you purchase or install a server, ensure that each server has two cards instead of one. For example, two dual-port HBA cards are preferred over one quad-port HBA card because you can spread the I/O and add redundancy.

5.3 iSCSI host attachment considerations

Isolating storage traffic from other network traffic is recommended.

Internet Small Computer System Interface (iSCSI) is a widely used protocol compatible with any Ethernet network. IBM Storage Virtualize 8.6 introduced performance enhancements, particularly for midrange and high-end systems. IBM Storage Virtualize 8.6 and later supports up to 1024 iSCSI hosts per I/O group on FlashSystem 9500 and 7300 systems.

Priority Flow Control (PFC) for iSCSI/iSER is supported on all supported Host Interface Card (HIC) options and, when used with all DCBX-enabled switches, can improve performance.

5.3.1 Connection basics

Symmetrical cabling is required. Port x on each node within an I/O group must connect to the same network. A maximum of four ports per host per portset is allowed. All ports must support identical speeds and be connected to SAN switches via optics that ensure consistent speed connectivity.

5.3.2 Performance tuning

Some attributes and host parameters can affect iSCSI performance:

- ▶ Transmission Control Protocol (TCP) Delayed ACK
- ▶ Ethernet jumbo frame
- ▶ Optimizing and using all iSCSI ports
- ▶ Priority flow control (PFC) setting and bandwidth allocation for iSCSI on the network

Disable the TCP delayed acknowledgment feature

To disable this feature, refer to the following OS or platform documentation:

- ▶ For VMware, see [ESX/ESXI hosts might experience read or write performance issues with certain iSCSI storage arrays](#).
- ▶ For Windows, see [Slow performance occurs when you copy data to a TCP server by using a Windows Sockets API program](#).
- ▶ For Linux (RHEL), see [How do I control TCP delayed ACK and delayed sending?](#).

To view this content you must login with a Red Hat customer or partner account.

- ▶ For Linux (Other), use the IP utility to enable quickack on the host side such as in the following example:

```
ip route show; ip route change 192.168.140.0/24 dev eth7 proto kernel scope
link src 192.168.140.55 quickack 1
```

The primary symptom of this issue is that read performance is significantly lower than write performance. Transmission Control Protocol (TCP) delayed acknowledgment is a technique that is used by some implementations of the TCP to improve network efficiency. However, in this scenario where the number of outstanding I/Os is 1, the technique can significantly reduce I/O performance.

In essence, several ACK responses can be combined into a single response, which reduces protocol overhead. As described in RFC 1122, a host can delay sending an ACK response by up to 500 ms. Additionally, with a stream of full-sized incoming segments, ACK responses must be sent for every second segment.

Tip: Field experience shows us that enforcement of this TCP parameter is stricter when attaching to a FlashSystem at code level 8.7.0.0 or higher. After an upgrade, hosts with TCP delayed ack enabled might exhibit higher read response time compared to older code levels.

Enable jumbo frames for iSCSI

Jumbo frames are Ethernet frames with a size in excess of 1,500 bytes. The maximum transmission unit (MTU) parameter is used to measure the size of jumbo frames.

Ethernet frames consist of a header, payload and CRC. The header and CRC portion of the frame is considered overhead. Increasing the MTU allows systems to send a larger payload with the same or lower number of Ethernet frames, which can improve efficiency. This is dependent on the network's ability to send the packet without fragmentation from end to end.

If packets are fragmented while traversing a network, then they must be reassembled at the point of delivery. This additional processing negates the benefit of jumbo frames.

Restriction: Systems with RoCE Adapters had a restriction that they could not have MTU greater than 1500 on code levels 8.6.0.0, 8.6.0.1 and 8.6.1. This restriction was partially lifted in code level 8.6.0.2 and 8.6.2.0. MTU can be set to 9000 if the port is being used for iSCSI only.

The network must support end-to-end jumbo frames to be effective. To verify, send a ping packet to be delivered without fragmentation to verify that the network supports jumbo frames.

Checking MTU delivery in Windows

For Windows, you can use the following example:

```
ping -t <iscsi target ip> -S <iscsi initiator ip> -f -l <new mtu size - packet overhead (usually 36, might differ)>
```

The following command is an example of a command that is used to check whether a 9000-bytes MTU is set correctly on a Windows system:

```
ping -t -S 192.168.1.117 192.168.1.217 -f -l 8964
```

The following output is an example of a successful reply:

```
192.168.1.217: bytes=8964 time=1ms TTL=52
```

Checking MTU delivery in Linux

For Linux, you can use the following example:

```
ping -l <source iscsi initiator ip> -s <new mtu size> -M do <iscsi target ip>
```

Checking MTU delivery in ESXi

► For ESXi, you can use the following example:

```
ping <iscsi target ip> -I <source iscsi initiator ip> -s <new mtu size> -d
```

These commands send a ping with a packet size equal to the configured MTU with the *do not fragment* bit set. If the packet cannot be propagated end to end without fragmentation, then the response to the ping is a time out with a “Message too long” response.

Verify the switch's port statistics where initiator or target ports are connected to make sure that packet drops are not high. Review network architecture to avoid any bottlenecks and oversubscription. The network needs to be balanced to avoid any packet drop. Packet drop significantly reduces storage performance.

Optimize and utilize all iSCSI ports

In code levels preceding 8.6.0.0, achieving optimal FlashSystem resource usage necessitated the engagement of all iSCSI ports. The architectural enhancements introduced in code level 8.6.0.0 now provide balanced system resource utilization, even when a reduced subset of iSCSI ports manages the majority of I/O operations.

Maintaining a balanced host workload distribution across all iSCSI ports is still considered a best practice to mitigate per-port Ethernet bandwidth saturation. Configure subnets equal to the quantity of iSCSI ports per system node, with each node port assigned a unique subnet IP address to ensure uniformity across all nodes. Example 5-3 displays an ideal configuration:

Example 5-3 Ideal configuration

Node 1
Port 1: 192.168.1.11
Port 2: 192.168.2.21
Port 3: 192.168.3.31
Node 2:
Port 1: 192.168.1.12
Port 2: 192.168.2.22
Port 3: 192.168.3.33

Avoid imbalanced host distribution across ports, such as 50 hosts logged into port 1 and only five into port 2.

Use proper subnetting to achieve a balance between the number of sessions and redundancy.

Ensure that proper bandwidth is given to iSCSI on the network

You can divide the bandwidth among the various types of traffic. It is important to assign proper bandwidth for good performance. To assign bandwidth for iSCSI traffic, first enable the priority flow control for iSCSI.

Priority flow control (PFC) is an Ethernet standard that supports the ability to select the priority of different types of traffic within the network.

With PFC, administrators can reduce network congestion by slowing or pausing certain classes of traffic on ports, thus providing better bandwidth for more important traffic. The Storage Virtualize system supports PFC on various supported Ethernet-based protocols on three types of traffic classes: system, host attachment, and storage traffic.

You can configure a priority tag for each of these traffic classes. The priority tag can be any value 0–7. You can set identical or different priority tag values to all these traffic classes. You can also set bandwidth limits to ensure a class of service for these traffic classes by using the Enhanced Transmission Selection (ETS) setting on the network.

To use PFC and ETS, ensure that the following tasks are completed:

- ▶ Ensure that ports support 10 Gb or higher bandwidth to use PFC settings.
- ▶ Configure VLAN on the system to use PFC capabilities. To configure VLAN settings, see [Modifying VLAN for priority flow control](#).
- ▶ Ensure that the same VLAN settings are configured on the all entities, including all switches between the communicating end points.
- ▶ Configure class of service (COS) values (priority tag values) for host attachment, storage, or system traffic by using the [chsystemethernet](#) command.
- ▶ To enable priority flow control for host attachment traffic on a port, ensure that the host flag is set to **yes** on the configured IP address on that port.
- ▶ To enable priority flow control for storage traffic on a port, ensure that storage flag is set to **yes** on the configured IP address on that port.
- ▶ On the switch, enable Data Center Bridging Exchange (DCBx). DCBx enables switch and adapter ports to exchange parameters that describe traffic classes and PFC capabilities. For these steps, check your switch documentation for details.

For more information about iSCSI performance, see [iSCSI performance analysis and tuning](#).

5.4 iSCSI extensions for RDMA host attachment considerations

For host access, iSCSI Extensions for RDMA (iSER) is not supported on the following platforms:

- ▶ IBM Storage FlashSystem 7300, 9500
- ▶ SVC SV3, which includes no support for both RoCE and iWARP

If you are using iSER host attachment on existing IBM Storage Virtualize storage, these systems continue to support both RoCE and iWARP on IBM Storage Virtualize code levels 8.6.0 and 8.7.0

Priority Flow Control for iSCSI or iSER is supported on earlier FlashSystem storage systems with all supported RoCE and iWARP adapter options with all DCBX enabled switches.

Note: iSER host attachment is not supported on code level 8.7.1 and later.

5.5 NVMe over Fibre Channel host attachment considerations

IBM Storage Virtualize supports a single host initiator port that uses SCSI and Fibre Channel-Nonvolatile Memory Express (FC-NVMe) connections to the storage. Enable NPIV on the FlashSystem or SVC and zone the NVMe FC ports. NVMe/FC is not used for external virtualization. Use different zones for SCSI/FC and NVMe/FC for an easier management and overview. A volume can be accessed by using only a single protocol (SCSI or NVMe), not both.

If you want to use both type of volumes from one host, for example for migration, then create two host definitions for the host, one for SCSI and one for NVMe:

- ▶ Host1scsi-LUN1-FC(SCSI)
- ▶ Host1nvme-LUN2-FC(NVMe)

Note: As a best practice, use only one protocol from a host.

Asymmetric Namespace Access was added to the FC-NVMe protocol standard, which gives it functions that are similar to ALUA. As a result, FC-NVMe can now be used in stretched clusters. Consider NVMe over FC target limits when you plan and configure the hosts.

Include the following points in your planning:

- ▶ An NVMe host can connect to four NVMe controllers on each system node. The maximum per node is four with an extra four in failover.
- ▶ Zone up to four ports in a single host to detect up to four ports on a node. To allow failover and avoid outages, zone the same or additional host ports to detect an extra four ports on the second node in the I/O group.
- ▶ A maximum of 64 FC-NVMe hosts can log in to a single IBM Storage FlashSystem node port. For more information about FC-NVMe enhanced scaling and system requirements, see [V8.7.3.x Configuration Limits for IBM FlashSystem and SAN Volume Controller](#).

By using IBM Storage Virtualize 8.7.3.x, you can define a maximum of 256 NVMe hosts per system, which is a significant increase compared to version 8.6.0. IBM Storage Virtualize code does not monitor or enforce these limits, so administrators must ensure that the limits are not exceeded.

For more information about using NVMe hosts with FlashSystem, see [NVMe over Fibre Channel Host attachments](#).

Note: Do not map the same volumes to SCSI and NVMe hosts concurrently, even one after the other. Also, take care not to add NVMe hosts and SCSI hosts to the same host cluster.

5.6 NVMe over Ethernet host attachment considerations

It is recommended to separate Storage traffic from other network traffic.

Asymmetric Namespace Access was added to the NVMe over RDMA and NVMe over TCP protocol standards, which gives it functions that are similar to Asymmetric Logical Unit Access (ALUA). As a result, FC-NVMe can now be used in stretched clusters.

iSCSI, iWarp, iSER, NVMe/RDMA and NVMe/TCP can coexist on the same port with different host mapping types. This includes the ports that are used for replication. A single host working with multiple protocols is not supported. Different hosts can have different protocols working with the same storage port.

Note: Do not map the same volumes to SCSI and NVMe hosts concurrently, even one after the other. Also, do not add NVMe hosts and SCSI hosts to the same host cluster.

Tip: IBM Storage Virtualize uses discovery port 4420 for all RDMA protocol instead of 8009.

5.6.1 NVMe over RDMA host attachment considerations

IBM Storage Virtualize 8.7.3 allows a maximum of 512 NVMe hosts per system and 256 hosts per I/O group, if no other types of hosts are attached. IBM Storage Virtualize code does not monitor or enforce these limits, so administrators must ensure that the limits are not exceeded. RoCE v2 is a high performance network protocol based on UDP with RDMA.

RDMA can reduce the CPU load of the server, compared with other Ethernet protocols. One limitation of existing NVMe/RoCE is that it requires special Ethernet infrastructure (lossless Ethernet / DCB). Configuration is dependent on switch vendor and model. RoCE capable host adapters are also required. For more information about what is supported, see [IBM System Storage Interoperation Center \(SSIC\)](#). RoCE v2 is routable but is typically more efficient when it is used in layer 2 networks.

Restriction: Systems with ROCE Adapters cannot have MTU greater than 1500. when they are used for NVMe over RDMA host attachment.

5.6.2 NVMe over TCP host attachment considerations

For code level 8.6.0 and later, NVMe over TCP is supported.

NVMe over TCP needs more CPU resources than protocols that use RDMA. NVMe/TCP is a ubiquitous transport in which NVMe performance is not constrained by the data center infrastructure. Each NVMe/TCP port on FlashSystem supports multiple IPs and multiple VLANs. NVMe/TCP is supported on FlashSystem platforms that are installed with RoCE V2 25 Gb/s or 100 Gb/s adapters.

NVMe-TCP can generally work with any switch and is routable. For more information about operating system support and multipathing, see [IBM System Storage Interoperation Center \(SSIC\)](#).

For more information about the use of NVMe hosts with FlashSystem, see [V8.7.3.x Configuration Limits and Restrictions for IBM Storage FlashSystem 9500](#).

Restriction: Systems with RoCE adapters that are being used for NVMe over TCP host attachment cannot have MTU greater than 1500 on code levels earlier than 8.7.2.0. The restriction was lifted in code level 8.7.2.0.

5.7 Portsets

Portsets are groupings of logical addresses that are associated with specific traffic types. IBM Storage Virtualize version 8.7.3 systems support both IP and FC portsets for host attachment, back-end storage connectivity, and replication traffic.

The IBM Storage Virtualize IP object model introduced a feature that is named the *portset*. The *portset object* is a group of logical addresses that represents a typical IP function and traffic type. Portsets can be used for multiple traffic types, such as host attachment, back-end storage connectivity (iSCSI only), or IP replication.

A system can have a maximum of 72 portsets, which is a collective maximum limit for FC and Ethernet portsets. Code level 8.6.2.0 also added a feature to create high-speed IP replication partnerships by using RDMA with supported adapters. The maximum number of high-speed

portsets per system is 6. A portset can be of the host attach, remote copy, or storage type. The default portset is the host attach type. A portset of a specific type can be used for only that function. For example, a host attach type portset cannot be used for a remote copy partnership.

To see the FC and IP portsets, run the **lsportset** command, as shown in Example 5-4

Example 5-4 The lsportset command

```
svcinfolspportset
```

id	name	type	port_count	host_count	lossless	owner_id	owner_name	port_type	is_default
0	portset0	host	0	0				ethernet	yes
1	portset1	replication	0	0				ethernet	no
2	portset2	replication	0	0				ethernet	no
3	portset3	storage	0	0				ethernet	no
4	PortSet16	host	1	1	yes	0	Bank_Gr_Owngrp	fc	no
5	portset32	host	1	1	yes	1	Health_Gr_Owngrp	fc	no
64	portset64	host	6	4	yes			fc	yes

5.7.1 IP multitenancy

IP support for all IBM Storage Virtualize products previously allowed only a single IPv4 and IPv6 address per port for use with Ethernet connectivity protocols iSCSI and iSER.

For version 8.4.2 and later, IBM Storage Virtualize supports an increased per port limit to 64 IP addresses (IPv4, IPv6, or both). The increase of IP definitions also increased the virtual local area network (VLAN) limits, which can be done per IP address or as needed.

The object-based access control (OBAC) model, specifically designed for per-tenant administration and partitioning in multitenant cloud environments, is also implemented in the Ethernet configuration management system.

The following commands can be used to manage an IP or Ethernet configuration:

- ▶ **lsportset**
- ▶ **mkportset**
- ▶ **chportset**
- ▶ **rmportset**
- ▶ **lsip** (**lsportip** deprecated)
- ▶ **mkip** (**cfgportip** deprecated)
- ▶ **rmkip** (**rmportip** deprecated)
- ▶ **lsportethernet** (**lsportip** deprecated)
- ▶ **chportethernet** (**cfgportip** deprecated)
- ▶ **mkhost** (with the parameter **-portset** to bind the host to the portset)
- ▶ **chost** (with the parameter **-portset** to bind the host to the portset)

A host can access storage through the IP addresses that are included in the portset that is mapped to the host. The process to bind a host to a portset includes the following steps:

1. Create the portset.
2. Configure the IP addresses with the portset.
3. Create a host object.
4. Bind the host to the portset.
5. Discover and log in from the host.

IP portsets can be added by using the management GUI or the command-line interface (CLI). You can configure portsets by using the GUI and selecting **Settings** → **Network** → **Portsets**.

After the portsets are created, IP addresses can be assigned by using the management GUI or the CLI. You can configure portsets by using the GUI and selecting **Settings** → **Network** → **Ethernet Ports**.

Example 5-5 shows the output of the **lsip** command.

Example 5-5 Output of using the lsip command

```
svcinfo lsip
```

id	node_id	node_name	port_id	portset_id	portset_name	IP_address	prefix	vlan	gateway
owner_id	owner_name								
0	1	node1	1	0	portset0	10.0.240.110	24		10.0.240.9
1	1	node1	1	1	portset1	10.0.240.110	24		10.0.240.9
2	2	node2	1	0	portset0	10.0.240.111	24		10.0.240.9
3	2	node2	1	1	portset1	10.0.240.111	24		10.0.240.9

5.7.2 Fibre channel portset

FC portsets are supported in IBM Storage Virtualize version 8.5 and later for port management and host access. An FC portset is a group of FC I/O ports. A host must be associated to a portset and can access storage through the associated portset only. This configuration is recommended for larger configurations with host counts of 256 and greater. Conventionally, host access is managed by FC zoning, which might be complex and result in unbalanced resource usage. When used with good zoning practices, portsets provide better management and resource usage in configurations with host counts of 256 and greater.

Two modes of portsets are available:

1. Legacy mode. On a new installation or when upgrading to version 8.6.0.x, all FC ports are added to a default portset. All existing host objects are automatically associated with the default portset. This mode is useful for smaller configurations and does not require portset knowledge.
2. User-defined portsets. Recommended for larger configurations. Portsets can be defined by a user as part of the initial configuration. All host objects are limited by their associated portset for storage access.

When you use the FC portset feature, each FC I/O port can be added to multiple FC portsets. However, a host can be added to only one FC portset. Every portset can support up to four FC I/O ports.

Each portset is identified by a unique name. Portset 0 is an Ethernet default portset, and portset 64 is a default FC portset that is configured when the system is created or updated. Portsets 1 and 2 are replication portsets. Portset 3 is a storage portset.

To see the host login for each FC port, run the **lstargetportfc** command.

Example 5-6 The lstargetportfc command

svcinfo lstargetportfc											
id	WWPN	WWNN	port_id	owning_node_id	current_node_id	nportid	host_io_permitted	virtualized	protocol	fc_io_port_id	
portset_count	host_count	active_login_count									
1	5005076812110967	5005076812000967	1	1	1	060600	no	no	scsi	1	0
0	3										
2	5005076812150967	5005076812000967	1	1	1	060601	yes	yes	scsi	1	1
6	1										
3	5005076812190967	5005076812000967	1	1	1	060602	yes	yes	nvme	1	1
0	0										
4	5005076812120967	5005076812000967	2	1	1	070600	no	no	scsi	2	0
0	3										
5	5005076812160967	5005076812000967	2	1	1	070601	yes	yes	scsi	2	1
6	1										

6	50050768121A0967	5005076812000967	2	1	1	070602	yes	yes	nvme	2	1
0	0										
7	5005076812130967	5005076812000967	3	1	1	080600	no	no	scsi	3	0
0	1										
8	5005076812170967	5005076812000967	3	1	1	080601	yes	yes	scsi	3	1
6	0										
9	5005076812180967	5005076812000967	3	1	1	080602	yes	yes	nvme	3	1
0	0										
10	5005076812140967	5005076812000967	4	1	1	090600	no	no	scsi	4	0
0	1										
11	5005076812180967	5005076812000967	4	1	1	090601	yes	yes	scsi	4	1
6	0										
12	50050768121C0967	5005076812000967	4	1	1	090602	yes	yes	nvme	4	1
0	0										

The output shows details about each FC port, associated portset count, host count, and active login counts. If some FC ports have a higher number of active logins, they can cause an unbalanced performance.

Note: Event ID 088007 “*Fibre Channel I/O port has more than recommended active login*” is logged when there are more than 256 active logins on all or any of the worldwide port names (WWPNs) on the FC I/O port on any node. This event informs the customer that an FC I/O port is serving I/O on logins that are more than the recommended limit, and the system is being under-utilized because the load is not distributed uniformly across the FC I/O ports and nodes. The event is cleared when an administrator fixes the zoning such that the total active login count of the FC I/O port becomes less than or equal to 256.

Generally for FC, the host to storage connection is controlled by SAN zoning. A portset helps to set a rule on storage layer to avoid many FC logins to the same port while other ports remain idle. Misconfiguring portsets and hosts or wrong ports that are used in zones result in the Event ID 064002 being logged in the FlashSystem event log.

Note: Event ID 064002 “*Host and Fibre Channel port must be in same portset.*” is logged when the host tries to log in to a port that is associated with a different portset. A login is detected from the host to a storage port that is not part of the same portset of which the host is a part. For an FC host, check “Fibre Channel Connectivity” for the host with a state of Blocked. For an NVMe host, check “NVMe Connectivity” for a host with a state of Invalid. Add storage port WWPNs that are assigned to the correct portset on the host to the storage zone and remove the wrong ones. This action automatically clears the event.

For multi-tenant configuration portsets, separate the FC ports to the tenants. Example 5-4 on page 101 shows two portsets that are assigned to different ownership groups.

5.7.3 Portsets considerations and limitations

Consider the following points about portsets:

- ▶ Multiple hosts can be mapped to a single portset.
- ▶ A single host cannot be mapped to multiple portsets.
- ▶ A system can have a maximum of 72 portsets.
- ▶ IP addresses and FC ports can belong to multiple portsets.
- ▶ Port masking is used to enable or disable each port per feature for specific traffic types (host, storage, and replication).
- ▶ Portset 0, Portsets 1 and 2, Portset 3, Portset 64, and the replication portset are predefined.

- ▶ When an IP address or host is configured, a portset must be specified.
- ▶ Portset 0 is the default portset for iSCSI, and Portset 64 is the default portset for FC that is automatically configured when the system is updated or created and cannot be deleted.
- ▶ Administrators can use portset 0 and portset 64 to continue to use an original configuration that does not require multi-tenancy.
- ▶ Portset 3 is used for iSCSI back-end storage virtualization.
- ▶ After an update to the 8.5.x code or later, all configured FC host objects are automatically mapped to Portset 64, and iSCSI host objects are mapped to Portset 0.
- ▶ Unconfigured logins are rejected upon discovery.
- ▶ The internet Storage Name Service (iSNS) function registers IP addresses in Portset 0 only with the iSNS server.
- ▶ Each port can be configured with only one unique routable IP address (gateway-specified).

5.8 Host multipathing

Each host mapping associates a volume with a host object and allows all HBA ports in the host object to access the volume. You can map a volume to multiple host objects.

5.8.1 Path management

When a mapping is created, multiple paths normally exist across the SAN fabric from the hosts to the IBM Storage Virtualize system. Most operating systems present each path as a separate storage device. Therefore, multipathing software is required on the host. The multipathing software manages the paths that are available to the volume, presents a single storage device to the operating system, and provides failover if a path is lost.

If your IBM Storage Virtualize system uses NPIV, path failures that occur because of an offline node are masked from host multipathing.

Note: When you use NPIV and an SVC cluster with site awareness and without Hot Spare nodes, the remaining node in the same I/O group takes over the addresses. Be aware of the potential impact on inter-data center traffic.

5.8.2 Path selection

I/O for a specific volume is handled exclusively by the nodes in a single I/O group. Although both nodes in the I/O group can service the I/O for the volume, the system has a *preferred node*, which during normal operations it uses on a consistent basis. The primary reasons for using a preferred node is to optimize efficiency at various points in the I/O stack.

During volume creation, an I/O group and preferred node are defined, and optionally, the administrator can specify these settings. The volume's owner node defaults to the preferred node when both nodes are active.

With IBM HyperSwap®, configuration nodes in multiple I/O groups can potentially service I/O for the same volume, and site IDs are used to optimize the I/O routing.

IBM Storage Virtualize uses Asymmetric Logical Unit Access (ALUA), and most multipathing drivers support this access method. Therefore, the multipathing driver gives preference to paths to the preferred node. Most modern storage systems use ALUA.

Note: Some competitors claim that ALUA means that IBM Storage Virtualize is effectively an active-passive cluster. However, *both* nodes in IBM Storage Virtualize can and do service I/O concurrently.

In ALUA terminology, paths to FlashSystem target ports on the preferred node are known as *Active Optimized* (AO) paths, and paths to target ports on the non-preferred node are known as *Active Non-optimized* (ANO) paths. This is on a per volume basis.

If a host is configured to submit I/O down AO and ANO paths for FlashSystem volumes, those I/Os complete successfully. Section 5.13.2, “Scenario - Override ALUA and use all paths” on page 120, explains why this is potentially sub optimal.

5.9 I/O queues and queue depths

Host operating system and HBA software must have a way to fairly prioritize I/O to the storage. The host bus might run faster than the I/O bus or external storage. Therefore, you must have a way to queue I/O to the devices. Each operating system and host adapter use unique methods to control the I/O queue.

The I/O queue can be controlled by using one of the following unique methods:

- ▶ Host adapter-based.
- ▶ Memory and thread resources-based.
- ▶ Based on the number of commands that are outstanding for a device.

Queue depth is used to control the number of concurrent operations that occur on different storage resources. Queue depth is the number of I/O operations that can be run in parallel on a device.

Queue depths apply at various levels of the system:

- ▶ Disk or flash
- ▶ Storage controller
- ▶ Per volume and HBA on the host

For example, each IBM Storage Virtualize node has a queue depth of 10,000. A typical disk drive operates efficiently at a queue depth of 8. Most host volume queue depth defaults are approximately around 32.

There is no fixed rule available for setting a queue-depth value per host HBA or per volume. The requirements for your environment are driven by the intensity of each workload.

For more information, see [Queue depth in Fibre Channel hosts](#). The documentation suggests a good starting point of an HBA queue depth of 32 for most hosts or 128 for hosts that are significantly busier.

Ensure that one application or host cannot use the entire controller queue. However, if you have a specific host application that requires the lowest latency and highest throughput, consider giving it a proportionally larger share than others.

Consider the following points:

- ▶ A single IBM Storage Virtualize FC port accepts a maximum concurrent queue depth of 2048.
- ▶ A single IBM Storage Virtualize node accepts a maximum concurrent queue depth of 10,000. After this depth is reached, it reports a full status for the queue.

Tip: An enhancement in code level 8.5.1 and later introduces queue depth fairness rules. If queue depth usage is > 80% and if one host is using a disproportionately high share of queue depth (based on the number of hosts using the same target ports), then the system returns queue full to this host. For this type of extreme use case, it is better to dedicate a set of ports for the host with the high workload.

For more information about queue depths, see the following IBM Documentation:

- ▶ [Queue depth in Fibre Channel hosts](#)
- ▶ [Queue depth in iSCSI hosts](#)
- ▶ [Queue depth in iSER hosts](#)

Note: NVMe hosts don't require explicit queue depth configuration. Architectural advantages with NVMe ensure that queue depth is negotiated in-band automatically per queue pair.

5.10 Host clusters

IBM Storage Virtualize supports host clusters. This feature allows multiple hosts to access the same set of volumes.

Volumes that are mapped to that host cluster are assigned to all members of the host cluster with the same SCSI ID. A typical use case is to define a host cluster that contains all the WWPNs that belong to the hosts that are participating in a host operating system-based cluster, such as IBM PowerHA, Microsoft Cluster Server (MSCS), or VMware ESXi clusters.

5.10.1 Managing host clusters

The following commands can be used to manage host clusters:

- ▶ `lshostcluster`
- ▶ `lshostclustermember`
- ▶ `lshostclustervolumemap`
- ▶ `addhostclustermember`
- ▶ `chhostcluster`
- ▶ `mkhost` (with parameter `-hostcluster` to create the host in one cluster)
- ▶ `mkhostcluster`
- ▶ `mkvolumehostclustermap`
- ▶ `rmhostclustermember`
- ▶ `rmhostcluster`
- ▶ `rmvolumehostclustermap`

Host clusters can be added by using the GUI. By using the GUI, the system assigns the SCSI IDs for the volumes. You can also manually assign them. For ease of management purposes, use separate ranges of SCSI IDs for hosts and host clusters.

For example, you can use SCSI IDs 0–99 for non-cluster host volumes, and greater than 100 for the cluster host volumes. When you choose the **System Assign** option, the system automatically assigns the SCSI IDs starting from the first available in the sequence.

If you choose **Self Assign**, the system enables you to select the SCSI IDs manually for each volume. On the right side of the window, the SCSI IDs that are used by the selected host or host cluster are shown in Figure 5-1.

Map Volumes to HDMCENTEX_Cluster: Select SCSI IDs

Select SCSI ID these mappings will be placed on:

Name	SCSI ID	↓	Caching I/O Group ID
test_0	1	↑ ↓	0
test_1	2	↑ ↓	0
test_2	3	↑ ↓	0
test_3	4	↑ ↓	0
test_4	5	↑ ↓	0
test_5	6	↑ ↓	0

These SCSI IDs are already occupied:

Type of Mapping	SCSI ID	↓
Shared	0	
Private	13	
Private	14	

Buttons: Cancel, ◀ Back, Next ▶

Figure 5-1 SCSI ID assignment on volume mappings

Note: Although extra care is always recommended when dealing with hosts, with IBM Storage Virtualize, you cannot join a host into a host cluster if it includes a volume mapping with a SCSI ID that also exists in the host cluster:

```
IBM_2145:ITS0-SVCLab:superuser>addhostclustermember -host ITS0_HOST3
ITS0_CLUSTER1
CMMVC9068E Hosts in the host cluster have conflicting SCSI IDs for their
private mappings.
IBM_2145:ITS0-SVCLab:superuser>
```

5.10.2 Persistent reservations

To prevent hosts from sharing storage inadvertently, they must follow a storage reservation mechanism. The mechanisms for restricting access to IBM Storage Virtualize volumes use the SCSI-3 persistent reserve commands or the SCSI-2 reserve and release commands.

The host software uses several methods to implement host clusters. These methods require sharing the volumes on IBM Storage Virtualize between hosts. To share storage between hosts, the cluster must maintain control over accessing the volumes. Some clustering software uses software locking methods.

You can choose other methods of control by directing the clustering software or the device drivers to use the SCSI architecture reserve or release mechanisms. The multipathing

software can change the type of reserve that is used from an earlier reserve to persistent reserve, or remove the reserve.

Persistent reserve refers to a set of SCSI-3 standard commands and command options with which SCSI initiators can establish, preempt, query, and reset a reservation policy with a specified target device. The functions that are provided by the persistent reserve commands are a superset of the original reserve or release commands.

The persistent reserve commands are incompatible with the earlier reserve or release mechanism. Also, target devices can support only reservations from the earlier mechanism or the new mechanism. Attempting to mix persistent reserve commands with earlier reserve or release commands results in the target device returning a reservation conflict error.

Earlier reserve and release mechanisms (SCSI-2) reserved the entire LUN (volume) for exclusive use down a single path. This approach prevents access from any other host or even access from the same host that uses a different host adapter. The persistent reserve design establishes a method and interface through a reserve policy attribute for SCSI disks. This design specifies the type of reservation (if any) that the operating system device driver establishes before it accesses data on the disk.

Tip: Systems configured for policy-based high availability do not support persistent reservation commands if either system is on code level 8.7.2.

This restriction is removed when both policy-based high availability systems have been upgraded to code level 8.7.3.

5.11 Multipath configuration and optimization

For the complete host attachment instructions, refer to the following IBM Documentation links:

- ▶ [SCSI Fibre Channel host attachments](#)
- ▶ [iSCSI Ethernet host attachment](#)
- ▶ [NVMe over Fibre Channel host attachments](#)
- ▶ [NVMe over RDMA and NVMe over TCP host attachments](#)
- ▶ [iSER Ethernet host attachment \(for older FlashSystem products on code levels lower than 8.7.1\)](#)

5.11.1 AIX hosts

This section discusses tunable performance parameters for IBM AIX® hosts.

Multipathing support

Subsystem Device Driver Path Control Module (SDDPCM) is no longer supported. Use the default AIX PCM. For more information, see [The Recommended Multi-path Driver to use on IBM AIX and VIOS When Attached to SVC and Storwize storage](#).

AIX configuration recommendations

These device settings can be changed by using the `chdev` AIX command:

- ▶ `reserve_policy=no_reserve` (per hdisk)

The default reserve policy is **single_path** (SCSI-2 reserve). Unless a specific need exists for reservations, use **no_reserve**.

- ▶ **algorithm=shortest_queue** (per hdisk)

If migrating from SDD PCM, AIX defaults to **fail_over**. You cannot set the algorithm to **shortest_queue** unless the reservation policy is **no_reserve**.

- ▶ **queue_depth=32** (per hdisk)

The default queue depth is **20**. IBM recommends **32**.

- ▶ **rw_timeout=30** (per hdisk)

The default for SDD PCM is **60**. For AIX PCM, the default is **30**. IBM recommends **30**.

- ▶ **fc_err_recov=fast_fail** (per Fibre Channel protocol adapter - fscsi)

If the FC adapter driver detects a link event, such as a lost link between a storage device and a switch, the FC adapter driver waits a short period of approximately 15 seconds so that the fabric can stabilize. At that point, if the FC adapter driver detects that the device is not on the fabric, it begins failing all I/Os at the adapter driver. Any new I/O or future retries of the failed I/Os are failed immediately by the adapter until the adapter driver detects that the device rejoined the fabric.

Fast I/O Failure is useful in situations where multipathing software is used. Setting the **fc_err_recov** attribute to **fast_fail** can decrease the I/O fail times because of link loss between the storage device and switch. This would support faster failover to alternate paths.

- ▶ **dyntrk=yes** (per Fibre Channel protocol adapter - fscsi)

With dynamic tracking of Fibre Channel (FC) devices enabled, the FC adapter driver detects N_Port ID changes. Upon detection, the driver seamlessly reroutes traffic to the device's new address without interrupting online operations. N_Port ID changes can arise from various events, including port reassignment between a switch and storage device, inter-switch link (ISL) connections, and NPIV failover on FlashSystems.

- ▶ **max_xfer_size= x** (per Fibre Channel adapter - fcs)

Where *x* represents a system-dependent value within the permissible range. On systems experiencing a high load, hosts might encounter direct memory access (DMA) resource limitations. This can manifest as errors indicating HBA I/O request activation failures on the initial attempt or as unexpected performance degradation without logged errors. Consultation with AIX support is recommended for guidance on implementing this change.

5.11.2 Virtual I/O Server hosts

This section discusses tunable performance parameters for Virtual I/O Server (VIOS) hosts.

Multipathing support

Subsystem Device Driver Path Control Module (SDDPCM) is no longer supported. Use the default AIX PCM.

For more information, see [The Recommended Multi-path Driver to use on IBM AIX and VIOS When Attached to SVC and Storwize storage](#). For information about VIOS SAN Boot or dual VIOS configurations, see [SSIC](#).

VIOS configuration recommendations

These device settings can be changed by using the *chdev* AIX command:

- ▶ **reserve_policy=single_path** (per hdisk)

- ▶ The default reserve policy is **single_path** (SCSI-2 reserve).
- ▶ **algorithm=fail_over** (per hdisk)
If coming from SDD PCM, AIX defaults to **fail_over**.
- ▶ **queue_depth=32** (per hdisk)
The default queue depth is **20**. IBM recommends **32**.
- ▶ **rw_timeout=30** (per hdisk)
The default for SDD PCM is **60**. For AIX PCM, the default is **30**. IBM recommends **30**.
- ▶ **fc_err_recov=fast_fail** (per Fibre Channel protocol adapter - fscsi)
If the FC adapter driver detects a link event, such as a lost link between a storage device and a switch, the FC adapter driver waits a short period of approximately 15 seconds so that the fabric can stabilize. At that point, if the FC adapter driver detects that the device is not on the fabric, it begins failing all I/Os at the adapter driver. Any new I/O or future retries of the failed I/Os are failed immediately by the adapter until the adapter driver detects that the device rejoined the fabric.

Fast I/O Failure is useful in situations where multipathing software is used. Setting the **fc_err_recov** attribute to **fast_fail** can decrease the I/O fail times because of link loss between the storage device and switch. This would support faster failover to alternate paths.
- ▶ **dyntrk=yes** (per Fibre Channel protocol adapter - fscsi)
When dynamic tracking of FC devices is enabled, the FC adapter driver detects when the Fibre Channel N_Port ID of a device changes. The FC adapter driver then reroutes traffic destined for that device to the new address while the devices are still online. Events that can cause an N_Port ID to change include moving a cable between a switch and storage device from one switch port to another, connecting two separate switches using an inter-switch link (ISL), and NPIV failover at the FlashSystem level.
- ▶ **max_xfer_size= x** (per Fibre Channel adapter - fcs)
Where *x* is in the range of allowed values. The allowed range is system dependent.

On a heavily loaded system, if you see errors that indicate that the HBA was unable to activate an I/O request on the first attempt or if you see lower than expected performance with no errors being logged, a host could be low on direct memory access (DMA) resources. It is advisable to consult VIOS support for guidance on this change.

5.11.3 Microsoft Windows hosts

This section discusses tunable performance parameters for Microsoft Windows hosts, including Microsoft Hyper-V.

Multipathing support

For multipathing support, use Microsoft multipath I/O (MPIO) with Microsoft Device Specific Module (MSDSM), which is included in the Windows Server operating system. The older Subsystem Device Driver Device Specific Module (SDDDSM) is no longer supported. For more information, see [IBM Storage Virtualize Multipathing Support for AIX and Windows Hosts](#).

The Windows multipathing software supports the following maximum configuration:

- ▶ A maximum of eight paths to each volume
- ▶ A maximum of 2048 volumes per windows server or host
- ▶ A maximum of 512 volumes per Hyper-V host

Windows and Hyper-V configuration recommendations

Ensure that the following components are configured:

- ▶ Operating system service packs and patches and clustered-system software
- ▶ HBAs and HBA device drivers
- ▶ Multipathing drivers (MSDSM)
- ▶ Change the disk I/O timeout value to **60** in the Windows registry as per Example 5-7.

Example 5-7 Edit registry to change disk timeout to 30 seconds

Key:

HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\Disk

Parameter:

TimeOutValue=60

5.11.4 Linux hosts

This section discusses tunable performance parameters for Linux hosts.

Multipathing support

IBM Storage Virtualize supports Linux hosts that use native Device Mapper-Multipathing (DM-MP) and native multipathing support.

Note: Occasionally, storage administrators modify parameters in the `multipath.conf` file to address some perceived shortcoming in the DM-MP configuration. These modifications can create unintended and unexpected behaviors. The recommendations that are provided in IBM Documentation are optimal for most configurations.

Linux configuration recommendations

Consider the following points about configuration settings for Linux:

- ▶ Modify `multipath.conf` and add or adjust the configuration stanzas. For each Linux distribution and releases within a distribution, refer to the default settings under `/usr/share/doc/device-mapper-multipath.*` for Red Hat and `/usr/share/doc/packages/multipath-tools` for Novell SUSE. Ensure that the entries added to `multipath.conf` match the format and syntax for the required Linux distribution. Use the `multipath.conf` file from your related distribution and release only. Do not copy the `multipath.conf` file from one distribution or release to another.
 - For some operating system levels, the **polling_interval** must be located under defaults instead of under device settings. If **polling_interval** is present in the device section, disable the line by making it a comment **polling_interval**.

Example 5-8 shows how to configure the **polling_interval** in `multipath.conf`. This is the interval between two path checks in seconds. For properly functioning paths, the interval between checks gradually increases to **max_polling_interval**.

Example 5-8 Add polling_interval to the defaults stanza

Under Device Section

```
# polling_interval 5,
```

Under Defaults Section

```
defaults {  
    user_friendly_names yes
```

```
        polling_interval 5
    }
```

- Red Hat Linux versions 7.x, 8.x, and 9.x sample `multipath.conf`

Example 5-9 shows a sample `multipath.conf` file for supported Red Hat Linux distributions.

Example 5-9 Red Hat Linux multipath.conf

```
vendor "IBM"
    product "2145"
    path_grouping_policy "group_by_prio"
    path_selector "service-time 0" # Used by Red Hat 7.x
    prio "alua"
    path_checker "tur"
    failback "immediate"
    no_path_retry 5
    rr_weight uniform
    rr_min_io_rq "1"
    dev_loss_tmo 120
```

- SUSE Linux Versions 12 and 15

Example 5-10 shows a sample `multipath.conf` file for supported SUSE Linux distributions.

Example 5-10 SUSE Linux multipath.conf

```
vendor "IBM"
    product "2145"
    path_grouping_policy "group_by_prio"
    path_selector "service-time 0"
    prio "alua"
    path_checker "tur"
    failback "immediate"
    retain_attached_hw_handler "yes"
    no_path_retry 5 # or no_path_retry "fail"
    fast_io_fail_tmo 5
    rr_min_io 1000
    rr_min_io_rq 1
    rr_weight "uniform"
```

- Ubuntu

Example 5-11 shows a sample Ubuntu Linux `multipath.conf`.

Example 5-11 Ubuntu Linux multipath.conf

```
vendor "IBM"
    product "2145"
    path_grouping_policy "group_by_prio"
    path_selector "service-time 0"
    prio "alua"
    path_checker "tur"
    failback "immediate"
    no_path_retry 5 # or no_path_retry "fail"
    retain_attached_hw_handler "yes"
    fast_io_fail_tmo 5
```

```
rr_min_io 1000
rr_min_io_rq 1
rr_weight "uniform"
```

- ▶ Adjust SCSI inquiry timeout to **70** seconds. This is to ensure that the host restores the paths if they are lost during an upgrade or other hardware maintenance that results in a node being offline. This is mainly applicable to non-NPIV configurations.

- For RHEL6, RHEL7 and RHEL8 servers, follow these steps:

- To make the change permanent, edit `/etc/sysconfig/grub` and add to the `GRUB_CMDLINE_LINUX` line as per Example 5-12.

Example 5-12 Set SCSI inquiry timeout to 70 seconds.

```
scsi_mod.inq_timeout=70
```

- Rewrite the boot record as per Example 5-13.

Example 5-13 Rewrite boot record to make the change persistent

```
grub2-mkconfig -o /etc/grub2.cfg
```

- For SLES12 and SLES15 servers, follow these steps:

- To make the permanent change, edit `/etc/default/grub` and add to the `GRUB_CMDLINE_LINUX_DEFAULT` line as per Example 5-14.

Example 5-14 Set SCSI inquiry timeout to 70 seconds

```
scsi_mod.inq_timeout=70
```

- Rewrite the boot record as per Example 5-15

Example 5-15 Rewrite boot record to make the change persistent

```
grub2-mkconfig -o /boot/grub2/grub.cfg
```

- ▶ Set the SCSI command timeout to 120 seconds. This is to ensure that the host restores the paths if they are lost during an upgrade or other hardware maintenance that results in a node being offline. This is mainly applicable to non-NPIV configurations.

- Create a new udev rule as per Example 5-16 to set the SCSI command timeout to 120 seconds.

Example 5-16 Create a new udev rule

```
# Set SCSI command timeout to 120s (default == 30 or 60) for IBM 2145
devices
SUBSYSTEM=="block", ACTION=="add",
ENV{ID_VENDOR}=="IBM",ENV{ID_MODEL}=="2145", RUN+="/bin/sh -c 'echo 120
>/sys/block/%k/device/timeout'"
```

- Reload the udev rules as per Example 5-17 to avoid rebooting the hosts.

Example 5-17 Reload udev rules

```
udevadm control -R
/sbin/udevadm trigger --type=devices --action=add
```

- Verify that the SCSI command timeout has been set correctly as per Example 5-18.

Example 5-18 Verify that SCSI command timeout is now 120 seconds

```
cat /sys/block/sdX/device/timeout (where X is each 2145 block device path)
```

Note: Changes to `multipath.conf` on a running system might require the multipath service to be reloaded and scanned for configurations changes. Some device specific changes also require the device to be flushed.

► **service multipathd reload**

The reload process sends a HUP signal to the multipathd daemon, which causes it to re-read the configuration file. Device-mapper-multipath functions without the multipathd daemon running, however without multipathd, path failures are not detected or handled.

► **multipath -v2**

The **multipath -v2** command prints out multipathed paths that show which devices are multipathed.

5.11.5 IBM i hosts support

This section describes support and considerations for IBM i hosts.

Multipathing support and IBM i configuration recommendations

Multipathing support and IBM i configuration includes the following recommendations:

- Ensure that the system meets the minimum interoperability requirements.
- Zone the initiator ports to the target ports to achieve maximum redundancy without exceeding the maximum path count of 8.
- Section 2.3.9, “Recommendations based on field experience” on page 39 discusses a specific requirement on IBM i hosts that are attached with VFC adapters. For hosts with more than 8 VFC adapters it might be necessary to create pseudo host objects to keep the path count to 8.
- If pseudo host objects were created, map an equal number of volumes to each pseudo host to make up the full complement of volumes.
- Work with IBM i Support to determine whether enabling IBM i Journal cache benefits specific write workloads. For more information, see [Basics on Journal Cache](#).
- Work with IBM i Support to determine whether enabling IBM i Expert cache will benefit specific read workloads. For more information, see [Expert Cache - How It Works](#).
- IBM i hosts require a special procedure for attachment to FlashSystem with NPIV enabled. Code level 8.7.0 makes the use of NPIV mandatory. It is therefore necessary to upgrade to an intermediate level of 8.6.0.4 (or a more recent 8.6.0 PTF) first, to make these special configuration changes. These changes are required before the FlashSystem can be updated to code level 8.7.0.0 or higher. For the documented procedure, see [IBM i Systems attached to FlashSystem or SVC must be converted to use FlashSystem NPIV before upgrading to 8.7 or higher](#).

5.11.6 Oracle Solaris hosts support

This section describes support and considerations for Oracle hosts.

Multipathing support

IBM Storage Virtualize supports multipathing for Oracle Solaris hosts through Oracle Solaris MPxIO. Symantec Veritas Volume Manager Dynamic Multi-Pathing (DMP), and the Native Multipathing Plug-in (NMP) were supported for older FlashSystem code levels.

Solaris MPxIO configuration recommendations

IBM Storage Virtualize software supports load balancing of the MPxIO software, which can be done by using the following command:

```
svctask mkhost -name new_name_arg -hbawwpn wwpn_list
```

Note: Starting with IBM Storage Virtualize version 8.1, a host type of `tpgs` is no longer mandatory. The behavior is now the same as with a host type of `generic`.

To complete your configuration, complete the following steps:

1. Install the latest Solaris host patches.
2. Copy the `/kernel/drv/scsi_vhci.conf` file to the `/etc/driver/drv/scsi_vhci.conf` file.
3. Set the `load-balance="round-robin"` parameter.
4. Set the `auto-failback="enable"` parameter.
5. Comment out the `device-type-scsi-options-list = "IBM 2145", "symmetric-option"` parameter.
6. Comment out the `symmetric-option = 0x1000000` parameter.
7. Restart hosts or run `stmsboot -u` based on the host level.
8. Verify changes by running `luxadm display /dev/rdisk/cXtYdZs2`, where `cXtYdZs2` is your storage device.
9. Verify that the preferred node paths are primary and online, and that the non-preferred node paths are secondary and online.

5.11.7 VMware ESXi server hosts

This section describes considerations for VMware hosts.

To determine the various VMware ESXi levels that are supported, see the [SSIC](#).

We recommend using VMware Tools on virtual machines for enhanced functionality and integration. You can use the VMware plugin, whether the newer native plugin or through IBM Spectrum Connect, to simplify management tasks. For example, the plugin enables features such as aligning volume names on VMware with those on IBM FlashSystem, ensuring consistency, and easier administration.

There are some limitations by VMware on NVMe volumes:

- ▶ [Requirements and Limitations of VMware NVMe Storage VMware 7](#)
- ▶ [Requirements and Limitations of VMware NVMe Storage VMware 8](#)

Multipathing support

VMware features a built-in multipathing driver that supports IBM Storage Virtualize ALUA-preferred path algorithms.

For information about limits and a complete list of maximums, see [VMware Configuration Maximums](#).

VMware configuration recommendations

For more information about specific configuration best practices for VMware refer to the following IBM Documentation links:

- ▶ [Configuring the ESXi operating system](#)
- ▶ [VMware iSCSI host attachment](#)
- ▶ [VMware ESXi installation and configuration for NVMe over Fibre Channel hosts](#)
- ▶ [NVMe over RDMA and NVMe over TCP host attachments](#)

Section 2.2, “VMware workloads” on page 19 also provides useful guidance on this topic.

Consider and verify the following settings:

- ▶ The storage array type plug-in is ALUA (VMW_SATP_ALUA).
- ▶ Path selection policy is RoundRobin (VMW_PSP_RR).
- ▶ Set the XCopy transfer size to 4096.
- ▶ The recommended number of paths per volume is four. It is recommended not to exceed 8. In a HyperSwap configuration, you can use eight, up to 16 paths per volume is supported.
- ▶ Change the RoundRobin IOPS from 1000 to 1 so that I/Os are evenly distributed across as many ports on the system as possible. For more information about how to change this setting, see [Adjusting Round Robin IOPS limit on VMware knowledge base](#).
- ▶ For Ethernet attachment, such as NVMe over TCP, use a dedicated pair of Ethernet ports. Configure these ports with dedicated vSwitches and VMkernel ports to ensure optimal performance and isolation. This setup reduces network congestion and improves the reliability and efficiency of the data transfer.

Note: In code levels earlier than 8.7.0.0, VMware cannot identify the drive technology type and the flash based datastores might be marked as HDD datastores. Code level 8.7.0.0 introduced a change that identifies all volumes as flash to VMware hosts.

5.12 Network considerations

This section addresses network considerations.

5.12.1 Ethernet connectivity

These are some factors that require consideration when planning a host attachment that uses Ethernet as a medium:

- ▶ Routing considerations
- ▶ Configuring VLAN
- ▶ System IP addresses
- ▶ Priority flow control

- Modifying VLAN for priority flow control
- Planning for a direct-attached iSCSI configuration

For more information, see [Ethernet connectivity](#).

Table 5-1 lists the TCP/UDP port usage for host attachment, replication, and IP quorum. This information can be used to plan for firewall configuration.

The table is a subset of the complete version. For the complete version, see [Planning management connections](#). It also includes firewall requirements for management functions.

Table 5-1 Firewall requirements

Service	Traffic direction	Protocol	Port
iSCSI	Inbound	TCP	3260
iSCSI iSNS	Outbound	TCP	3260
IP-based RDMA replication/high availability	Inbound and Outbound	TCP	4791, 21451, 21452, and 21455
IP-based RDMA replication/high availability	Inbound and Outbound	UDP	4791, 21451, 21452, and 21455
IP Partnership management IP communication	Inbound and Outbound	TCP	3260
IP Partnership data path connections ^a	Inbound and Outbound	TCP	3265
REST API access and replication management	Inbound	TCP	7443
NVMe over RDMA	Inbound and Outbound	UDP over RoCEv2	4420
NVMe over TCP	Inbound and Outbound	TCP	4420
IP quorum application	Inbound and Outbound	TCP	1260

a. Ports must be open for IP addresses that are configured for replication and not the system IP address.

5.12.2 Fibre channel connectivity

Consider the following factors when planning a host attachment that uses Fibre Channel as a medium:

- Fabric Device Management Interface (FDMI)
- Fibre Channel port numbers and worldwide port names
- N-Port ID Virtualization
- Planning for connectivity
- Zoning examples
- Fibre Channel port masking
- Fibre Channel SAN configuration details
- System configuration by using SAN fabrics with long-distance fiber connections

- Determining and changing the Fibre Channel network speed

For more information, see [Fibre channel connectivity](#).

Tip: FDMI capability can be quite useful when performing interoperability checks for hosts and backend storage. The `lsfabricport` command provides useful information about host and backend storage Fibre Channel port properties.

5.13 Impact of path selection on performance and availability

The section discusses the impacts of path selection and path failure on performance and availability.

This section uses a Red Hat Linux host as an example.

5.13.1 Scenario - Recommended configuration

This section describes the recommended configuration for a Red Hat Linux host.

Multipath.conf

Example 5-19 shows the recommended multipath configuration for a Red Hat host.

Example 5-19 Red Hat multipath.conf devices stanza

```
devices {
    device {
        vendor "IBM"
        product "2145"
        path_grouping_policy "group_by_prio"
        path_selector "service-time 0" # Used by Red Hat 7.x
        prio "alua"
        path_checker "tur"
        failback "immediate"
        no_path_retry 5
        rr_weight uniform
        rr_min_io_rq "1"
        dev_loss_tmo 120
    }
}
```

FlashSystem volumes and paths

Example 5-20 shows the output of the command `multipath -ll`.

Example 5-20 Output of multipath -ll

```
# multipath -ll
mpathaes (36005076810db85df0000000000000000) dm-2 IBM,2145
size=10G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
|-+- policy='service-time 0' prio=50 status=active
|  |- 6:0:4:0 sdc 8:32 active ready running
|  |- 6:0:5:0 sdg 8:96 active ready running
|  |- 7:0:3:0 sdi 8:128 active ready running
```

```

| ~- 7:0:13:0 sdo 8:224 active ready running
~+- policy='service-time 0' prio=10 status=enabled
| - 6:0:0:0 sda 8:0 active ready running
| - 6:0:3:0 sde 8:64 active ready running
| - 7:0:9:0 sdm 8:192 active ready running
| ~- 7:0:4:0 sdk 8:160 active ready running

```

The output of the command provides the following information:

- ▶ There are 8 paths to a 10 GiB volume.
- ▶ The device prefers alua.
- ▶ The paths are separated into 2 priority groups.
prio=50 describes the active optimized paths and prio=10 describes the active non-optimized paths.
- ▶ The path selector algorithm is service-time 0.

Path selection

Figure 5-2 shows that I/O to multipathed device dm-2 is being sent down active optimized paths only and peak host CPU utilization is < 25%.

This is the optimal configuration because reads and writes arrive on the same node, and the configuration provides the best potential for read cache hits. This, in turn, reduces the workload to the backend. Fewer active paths also reduces CPU usage on the host.

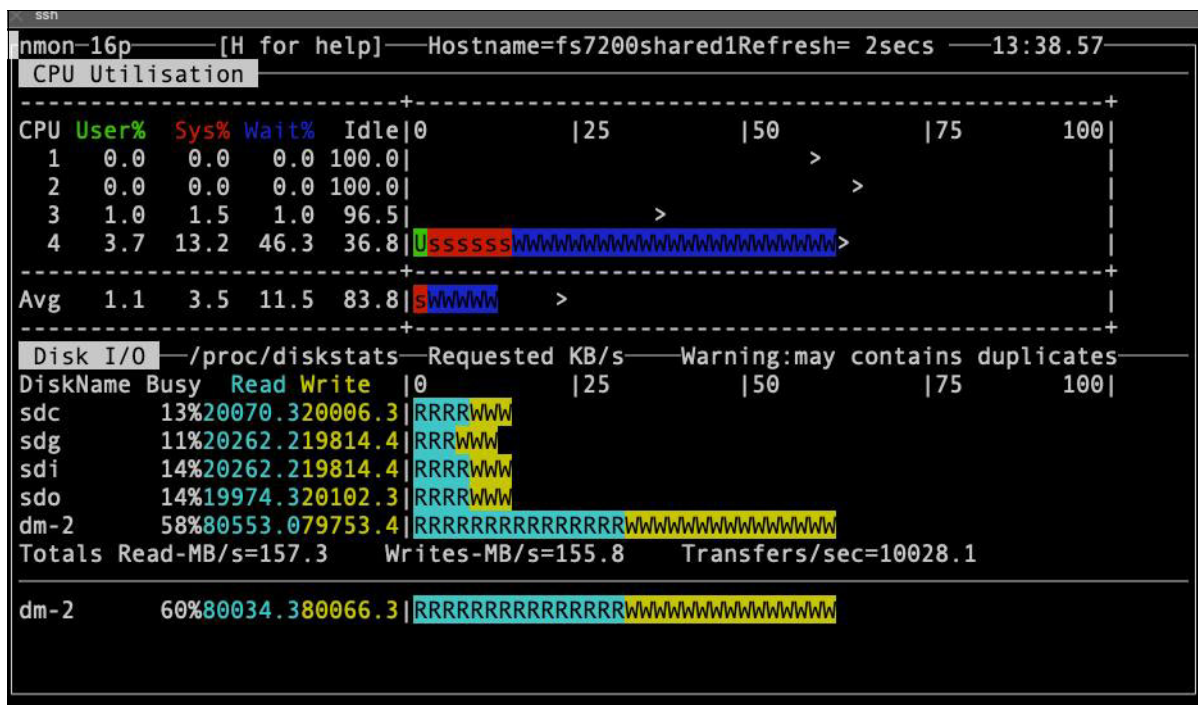


Figure 5-2 I/O is only being sent down active optimized paths for device dm-2

5.13.2 Scenario - Override ALUA and use all paths

This section describes a scenario that involves overriding ALUA and using all paths.

Multipath.conf

Example 5-21 shows that `path_grouping_policy` has been changed to `multibus`. This groups all paths in the same priority group.

Example 5-21 Red Hat multipath.conf devices stanza - Override ALUA

```
devices{
  device{
    vendor "IBM"
    product "2145"
    path_grouping_policy "multibus" <<<
    path_selector "service-time 0" # Used by Red Hat 7.x
    prio "alua"
    path_checker "tur"
    failback "immediate"
    no_path_retry 5
    rr_weight uniform
    rr_min_io "1000"
    rr_min_io_rq "1"
    dev_loss_tmo 120
  }
}
```

FlashSystem volumes and paths

Example 5-22 shows the output of `multipath -ll`, all paths are now in the same priority group.

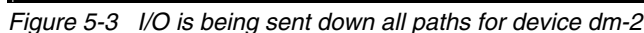
Example 5-22 All paths are in the same priority group

```
# multipath -ll
mpathaes (36005076810db85df00000000000000000) dm-2 IBM,2145
size=10G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
`-+- policy='service-time 0' prio=30 status=active
   |- 6:0:0:0 sda 8:0 active ready running
   |- 7:0:9:0 sdm 8:192 active ready running
   |- 6:0:3:0 sde 8:64 active ready running
   |- 7:0:4:0 sdk 8:160 active ready running
   |- 6:0:4:0 sdc 8:32 active ready running
   |- 7:0:3:0 sdi 8:128 active ready running
   |- 6:0:5:0 sdg 8:96 active ready running
   `-- 7:0:13:0 sdo 8:224 active ready running
```

The output provides the following information:

- ▶ There are 8 paths to a 10 GiB volume.
- ▶ The device prefers `alua`.
- ▶ All 8 paths are in the same priority group.
- ▶ The path selector algorithm is `service-time 0`

Figure 5-3 shows that I/O to the multipathed device dm-2 is sent down all 8 paths and peak host CPU usage is > 25%.



This might seem more favorable compared to Figure 5-2 on page 119 because more paths are being used. However I/O is being sent to both nodes, so it potentially reduces the opportunity for read cache hits by 50%. Read cache misses drive backend workload which is less efficient. Additionally, host CPU usage is slightly higher because the host has to manage a larger set of active paths. Also, the CPU usage might increase when more volumes are mapped.

The scenario in this section describes a single target port that is offline.

The configuration in Example 5-23 is the same configuration as Example 5-19 on page 118.

```
devices {
    device {
        vendor "IBM"
        product "2145"
        path_grouping_policy "group_by_prio"
        path_selector "service-time 0" # Used by Red Hat 7.x
        prio "alua"
        path_checker "tur"
        failback "immediate"
        no path retry 5
    }
}
```


After the offline ports are recovered, the path states are updated, and I/O resumes on all active optimized paths without any disruption.

5.13.4 Scenario of a FlashSystem node offline

This section describes a scenario involving a FlashSystem node going offline.

Multipath.conf

The configuration in Example 5-25 is the same configuration as Example 5-19 on page 118.

Example 5-25 Red Hat multipath.conf devices stanza

```
devices {
    device {
        vendor "IBM"
        product "2145"
        path_grouping_policy "group_by_prio"
        path_selector "service-time 0" # Used by Red Hat 7.x
        prio "alua"
        path_checker "tur"
        failback "immediate"
        no_path_retry 5
        rr_weight uniform
        rr_min_io_rq "1"
        dev_loss_tmo 120
    }
}
```

FlashSystem volumes and paths

A node was placed in service state as shown in Example 5-26.

Example 5-26 Node canister 01-2 in service state

```
sainfo lsservicenodes
panel_name cluster_id      cluster_name      node_id node_name relation
node_status error_data
01-1      0000020436E177C0 fs7200shared1-cl 1      node1      local      Active
01-2      0000020436E177C0 fs7200shared1-cl 2      node2      partner   Service
690
```

Example 5-27 shows that all paths remained online.

Example 5-27 path status does not show the offline node

```
# multipath -ll
mpathaes (36005076810db85df00000000000000000) dm-2 IBM,2145
size=10G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
| | 6:0:0:0 sda 8:0 active ready running
| | 6:0:3:0 sde 8:64 active ready running
| | 7:0:9:0 sdm 8:192 active ready running
| | 7:0:4:0 sdk 8:160 active ready running
|+- policy='service-time 0' prio=10 status=enabled
| | 6:0:4:0 sdc 8:32 active ready running
| | 7:0:3:0 sdi 8:128 active ready running
```

```

|- 6:0:5:0 sdg 8:96 active ready running
^- 7:0:13:0 sdh 8:112 active ready running

```

Path selection

Figure 5-5 shows that I/O continued on the active optimized paths.

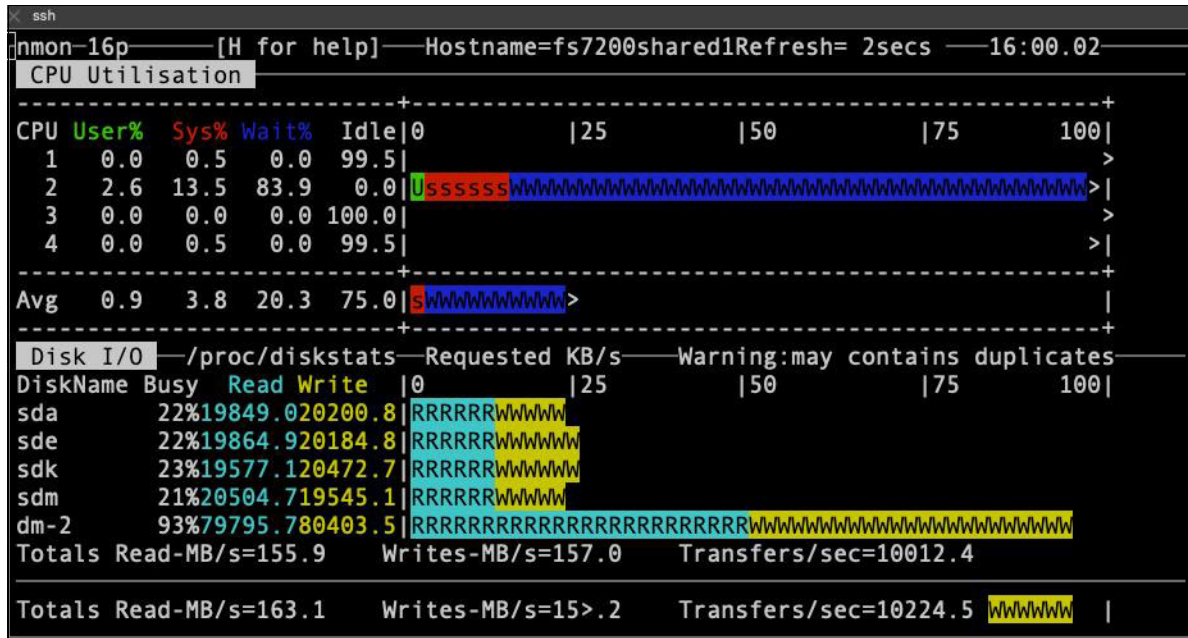


Figure 5-5 I/O continues on active optimized paths

However, as shown by the IBM Spectrum Control statistics in Figure 5-6, all I/O moved to the partner FlashSystem node at that time.

This is working as designed, the NPIV target ports failed over to the partner node. The host is unaffected by the NPIV failover and still sends I/O down the active, optimized paths for the multipath device dm-2.

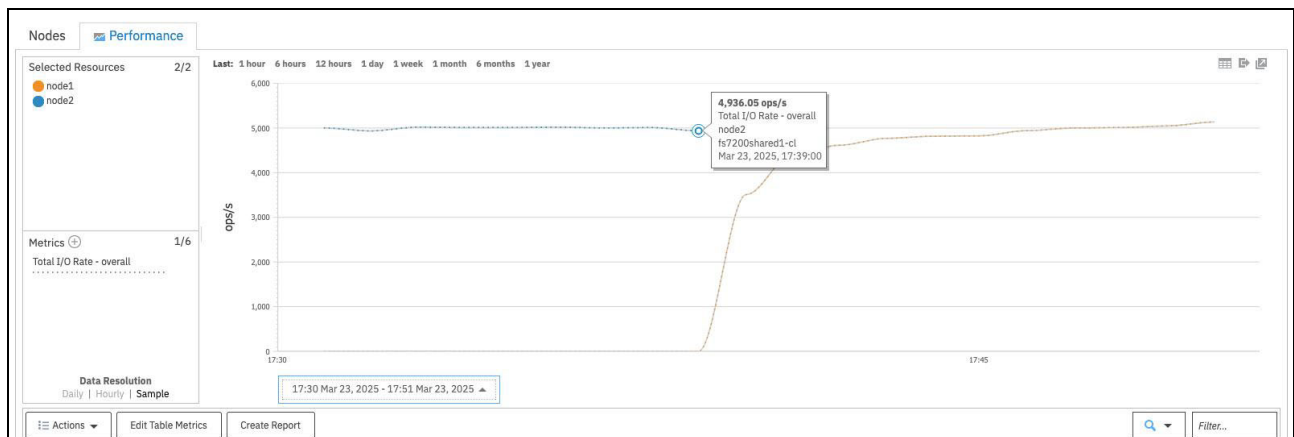


Figure 5-6 NPIV failover

Figure 5-7 shows confirmation of NPIV failover, with the owning node different from current node for some ports. To view the panel, select **Settings** → **Network** → **Fibre Channel Ports**.

Fibre Channel Ports									
Each port can connect to hosts and storage systems and can be used for remote copy traffic. You can use portsets to provide a grouping of logical addresses used for specific types of traffic.									
Actions	Display WWPN without notation		Default		Contains		Filter		
Port	Owning Node	WWPN	Host Count	Portset Count	Login Count	Host IO Permitted	Virtualized	Current Node	Protocol
1(Upper)		500507681013BBE0	0	0	13	No	☒ No	1	SCSI
2(Lower)		5005076810137FE2	0	0	0	No	☒ No		SCSI
1(Upper)		500507681017BBE0	1	1	2	Yes	☒ Yes	1	SCSI
2(Lower)		5005076810177FE2	1	1	2	Yes	☒ Yes	1	SCSI
1(Upper)		500507681018BBE0	0	1	0	Yes	☒ Yes	1	NVMe
2(Lower)		5005076810187FE2	0	1	0	Yes	☒ Yes	1	NVMe

Figure 5-7 FlashSystem GUI view of `lstartgetportfc`

5.13.5 Scenario to mitigate large transfer size workloads with multipath configuration

Workloads with a large transfer size can often have increased latency. In most cases, this does not really impact applications because throughput is usually a more appropriate measure of large block sequential workloads. However, there are specific circumstances where large transfer size and low latency is a requirement. It is not always possible to tune the application to adjust transfer size to mitigate for increased latency.

Linux hosts can take advantage of multipath configuration parameters to limit the transfer size for specific volumes.

Multipath.conf

Example 5-28 shows a `multipath.conf` file with the optional `multipaths` stanza that contains a `wwid` parameter and a `max_sectors_kb` parameter.

`wwid` is valid in the `multipaths` stanza to identify a specific multipathed device. The `max_sectors_kb` parameter is valid in the `defaults`, `devices` and `multipaths` stanzas. `max_sectors_kb` limits the transfer size of I/Os to the value specified in KB. Its scope depends on the stanza in which stanza it is specified. Specifying it in `devices` makes it applicable to all FlashSystem multipathed devices. Specifying it in `multipaths` makes it applicable to a specific FlashSystem multipath device.

The configuration parameters in Example 5-28 are intended to limit the transfer size of I/O to multipathed device `dm-2` to 64 KB.

Example 5-28 Red Hat `multipath.conf` `devices` and `multipaths` stanza

```
devices{
    device{
        vendor "IBM"
        product "2145"
        path_grouping_policy "group_by_prio"
        path_selector "service-time 0" # Used by Red Hat 7.x
        prio "alua"
```

```

        path_checker "tur"
        failback "immediate"
        no_path_retry 5
        rr_weight uniform
        rr_min_io "1000"
        rr_min_io_rq "1"
        dev_loss_tmo 120
    }
}

multipaths {
    multipath {
        wwid "36005076810db85df0000000000000000" <<
        max_sectors_kb "64" <<
    }
}

```

FlashSystem volumes and multipaths

Example 5-29 shows that there are 2 FlashSystem volumes mapped to this Linux host.

Example 5-29 multipathed devices dm-2 and dm-3 are mapped to this host

```

# multipath -ll
mpathaes (36005076810db85df0000000000000000) dm-2 IBM,2145
size=10G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
|  |- 6:0:4:0 sdc 8:32 active ready running
|  |- 6:0:5:0 sdg 8:96 active ready running
|  |- 7:0:3:0 sdi 8:128 active ready running
|  ~- 7:0:13:0 sdh 8:112 active ready running
|+- policy='service-time 0' prio=10 status=enabled
|  |- 6:0:0:0 sda 8:0 active ready running
|  |- 6:0:3:0 sde 8:64 active ready running
|  |- 7:0:9:0 sdm 8:192 active ready running
|  ~- 7:0:4:0 sdk 8:160 active ready running
mpathaer (36005076810db85df0000000000000001) dm-3 IBM,2145
size=10G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
|  |- 6:0:0:1 sdb 8:16 active ready running
|  |- 6:0:3:1 sdf 8:80 active ready running
|  |- 7:0:9:1 sdn 8:208 active ready running
|  ~- 7:0:4:1 sdl 8:176 active ready running
|+- policy='service-time 0' prio=10 status=enabled
|  |- 6:0:4:1 sdd 8:48 active ready running
|  |- 6:0:5:1 sdp 8:240 active ready running
|  |- 7:0:3:1 sdj 8:144 active ready running
|  ~- 7:0:13:1 sdo 8:224 active ready running

```

Path selection

Figure 5-8 shows that I/O is being sent down the active optimized paths for multipathed devices dm-2 and dm-3.

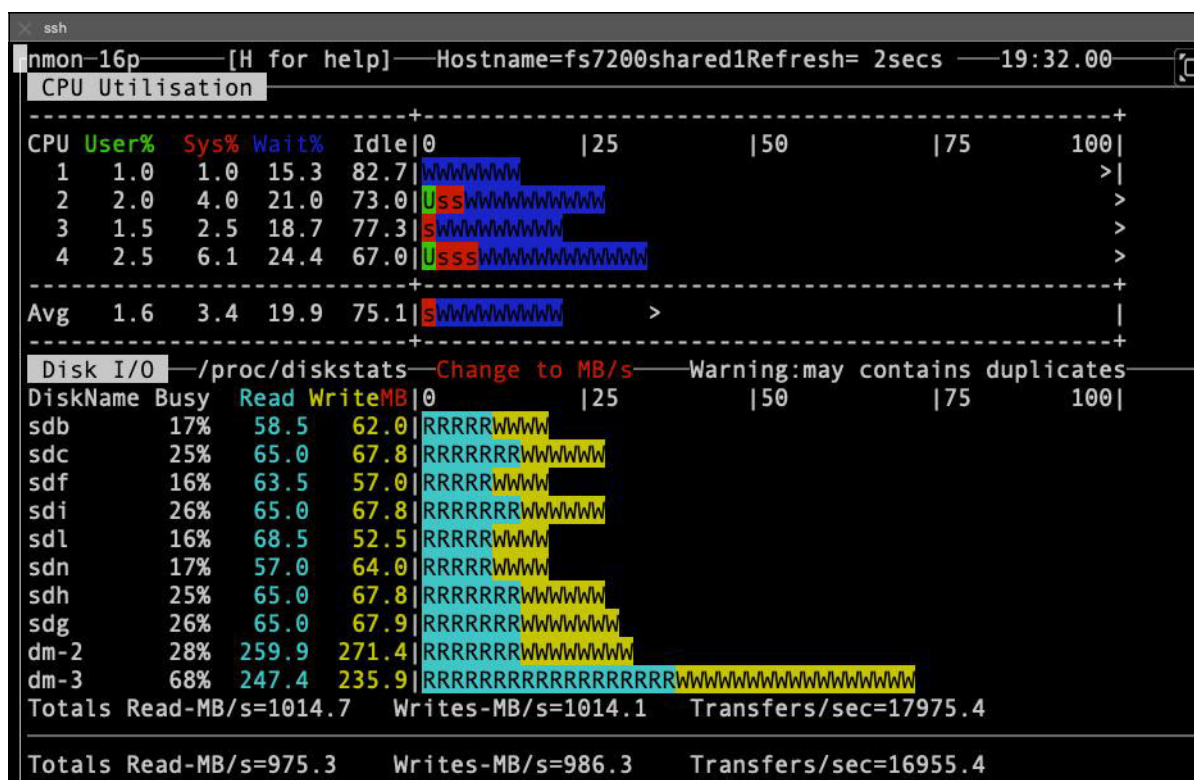


Figure 5-8 I/O for dm-2 and dm-3

Example 5-30 describes the Vdbench configuration file that is driving workload to devices dm-2 and dm-3.

Example 5-30 Vdbench configuration file

```
sd=sd1,lun=/dev/dm-2,openflags=o_direct,threads=1
sd=sd2,lun=/dev/dm-3,openflags=o_direct,threads=1
wd=wd1,sd=sd*,xfersize=1024k,rdpct=50
rd=run1,wd=wd*,iorate=1000,elapsed=3600,interval=1
```

Both devices are subjected to the same workload definition from Vdbench:

- ▶ read/write percentage = 50/50
- ▶ transfer size = 1024 KB = 1 MiB
- ▶ combined I/O rate of 1000 IOPS

Figure 5-9 on page 128 shows Spectrum Control statistics for volume V2, which corresponds to device dm-3 on the host. It shows a workload of 500 IOPS at a transfer size of 1024 KB.

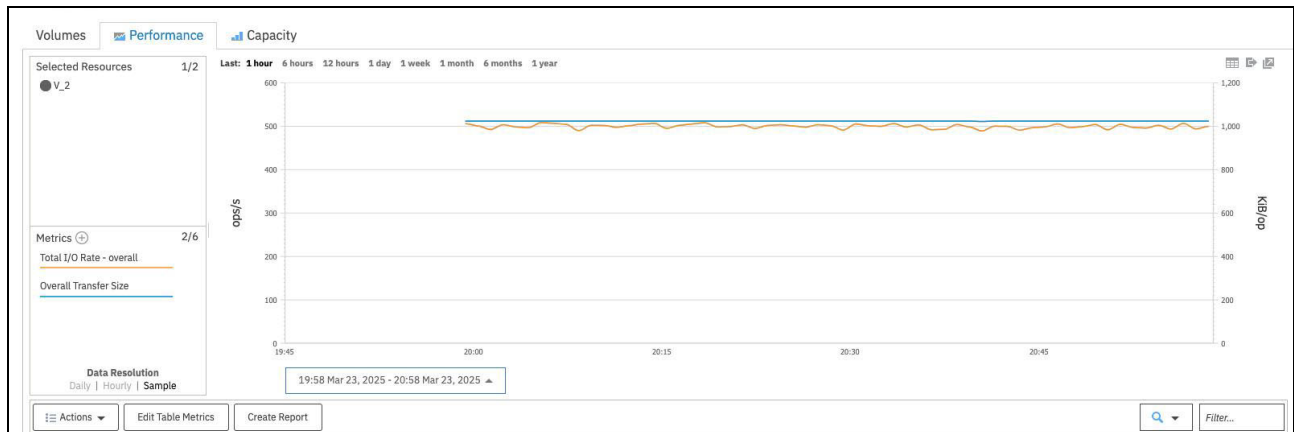


Figure 5-9 Volume V2 (dm-3) statistics matches the Vdbench configuration

Figure 5-10 shows Spectrum Control statistics for volume V1, which corresponds to device dm-2 on the host. It shows a workload of 8000 IOPS at a transfer size of 64 KB.

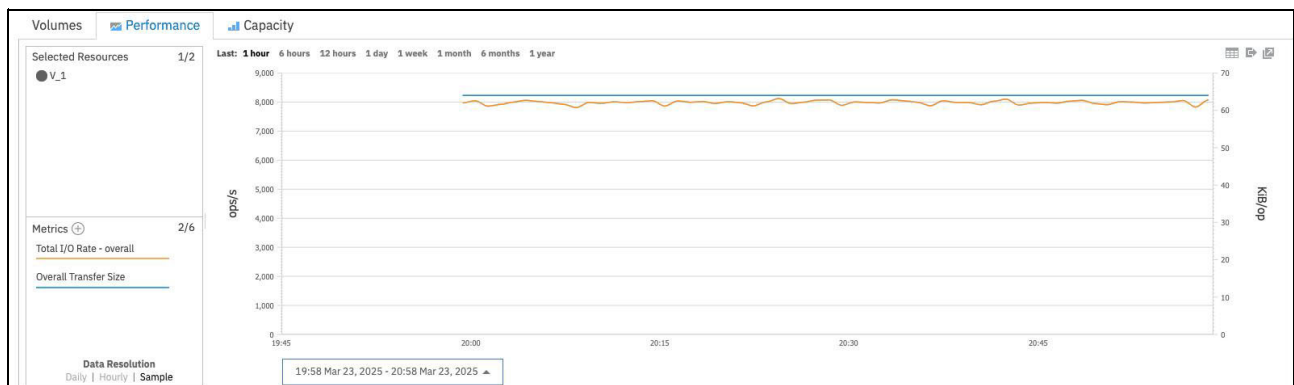


Figure 5-10 Volume V1 (dm-2) statistics shows the transfer size limited to 64KB and commensurate increase in I/O rate

Figure 5-11 shows the difference in response time between volumes V1 (dm-2) and V2 (dm-3). This confirms that configuring max_sectors_kb is an effective method of limiting transfer size on a per volume basis to reduce response time.

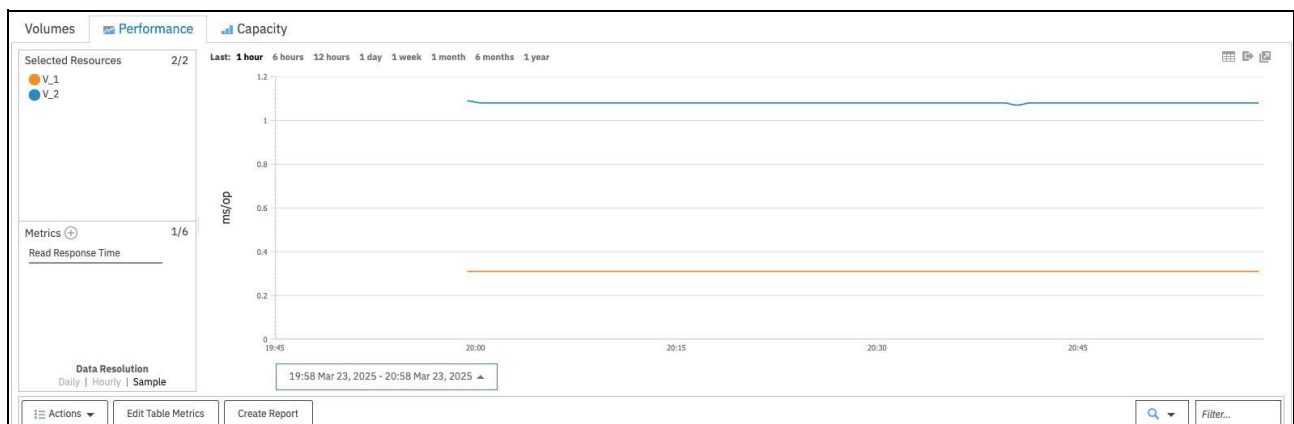


Figure 5-11 Volume V1 has lower response time

5.13.6 Scenario of fabric and hosts that support Fabric Performance Impact Notification

This section describes the fabric and hosts that support the Fabric Performance Impact Notification (FPIN) scenario.

What is FPIN?

Brocade Fabric Performance Impact Notification (FPIN) is a feature introduced in Brocade FOS v9.0. It is available on Brocade Gen6 and Gen7 switches. With this feature the switch can detect issues on a fabric, such as congestion or physical link issues, and then notify the affected devices that have registered for these notifications. FPIN functions in a similar mechanism to RSCN. With RSCN, the fabric can send notifications to devices when a device they are zoned to is going offline. The devices that receive these notifications can then proactively take steps such as path failover rather than have to react to a path being down. FPIN provides a means to notify devices of link issues and other issues with a connection or possibly a path through the fabric.

For both RSCN and FPIN, a device must register with fabric services to receive these notifications. The new Brocade Gen7 hardware can send hardware or software signal notifications. Gen6 can only send software notification. Both the hardware and software notifications require FOS v9.0 on the switches.

Hardware signals can be sent from the switch to the adapter in the device. The adapter itself can then decide what to do about the notification. Software signals are sent higher up in the Fibre-Channel stack, and the adapter driver decides how to handle the notification.

One advantage to notifications in hardware is reaction time. The adapter can process the notifications and react more quickly than the driver can. Another is that the hardware-based notification is a Fibre-Channel primitive. This means that even if buffer credits are depleted the signal can still be delivered to the device on the other end of the link. Primitives are not frames, so they do not need buffer credits to be sent. The software layer signal is an ELS frame, so it can be affected by buffer credit depletion and other link congestion. Whether the signal sent is hardware or software, how the devices handle the notifications is up to the vendor of the adapter. Some log the notification; some take action. The action that an adapter takes is also vendor specific.

FPIN can alert devices about these events:

- ▶ End Device Congestion
- ▶ Device Link Integrity (CRC)
- ▶ Frame Drops

At the time of writing, the following vendors support FPIN:

- ▶ Linux multipath in RHEL 8.2.
- ▶ Emulex supports Congestion and Link Integrity notifications on Linux.
- ▶ Marvell registers for FPIN and logs the notifications. These could be used as a source of log data for troubleshooting.
- ▶ AIX 7.2.5.x and 7.3.1.x registers for Link Integrity and Congestion notifications
AIX 7.2 provides foundational FPIN support by treating affected paths as degraded.
AIX 7.3 enhances this functionality with features like I/O throttling and host-configurable FPIN settings for better performance optimization and control.
- ▶ VMware ESXi 8.0 update 3 introduces support for Fabric Performance Impact Notifications Link Integrity (FPIN-LI).

Relationship of FPIN to host multipathing

The host must take action based on the fabric notification.

For example, when you use Linux, if a fabric notification is received for a congested target port, then paths to that target port are moved to the marginal state and marginal path group. Additional configuration parameters in `multipath.conf` describe how multipathing handles marginal paths. One of the options might be to stop using the path for a defined period. This manifests as an imbalance in host to storage traffic over different paths.

For more information about how different versions of AIX and VIOS respond to fabric notifications, see [FPIN Support Requirements and Key Considerations for AIX and PowerVM Virtual I/O Server](#).

Conclusion

FPIN is a valuable feature, but it is important for storage administrators and server administrators to know that it exists and whether it is enabled. Some symptoms that manifest on hosts and storage might be a direct result of a fabric notification.

5.13.7 Scenario of policy-based high availability single system failure

This section describes a FlashSystem failure within a policy-based high availability configuration.

Multipath.conf

The configuration in Example 5-31 is the same configuration as in Example 5-19 on page 118.

Example 5-31 Red Hat multipath.conf devices stanza

```
devices {
    device {
        vendor "IBM"
        product "2145"
        path_grouping_policy "group_by_prio"
        path_selector "service-time 0" # Used by Red Hat 7.x
        prio "alua"
        path_checker "tur"
        failback "immediate"
        no_path_retry 5
        rr_weight uniform
        rr_min_io_rq "1"
        dev_loss_tmo 120
    }
}
```

FlashSystem volumes and paths before failure

Figure 5-12 on page 131 shows that the high availability status of the storage partition is optimal.

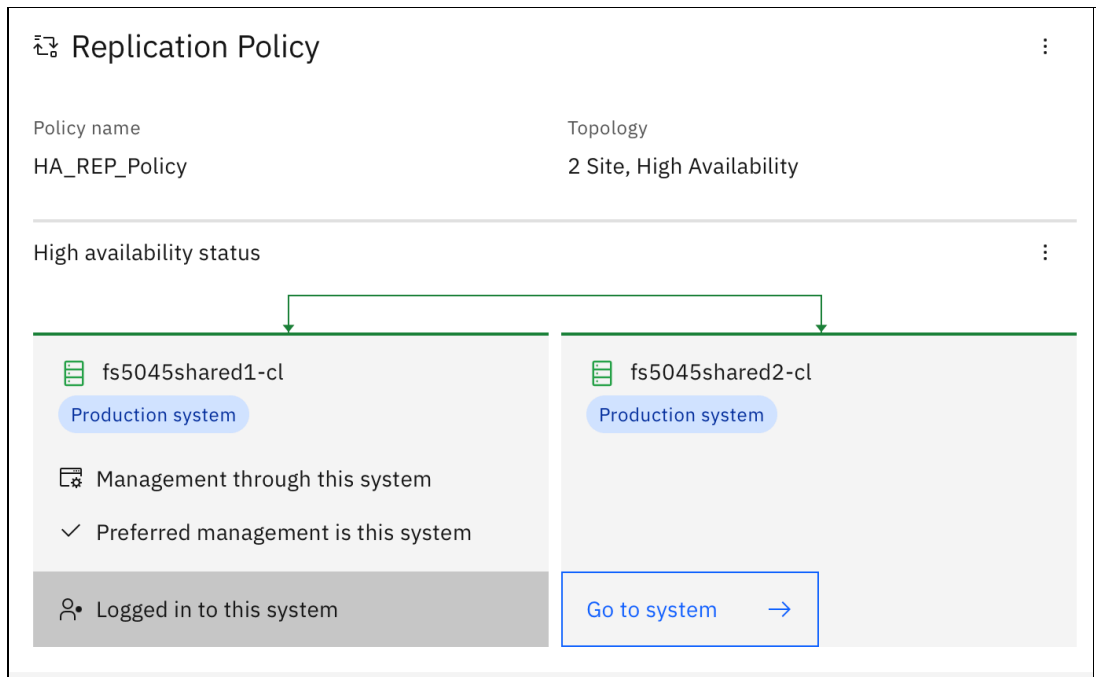


Figure 5-12 Status of storage partition is optimal

Example 5-32 shows 4 AO paths and 12 ANO paths for a single volume that is mapped to a host. Four AO paths are to the preferred node of the system in the same site. Four ANO paths are to the non-preferred node of the system in the same site. Eight ANO paths are to both nodes in the system at the alternate site.

Example 5-32 Multipathed device dm-2 is mapped to this host

```
# multipath -ll
mpathe (36005076380c886cf7800000000000000) dm-2 IBM,2145
size=50G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
| | - 6:0:7:0 sdd 8:48 active ready running <-- fs5045shared1-cl
| | - 6:0:6:0 sdb 8:16 active ready running <-- fs5045shared1-cl
| | - 7:0:5:0 sdm 8:192 active ready running <-- fs5045shared1-cl
| | - 7:0:7:0 sdn 8:208 active ready running <-- fs5045shared1-cl
|+- policy='service-time 0' prio=10 status=enabled
| | - 6:0:10:0 sde 8:64 active ready running <-- fs5045shared2-cl
| | - 6:0:11:0 sdg 8:96 active ready running <-- fs5045shared2-cl
| | - 6:0:14:0 sdf 8:80 active ready running <-- fs5045shared2-cl
| | - 6:0:15:0 sdh 8:112 active ready running <-- fs5045shared2-cl
| | - 6:0:5:0 sdc 8:32 active ready running <-- fs5045shared1-cl
| | - 6:0:1:0 sda 8:0 active ready running <-- fs5045shared1-cl
| | - 7:0:1:0 sdi 8:128 active ready running <-- fs5045shared1-cl
| | - 7:0:8:0 sdo 8:224 active ready running <-- fs5045shared2-cl
| | - 7:0:9:0 sdl 8:176 active ready running <-- fs5045shared2-cl
| | - 7:0:11:0 sdk 8:160 active ready running <-- fs5045shared2-cl
| | - 7:0:13:0 sdp 8:240 active ready running <-- fs5045shared2-cl
| | - 7:0:4:0 sdj 8:144 active ready running <-- fs5045shared1-cl
```

Note: In code level 8.6.1.0, paths to the system in the alternate site are listed in *standby* state. This is changed in code level 8.7.0.0.

Figure 5-13 shows I/O for multipathed device dm-2 being sent down the AO paths to fs5045shared1-cl.

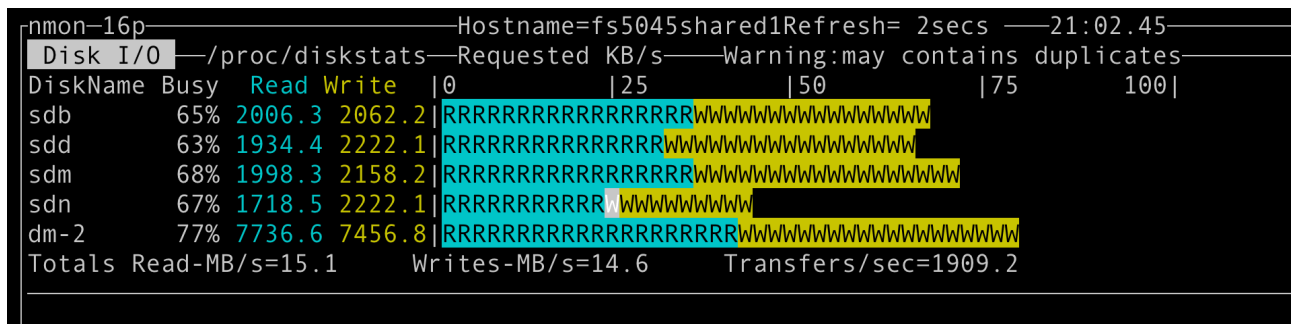


Figure 5-13 I/O being sent down AO paths for dm-2

FlashSystem volumes and paths after failure

Figure 5-14 shows that the high availability status of the storage partition has changed to Suspended after simulating a failure on `fs5045shared1-c1`.

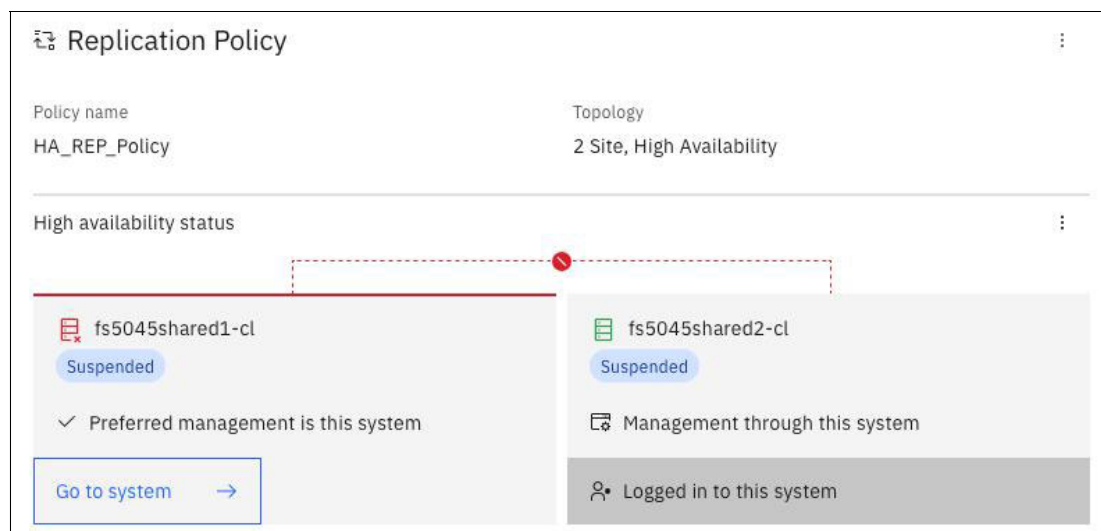


Figure 5-14 High availability status is suspended

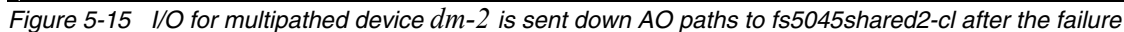
Example 5-33 shows that all paths to `fs5045shared1-cl` are failed. Eight paths to `fs5045shared2-cl` are active, and 4 of them are active optimized.

Example 5-33 Paths to fs5045shared2-cl are in a failed state

```
# multipath -ll
mpathe (36005076380c886cf78000000000000000) dm-2 IBM,2145
size=50G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
| | - 6:0:7:0 sdd 8:48 failed faulty running <-- fs5045shared1-cl
| | - 6:0:6:0 sdb 8:16 failed faulty running <-- fs5045shared1-cl
| | - 7:0:5:0 sdm 8:192 failed faulty running <-- fs5045shared1-cl
| | - 7:0:7:0 sdn 8:208 failed faulty running <-- fs5045shared1-cl
| | - 6:0:10:0 sde 8:64 active ready running <--fs5045shared2-cl
| | - 6:0:11:0 sdg 8:96 active ready running <--fs5045shared2-cl
| | - 7:0:8:0 sdo 8:224 active ready running <--fs5045shared2-cl
```

Path selection after failure

Note: The usage is higher after the failure because the fs5045shared2-cl has nearline drives compared to 10K rpm enterprise drives on fs5045shared1-cl.





Performance monitoring and analysis

Performance monitoring and analysis is a critical aspect and is vital for any IT infrastructure. It defines the baseline for a comprehensive approach to ensure optimal functioning and efficiency.

This chapter outlines FlashSystem performance monitoring and analysis techniques. The chapter describes performance metrics, statistic files, and real-time monitoring capabilities by using IBM Spectrum Control, IBM Storage Insights, and the built-in FlashSystem tools.

Additionally, the chapter provides an overview of analytics tools for understanding performance trends, predictive analysis, and optimization recommendations.

To address performance bottlenecks, the chapter also includes troubleshooting and optimization methods, which include workload adjustments and hardware upgrades.

Additionally, the chapter details integration methods with other IBM and third-party tools for comprehensive performance monitoring and analysis. This chapter has the following sections:

- ▶ 6.1, “Performance metrics and statistics files” on page 136
- ▶ 6.2, “Real-time monitoring tools” on page 137
- ▶ 6.3, “Predictive features” on page 148
- ▶ 6.3.3, “Troubleshooting and optimization” on page 149
- ▶ 6.4, “Integration and compatibility” on page 151

This chapter delves into the core concepts, available tools, and recommended best practices for effective performance monitoring and analysis. For a practical case study on issue isolation utilizing IBM Storage Insights, refer to Chapter 6, “Performance monitoring and analysis” on page 135.

6.1 Performance metrics and statistics files

FlashSystem storage systems have the ability to store a maximum of 16 files in a directory at any one time for each statistics file type. If the maximum of 16 generations of each file type is reached, the 17th file overwrites the oldest one. The time interval between the gathering of statistics is 1–60 minutes in increments of 1 minute.

The **startstats** command modifies the interval at which per-node statistics for volumes, managed disks (MDisks), and nodes are collected.

Tip: IBM Storage Insights uses a default performance monitoring interval of 5 minutes.

- ▶ Set the statistics collection interval on the IBM Storage Virtualize storage system to a value less than or equal to the current performance monitoring interval set in IBM Storage Insights.
- ▶ Configure the IBM Storage Virtualize statistics collection interval to a value that divides evenly into the IBM Storage Insights monitoring interval.

With IBM Storage Insights set to its default 5-minute monitoring interval, use the **startstats** command on the storage system to configure your storage system's statistics collection for either 1 minute or 5 minutes. Deviating from these intervals can lead to missing data samples.

Example 6-1 shows an example of the **startstats** command to change the statistics interval.

Example 6-1 command to change the statistic interval

```
IBM_FlashSystem:FS9500:superuser>startstats -interval 1
```

Performance statistics files abbreviation:

- ▶ Nv_stats is [N]ode [v]olume data
- ▶ Nm_stats is [N]ode [m]disk data
- ▶ Nn_stats is [N]ode [n]ode data
- ▶ Nd_stats is [N]ode [d]rive data
- ▶ Ng_stats is [N]ode [g]arbage collection data

Example 6-2 shows an example output of the **lsdumps** command to list all available statistics files on a FlashSystem.

Example 6-2 lsdumps command to display a list of files in a particular dumps directory

```
IBM_FlashSystem:FS9500:superuser>lsdumps -prefix /dumps/iostats
id  filename
0   Nv_stats_XXXXXXX-1_250212_092543
1   Nn_stats_XXXXXXX-1_250212_092543
2   Nd_stats_XXXXXXX-1_250212_092543
3   Nm_stats_XXXXXXX-1_250212_092543
4   Ng_stats_XXXXXXX-1_250212_092543
5   Nv_stats_XXXXXXX-2_250212_093043
6   Nd_stats_XXXXXXX-2_250212_093043
7   Nn_stats_XXXXXXX-2_250212_093043
8   Nm_stats_XXXXXXX-2_250212_093043
9   Nv_stats_XXXXXXX-1_250212_093043
```

```
10 Ng_stats_XXXXXXX-2_250212_093043
11 Nm_stats_XXXXXXX-1_250212_093043
12 Ng_stats_XXXXXXX-1_250212_093043
...
```

Note: XML metric data is stored as cumulative counters, necessitating external tools for proper interpretation. These tools typically calculate interval values by subtracting the counter value at time *t* from the value at time *t*+1.

6.2 Real-time monitoring tools

This section describes various methods for conducting real-time performance monitoring of FlashSystem. It encompasses both traditional command-line interface (CLI) and graphical user interface (GUI) approaches. The section also includes more advanced tools such as IBM Spectrum Control and IBM Storage Insights.

6.2.1 Onboard real-time monitoring tools

This section describes real-time monitoring tools of Storage Virtualize.

Command-line interface

The **lsnodestats**, **lsnodecanisterstats**, and **lssystemstats** commands continue to report latency in milliseconds only. The latency was reported as an integer value before code level 8.5. For code level 8.5 and later, it is reported with three decimal places of granularity. This format makes it possible to monitor variations in response time that are less than 1 ms, as shown in Example 6-3.

Example 6-3 lssystemstats command to display the most recent values of all node statistics

```
IBM_FlashSystem:FS9500:superuser>lssystemstats
...
fc_mb           197           197       250318064422
fc_io           50740        50740       250318064422
...
```

Note: Errors on the physical connection can impact performance as well, therefore the **lsportstats** command can be used to view the port transfer and failure counts and Small Form-factor Pluggable (SFP) diagnostics data that is recorded in the statistics file for a node.

Management GUI

In FlashSystem, real-time performance statistics provide short-term status information for your systems. The statistics are shown as graphs in the management GUI.

Dashboard

The Performance section on the Dashboard presents a graph visualizing either system or node performance. Users can toggle between system-level and node-level views by selecting **System** or **Node Comparison**. Data granularity can be adjusted to display performance in seconds, minutes, hours, or days. The graph updates every five seconds. Users can select from various available metrics to modify the graph's display. For metrics not shown on the Dashboard, navigate to the Performance page by using the navigation menu. Figure 6-1

illustrates the read and write bandwidth of a FlashSystem within the Dashboard's Performance section, as displayed in the management GUI.

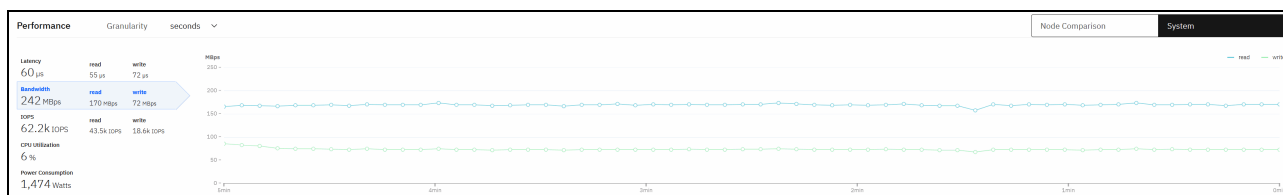


Figure 6-1 Performance graph of the FlashSystem dashboard

Monitoring performance statistics

In the management GUI select **Monitoring** → **Performance Statistics** to view real-time performance statistics which can offer insights into the system's immediate status. Users can monitor bandwidth usage for volumes, protocols, and MDisk, as well as overall System CPU usage.

System or Node statistics are presented graphically to assess the system's performance health. These graphs, categorized into IOPS, Bandwidth, Latency, and Utilization, track performance trends, enabling the detection of deviations from stable values or discrepancies between related metrics. Such anomalies can be further investigated using performance diagnostic tools.

The MDisk and Volumes section displays metrics including Read/Write IOPS, Read/Write Bandwidth, and Read/Write Latency. These metrics aid in evaluating the overall performance health of volumes and MDisk. Unexpected results might indicate configuration errors, system faults, or connectivity issues.

The System CPU (total) chart illustrates current CPU usage and usage peaks. Although isolated spikes might not signify performance degradation, sustained high usage (above 90%) coupled with high I/O input suggests potential system overload.

The Protocols chart showcases configurable protocol types, which vary based on system models and installed adapters. Data point availability can differ across protocols. This section details the function and performance impact of each protocol type: Fibre Channel, iSCSI, SAS, IP Replication, and IP Replication (Compressed).

Note: For code release 8.7 and later, the granularity in the performance charts can be changed to seconds, minutes, hours, and days. With this setting, a maximum time period of 60 days can be covered by the charts. Only average values can be achieved.

Figure 6-2 on page 139 illustrates the graphs of MDisk and Protocols in the Monitoring section of the GUI and includes a write spike that is triggered by disabling the write cache of a specific volume.

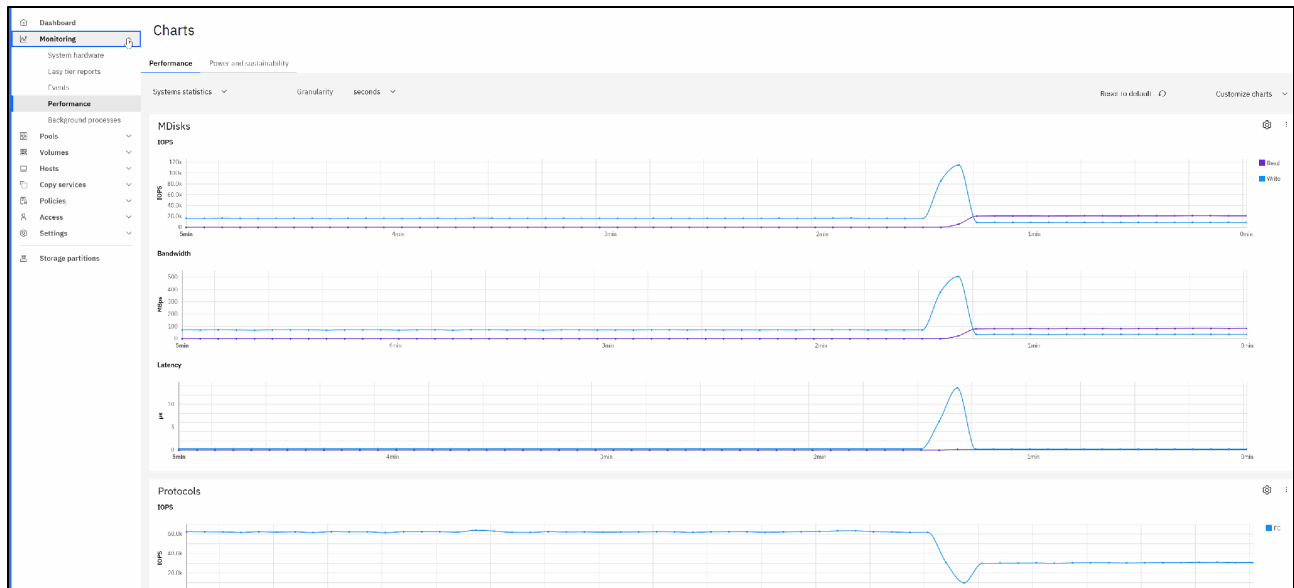


Figure 6-2 Performance metrics from the management GUI in the Monitoring section

6.2.2 IBM Spectrum Control

IBM Spectrum Control is an on-premises storage management solution that is designed to manage and optimize storage resources across heterogeneous storage environments. It provides a unified view of storage asset, so that administrators can efficiently monitor, manage, and optimize their storage infrastructure.

IBM Spectrum Control integrates seamlessly with FlashSystem and offers advanced features such as real-time performance monitoring, capacity planning, and automated storage tiering. Administrators can use it to take advantage of the speed of flash storage while ensuring optimal utilization and cost-efficiency.

Overview

The Overview section of the navigation pane contains metrics on the activity of your storage resources. Depending on the selected chart, the interactive graphs provide an initial overview of the most heavily used resources from the past 24 hours.

Figure 6-3 on page 140 contains the Overview page of a FlashSystem in IBM Spectrum Control and illustrates the Overall System Activity and Most Active Volumes, Nodes and Pools. The metrics show a 70% Read and 30% Write I/O workload increase of one Pool, but with no significant changes in Response Time or CPU Utilization.

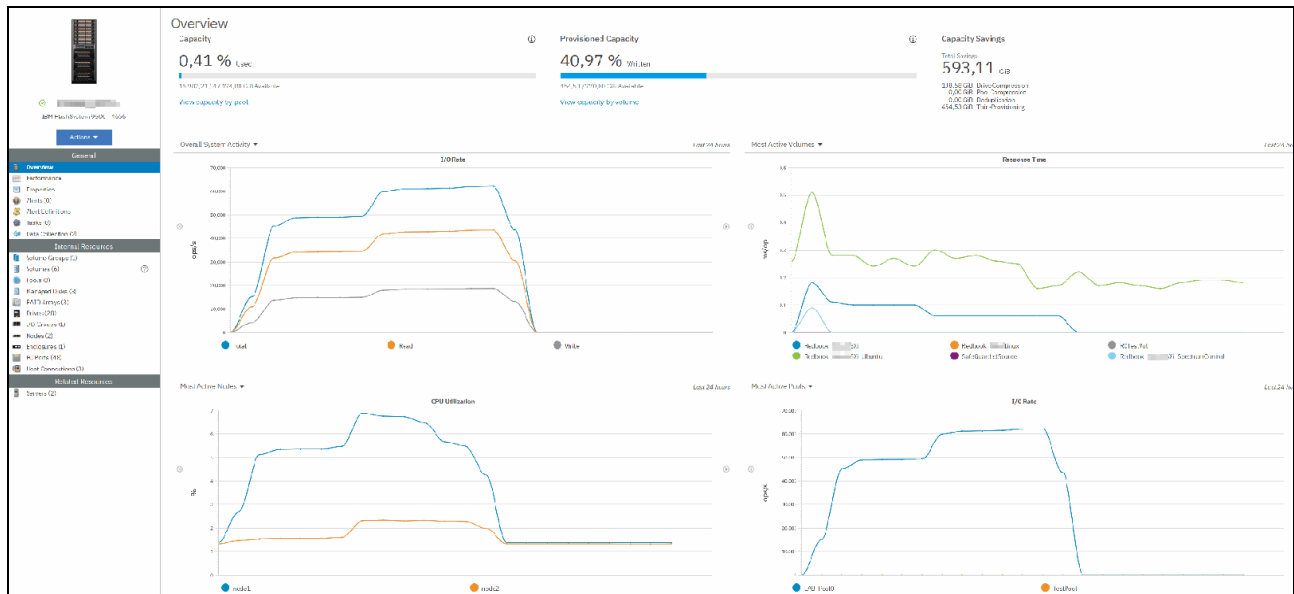


Figure 6-3 Overview page of a FlashSystem in IBM Spectrum Control

Performance

The Performance section of the navigation pane provides comprehensive insights into the operational efficiency and health of your FlashSystem.

Administrators can access a large amount of real-time data and historical trends that are related to key performance indicators (KPIs). These KPIs include metrics such as IOPS, latency, throughput, and response time, which are critical for understanding system behavior and identifying potential performance bottlenecks.

The section offers interactive graphs and tables that visually represent these metrics, making it easier to interpret complex performance data.

Key performance indicators

Key performance indicators (KPIs) show performance metrics collected over 24 hours. These charts also include guideline values for performance, which were established by monitoring, measuring, analyzing, and stress testing the performance of FlashSystem systems.

The key performance indicators include the following metrics:

- ▶ Read Response Time by I/O Group
- ▶ Write Response Time by I/O Group
- ▶ Max Cache Fullness by Pool
- ▶ Node Utilization Percentage by Node
- ▶ Overall Port Bandwidth Percentage by Port
- ▶ System CPU Utilization by Node
- ▶ Port-to-Local Node Send Response Time by Node
- ▶ Port-to-Remote Node Send Response Time by Node
- ▶ Port Send Delay Time

The charts compare the performance metrics that were collected each hour over the last 24 hours with the guideline values. The date can be changed in the calendar, which is next to the title of the page, so that you can compare current values with historical values.

For more information of all the KPIs, see [IBM Storage Virtualize guideline values for key performance indicators](#).

Figure 6-4 contains the overview page of the KPIs. The guideline values are marked in orange.



Figure 6-4 Key performance indicators (KPIs) in IBM Spectrum Control

Performance view of internal resources

Internal resources for FlashSystem in IBM Spectrum Control contains the block storage resources, which are monitored by the IBM Spectrum Control software stack. These resources can include various components of FlashSystem like Volume Groups, Volumes, Managed Disks, RAID Arrays, Drives, I/O Groups, Nodes, Enclosures, FC Ports and Host Connections.

Volume Groups, Volumes, Pools, Managed Disks, Drives, I/O Groups, Nodes, FC Ports and Host Connections contains a Performance tab, which can be used for a comprehensive analysis of the performance metrics.

Real-time and historical performance data, including metrics like I/O per second (IOPS), throughput, latency, response time, and more can be selected for each individual resource to identify a performance bottleneck.

Figure 6-5 on page 142 contains an example graph of the performance tab from the Host Connections, which illustrates a workload increase to a host with a slight increase of the Write Response Time. The Start and the End Time was filtered to a specific time frame.

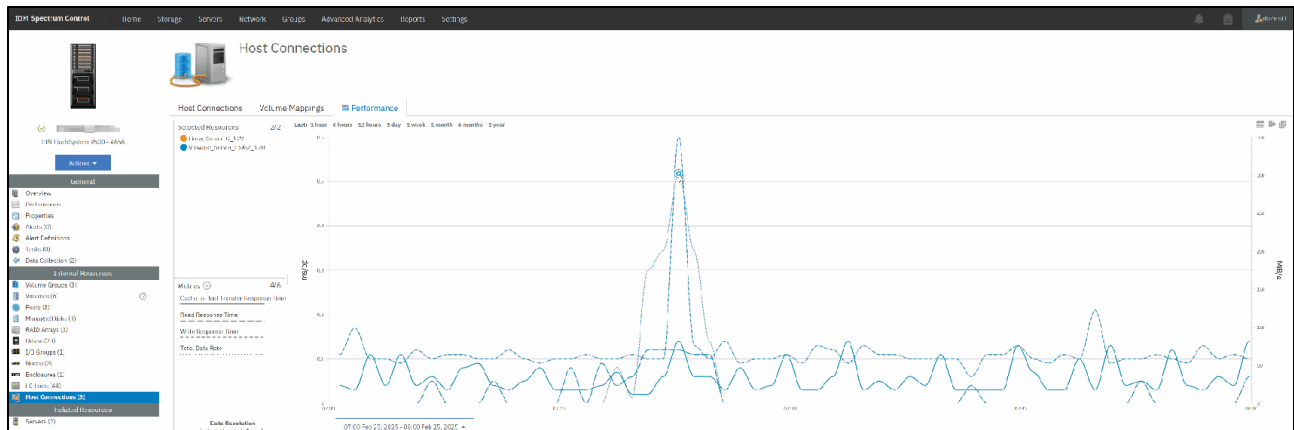


Figure 6-5 Host Connections in IBM Spectrum Control

For more information about all the FlashSystem performance metrics that are monitored by IBM Spectrum Control, see [Performance metrics for resources that run IBM Storage Virtualize](#).

Performance view from the Actions menu

By using IBM Spectrum Control, you can view the individual performance for each resource in a separate window. The Actions menu is either available for FlashSystem in the main web-based GUI or in the performance tab of each individual selected internal resource.

Figure 6-6 on page 143 shows the main web-based GUI from IBM Spectrum Control and the system performance of a FlashSystem in a separate window.

As an example, Figure 6-7 on page 143 contains the performance filtered to the Fibre Channel ports of selected resource node1.

Note: The discrepancy between peaks in IBM Spectrum Controls performance graphs and the corresponding max values in the table below can be attributed to the way data is aggregated and displayed. A description for the discrepancy is documented in section 6.2.3, “IBM Storage Insights” on page 144.

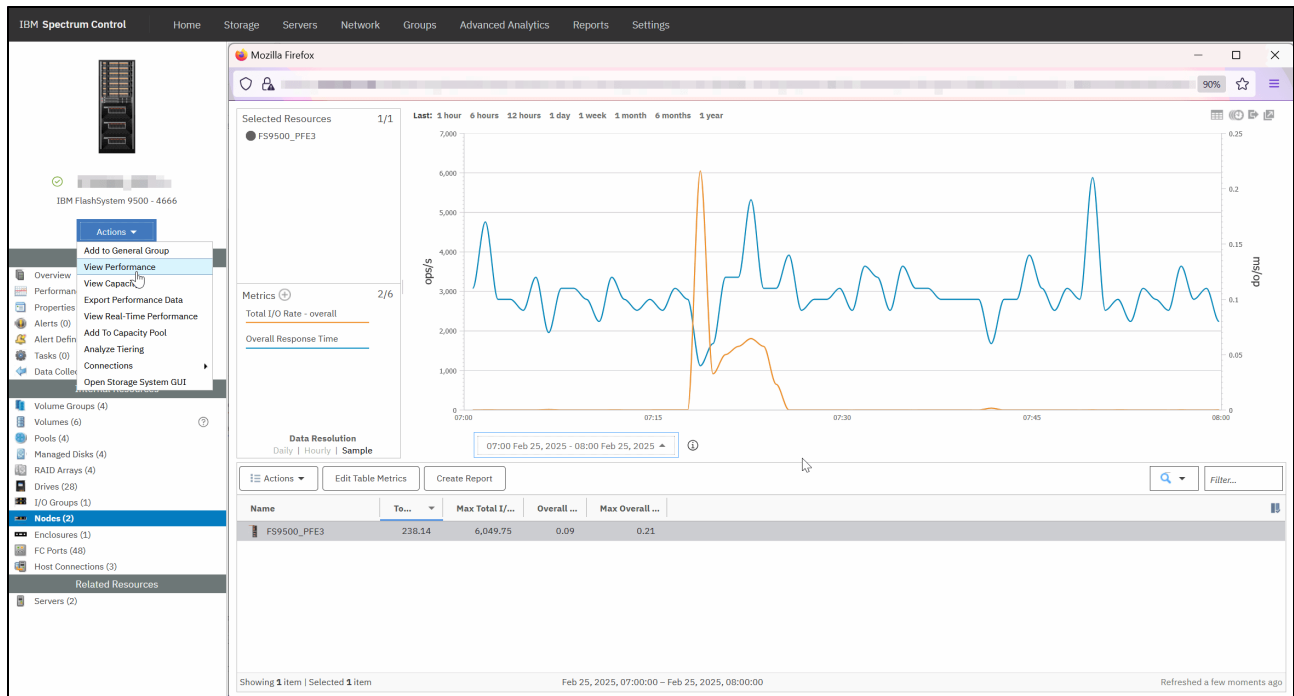


Figure 6-6 FlashSystem system performance view in IBM Spectrum Control

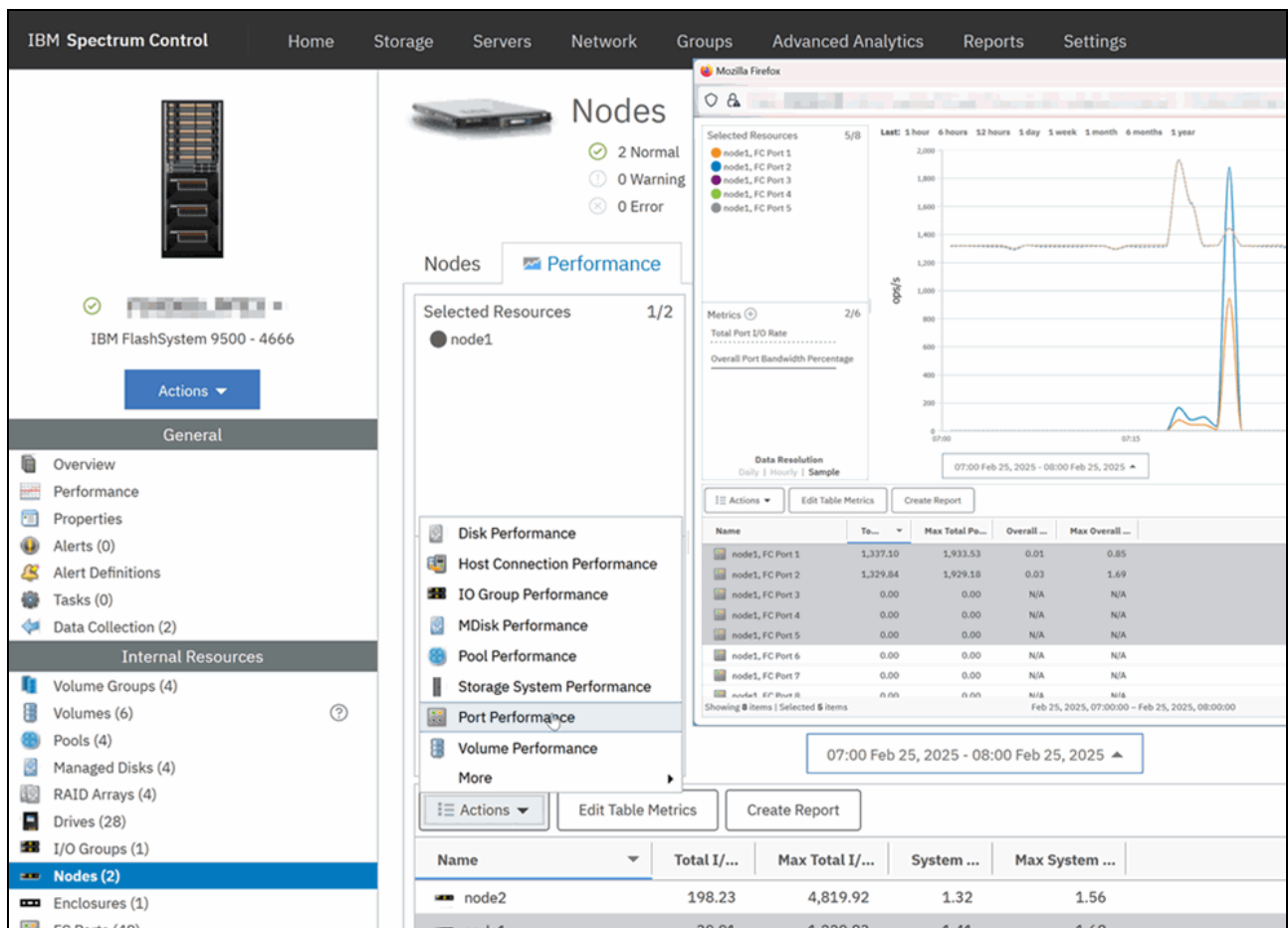


Figure 6-7 Performance view of Fibre Channel ports of one FlashSystem node in IBM Spectrum Control

Alerts

Alerting is vital for maintaining system health, optimizing performance, ensuring compliance, and facilitating proactive management.

For FlashSystem, there are two reasons to implement performance alerts:

1. Proactive problem resolution
2. Performance tuning

With IBM Spectrum Control, proactive problem resolution alerts can be defined, to notify you about potential issues before they escalate into major problems. By addressing the problems promptly, you can help avoid system downtime.

Performance bottleneck alerts can be highlighted for performance tuning. These types of alerts contains input to fine-tune your system for better performance. This can involve adjusting configurations, reallocating resources, or upgrading hardware.

For more information, see [Defining alerts](#) and [Alerts and alert policies](#).

Note: Alert notifications can be sent to email addresses, SNMP, IBM Netcool® / OMNibus or UNIX Syslog.

Exporting performance data

To effectively address performance issues with FlashSystem, it is often necessary to export performance data. This exported data can be invaluable when seeking assistance from IBM Support to diagnose and resolve any issues.

For information about exporting performance data see the following documentation:

- ▶ [Exporting performance data by using the GUI](#)
- ▶ [Exporting performance data by using a script](#)

Note: Advanced export is required for volume data if the time range for the export is greater than 12 hours and the export includes data that was collected at 1-minute intervals.

6.2.3 IBM Storage Insights

IBM Storage Insights is a cloud-based storage management service that is designed to provide real-time monitoring, analysis, and insights into the performance and health of FlashSystem storage arrays.

This tool allows administrators to efficiently manage their storage resources, identify potential issues before they escalate, and optimize system performance. It offers features such as capacity and performance trending, capacity planning, and alerting, which are all accessible through a web interface.

Note: With a Spectrum Control license, you can access Storage Insights Pro. This allows the use of the cloud-based Artificial Intelligence for IT Operations (AIOps) and many more features to monitor FlashSystem and other storage products.

Overview

The Overview section of the navigation pane in IBM Storage Insights is very similar to IBM Spectrum Control. All metrics contain the activity of your storage resources. Depending

on the selected chart, the interactive graphs provide an initial overview of the most heavily used resources from the past 24 hours.

Figure 6-8 contains the Overview page of a FlashSystem in IBM Storage Insights and illustrates the Overall System Activity and Most Active Volumes, Nodes and Pools. The metrics shows a 70% Read and 30% Write I/O workload increase of one Pool, but with no significant changes in Response Time or CPU Utilization.

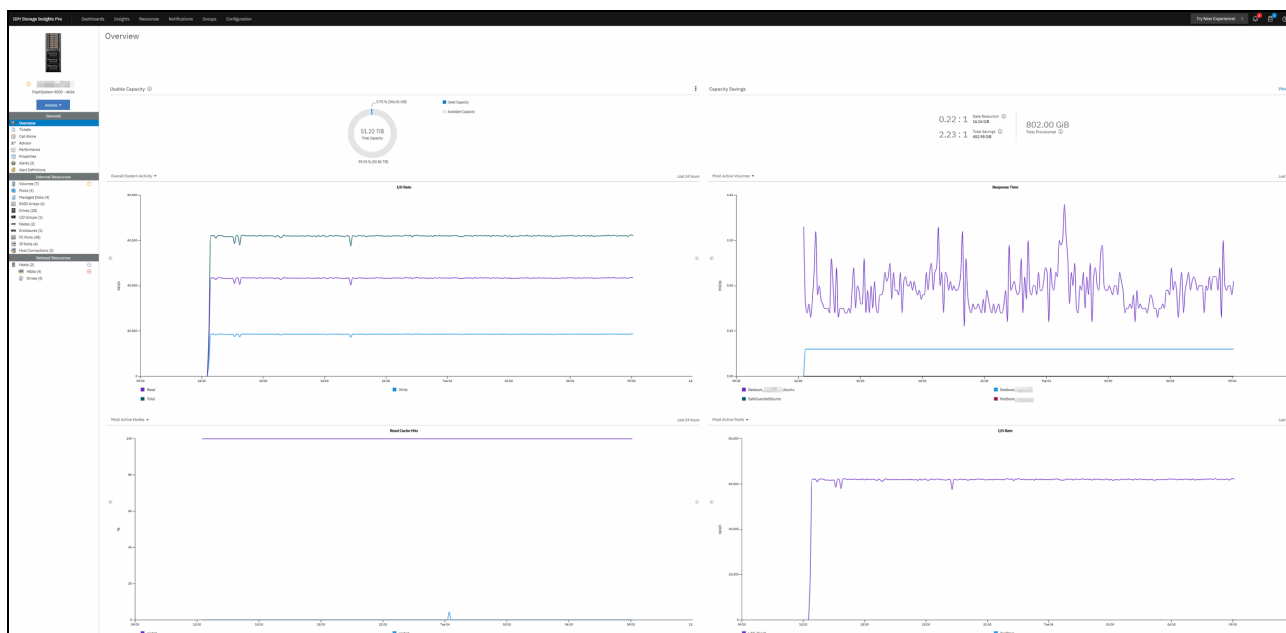


Figure 6-8 Overview page of a FlashSystem in IBM Storage Insights

Performance

The Performance tab in the navigation pane of IBM Storage Insights provides the same comprehensive insights into the operational efficiency and health of your FlashSystem as IBM Spectrum Control.

Administrators can access real-time data and historical trends that are related to key KPIs. These KPIs include metrics such as IOPS, latency, throughput, and response time, which are critical for understanding system behavior and identifying potential performance bottlenecks.

All KPIs have the same explanation as documented in section “Key performance indicators” on page 140.

Performance view of Internal Resources

Within the IBM Storage Insights tenant, Internal Resources for FlashSystem serves as a comprehensive monitoring tool for block storage resources. These resources encompass different FlashSystem components, which include Volume Groups, Volumes, Managed Disks, RAID Arrays, Drives, I/O Groups, Nodes, Enclosures, FC Ports, and Host Connections.

Each of these components is equipped with a Performance tab, which can provide a detailed analysis of performance metrics. This tab provides both real-time and historical performance data so that users can select specific metrics such as IOPS, throughput, latency, and more. These metrics are instrumental in identifying performance bottlenecks within the system.

Figure 6-9 on page 146 illustrates a graph from the performance tab of the internal resource Nodes. The graph highlights an imbalance of the workload between both nodes.

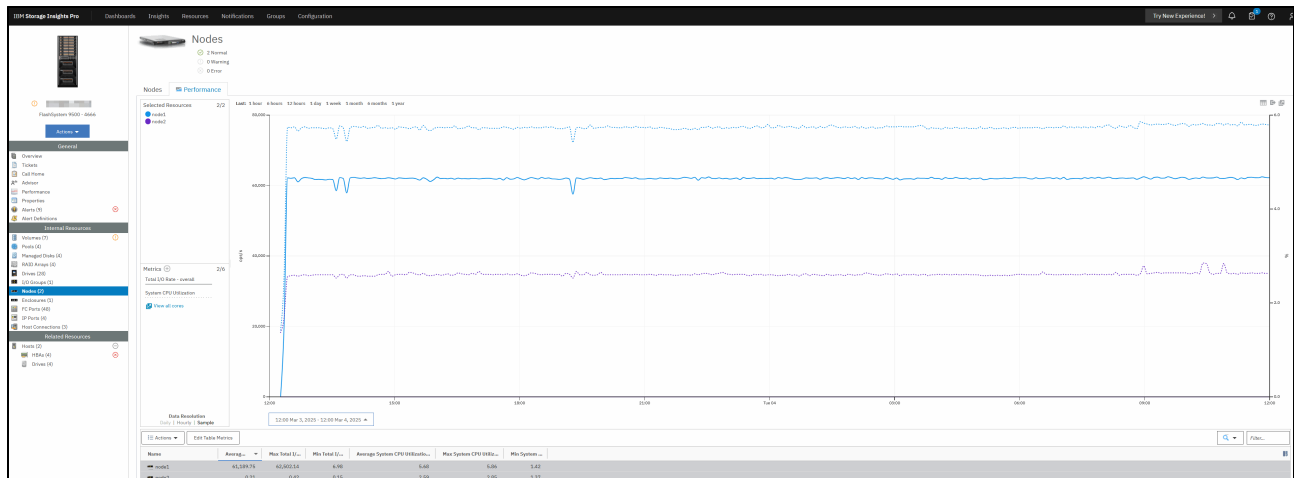


Figure 6-9 Performance tab of Internal Resource - Nodes in Storage Insights

For an explanation of all the performance metrics of FlashSystem storage that are monitored by IBM Storage Insights, see [Performance metrics for resources that run IBM Storage Virtualize](#).

Performance view from the Actions menu

You can use IBM Storage Insights to view the performance of each individual resource in a dedicated window. This function provides a detailed, focused perspective on the performance metrics of specific components within your FlashSystem.

The Actions menu is accessible either in the main web-based GUI view for the system or the performance tab of each individually selected internal resource.

Figure 6-10 shows the main web-based GUI from IBM Storage Insights and the system performance of a FlashSystem in a separate window.

The chart in Figure 6-11 on page 147 contains the performance filtered to volumes of selected resource node1 (preferred node).

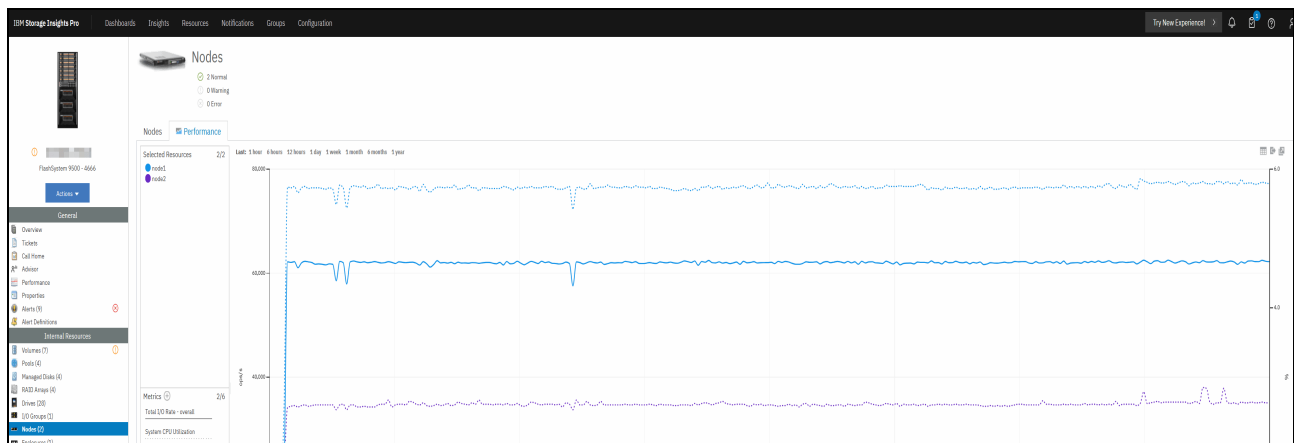


Figure 6-10 FlashSystem system performance view in IBM Storage Insights

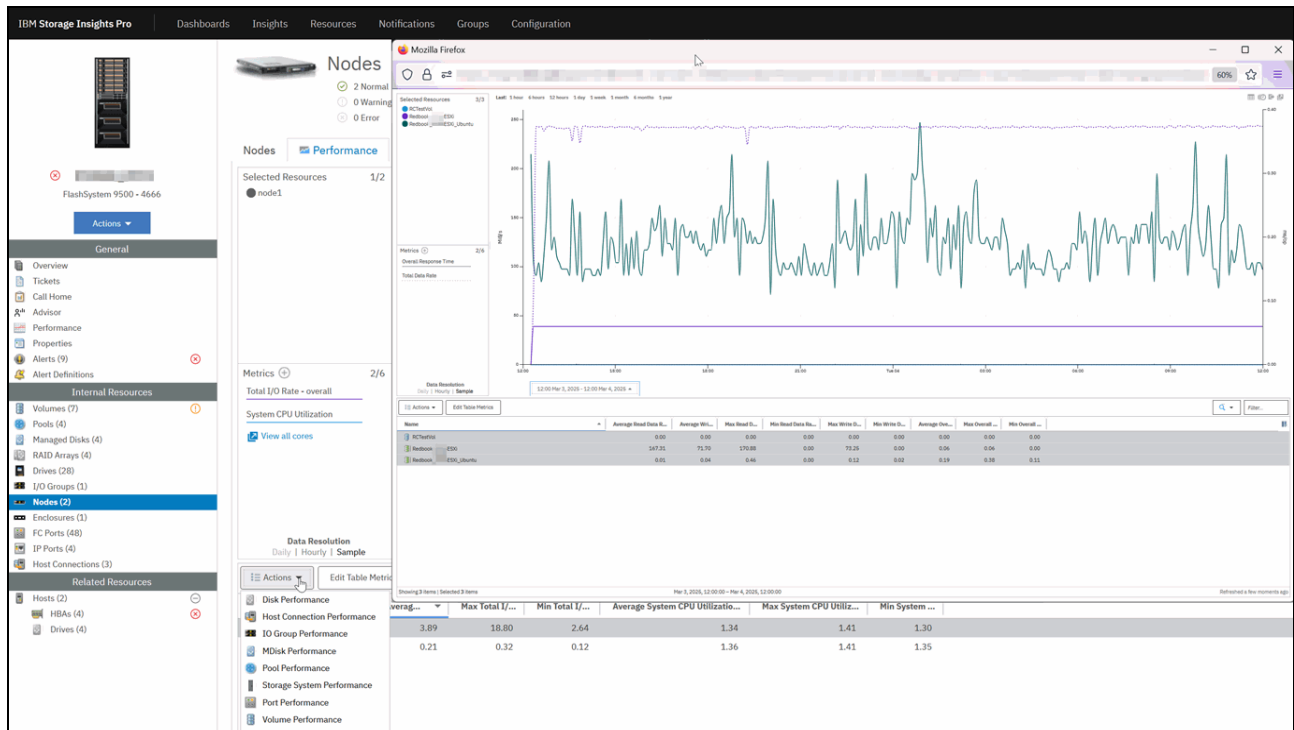


Figure 6-11 Volume performance of all volumes with node1 as selected resource in IBM Storage Insights

The discrepancy between peaks in IBM Storage Insights performance graphs and the corresponding max values in the table below can be attributed to the way data is aggregated and displayed.

Fo Exporting performance data

To effectively address performance issues with FlashSystem, it is often necessary to export performance data. This exported data can be invaluable when seeking assistance from IBM Support to diagnose and resolve any issues. For more information, see [Exporting and uploading performance data for storage systems](#) and [Exporting performance data by using the GUI](#).

Note: For IBM Storage Insights Pro users, the option exists to upload performance data directly to a support ticket linked to the storage system.

Note: Advanced export is required for volume data if the time range for the export is greater than 12 hours and the export includes data that was collected at 1-minute intervals.

For more information on IBM Storage Insights refer to the following publications:

- *Maximize Your Storage Investment: Leverage IBM Storage Insights for Observability and Artificial Intelligence for IT Operations*, [SG24-8584](#)
- *Intelligent Storage Management with Artificial Intelligence for IT Operations (AIOps)*, [REDP-5755](#)
- *Ecosystem Integration with IBM Storage Insights*, [REDP-5754](#)

6.3 Predictive features

IBM Spectrum Control Advanced Analytics and IBM Storage Insights Advisor are both components developed for managing and optimizing resources in storage environments.

6.3.1 Spectrum Control Advanced Analytics

Advanced Analytics is an embedded component of the IBM Spectrum Control software, which provides advanced analytics capabilities to understand and optimize the storage environment. It uses historical data to provide insights into trends, capacity planning, performance, and more. It can help identify underutilized or over-provisioned storage, predict future capacity needs, and optimize your storage infrastructure.

The Spectrum Control Advanced Analytics contains four categories:

1. [Cloud Configuration](#)
2. [Provisioning storage](#)
3. [Storage Optimization](#)
4. [Storage Reclamation](#)

6.3.2 Storage Insights Advisor

The Advisor feature of Storage Insights is a cloud-based service, which uses predictive analytics and machine learning to monitor and optimize your storage environment. It provides insights and recommendations to improve performance, capacity, and cost efficiency. It can predict potential issues before they occur, allowing for proactive management of storage resources.

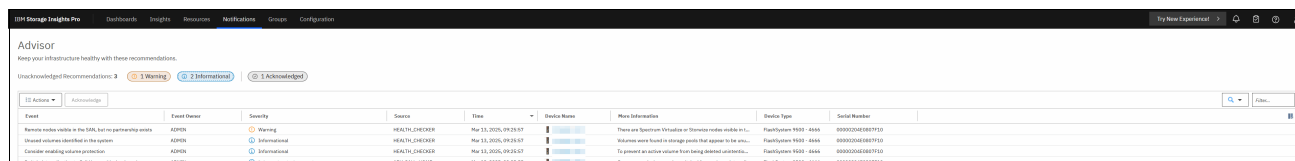
Note: IBM Storage Advisor is based on IBM's best practices and is continually refined and developed to ensure optimal health state.

These best practices are kept up to date and get adjusted periodically to exploit new features and functions that based on field experience and feedback.

Within the web-GUI of IBM Storage Insights, the Advisor can be opened either in main menu bar: **Notifications** → **Advisor** or from the **Resources** menu.

To open the Advisor from the Resources menu, open the Block Storage Systems. Right-click a storage system in the list and click **View Details**. Click **Advisor** in the General section.

Figure 6-12 contains Advisor recommendations for a FlashSystem storage system from the web GUI of IBM Storage Insights.



Event	Event Owner	Severity	Source	Time	Status Name	More Information	Serial Number
Remote nodes visible in the SAN, but no partnership exists	ADMIN	Warning	HEALTH_CHECKER	Mar 13, 2020, 09:25:57	Warning	There are Spectrum Virtualize or Storwize nodes visible in...	FlashSystem 9301 - 4845
Unread volumes identified in the system	ADMIN	Informational	HEALTH_CHECKER	Mar 13, 2020, 09:25:57	Informational	Volumes were found in storage pools that appear to be abn...	FlashSystem 9301 - 4845
Consider enabling system protection	ADMIN	Informational	HEALTH_CHECKER	Mar 13, 2020, 09:25:57	Informational	To prevent an issue where data being deleted incorrectly...	FlashSystem 9301 - 4845
Switch data collection to SAN from host-based method	ADMIN	Informational	ADMIN_PAGE	Mar 13, 2020, 09:25:57	Informational	One or more devices can be monitored from using a SAN-side...	FlashSystem 9301 - 4845

Figure 6-12 IBM Storage Insights Advisor recommendations for a FlashSystem

6.3.3 Troubleshooting and optimization

This section includes details about the general support that is provided for diagnosing and resolving FlashSystem performance issues or bottlenecks. Although these systems are typically fast and efficient, they can occasionally encounter performance hurdles. Checklists and key questions are described to identify, understand, and rectify these issues, ensuring optimal performance and reliability.

In the realm of performance analysis, particularly within the I/O stack, two key components stand out: upper cache and lower cache.

The upper cache, often closer to the application layer, handles data requests and responses at a higher level of abstraction. Conversely, the lower cache, situated nearer to the physical storage devices, manages data at a more granular level.

Understanding the intricacies of these caches is crucial for optimizing system performance and troubleshooting potential bottlenecks.

Note: In IBM Spectrum Control and IBM Storage Insights, the upper cache is referred to as volume cache (VC) and the lower cache as volume copy cache (VCC).

Troubleshooting considerations and recommendations

Although FlashSystems are virtualized, their storage performance is limited by the capabilities of the underlying back-end storage (for example, the FlashSystem array or externally virtualized MDisks). The majority of performance problems stem from overloading the system.

As a best practice, remember these points when troubleshooting performance issues:

- ▶ Monitor storage pools (mdisk groups) regularly. These are the best place to see both front-end (upper cache) and backend (lower cache) latency.
 - If latency is high on MDisks (arrays), the system is likely being overloaded.
 - If latency is high on volumes (vdisks) but mdisk latency is good:
 - Ensure you're comparing performance metrics with consistent transfer sizes.
 - Review any active replications or copy services because these can significantly impact the load through copying or mirroring operations.
 - Verify node cache statistics, whether the cache or partition is full, the percentage of flushes, write-through delays, and other relevant metrics.
 - Verify that node latencies, such as node to node writes to local nodes are less than 0.5 ms.
 - Verify fabric statistics, for example, ports, buffer credit (congestion) issues.
- ▶ Note that data is typically a 1-minute average and potentially longer. Although bursty workloads are generally undesirable, their impact can be masked by these averages.
- ▶ Although peak values can seem alarming, they represent the single slowest I/O in the observed interval. It is important to avoid overreacting to these unless they are consistently above 500 ms on volumes or substantially elevate the average, which reduces the margin between the average and the peak.
- ▶ Be cautious about focusing too much on statistics:
 - Create a clear problem description of the performance issue you are analyzing.
 - Do not look for issues where there are none.

6.3.4 Optimization considerations and recommendations

Although FlashSystems are designed to handle diverse workloads when initially sized correctly, performance can sometimes fall short of expectations, such as from predictions of IBM Storage Modeler or StorM predictions. In such cases, the initial step is to contact IBM Support to determine if any underlying technical issues or code defects are the cause. Following this, consider the following recommendations to enhance system performance.

Identifying the nature of the workload and the most probable bottleneck is essential. Bottlenecks can occur in various components of the FlashSystem or across different layers of the I/O stack. If the system is operating within its designed capacity (based on initial sizing), resolving the bottleneck can improve overall performance. However, if the overall workload is pushing the system to its limits, addressing one bottleneck might simply reveal another, so overall performance gain might be minimal. The exception to this is if the newly found bottleneck has a less significant impact than the previous one.

Each workload can be labeled with a classification:

- ▶ IOPS driven
- ▶ Bandwidth driven
- ▶ CPU bound
- ▶ Memory bound

Depending on the nature and type of workloads, bottlenecks first appear at different places. Use the following list to determine as an example checklist:

- ▶ Is the workload driven by IOPS?
 - Has the system initially been sized on capacity only?
 - Serial Attached SCSI (SAS) spindle counts. The more spindles and modules, the more parallelism, which can create higher throughput.
 - Nearline Serial Attached SCSI (NL-SAS) OLTP performance expectations are low because this drive technology is better suited for streaming I/O.
- ▶ Is the workload driven by bandwidth (MBPS)?
 - Has the system initially been sized on capacity only?
 - NVMe module memory bandwidth and number of adapters and ports count.
 - Does the system have a sufficient number of ports to avoid overloading individual ports?
- ▶ Is the workload CPU bound?
 - Can individual CPUs be identified to show usage of > 90%?
 - Could outdated advanced functions, such as legacy remote copy be the cause? Upgrading to more recent replication methods might then improve performance.
 - Spreading such workloads across more volumes might help (increasing parallelism).
- ▶ Is the workload cache/memory bound?
 - The primary purpose of cache is to minimize application I/O latency.
 - Cache remains relevant with modern solid-state/flash technology, so ensure the system has a sufficient memory to maximize memory bandwidth usage.
 - For read I/Os:
 - If data is not in cache, then reading from media can increase latency.
 - A larger read cache can improve the probability of data being in cache.

- For write I/Os:
 - All writes are “written to cache” and there are cache-friendly workloads overwriting blocks in cache before they get destaged to media. This is called “write efficiency” or “dirty writes (LCA)” in Storage Insights.
 - Throughput of write cache is more important than size of write cache.
 - If throughput of write cache and media are the same, then the net effect is an improvement in latency.
 - Once achieved, having a bigger write cache doesn’t make the system faster.
- Are you using advanced features?
 - Replication:
 - DR requires 2 ports per node.
 - HA requires 2 ports per node.
 - Multipathing drivers add latency if the number of paths increase.
 - There are limited logins per port, so consider no more than 4 paths per host.
 - Data reduction:
 - Data reduction pools uses a *chatty* node-to-node protocol. This typically occurs over the PCIe bus on FlashSystem. For SAN Volume Controller, ensuring the lowest latency for these connections is crucial.
 - Consider using compression in only Flash Core Modules because compression does not affect latency and performance.
- Additional points to consider when evaluating port sufficiency:
 - For SVC or externally virtualizing FlashSystem, host and disk can share traffic on the same ports, bi-directional nature of virtualization I/O (for example, write in = write out).
 - Ensure the port count adequately supports both the required throughput and the necessary Fibre Channel (FC) logins per port.
 - For replication purposes, it is recommended to dedicate node ports.
 - Therefore, ensure you have a sufficient number of Host Bus Adapters (HBAs) or ports per node (typically 2 ports per node for replication). Also, try to match this port count with your link bandwidth, or at least, size the port count close to your peak replicated volume write bandwidth.

6.4 Integration and compatibility

FlashSystem collects performance statistics in XML files (see also 6.1, “Performance metrics and statistics files” on page 136), which can be copied off the system to be fed into IBM or 3rd-party tools for further analysis and visualization.

IBM Storage Insights is strongly recommended for performance inspection and as a central monitoring and alerting platform, which includes real-time threat detection. Other tools are also available for monitoring FlashSystem performance and building historical data.

For example, two common tools are [Prometheus](#) and [Grafana](#), both of which can be used with [spectrum-virtualize-exporter](#) for IBM Storage Virtualize systems.

This book's scope is to provide an understanding of the components needed to work with XML statistics and is not a detailed step-by-step guide for setting up these environments.

6.4.1 Integration in IBM Storage Insights

IBM Storage Insights is a cloud-based service that you can use to analyze, protect, and optimize your storage infrastructure. Every FlashSystem under warranty is entitled to be registered in Storage Insights. For more information about Storage Insights, see [Before you begin checklist for IBM Storage Insights](#).

For ongoing performance monitoring that includes historical statistics, consider Storage Insights Pro. For more information, see [IBM Storage Insights free vs IBM Storage Insights Pro](#).

Either way, starting with Storage Virtualize version 8.5.3, the need for a data collector is no longer required. Instead, on nodes with 128 GB or more memory, you can configure Call Home to send all statistics directly (or via a proxy server) from a FlashSystem to IBM. For more information, see [Data collector vs Call Home with cloud services](#).

6.4.2 Integration in IBM Spectrum Control

IBM Spectrum Control is an on-premises software solution that gathers data for deeper analytics and automation to help manage even more complex storage environments. IBM Spectrum Control might be preferable if you prefer to use on-premises tools to collect and inspect your performance data.

For an interactive demo and comparison of Storage Insights and Spectrum Control, see [IBM Storage Insights and IBM Spectrum Control](#).

Note: A Spectrum Control license also entitles you to use Storage Insights Pro.

For more information, see [Getting started with IBM Spectrum Control](#).

6.4.3 Integration with third-party performance monitoring tools

For a general guidance about how to pull XML performance data files from a FlashSystem, see [Performance data files](#).

At the time of writing, two methods exist:

1. By using the REST API. For more information, see [Storage Virtualize Performance Statistics](#).
2. With the help of Secure Shell's Secure Copy (scp).

The use of SCP for data collection necessitates scripting because each node generates independent statistical files, which must be aggregated onto the cluster's configuration node before retrieval. Example 6-4 shows a basic sample code snippet (for Korn shell) to accomplish this.

Note: The example assumes that you have configured ssh-key authentication to avoid prompts for a password.

Example 6-4 Korn Shell sample script to copy XML performance statistics off a FlashSystem

```
# get IDs of non-config nodes in cluster
```

```

ncn=$(ssh user@10.0.0.1 'svcinfolsnnode -nohdr -filtervalue "config_node=no" |
cut -d " " -f1)

# copy files that haven't been downloaded yet from all
# non-config nodes to config node
i=1
for n in $ncn; do
    files[i]=$(ssh user@10.0.0.1 "svcinfolsdumps -prefix /dumps/iostats
-nohdr -delim : $n"|cut -d: -f2)
    for m in ${files[i]}; do
        if [ ! -s $m ]; then
            ssh user@10.0.0.1 "svctask cpdumps -prefix
/dumps/iostats/$m $n"
        fi
    done
    i=i+1
done

# now download all new files from config node
allfiles=$(ssh user@10.0.0.1 "svcinfolsdumps -prefix /dumps/iostats -nohdr -delim
:"|cut -d: -f2)
for n in $allfiles; do
    if [ ! -s $n ]; then
        scp user@10.0.0.1:/dumps/iostats/$n .
    fi
done

```

Next, you can post-process these XML files with tools of your choice.



Isolating issues by using IBM Storage Insights

IBM Storage Insights is a powerful tool that is designed to accelerate the identification of storage environment issues. Its Alerting feature offers real-time monitoring and detection of problems across your storage, switches, and hosts. This chapter describes the process for troubleshooting Storage Insights alerts, introduces multi-conditional alerts, and explains how to create and manage alerts on groups of storage resources.

When Storage Insights generates an alert based on an alert definition, you can use Storage Insights to troubleshoot it. Storage Insights includes the following information and other aspects to assist in troubleshooting:

- ▶ Affected hosts
- ▶ The parts of the storage system that are impacted
- ▶ Affected volumes

An examination of the listed items can help you assess whether any of your applications or other production infrastructure is affected. Also, the Storage Insights performance data can help you understand why applications were impacted.

This chapter has the following sections:

- ▶ 7.1, “Identifying performance deviations” on page 156
- ▶ 7.2, “Multi-condition alerts” on page 160

In section 7.1, “Identifying performance deviations” on page 156 describes the workflow for starting with an alert and determining what resources were impacted by the alert and which ones were not.

The section 7.2, “Multi-condition alerts” on page 160 shows examples of multi-conditional alerts. The examples use cyclic redundancy check (CRC) errors and other link-level counters. CRC errors always indicate physical link issues. This section shows an example of adding other counters to detect whether CRC errors are occurring at a rate that can affect performance.

7.1 Identifying performance deviations

Figure 7-1 shows a partial list of alerts from IBM Storage Insights. In this example, you focus on the alert name of Write Cache Delay Percentage that is highlighted in the figure. This particular performance metric measures how many I/O operations were delayed because of space issues in the write cache.

Alerts					
Unacknowledged Alerts: 268					
12 Critical					
68 Warning					
188 Informational					
379 Acknowledged					
Actions Refresh					
Resources	Alert Name	Occurrence Time	Condition	Violation	Severity
2107.75KMP41	Status	Mar 1, 2025, 11:14:53	Error	Error	Warning
taco-vm.tuc.stglabs.ibm.com	Probe Status	Mar 1, 2025, 09:29:15	Error	Error	Critical
FS5300-1	Status	Mar 1, 2025, 04:22:55	Error	Error	Warning
Nemo	Used Capacity	Mar 1, 2025, 03:36:26	>= 75 % In ...	75.29 %	Warning
nsd002st002	Status	Mar 1, 2025, 03:07:21	Error	Error	Warning
loki2.tuc.stglabs.ibm.com	Status	Mar 1, 2025, 03:07:21	Error	Error	Warning
2107.75KMP41	Status	Mar 1, 2025, 02:14:55	Error	Error	Warning
krill.storage.tucson.ibm.com	Status	Mar 1, 2025, 02:06:17	Error	Error	Warning
node2	Write-cache Delay Percentage	Mar 1, 2025, 01:01:01	2 Conditions	Multiple	Warning
FS5300-1	Status	Feb 28, 2025, 16:52:54	Error	Error	Warning
Nemo	Used Capacity	Feb 28, 2025, 16:36:26	>= 75 % In ...	75.29 %	Warning
2107.75KMP41	Status	Feb 28, 2025, 15:14:53	Error	Error	Warning
Nemo	Used Capacity	Feb 28, 2025, 13:36:26	>= 75 % In ...	75.29 %	Warning
2107.75KMP41	Status	Feb 28, 2025, 13:14:54	Error	Error	Warning
000197903224	Performance Monitor Status	Feb 28, 2025, 12:59:20	Not Success...	Error	Warning
000197903224	Performance Monitor Status	Feb 28, 2025, 09:42:05	Not Success...	Error	Warning
taco-vm.tuc.stglabs.ibm.com	Probe Status	Feb 28, 2025, 09:29:15	Error	Error	Critical
Nemo	Used Capacity	Feb 28, 2025, 04:36:26	>= 75 % In ...	75.29 %	Warning

Figure 7-1 List of alerts

Complete the following steps:

1. Double-clicking the alert opens two windows:

- Figure 7-2 on page 157 shows more details about the alert, including the storage system that it was generated on. You can see that the error affected Node 2 in the storage system that is named FS9500-1.
- Figure 7-3 on page 157 provides some performance data about the alert. You can see that two conditions, that is, the write cache delay was above 3% but below 10%. (For information about multi-conditional alerts, see 7.2, “Multi-condition alerts” on page 160).

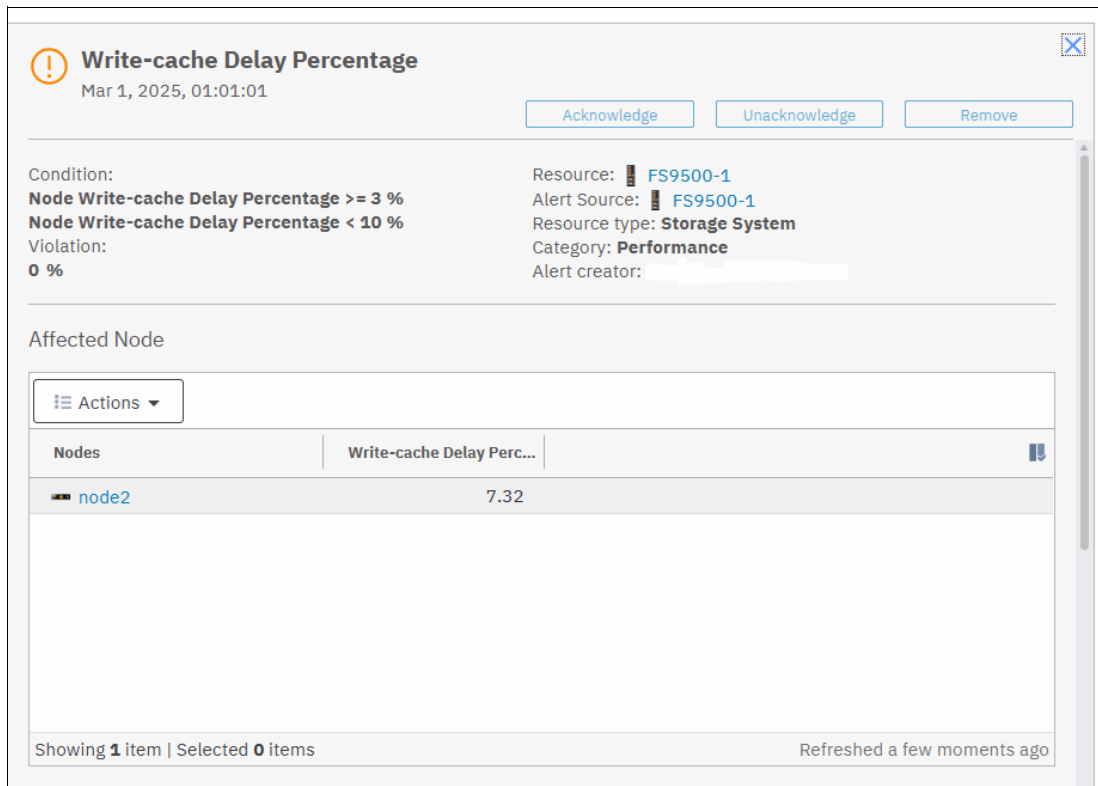


Figure 7-2 Alert details

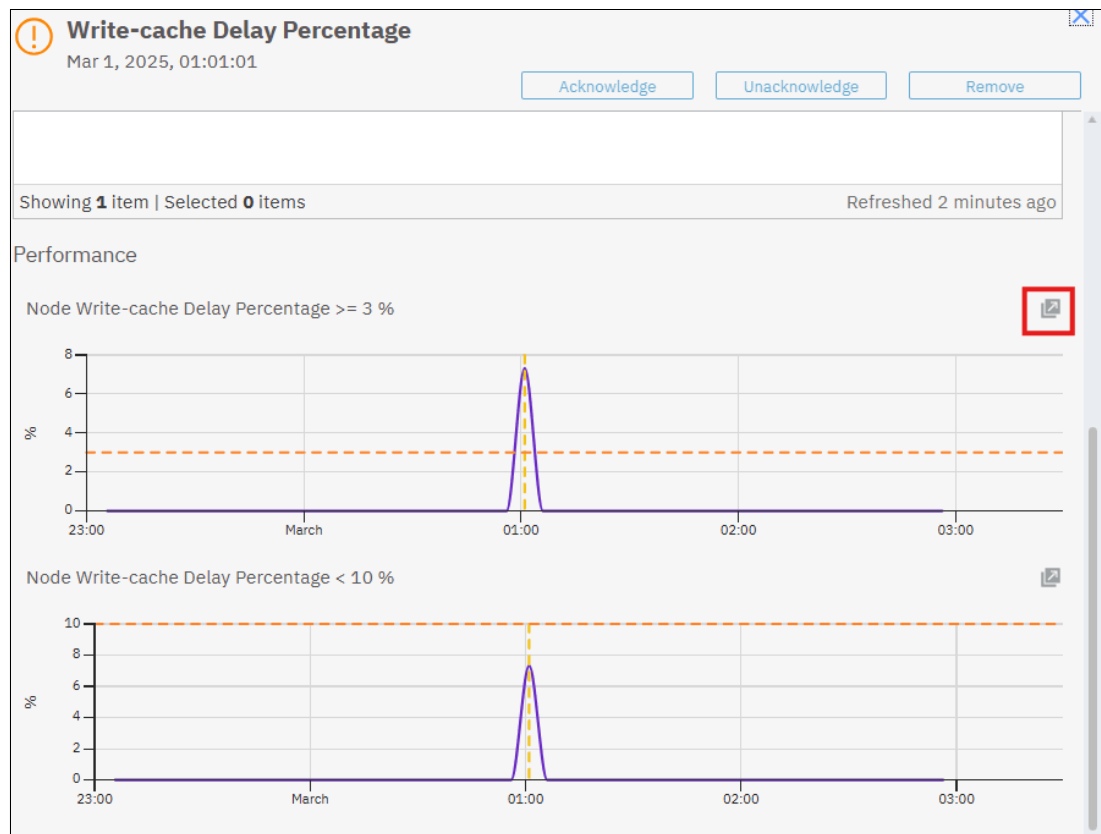


Figure 7-3 Alert performance data

- To investigate what might be causing the alert and what resources (hosts, volumes, and so forth.) that it might be affecting, click the pop-out arrow in the upper right that is highlighted in Figure 7-3 on page 157. The window that is shown in Figure 7-4 opens.

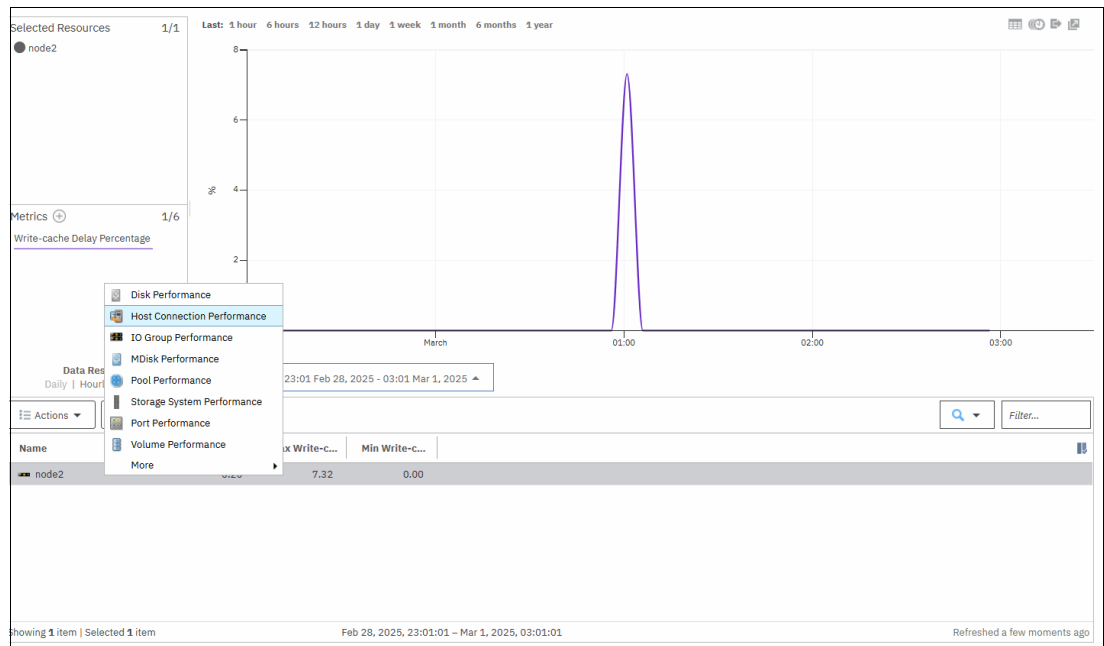


Figure 7-4 Node performance data

- Figure 7-4 shows the performance data for Node 2 because the alert was generated against Node 2. Write cache delays are potential delays of write operations. You can check to see whether there is a correlation to the write data rate.

Figure 7-5 shows same chart, but there is an added metric for the write data rate. You can see that the write cache delay and the write data rate both spiked at the time of the alert. So, there is a correlation to the amount of data being written to the node.

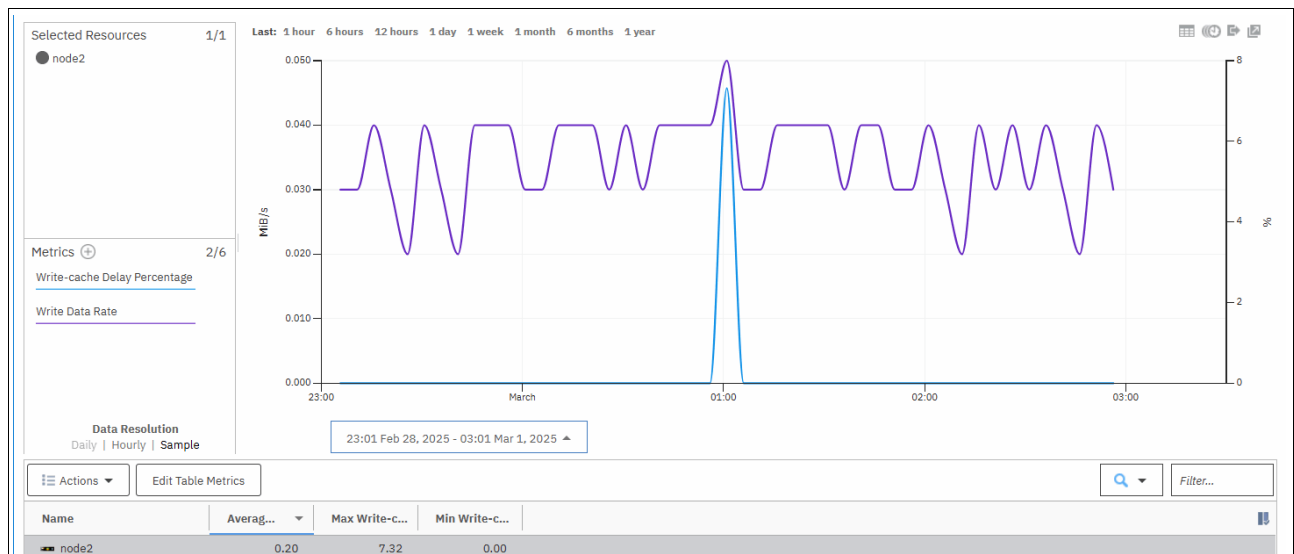


Figure 7-5 Node write data rate

- Now, you should see whether there are any other impacted resources. Right-click **node2** and select **Host Performance**, which opens the window in Figure 7-6.

Looking at the performance data, you see a correlation to the alert and the response time for the host named sds-esx4. You identified a resource that was potentially affected.



Figure 7-6 Host performance

Note: The period for the performance data in Figure 7-6 is the same as it was in Figure 7-4. IBM Storage Insights does this task automatically for any performance data that you are reviewing. So, if you set a custom period, you do not have to reset it for each new chart that you open.

- Now, view the volume performance, specifically the write data rate and response time to see whether any of the volumes were affected for the host. Figure 7-7 on page 160 shows the volume performance for the volumes on this host. The table in the figure is used to sort the volumes for the highest write data rates, and then those volumes are added to the chart. The data shows the write data rate and write response times for the busiest volumes. No correlation was found between the write data rate to any individual volume and the observed write data rate spike on the node. Therefore, you can conclude that the node's aggregate write data rate across all volumes caused the alert.

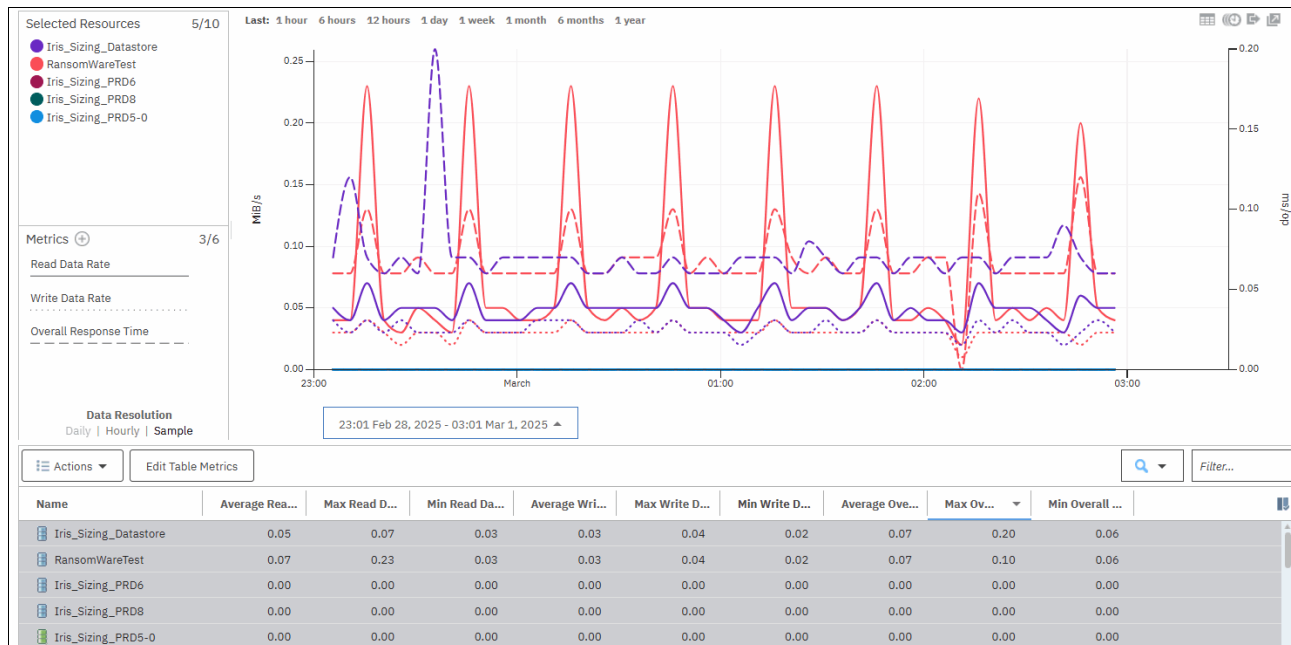


Figure 7-7 Volume performance

This section demonstrated using the workflow to troubleshoot an alert by using IBM Storage Insights. Although this alert did not impact production traffic, you can see how IBM Storage Insights is a valuable tool for troubleshooting performance problems.

7.2 Multi-condition alerts

An essential part of IBM Storage Insights Alerting is creating multi-conditional alerts. This section introduces multi-conditional alerts. We use an example use case of alerting about CRC errors. Physical link problems generate CRC errors, which are generated when data is altered between the sender and receiver transmissions. Data corruption during transmission typically results from physical layer issues, such as a faulty small form-factor pluggable transceiver or cabling problems. It is rarely the adapter in the attached device or the switch port. When CRC errors start, the link degrades over time until it is fixed, and the rate of errors increases until performance is affected.

CRC errors are relatively easy to detect, and when they are detected, easy to resolve. This example creates a multi-condition alert with CRC and another error counter to demonstrate an alert that indicates a problem likely affecting performance.

Figure 7-8 on page 161 shows an alert policy in IBM Storage Insights that is named DG Test.

Tip: It is best practice to define alerts in Alert Policies and not directly on a device because it makes it simpler to define consistent alerts across devices by defining alerts in a policy and then adding devices to that policy.

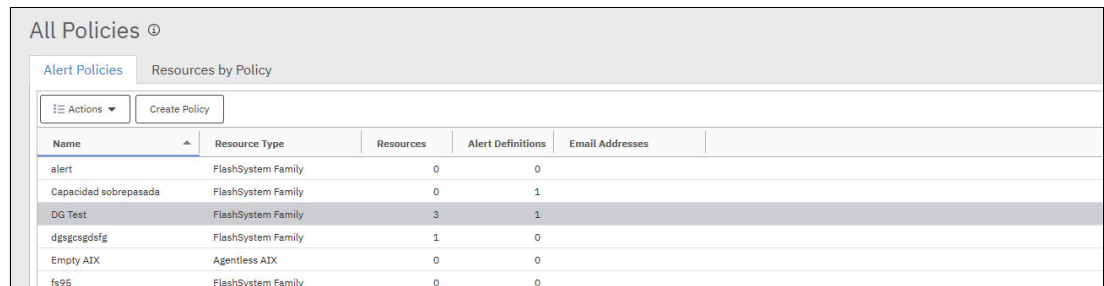
IBM Storage Insights provides several default policies for different storage system types. You cannot edit those policies, but you can copy them to a new policy and edit the new policy or you can create an empty policy, as shown in Figure 7-9 on page 161.

For more information alerts and alerting in IBM Storage Insights, see [Alerting](#),

7.2.1 Defining multi-condition alerts

To define multi-condition alerts, complete the following steps:

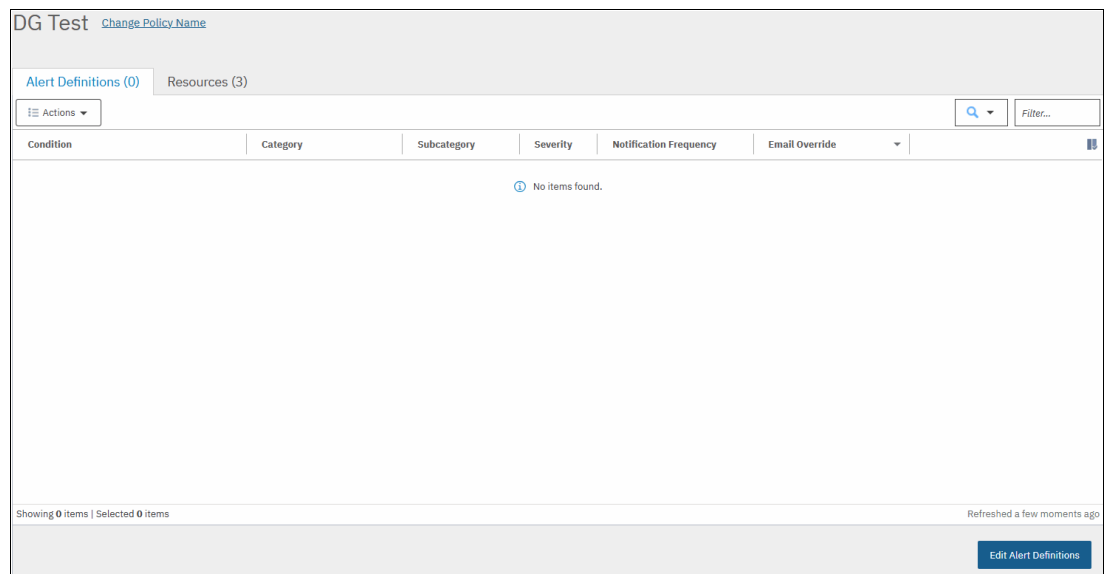
1. Open Alert Policies by selecting **Configuration** → **Alert Policies**. Figure 7-8 shows the Alert Policies list.



Name	Resource Type	Resources	Alert Definitions	Email Addresses
alert	FlashSystem Family	0	0	
Capacidad sobrepasada	FlashSystem Family	0	1	
DG Test	FlashSystem Family	3	1	
dgsgcsgdsfg	FlashSystem Family	1	0	
Empty AIX	Agentless AIX	0	0	
fs95	FlashSystem Family	0	0	

Figure 7-8 Alert Policies

2. Double-click a policy to open the alert policy and click **Edit Alert Definitions**, as shown in Figure 7-9.



Condition	Category	Subcategory	Severity	Notification Frequency	Email Override
No items found.					

Figure 7-9 Alert policy

3. You see the window that is shown in Figure 7-10. This window is where you add metrics to the alert. You can see an extensive list of possible metrics that are grouped by resource type, and for the Storage System the type of alert. For more information about these metrics, see [Alerting](#). To configure alerts for CRC errors, select **Performance**.

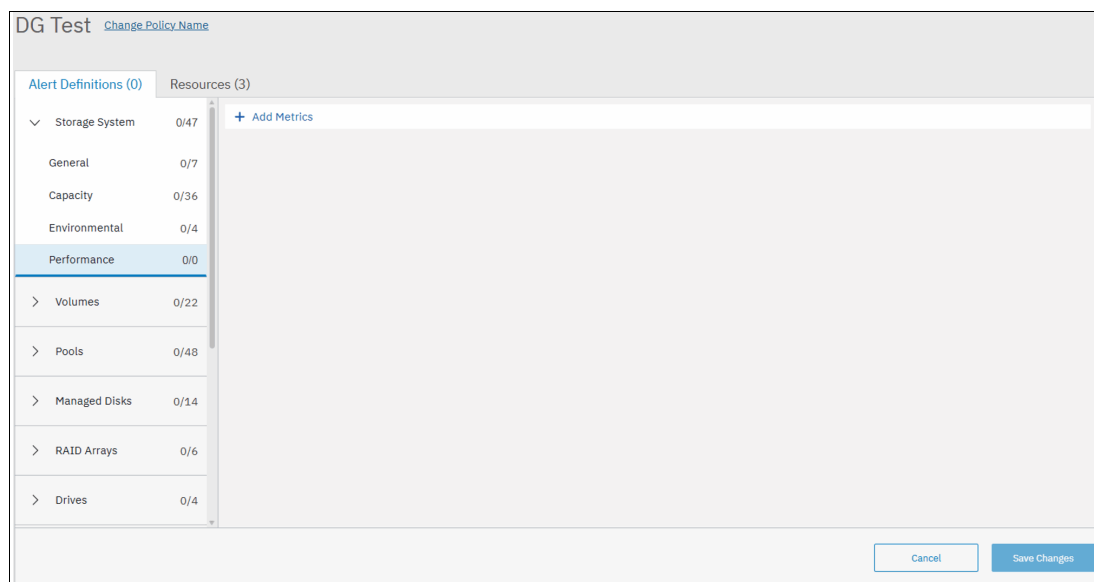


Figure 7-10 Add Metrics

4. Click **Add Metrics**. A window similar to Figure 7-11 opens. Here you can select the metrics that you want to add to the alert. It is possible to select multiple metrics.
5. The CRC error metric is found under **FC Port** → **Frame Errors**. To save the changes, click **Save**.

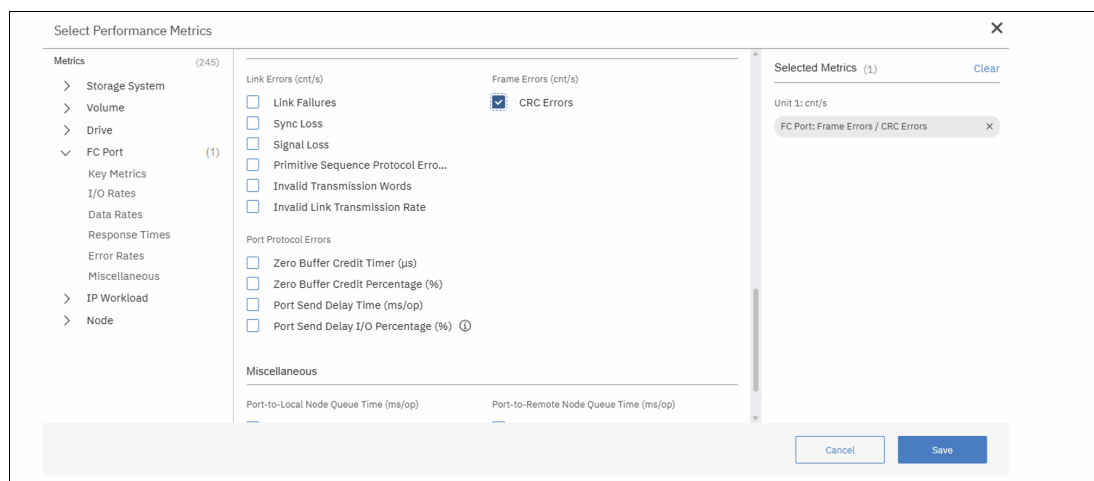


Figure 7-11 Selecting metrics

- You see a window similar to Figure 7-12. For this example, two metrics were added (CRC error rate and Zero Buffer Credit Percentage) because we want to know whether the CRC error rate is high enough to cause a loss of buffer credits, which is the first indicator of a performance problem.

The screenshot shows a configuration window for alert thresholds. It contains two sections, one for 'CRC Error Rate' and one for 'Zero Buffer Credit Percentage'. Each section has a 'View History' link, a 'Severity' dropdown (set to 'Warning'), and a 'View Additional Options' link. The 'CRC Error Rate' section has an 'Operator' of '>=' and a 'Value' of '0' with units 'cnt/s'. The 'Zero Buffer Credit Percentage' section has an 'Operator' of '>=' and a 'Value' of '2' with units '%'. Both sections have an 'Email Override' field with the placeholder text 'Separate entries with commas or spaces'. At the bottom right, there are 'Cancel' and 'Save Changes' buttons.

Figure 7-12 Alert thresholds

If the rate of CRC errors is causing buffers to stay at 0 for a high percentage of time, then performance is likely impacted. For more information about buffer credits and how they work on a storage network, see [Fibre Channel Buffer Credit Basics](#).

- After you set the alert thresholds to the values that you want, click **Save Changes**. You return to the Alert Policy window. You can either exit the configuration or create another alert.

Figure 7-13 on page 164 depicts another type of a multi-conditional alert. This alert uses multiple thresholds for a single CRC metric, triggering only within a specific range. In this case, an alert is sent when the error rate is 0–4 errors per second. Rates exceeding this threshold are ignored. This approach facilitates tiered alerting, where the alert serves as a warning, and a separate, higher-threshold alert (for example, more than 4 errors/second) signals a critical error. This use case can help administrators know how urgent the problem is.

Note: It is also possible to configure an alert with multiple performance metrics and multiple thresholds for each metric.

The screenshot shows a configuration window titled "Port: Error Rates". It contains two identical alert rule sections. Each section starts with a checked checkbox labeled "CRC Error Rate" and a "View History" link with minus and plus icons. Below this, there are three input fields: "Operator" (set to ">="), "Value" (set to "0"), and "Severity" (set to "cnt/s"). To the right of these fields are three severity icons: a red circle with an 'x', a yellow circle with an exclamation mark, and a blue circle with an 'i'. Below the severity icons is a logical connector "&" followed by another condition: "Value" (set to "4"), "Operator" (set to "<"), and "Severity" (set to "cnt/s"). Below each rule section is an "Email Override" text area with the placeholder "Separate entries with commas or spaces" and a "View Additional Options" link. At the bottom right of the window are "Cancel" and "Save Changes" buttons.

Figure 7-13 Multi-condition alerts

7.2.2 Alert definition and the Grouping feature

IBM Storage Insights alerting capabilities are fundamental for effective troubleshooting. Alerts are the first line of detection for performance and other issues. The Grouping feature strengthens alert definitions by enabling users to organize resources, such as storage systems, switches, and volumes, into logical collections.

The Grouping feature enables the creation of group-specific alerts, helping ensure consistent monitoring across resource types that might not be addressed by global alert policies. For example, volumes with FlashCore Module (FCM) back ends exhibit distinct performance profiles compared to traditional disk-based volumes. Creating groups based on these characteristics enables tailored alerts. Similarly, grouping volumes that are involved in replication facilitates the detection of performance degradations stemming from replication bottlenecks. Through this granular approach, IBM Storage Insights empowers administrators to pinpoint and address issues more efficiently.

For more information about how to create a group, see [Creating general groups in IBM Storage Insights Pro](#).

Grouping resources simplifies alert creation for items with shared, nonpolicy-compatible characteristics, such as volumes, hosts, or application or department-specific resources. Groups also function as a troubleshooting aid. For example, a replication group enables focused performance monitoring of relevant volumes.

To use the Grouping feature, complete the following steps:

1. After defining groups, create tailored alert definitions directly within them. To access the group settings, click **General Groups** in IBM Storage Insights, and then select a group to view its details (Figure 7-14). Click **Alert Definitions**.

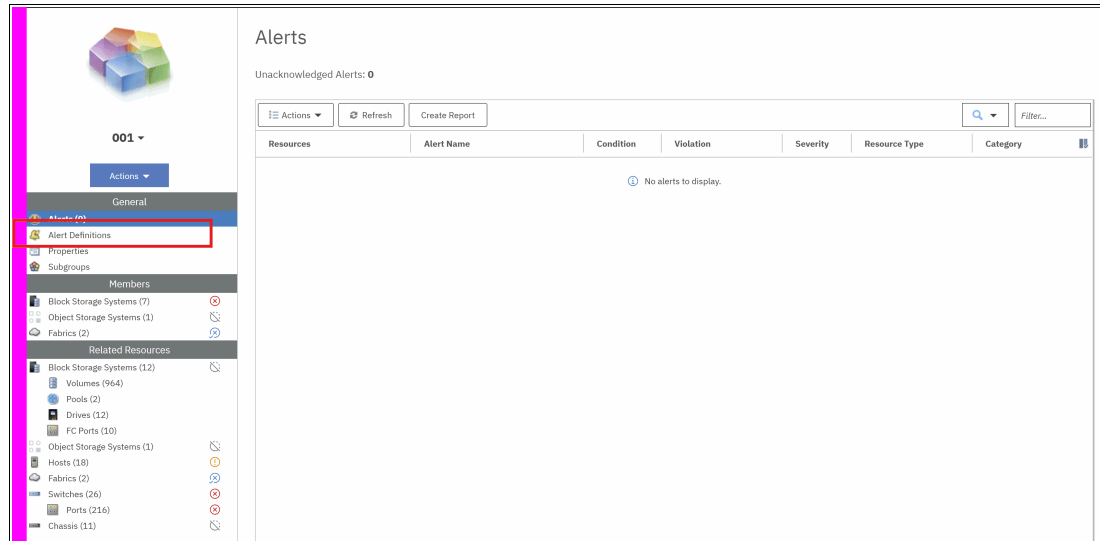


Figure 7-14 Group details

2. Click **Edit Alert Definitions** from the list of alert definitions that are shown in Figure 7-15.

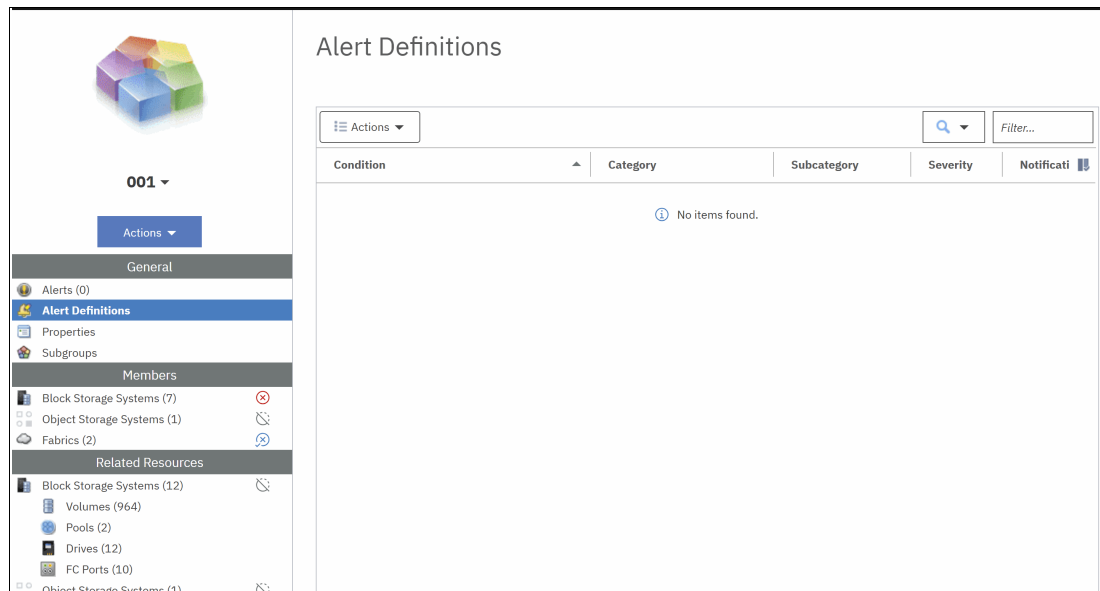


Figure 7-15 Alert definition list

3. In the Edit Alert Definitions page (Figure 7-16), you can create alerts for the group. You can also edit alerts or delete alerts. For more about creating alerts in general groups, see [Defining Alerts for General Groups](#).

Alert Definitions		
▼ Storage System0/48 General0/7 Capacity0/37 Environmental0/4 Performance0/0 > Object Storage System0/6 > Fabric0/1 Custom0/0	☐ Status- + ☐ Probe Status- + ☐ Performance Monitor Status- + ☐ Last Successful Probe- + ☐ Last Successful Monitor- + ☐ Firmware- + ☐ Warranty or Maintenance expires within the next- +	Resource Notifications Email Addresses ⓘ No resource addresses

Figure 7-16 Edit alert definitions

Troubleshooting performance issues by using artificial intelligence (AI) for support:

IBM Storage Insights uses AI-driven anomaly detection, providing IBM Remote Technical Support with advanced diagnostic capabilities. This feature empowers Remote Technical Support to quickly identify potential performance issues impacting production within your storage systems. This approach helps reduce the time to resolution for IBM Storage FlashSystem that is integrated with IBM Storage Insights.



Pauseless snapshots case study

This chapter discusses the phenomenon of high peak read and write response times during snapshot creation on code levels earlier than 8.7.0.0.

This chapter has the following sections:

- ▶ 8.1, “Description of the environment and configuration” on page 168
- ▶ 8.2, “Objective of the case study” on page 168
- ▶ 8.3, “Description of pauseless snapshot” on page 168
- ▶ 8.4, “Results” on page 168
- ▶ 8.5, “Conclusion” on page 170

8.1 Description of the environment and configuration

This issue was demonstrated in the following environment:

- ▶ FlashSystem 7200, initially running on code level 8.6.0.6 and updated to 8.7.0.3.
- ▶ The system was configured with 32 volumes in a volume group, all 32 volumes were mapped to a single host and Vdbench was used to drive a 50/50 read/write workload.
- ▶ IBM Copy Services Manager (CSM) 6.3.12.0 was used as an external scheduler to create volume group snapshots every 30 minutes.
- ▶ IBM Spectrum Control 5.4.13.0 was used to capture and monitor performance statistics from the FlashSystem 7200.

8.2 Objective of the case study

The objective of this case study is to demonstrate the difference in snapshot behavior on code levels earlier than 8.7.0.0 compared to code level 8.7.0.0 and later that have the “Pauseless Snapshot” feature.

The case study specifically covers peak read and write response times of volumes in a volume group when snapshots are triggered.

8.3 Description of pauseless snapshot

In previous implementations of snapshots in IBM Storage Virtualize, I/O was briefly paused in order to capture a crash-consistent image of the volume. If the volume was in a volume group and a snapshot of the entire volume group was triggered, then the system had to quiesce all of the volumes to take a mutually-consistent crash-consistent snapshot. The number of volumes in the volume group linearly increased the I/O pause associated with triggering the snapshot. IBM Storage Virtualize 8.7.0 and later improved this behavior by eliminating the linear scaling of the pause time as the number of volumes in the snapshot increased, which resulted in no noticeable spike in latency when triggering a snapshot. This also benefits policy-based replication when it is operating in the low-bandwidth cycling mode. To take advantage of this improvement the volumes must not be using Remote Copy, or be the source or target of an IBM FlashCopy® map.

8.4 Results

This section discusses the results and evidence for this case study.

8.4.1 Snapshots

Figure 8-1 on page 169 shows that eight snapshots were created by CSM. In chronological order, the first 4 were created while the FlashSystem 7200 was on code level 8.6.0.6, the latter 4 were triggered after the system was upgraded to code level 8.7.0.3.

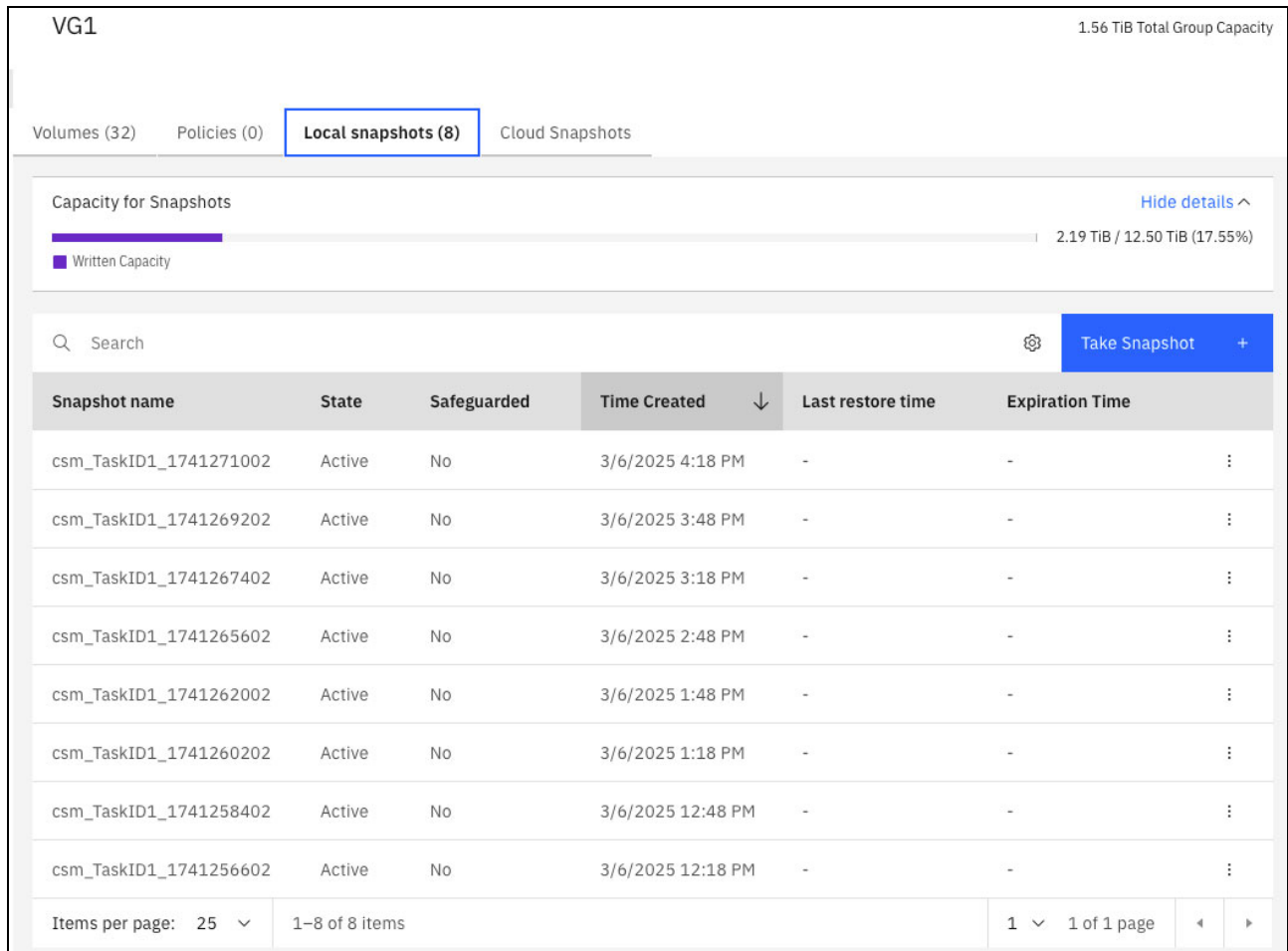


Figure 8-1 Eight snapshots were created for volume group VG1

8.4.2 Peak read and write response time on code level 8.6.0.6

Figure 8-2 shows that peak response time is > 1 second while creating snapshots on code level 8.6.0.6. Annotations 1 to 4 correlates with the creation of snapshots.

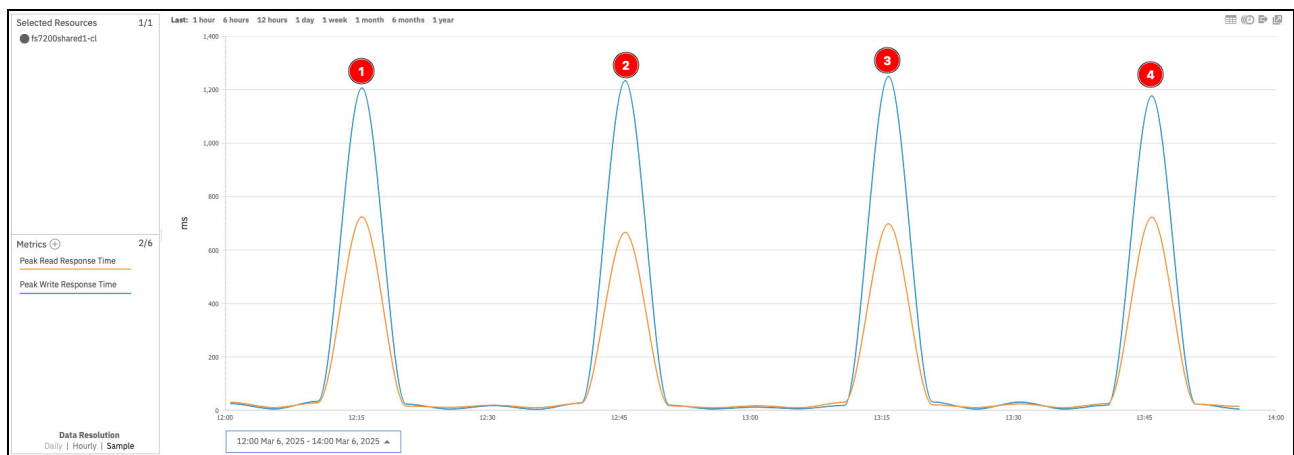


Figure 8-2 Peak response times exceeding 1 second while creating snapshots

8.4.3 Peak read and write response time on code level 8.7.0.3

Figure 8-3 shows that peak response time is < 40 ms while creating snapshots on code level 8.7.0.3. Annotations 5 to 8 correlates with the creation of snapshots.

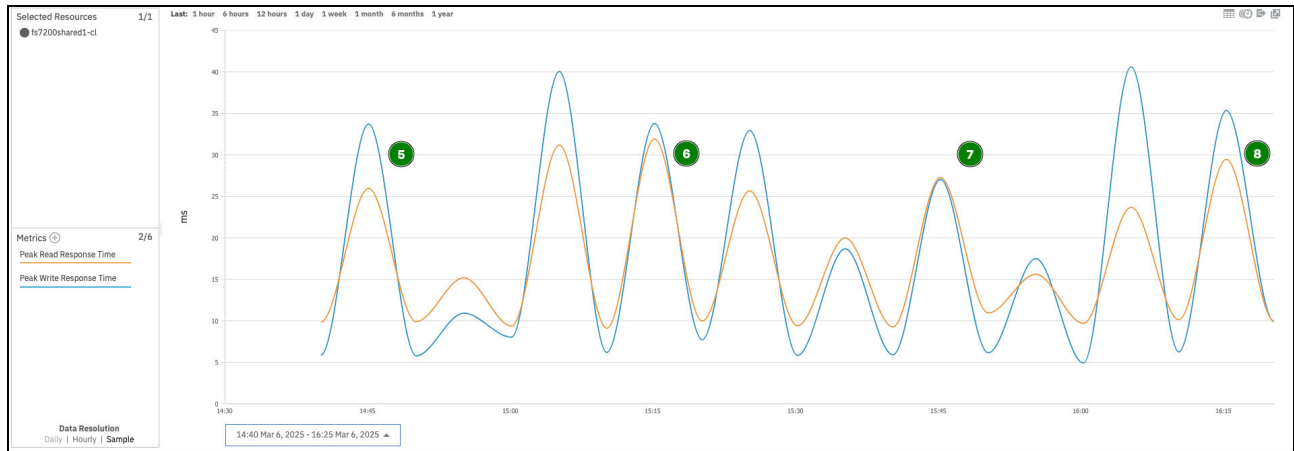


Figure 8-3 Peak response time is < 40 ms while creating snapshots

8.5 Conclusion

The pauseless snapshot feature in code level 8.7.0.0 and later benefits latency sensitive applications that use volume group snapshots or policy based replication.



A

Workload migration

Effective workload migration in storage virtualization requires careful consideration of performance impacts to ensure seamless transitions and maintain optimal application functionality. This appendix outlines best practices for this process.

Workload migration performance considerations

Workload migration involves transferring the entire ecosystem that supports business-critical applications, including their performance characteristics, access patterns, and dependencies. This process is critical for organizations looking to modernize their infrastructure, consolidate storage resources, or transition to cloud-integrated solutions.

Differences between storage migration and workload migration: Storage migration focuses on transferring data between systems. Workload migration prioritizes moving applications and their data, ensuring performance is maintained. This can also include migrating workloads to different storage pools within the same system.

Workload migration in the context of storage migration is not only about transferring data. It is about preserving and enhancing the performance characteristics of applications, such as latency, throughput, and IOPS. Poorly executed migrations can lead to performance bottlenecks, increased latency, and degraded user experiences, which can impact business operations. Therefore, performance optimization must be a central focus throughout the migration process. Performance monitoring and benchmarking play a crucial role in identifying and resolving potential bottlenecks before, during, and after the migration.

Because of this book's focus on performance best practices, Table A-1 describes the performance considerations associated with various workload migration options. Implementation details are outside the scope of this discussion; consult the following references for such information:

- *Block Storage Migration in Open Environments*, [SG24-8470](#)
- *Ensuring Business Continuity: A Practical Guide to Policy-Based Replication and Policy-Based High Availability for IBM Storage Virtualize Systems*, [SG24-8569](#)

Table A-1 Various workload migration options and associated performance considerations

Migration type	Advantages	Potential challenges	Performance considerations
Host based	► Only zoning and provisioning is required by storage admins. ► Migration of data is done by server/application administrators by using LVM features or third party tools.	► Host has to read and write all data. ► Migration tasks can consume host resources excessively causing application impact. ► Migration is likely to fail if medium errors are encountered.	► Host HBA bandwidth usage must be monitored during migration. ► Host CPU usage must be monitored during migration. ► Volume throttling can be used to limit migration bandwidth usage. ► ESXi Storage vMotion uses offloaded I/O within the same system to migrate VMs to new datastores. This reduces host HBA bandwidth utilization. Offloaded workloads can be throttled with offload throttles.

Migration type	Advantages	Potential challenges	Performance considerations
Non-disruptive volume migration. Important: This method is based on Remote Copy. IBM Storage Virtualize 8.7 is the final version that supports Remote Copy, specifically including inter-system Metro Mirror and Global Mirror compatibility. Policy-based replication was introduced in Storage Virtualize 8.5.2 and is a replacement for Remote Copy in previous versions of code (see “Policy-based replication migration type” below).	- FlashSystem is responsible for copying the data and releasing host resources to serve application I/O. - UUID of volume remains the same, requiring no host reconfiguration.	- Host operating system support is limited to hosts that support ALUA standby path. - Migration fails if medium errors are encountered.	Requires additional link bandwidth to avoid impacting existing replication workloads.
Extent migration migratevdisk addmdisk/rmmdisk migrateexts	- FlashSystem is responsible for copying the data and releasing host resources to serve application I/O. - This method is possible by migrating a volume copy to a new pool or by adding new MDisk to a pool and removing old MDisk. - This process occurs low down in the I/O stack, so it does not add workload to the DRP or cache layers. - Migration does <i>not</i> fail if medium errors are encountered; virtual medium errors are created on the target mdisk.	- Extent size between source and target pool must match. - Both pools must be standard pools. - This task cannot be cancelled after it is started. - The rate of data transfer cannot be adjusted.	Monitor system performance following the initiation of a migration. As migrations are non-interruptible, ensure adequate system resources before commencing further migrations.

Migration type	Advantages	Potential challenges	Performance considerations
Volume mirroring	▶ The FlashSystem is responsible for copying the data and releasing host resources to serve the application I/O. ▶ This method can be used regardless of extent size. ▶ This method can be used to migrate data into data reduction pools. ▶ Mirroring tasks can be stopped. ▶ Mirror sync rate can be adjusted to set the priority of migration tasks. ▶ Migration does <i>not</i> fail if medium errors are encountered; virtual medium errors are created on the target MDisk.	▶ This process occurs higher in the I/O stack which adds workload to the Space Efficient, DRP and lower cache layers of the I/O stack. ▶ Occupies twice the volume capacity until the old volume copy is deleted after the synchronization completes. ▶ Only supports two copies. If volume mirroring is already in use, this method cannot be used without removing an existing copy.	▶ Ensure that the aggregate bandwidth of the synchronizing volume mirrors does not overload the backend storage for the pool. Adjust mirror sync rates to mitigate overloading. ▶ Volume mirroring operates above data reduction pools (DRP) in the I/O stack. DRP thin, compressed and deduplicated volumes have lower single VDisk bandwidth limits compared to standard pools. Mirror sync rates that are set too high can cause high CPU core usage and high internode latency.
Storage partition migration	▶ Ability to move an entire application workload (storage partition) with this procedure. ▶ The simplicity of the procedure makes it feasible to perform frequent migrations to improve FlashSystem usage within a FlashSystem grid.	▶ The source and target systems must be in a FlashSystem grid. ▶ NVMe hosts are not supported.	Ensure that the target system has adequate resource availability to accommodate the additional workload.
Policy-based replication	▶ Policy-based replication facilitates data migration to a new system and preserves disaster recovery capabilities via remote copy. This functionality requires code level 8.6.0 or 8.7.0. ▶ Ability to prioritize migration workloads by using replication policies with different recovery point objectives.	UUID of volumes are different, which might require some host reconfiguration.	▶ Ensure that there is sufficient surplus link bandwidth to accommodate the additional replication workload. ▶ As volume groups complete their first full synchronization, increase their RPO alert threshold so that more priority is given to volume groups that are still synchronizing.

Abbreviations and acronyms

AI	artificial intelligence	IBM	International Business Machines Corporation
AIOPS	Artificial Intelligence for IT Operations	IOPS	input/output operations per second
ALUA	Asymmetric Logical Unit Access	IPL	initial program load
ANO	Active Non-optimized	IQNs	iSCSI qualified names
AO	Active Optimized	ISL	inter-switch link
AQDT	Adaptive Queue Depth Tuning	iWARP	internet wide-area RDMA protocol
ASPs	also called auxiliary storage pools	KAVG	kernel latency
BI	business intelligence	KB	kilobytes
BYO	bring your own	KPIs	Key performance indicators
CIC	Client Innovation Centre	LFF	large form factor
CLI	command-line interface	LPAR	logical partition
COS	Configure class of service	LPM	Live Partition Mobility
CRC	cyclic redundancy check	LPT	Logical to Physical Table
CSM	Copy Services Manager	LTS	long-term support
CSV	comma-separated value	LU	logical unit
DCBx	Data Center Bridging Exchange	LUN	logical unit number
DMA	direct memory access	LVM	Logical Volume Manager
DMP	Dynamic Multi-Pathing	MBPS	megabytes per second
DRAID	Distributed RAID	MDisks	Managed disks
DRET	Data Reduction Estimator Tool	MPIO	Microsoft multipath I/O
DRP	Data reduction pool	MRU	Most Recently Used
DWPD	Drive writes per day	MSCS	Microsoft Cluster Server
ECC	Error Correcting Code	MSDSM	Microsoft Device Specific Module
ETS	Enhanced Transmission Selection	MTU	maximum transmission unit
EUIs	extended unique identifiers	MVB	minimum viable business
FC	Fibre Channel	NDVM	nondisruptive volume move
FCM	FlashCore Module	NICs	network interface cards
FDMI	Fabric Device Management Interface	NL	Nearline
FPIN	Fabric Performance Impact Notification	NMP	Native Multipathing Plug-in
FS9500	FlashSystem 9500	NPIV	N-Port ID Virtualization
FSW	Full Stride Write	NQNs	names (WWPNs), NVMe qualified names
GUI	graphical user interface	NVMe	Non-Volatile Memory Express
HBA	Host Bus Adapter	NVMeoF	NVMe over Fabrics
HDDs	hard disk drives	OBAC	object-based access control
HIC	Host Interface Card	OLAP	Online analytical processing
HPP	High Performance Plug-in	OLTP	Online transaction processing
HSNs	Hot Spare Nodes	OS	operating systems
IASPs	independent auxiliary storage pools	PFC	Priority flow control
		PSA	Pluggable Storage Architecture

PSP	Path Selection Policy
QAVG	queue latency
RDMA	Remote Direct Memory Access
RI	read-intensive
RMW	read/modify/write
RR	Round Robin
RoCE	RDMA over Converged Ethernet
SANs	storage area networks

Related publications

The publications listed in this section are considered particularly suitable for a more detailed discussion of the topics covered in this book.

IBM Redbooks

The following IBM Redbooks publications provide additional information about the topic in this document. Note that some publications referenced in this list might be available in softcopy only.

- ▶ *Implementation Guide for IBM Storage FlashSystem and IBM SAN Volume Controller Updated for IBM Storage Virtualize Version 8.6*, [SG24-8542](#)
- ▶ *Performance and Best Practices Guide for IBM Storage FlashSystem and IBM SAN Volume Controller Updated for IBM Storage Virtualize Version 8.6*, [SG24-8543](#).
- ▶ *Ensuring Business Continuity: A Practical Guide to Policy-Based Replication and Policy-Based High Availability for IBM Storage Virtualize Systems*, [SG24-8569](#)
- ▶ *Maximize Your Storage Investment: Leverage IBM Storage Insights for Observability and Artificial Intelligence for IT Operations*, [SG24-8584](#)
- ▶ *Intelligent Storage Management with Artificial Intelligence for IT Operations (AIOps)*, [REDP-5755](#)

You can search for, view, download or order these documents and other Redbooks, Redpapers, Web Docs, draft and additional materials, at the following website:

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Online resources

These websites are also relevant as further information sources:

- ▶ FlashSystem 9x00 IBM Documentation - Version 8.7.X
https://www.ibm.com/docs/en/flashsystem-9x00/8.7.x_cd

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